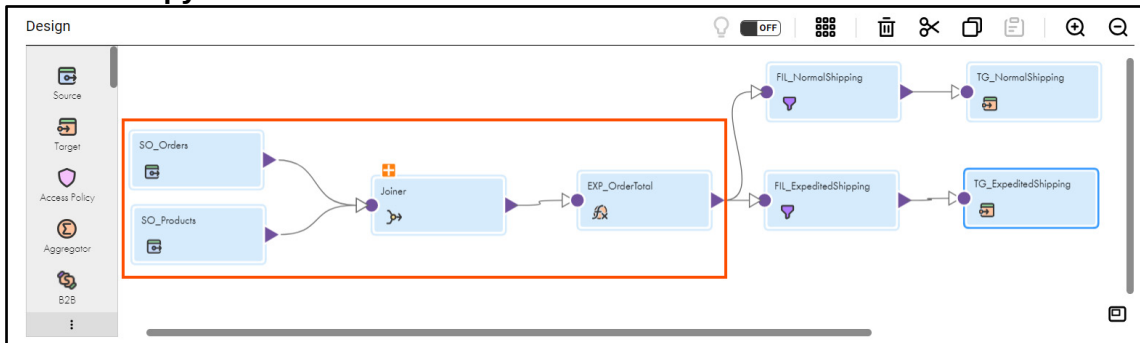
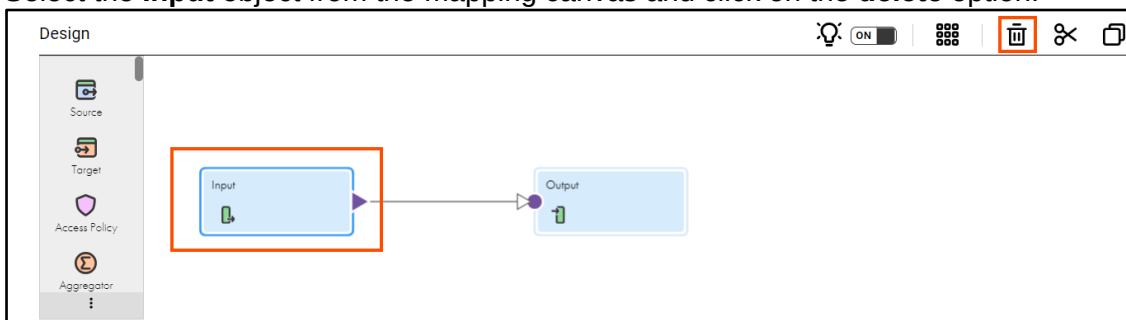


36. Click the **Copy** icon.



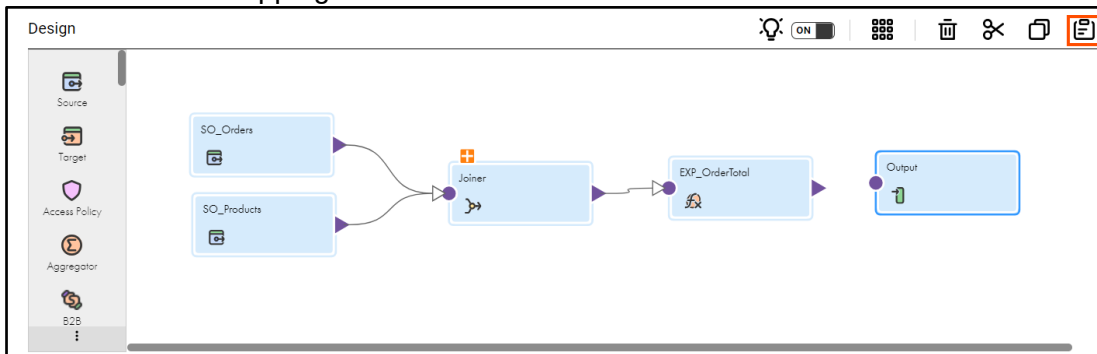
37. Go back to your Mapplet canvas.

38. Select the **Input** object from the mapping canvas and click on the **delete** option.



39. Click the **Paste** icon, on the mapping canvas.

40. You will see the mapping canvas as shown below:



41. Link the **EXP_OrderTotal** and the **Output**.

42. Select **Output** from your mapping canvas.

43. From the properties pane, click **Output Fields**.

44. To add a new Output Field, click .

45. Define the output fields, as shown in the table below:

Name	Type	Precision	Scale
ORDERID	integer	10	0
Product_Name	string	255	0
PRODUCTDESCRIPTION	string	255	0
Category	string	255	0
Total	integer	10	0

Properties |  Output

General Add output fields for the output data you want to pass to the downstream transformation. You must include at least one output field.

Incoming Fields

Output Fields

Field Mapping

☐ Output Field	Type	Precision	Scale
☐ ORDERID	integer	10	0
☐ Product_Name	string	255	0
☐ PRODUCTDESCRIPTION	string	255	0
☐ Category	string	255	0
☐ Total	integer	10	0

46. From the properties pane, click **Field Mapping**.

47. Match the fields as shown in the following table:

Note: Click and drag the fields to map them.

Incoming Field	Output Fields
ORDERID	ORDERID
Product_Name	Product_Name
Category	Category
Total	Total
PRODUCTDESCRIPTION	PRODUCTDESCRIPTION

Properties |  Output

General Field map options: Manual

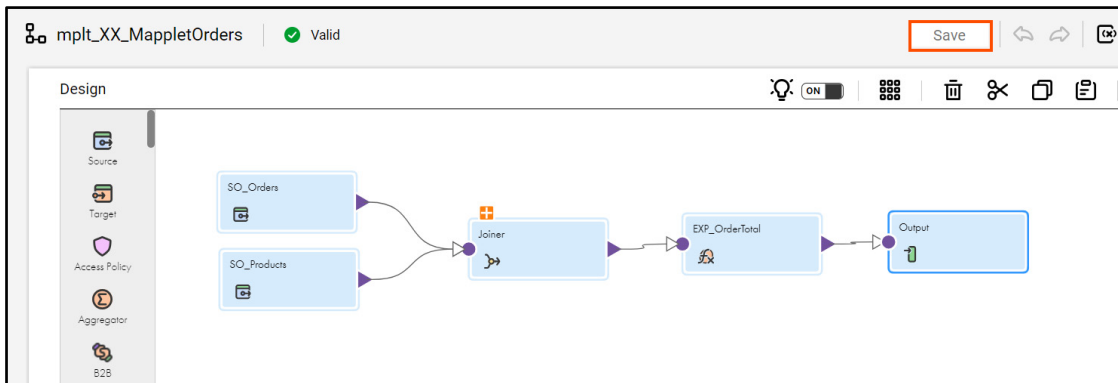
Incoming Fields Incoming Fields: (5 of 13 mapped) Find

Output Fields Output Fields: (5 of 5 mapped) Find Automap

Field Mapping

Field Name	Mapped Field
Total	Total
ORDERID	ORDERID
Product_Name	Product_Name
PRODUCTDESCRIPTION	PRODUCTDESCRIPTION
Category	Category

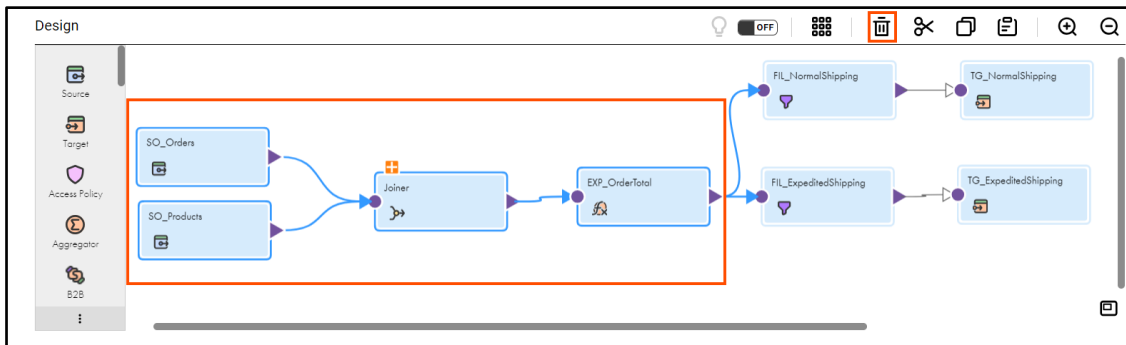
48. Click **Save** to save the Mapplet.



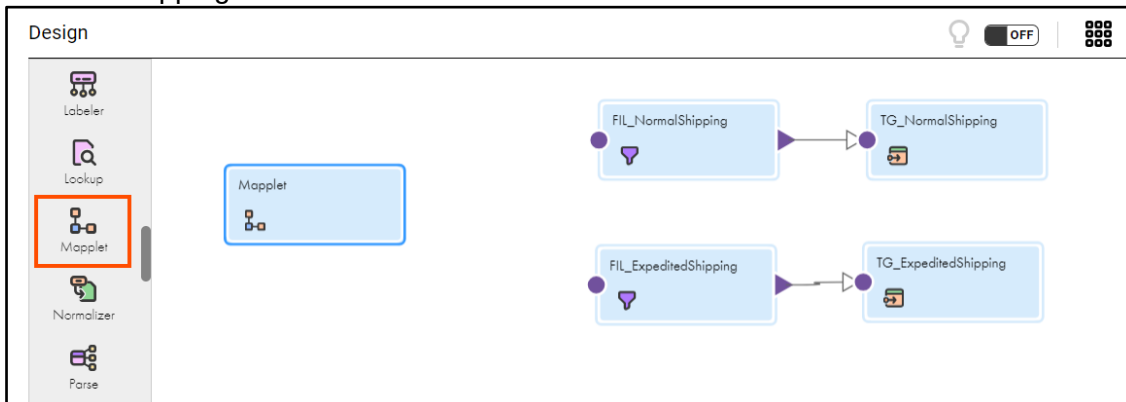
Mapping with Mapplet

49. Navigate to the **m_XX_ShippingFiles** mapping tab.

50. Now, select the **SO_Orders**, **SO_Products**, **Joiner**, and **EXP_OrderTotal** and click **Delete**.



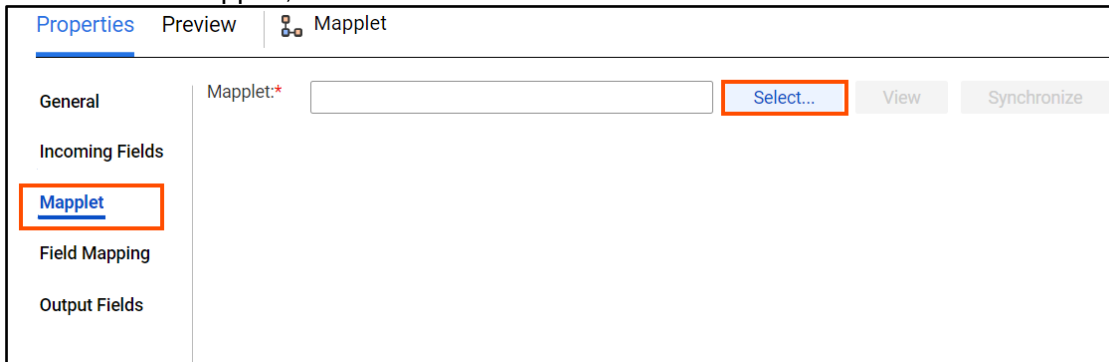
51. From the list of available transformations, drag and drop the **Mapplet** transformation onto the mapping canvas.



52. Select the **Mapplet** from the mapping canvas.

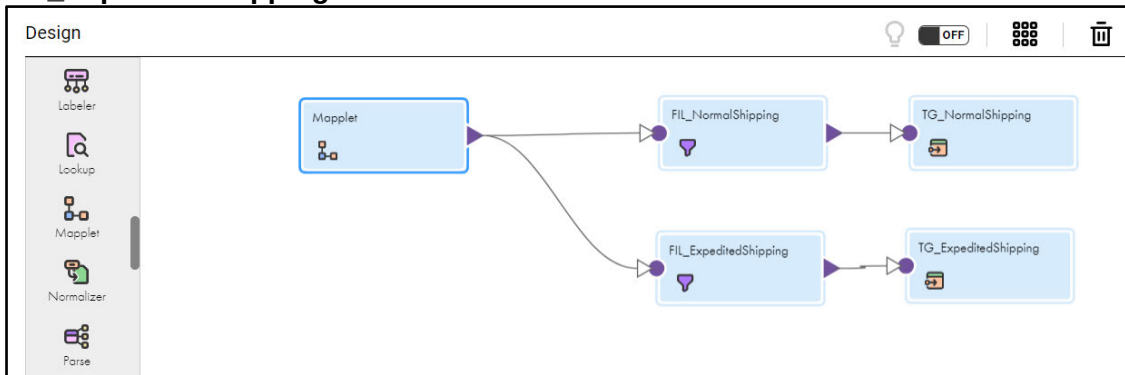
53. From the properties pane, click **Mapplet**.

54. To select the Mapplet, click **Select**.



55. From your working directory select the mapplet, **mplt_XX_MappletOrders**.

56. From the mapping canvas, link the Mapplet with the **FIL_NormalShipping** and **FIL_ExpeditedShipping**.



57. **Save** and **Run** the Mapping.



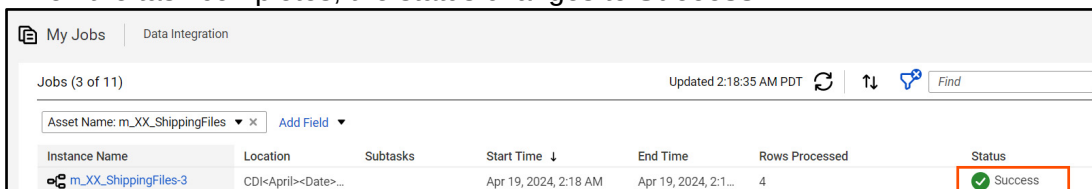
58. From the Runtime Environment drop down, select INFA-SERVER.

59. Click **Run**.

Monitor Status

60. Monitor the task status from the **My Jobs** page.

61. When the task completes, the status changes to **Success**.



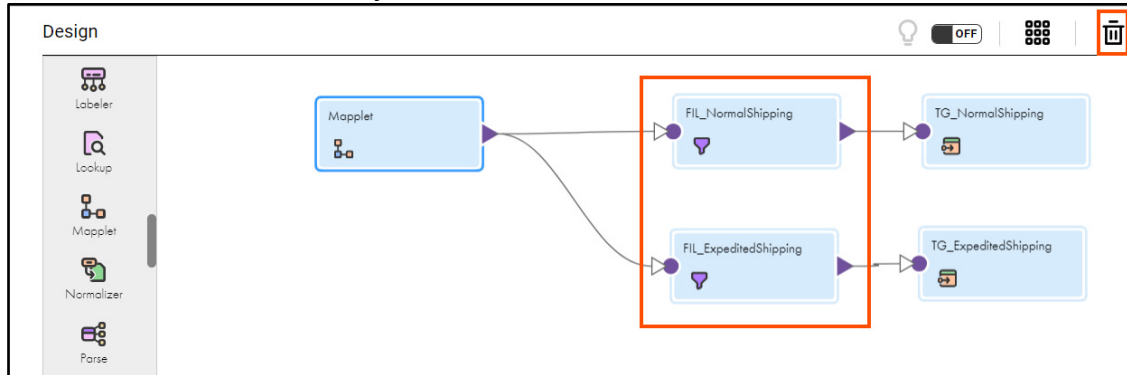
Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
m_XX_ShippingFiles-3	CDI<April><Date>...		Apr 19, 2024, 2:18 AM	Apr 19, 2024, 2:1...	4	Success

Note: You can refresh the job status if it does not change automatically.

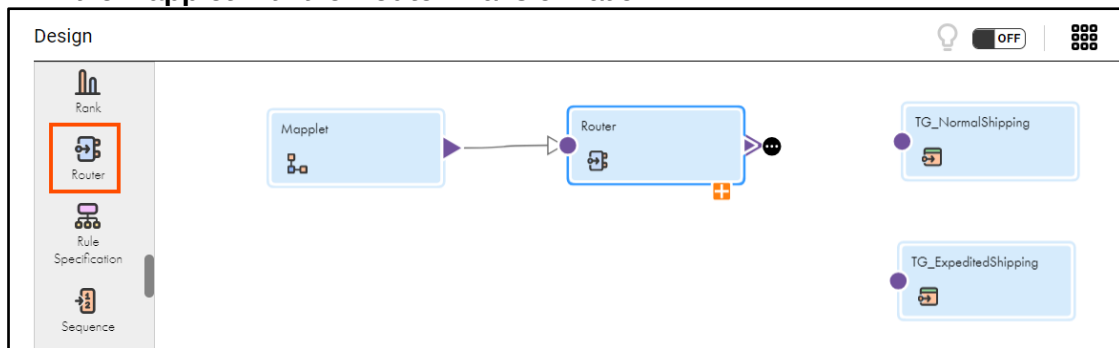
62. Optionally, you can view the preview data from the Target properties tab.

Mapping with Router Transformation

63. Navigate to the **m_XX_ShippingFiles** mapping tab.
64. From the mapping canvas, select **FIL_NormalShipping** and **FIL_ExpeditedShipping**.
65. Click **Delete** to delete the objects.



66. From the list of available transformations, drag and drop a **Router** transformation onto the mapping canvas.
67. **Link** the **Mapplet** with the **Router** Transformation.



68. In the General section of the Router properties, enter the Name as **Classify_Order_Size**.

Properties		Preview	Classify_Order_Size
General		Name:* Classify_Order_Size	
Incoming Fields		Description:	

69. From the properties pane, select **Output Groups**.
70. To add a new condition, click .

Properties		Preview	Classify_Order_Size				
General		Output Groups 					
Incoming Fields		<table border="1"> <thead> <tr> <th>Group Name</th> <th>Condition</th> </tr> </thead> <tbody> <tr> <td>DEFAULT1</td> <td></td> </tr> </tbody> </table>		Group Name	Condition	DEFAULT1	
Group Name	Condition						
DEFAULT1							
Output Groups							
Advanced							

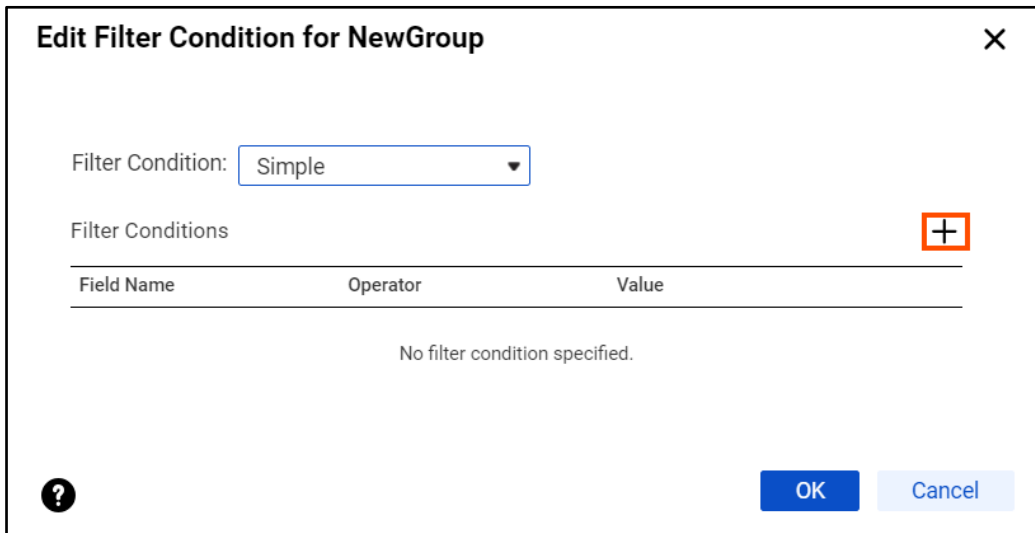
71. To enter the condition, click **Configure**.



Group Name	Condition
DEFAULT1	
NewGroup	Configure...

The Edit Filter Condition for NewGroup window appears.

72. Click the  icon.



Filter Condition: Simple

Filter Conditions

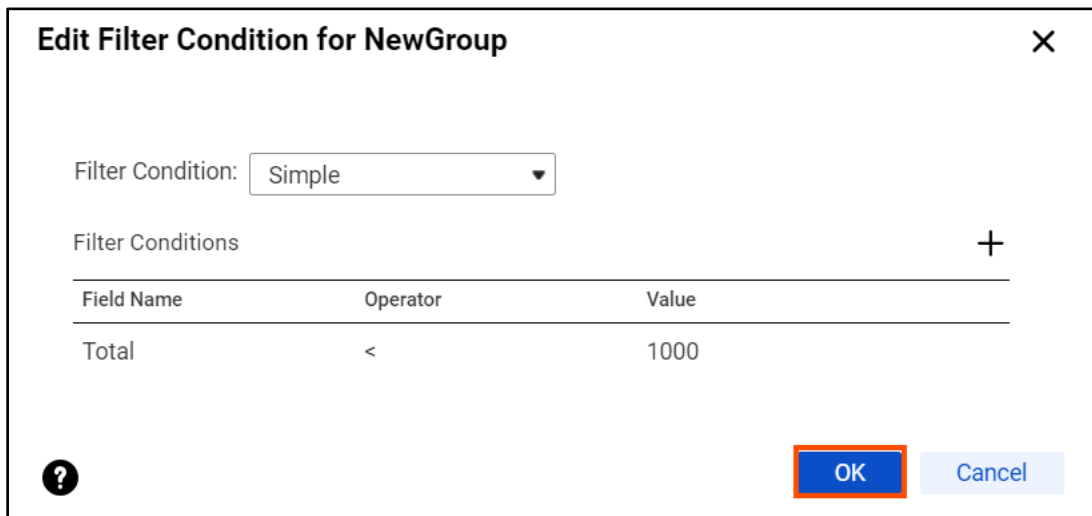
Field Name	Operator	Value
No filter condition specified.		



 OK Cancel

73. Enter the filter condition as shown in the table below and click **OK**.

Field Name	Operator	Value
Total	< (Less than)	1000



Filter Condition: Simple

Filter Conditions

Field Name	Operator	Value
Total	<	1000

 OK Cancel

74. Change the Group Name to **Normal_Shipping**.

75. To add another filter condition, click .

76. Click **Configure**.

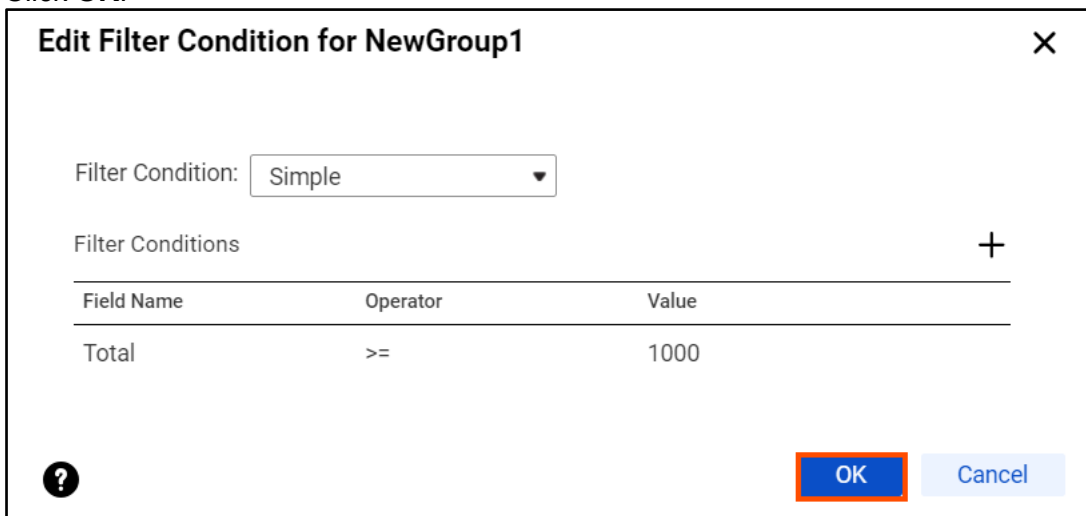


Group Name	Condition	
DEFAULT1		
Normal_Shipping	Total < '1000'	
NewGroup	Configure...	

77. Enter filter condition as shown in the table below:

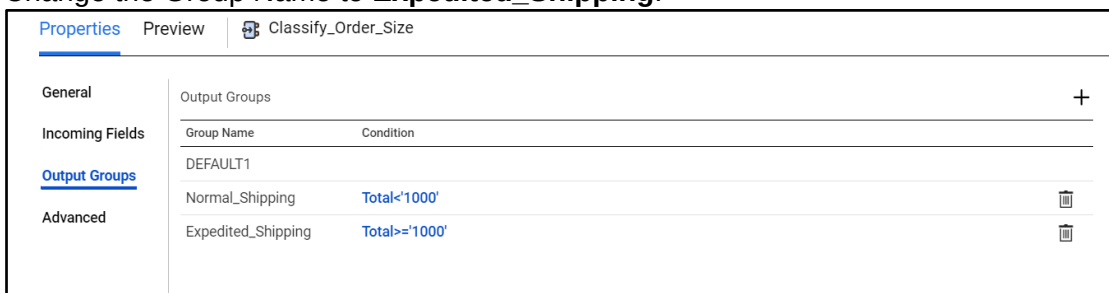
Field Name	Operator	Value
Total	>= (Greater than equal to)	1000

78. Click **OK**.



Field Name	Operator	Value
Total	>=	1000

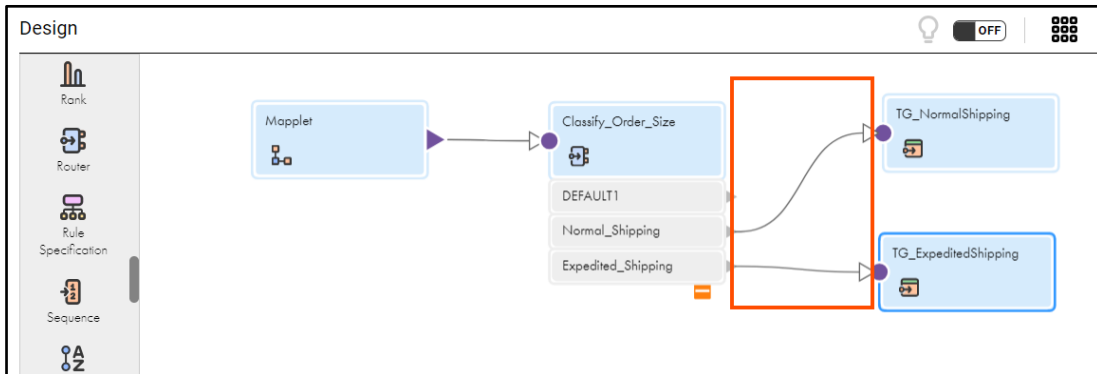
79. Change the Group Name to **Expedited_Shipping**.



Group Name	Condition	
DEFAULT1		
Normal_Shipping	Total < '1000'	
Expedited_Shipping	Total >= '1000'	

80. On the mapping canvas, click the plus icon on the Router transformation to expand the output groups.

81. Link the **Classify_Order_size** to the **TG_NormalShipping** and **TG_ExpeditedShipping** respectively.



82. **Save** and **Run** the Mapping.



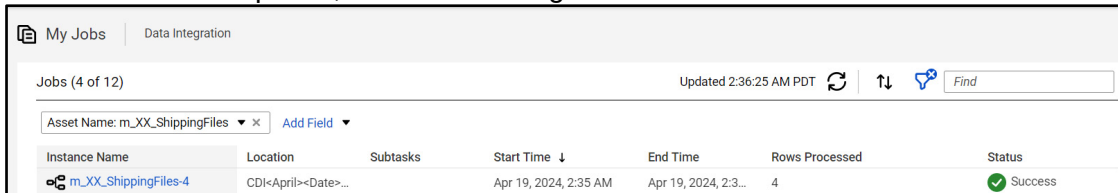
83. From the Runtime Environment drop down, select INFA-SERVER.

84. Click **Run**.

Monitor Status

85. To monitor the mapping status, navigate to the **My Jobs** page.

86. When the task completes, the status changes to **Success**.



Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
m_XX_ShippingFiles-4	CDI<April><Date>...		Apr 19, 2024, 2:35 AM	Apr 19, 2024, 2:3...	4	Success

87. Optionally, you can view the preview data from the Target properties tab.

88. Close the mapplet and mapping assets from the navigation pane.

This concludes the lab.

Module 8: Maplets

Lab 8-3: Creating a Mapping Using Existing Maplet and Aggregator Transformations

Overview:

A maplet is reusable transformation logic that you can use to transform source data before it is loaded into the target. After you create a maplet, you can add it to a Maplet transformation to use its transformation logic. You can use a maplet in another maplet.

You can select a maplet that you created or imported into the DI service.

In this lab, you will use the created Maplet in a mapping and finish the login using the aggregator transformation.

Objective:

- Use the existing Maplet
- Use the Aggregator transformation

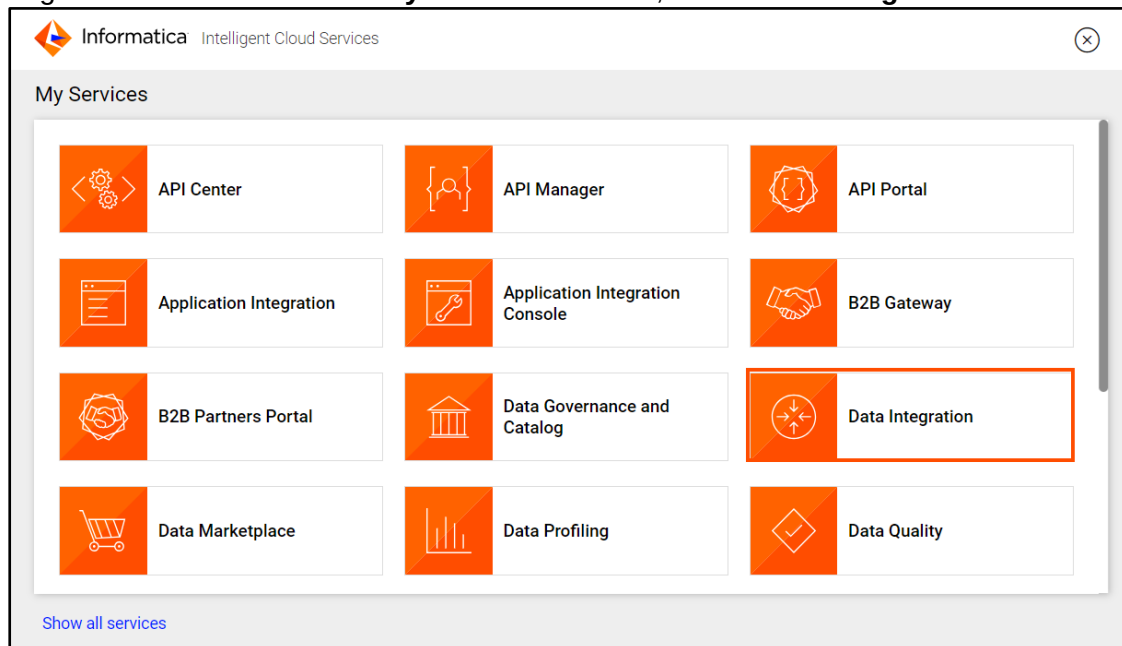
Duration:

20 minutes

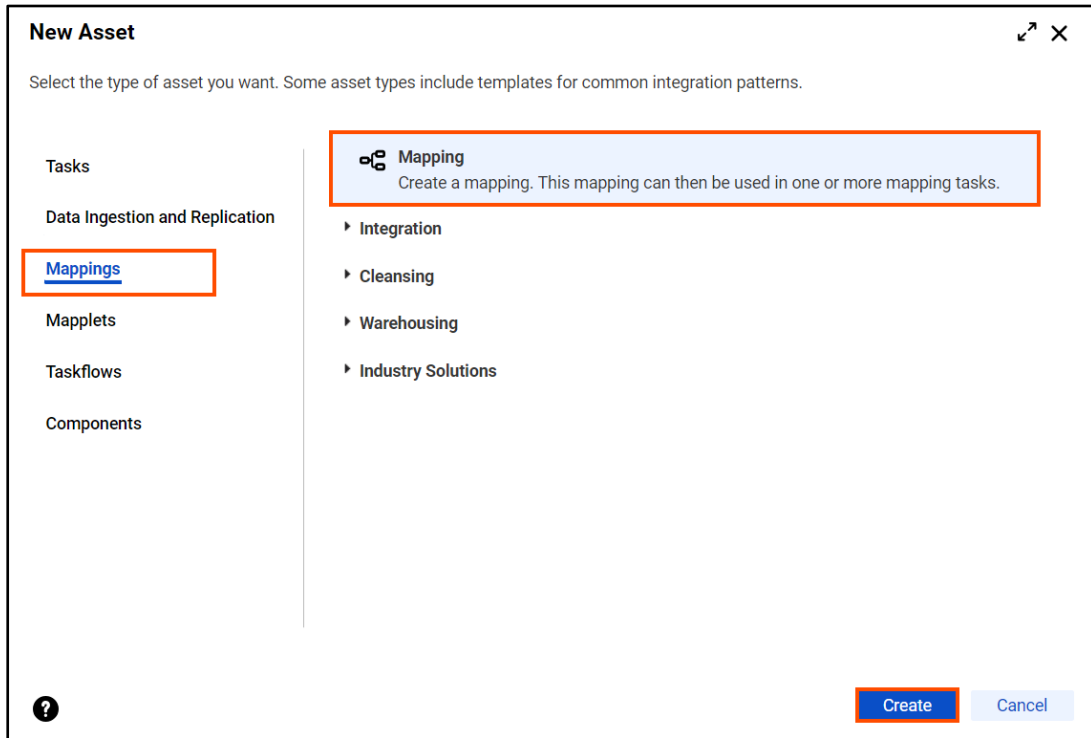
Tasks

Create Mapping

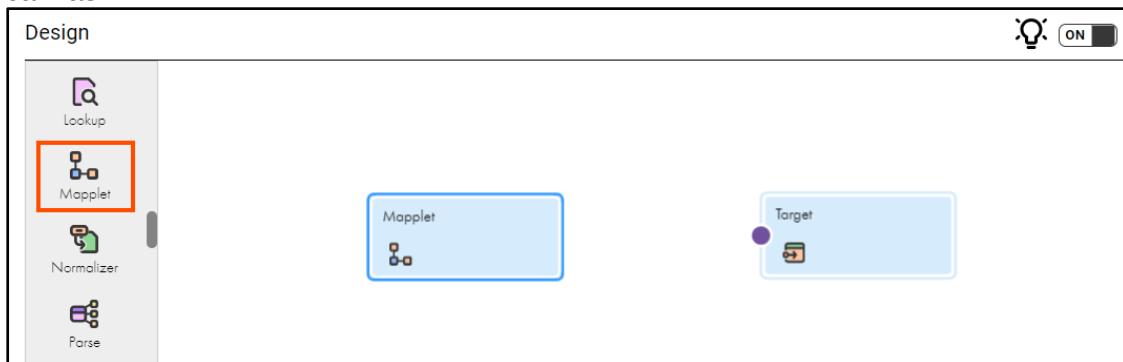
1. Login into IICS and from the **My Services** window, select **Data Integration**.



2. From the navigation pane, select **New**.
3. From the New Asset window, click the **Mappings** tab, and select **Mapping**.
4. Click **Create**.



5. In the Name field, enter **m_XX_SummarizedSales**.
6. Specify the location of your working directory.
7. From the mapping canvas, select the **Source** and click the **Delete** option.
8. From the list of available transformations, drag and drop a **Maplet** on the mapping canvas.



9. Select the **Maplet** from the mapping canvas.
10. From the properties pane, click **Maplet**.

11. From the Maplet, click **Select**.



Properties Preview Maplet

General Maplet:* **Select...** View Synchronize

Incoming Fields Description:

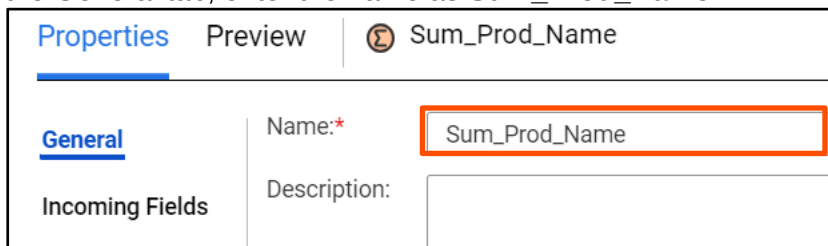
Maplet Type: Active ⓘ

Field Mapping

12. From your working directory, select the maplet **mplt_XX_MapletOrders**.

13. From the list of available transformations, drag and drop the **Aggregator** transformation between Maplet and Target on the mapping canvas.

14. With the Aggregator transformation being selected, navigate to the properties pane, in the General tab, enter the name as **Sum_Prod_Name**.

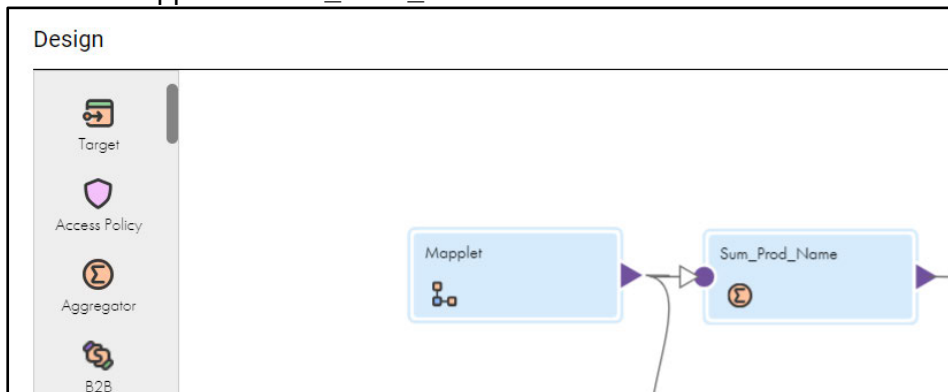


Properties Preview Sum_Prod_Name

General Name:* **Sum_Prod_Name**

Incoming Fields Description:

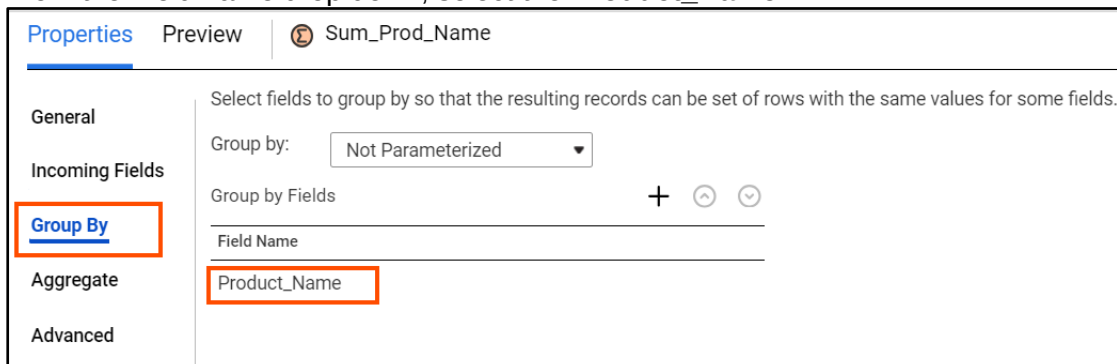
15. Link the Maplet to **Sum_Prod_Name**.



16. From the properties pane of Sum_Prod_Name, click **Group By**.

17. To add a new expression, click **+**.

18. From the Field name drop down, select the **Product_Name**.



Properties Preview Sum_Prod_Name

General Select fields to group by so that the resulting records can be set of rows with the same values for some fields.

Incoming Fields Group by: Not Parameterized

Group By Group by Fields + ^ v

Field Name

Product_Name

Aggregate

Advanced

19. From the properties pane, click **Aggregate**.

20. To add the new Aggregate field, click .

21. Enter the details as shown below:

Field Type	Name	Type	Precision	Scale
Output Field	Sum_Prod_Name	integer	10	0

22. Click **OK**.

New Field

×

Create new output field, variable field, input macro field or output macro field.

Field Type: Output Field

Name*: Sum_Prod_Name

Type*: integer

Precision*: 10

Scale: 0

Description:

?

OK

Cancel

23. Click **Configure**.

Properties

Preview

Sum_Prod_Name

General

Incoming Fields

Group By

Aggregate

Advanced

Create simple aggregate expressions. You can also use expression macros to create complex aggregate expressions.

☐ Allow additional fields and expressions during task creation

Aggregate

Field Name	Expression	Type	Precision	Scale
Sum_Prod_Name	Configure...	integer	10	0

24. In the Expression window, enter the following expression:

SUM(Total)

25. Click **OK**.

Field Expression: Sum_Prod_Name(integer, 10, 0) ✕

Configure the expression by adding fields, functions, and variables.

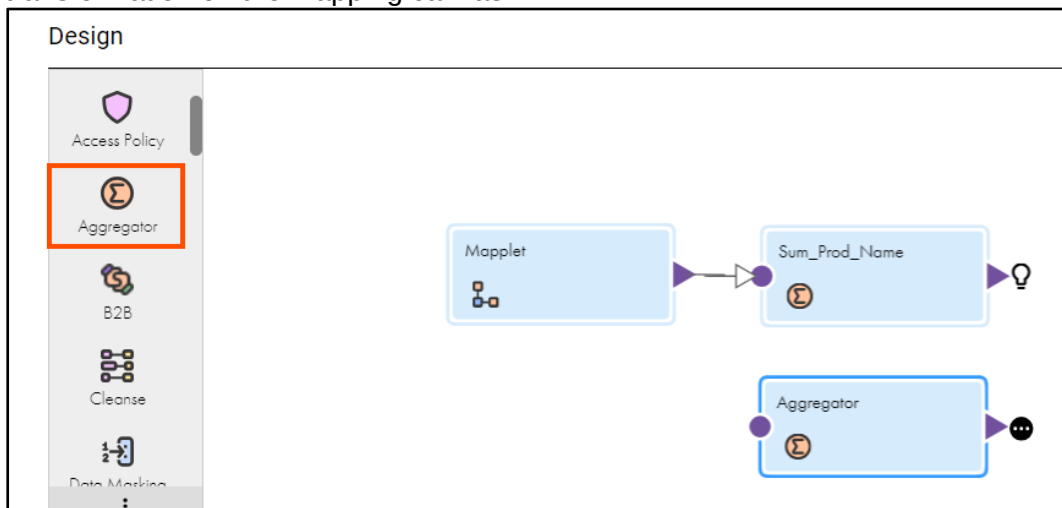
Expression: Not Parameterized

Fields	Expression
ORDERID	SUM(Total)
Product_Name	
PRODUCTDESCRIPTION	
Category	
Total	

Select a field to see detailed information about it.

? OK Cancel Validate

26. From the list of available transformations, drag and drop another aggregate transformation on the mapping canvas.



27. From the properties pane, in the Name field enter the name as **Sum_Prod_Category**.

Properties **Preview** Σ Sum_Prod_Category

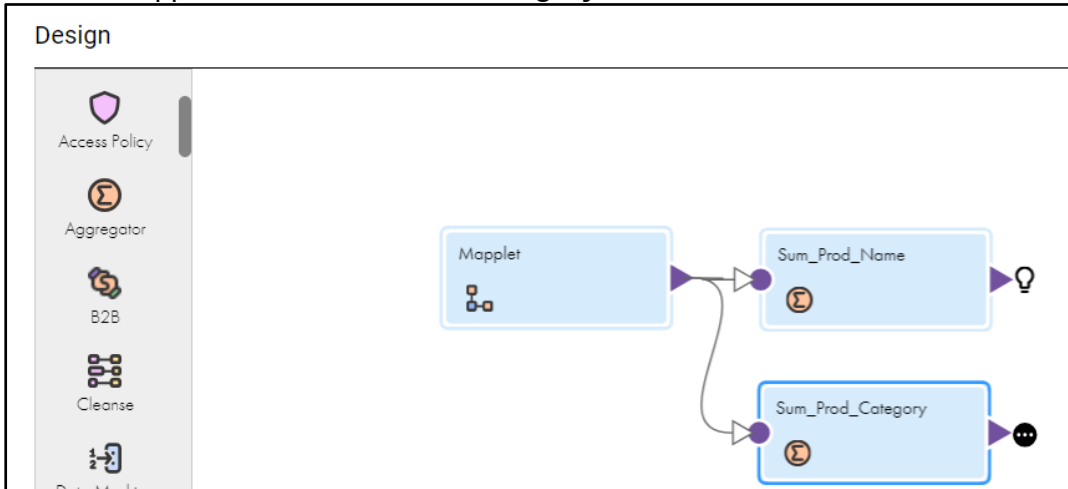
General

Incoming Fields

Name:* Sum_Prod_Category

Description:

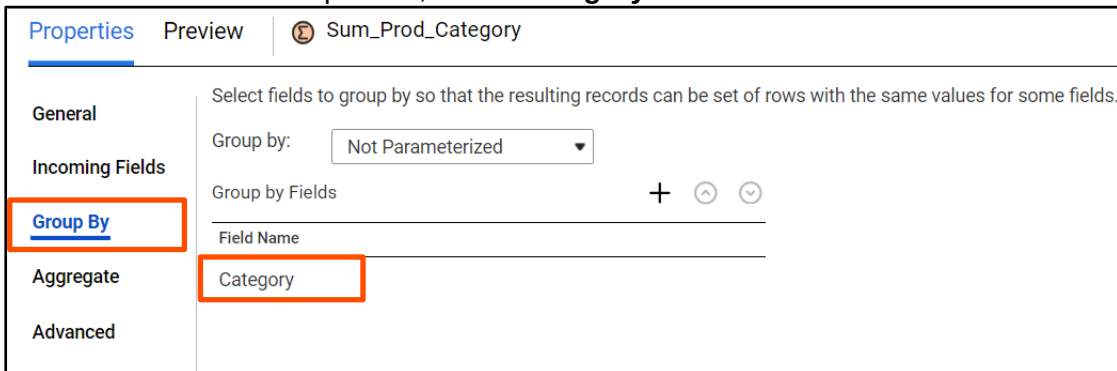
28. Link the Mapplet to the **Sum_Prod_Category** transformation.



29. From the properties pane of Sum_Prod_Category, click **Group By**.

30. To add a new expression, click .

31. From the Field name drop down, select **Category**.



Properties Preview Sum_Prod_Category

Select fields to group by so that the resulting records can be set of rows with the same values for some fields.

Group by: Not Parameterized

Group by Fields + ^ v

Field Name
Category

General Incoming Fields **Group By** Aggregate Advanced

32. From the properties pane, click on **Aggregate**.

33. To add the new Aggregate Field, click .

34. Enter the details as shown below:

Field Type	Name	Type	Precision	Scale
Output Field	Sum_Prod_Category	integer	10	0

35. Click **OK**.

New Field

×

Create new output field, variable field, input macro field or output macro field.

Field Type:

Output Field

Name:*

Sum_Prod_Category

Type:*

integer

Precision:*

10

Scale:

0

Description:

?

OK

Cancel

36. Click **Configure**.

Properties

Preview

Sum_Prod_Category

General

Incoming Fields

Group By

Aggregate

Advanced

Create simple aggregate expressions. You can also use expression macros to create complex aggregate expressions.

☐ Allow additional fields and expressions during task creation

Aggregate

Field Name	Expression	Type	Precision	Scale
Sum_Prod_Category	Configure...	integer	10	0

37. In the Expression window, enter the following expression:

SUM(Total)

38. Click **OK**.

Field Expression: Sum_Prod_Category(integer, 10, 0) ✕

Configure the expression by adding fields, functions, and variables.

Expression: Not Parameterized

Fields ▼ Expression Validate

ORDERID
Product_Name
PRODUCTDESCRIPTION
Category
Total

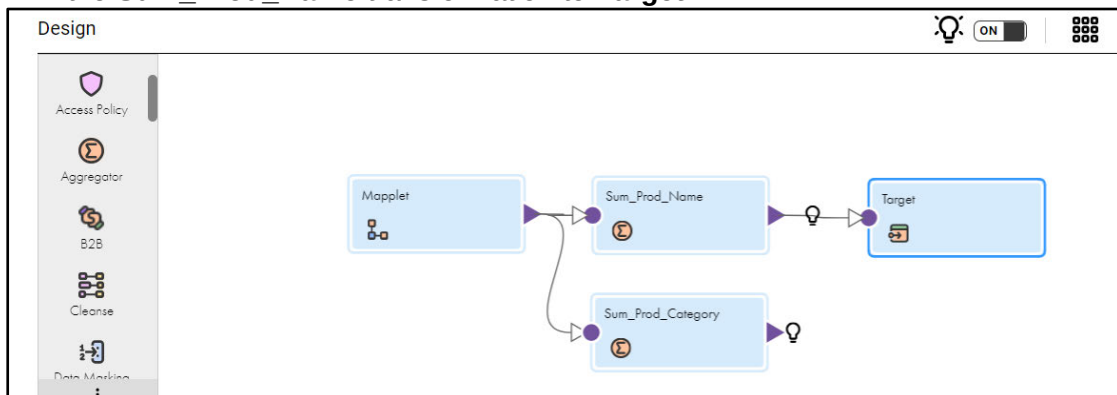
SUM(Total)

Select a field to see detailed information about it.

? OK Cancel

39. Select the **Target** from the mapping canvas.

40. Link the **Sum_Prod_Name** transformation to **Target**.



41. In the name field enter the name as **Total_Sales_Product**.

Properties Preview Total_Sales_Product

General

Name:*

Description:

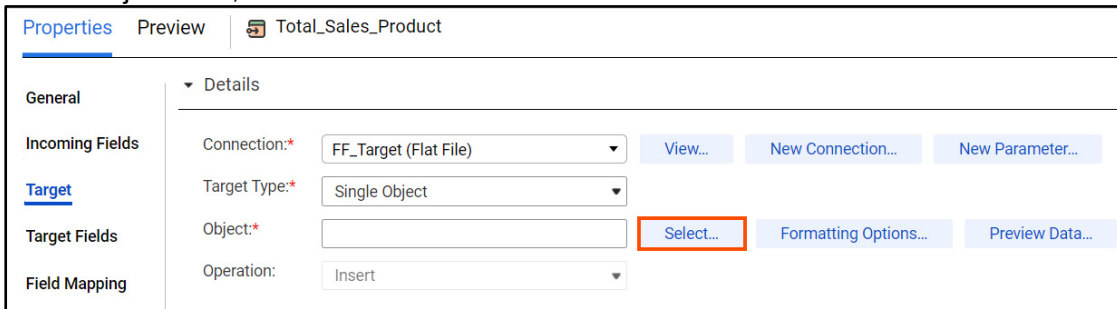
Incoming Fields

42. From the properties pane, select **Target**.

43. From the Connection drop down, select your **target Flat File** connection.

44. Retain the Target Type as **Single Object**.

45. In the Object field, click **Select**.

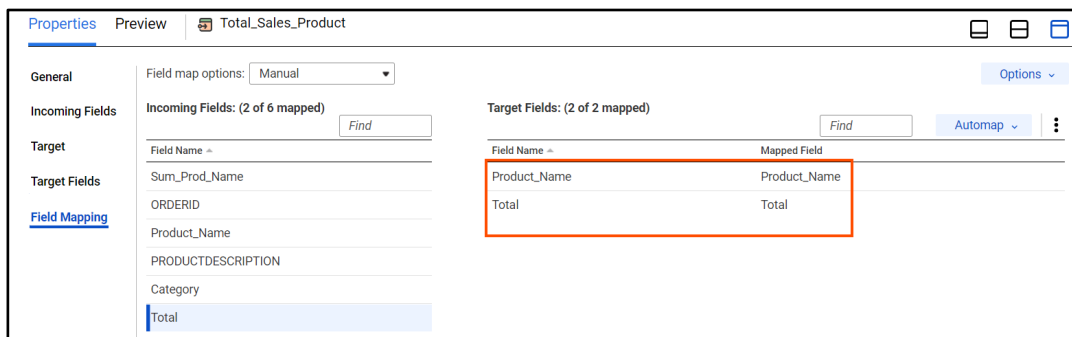


46. From the list, select the **Total_Sales_Product.csv**. Click **OK**.

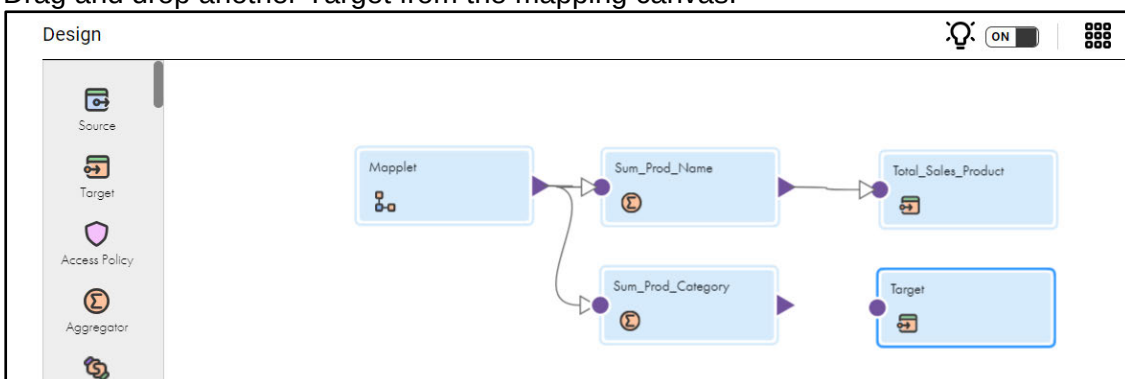
47. From the properties pane, select **Field Mapping**.

48. Map the fields as shown in the table below (If they are mapped already, verify, and proceed with the next step):

Incoming Field	Target Field
Product_Name	Product_Name
Total	Total



49. Drag and drop another Target from the mapping canvas.





50. From the general field, enter the name as **Total_Sales_Category**.

Properties Preview **Total_Sales_Category**

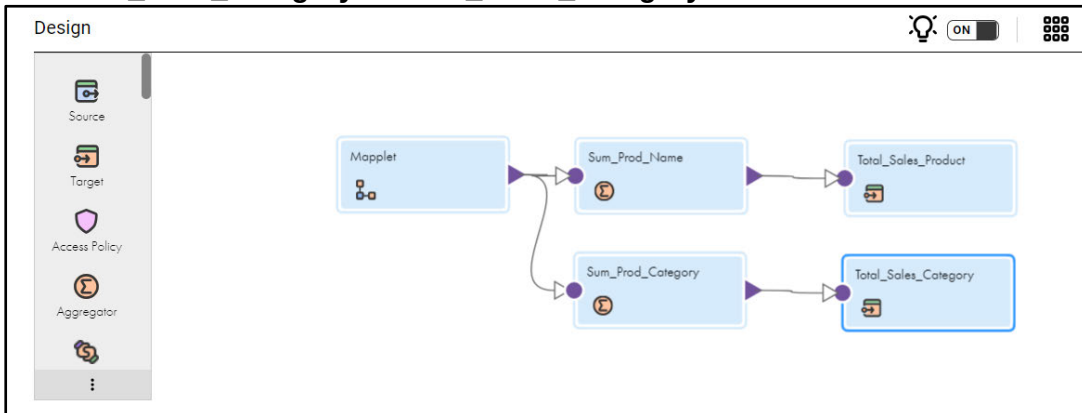
General

Name:* Total_Sales_Category

Description:

Incoming Fields

51. Link **Sum_Prod_Category** to **Total_Sales_Category**.



52. Select the **Total_Sales_Category** from the mapping canvas and from the properties pane, select **Target**.

53. From the Connection drop-down, select your **target Flat File** connection.

54. To select the source object from the Object field, click **Select**.

Properties Preview **Total_Sales_Category**

General

Incoming Fields

Target

Target Fields

Field Mapping

Details

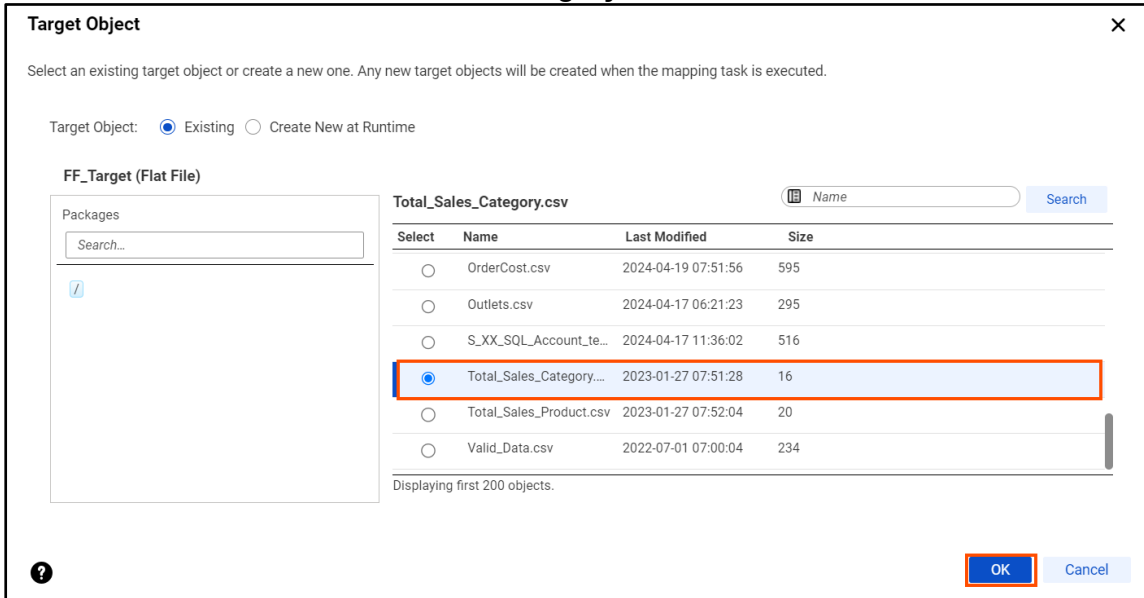
Connection:* FF_Target (Flat File) View... New Connection... New Parameter...

Target Type:* Single Object

Object:* Enter object name or click Select... **Select...** Formatting Options... Preview Data...

Operation: Insert

55. From the list, select the **Total_Sales_Category.csv**. Click **OK**.



Target Object

Select an existing target object or create a new one. Any new target objects will be created when the mapping task is executed.

Target Object: ☒ Existing ☐ Create New at Runtime

FF_Target (Flat File)

Packages

Search...

Total_Sales_Category.csv

Name Search

Select	Name	Last Modified	Size
☐	OrderCost.csv	2024-04-19 07:51:56	595
☐	Outlets.csv	2024-04-17 06:21:23	295
☐	S_XX_SQL_Account_te...	2024-04-17 11:36:02	516
☒	Total_Sales_Category...	2023-01-27 07:51:28	16
☐	Total_Sales_Product.csv	2023-01-27 07:52:04	20
☐	Valid_Data.csv	2022-07-01 07:00:04	234

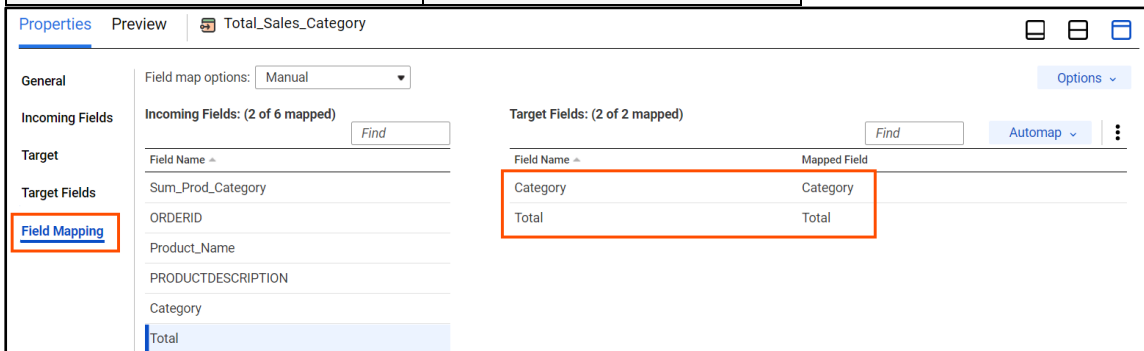
Displaying first 200 objects.

OK Cancel

56. From the properties pane, click **Field Mapping**.

57. Map the fields, as shown in the following table:

Incoming Fields	Target Fields
Category	Category
Total	Total



Properties Preview **Total_Sales_Category**

Field map options: Manual

Incoming Fields: (2 of 6 mapped) Find

Target Fields: (2 of 2 mapped) Find Automap

Field Name	Mapped Field
Category	Category
Total	Total

58. **Save** and **Run** the mapping.

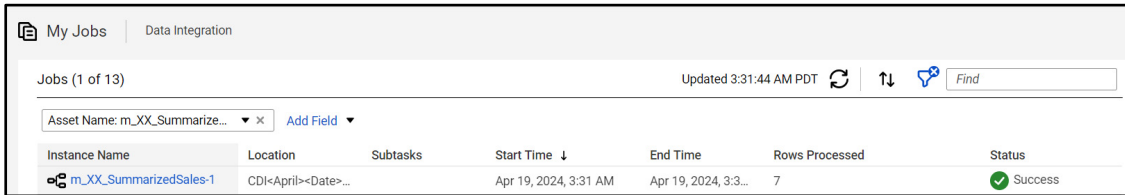
59. From the Runtime Environment drop down, select **INFA-SERVER**.

60. Click **Run**.

Monitor Status

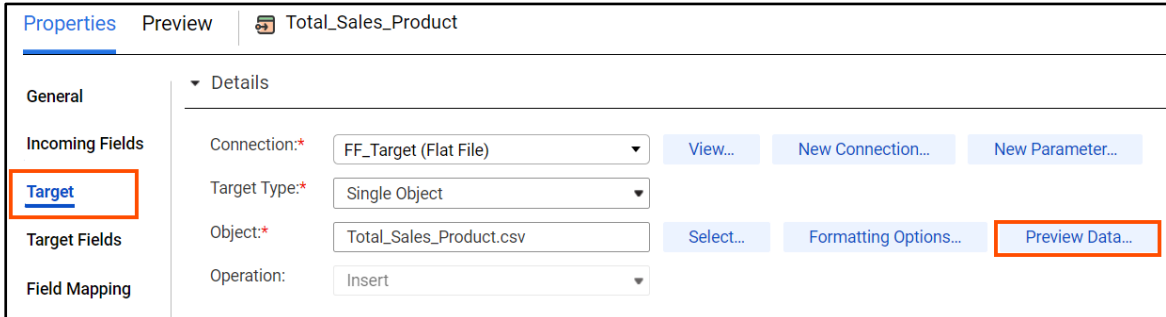
61. Monitor the task status from the **My Jobs** page.

62. When the task completes, the status changes to **Success**.



Instance Name	Location	Subtasks	Start Time ↓	End Time	Rows Processed	Status
m_XX_SummarizedSales-1	CDI<April><Date>...		Apr 19, 2024, 3:31 AM	Apr 19, 2024, 3:3...	7	Success

63. Optionally, preview the data on the targets.



Properties | Preview | **Total_Sales_Product**

General

Incoming Fields

Target

Target Fields

Field Mapping

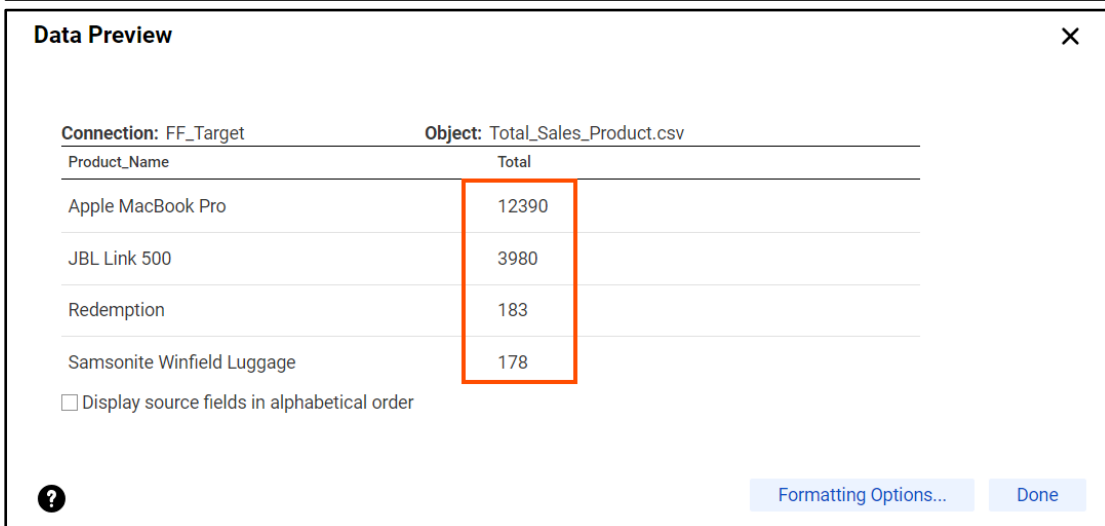
Details

Connection:* FF_Target (Flat File) [View...] [New Connection...] [New Parameter...]

Target Type:* Single Object

Object:* Total_Sales_Product.csv [Select...] [Formatting Options...] [Preview Data...]

Operation: Insert



Data Preview

Connection: FF_Target Object: Total_Sales_Product.csv

Product_Name	Total
Apple MacBook Pro	12390
JBL Link 500	3980
Redemption	183
Samsonite Winfield Luggage	178

☐ Display source fields in alphabetical order

[?] [Formatting Options...] [Done]

Data Preview ×

Connection: FF_Target **Object:** Total_Sales_Category.csv

Category	Total
Books	183
Electronics	3980
Luggage	178

☐ Display source fields in alphabetical order

? Formatting Options... Done

64. Close the asset from the navigation pane.

This concludes the lab.

Module 9: Mapping Parameterization

Lab 9-1: Performing Complete Parameterization

Overview:

Parameters are placeholders that represent values in a mapping. In IICS, you can make a mapping reusable with the help of parameters.

Objective:

- Create a completely parameterized mapping

Scenario:

Ruby wants to filter the Salesforce opportunities based on probability. However, creating a mapping every time an opportunity is added is time consuming. When she explained this concern to John, he recommends creating a completely parameterized mapping.

Duration:

30 minutes

Tasks

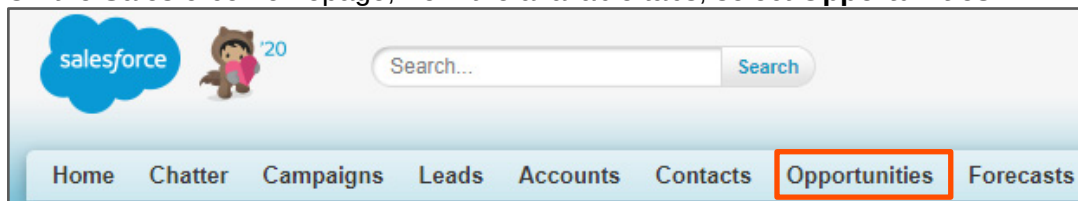
Create Opportunities in Salesforce

1. Log in to your Salesforce Developer account using your credentials.

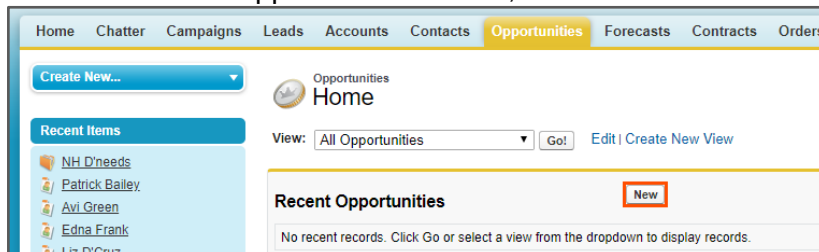
Note: You can use the below mentioned link to log in:

<https://login.salesforce.com/>

2. On the Salesforce homepage, from the available tabs, select **Opportunities**.



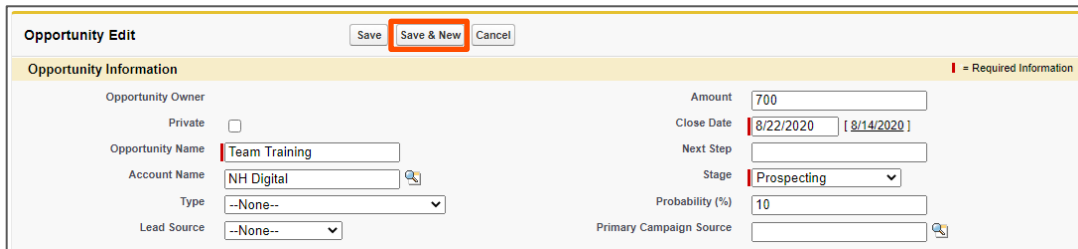
3. From the Recent Opportunities section, select **New**.



4. Enter the details, as shown in the table below:

Field Name	Values
Opportunity Name	Team Training
Account Name	NH Digital
Amount	700
Close Date	Choose a date greater than the current date
Stage	Prospecting
Probability	10

5. Click **Save & New**.



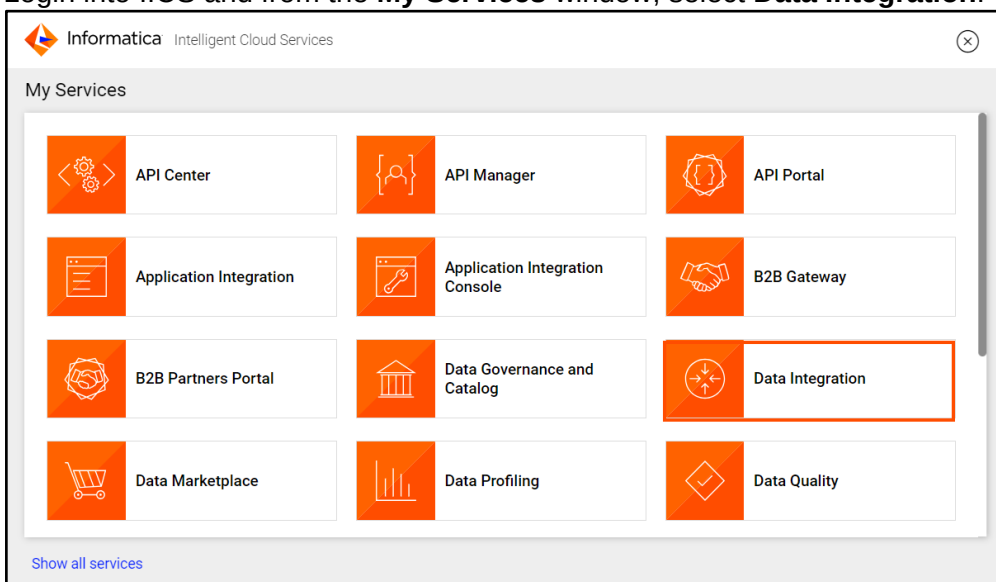
6. Similarly, create three or more opportunities in Salesforce. You can refer to the image below to create opportunities:

A	B	C	D	E	F
Opportunity Name	Account Name	Amount	Close Date	Probability	Stage
Stock Supply	NH Trends	1000000	Choose a date greater than the current date	75	Proposal/Price Quote
Service Contract Renewal	NH Everyday	95000	Choose a date greater than the current date	60	Id. Decision Makers
Annual Maintenance	NH Mart	50000	Choose a date greater than the current date	90	Negotiation/Review

7. After the last opportunity is added, click **Save**.

Create Mapping

8. Login into IICS and from the **My Services** window, select **Data Integration**.





9. Create a new Mapping and name it **m_XX_MappingParameters**. Replace XX with your student number.
10. Verify the asset location.
11. To configure the source, from the mapping canvas, click the **Source** transformation.
12. In the General section of the Source properties, enter Name as **SO_Salesforce**.

The screenshot shows the 'Properties' pane for the 'SO_Salesforce' source transformation. The 'General' tab is selected. The 'Name' field is populated with 'SO_Salesforce'. The 'Description' field is empty.

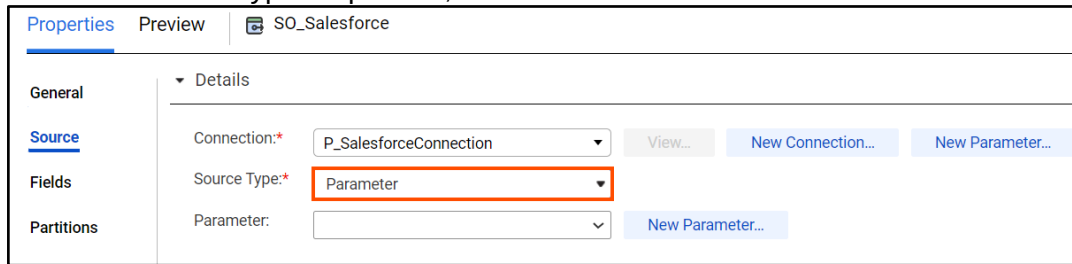
13. From the properties pane, click **Source**.
14. To create a new connection parameter, click **New Parameter**.

The screenshot shows the 'Properties' pane for the 'SO_Salesforce' source transformation. The 'Source' tab is selected. The 'Connection' dropdown is empty. The 'Source Typ...' dropdown is empty. The 'New Parameter...' button is highlighted with a red box.

15. Enter Name as **P_SalesforceConnection**, and Display Label as **Salesforce**.
16. Click **OK**.

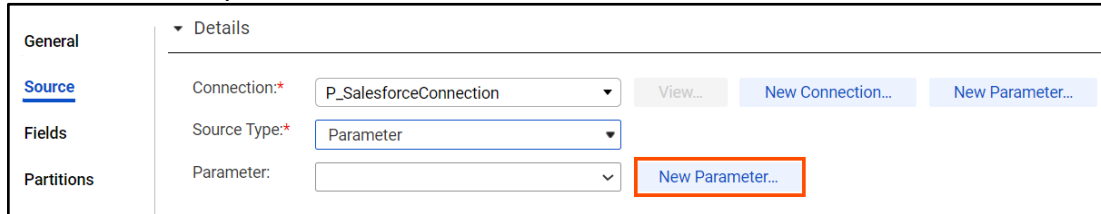
The screenshot shows the 'New Input Parameter' dialog box. The 'Name' field is 'P_SalesforceConnection' and the 'Display Label' is 'Salesforce'. The 'Type' is 'connection'. The 'Connection Type' is empty. The 'Allow parameter to be overridden at run time' checkbox is unchecked. The 'Default Value' field is empty. The 'OK' button is highlighted with a red box.

17. From the Source Type drop-down, select **Parameter**.



The screenshot shows the 'Properties' window for 'SO_Salesforce'. The 'Source' tab is selected. In the 'Details' section, the 'Source Type' dropdown menu is highlighted with a red box and contains the value 'Parameter'. Other fields include 'Connection' set to 'P_SalesforceConnection' and a 'Parameter' field with a 'New Parameter...' button next to it.

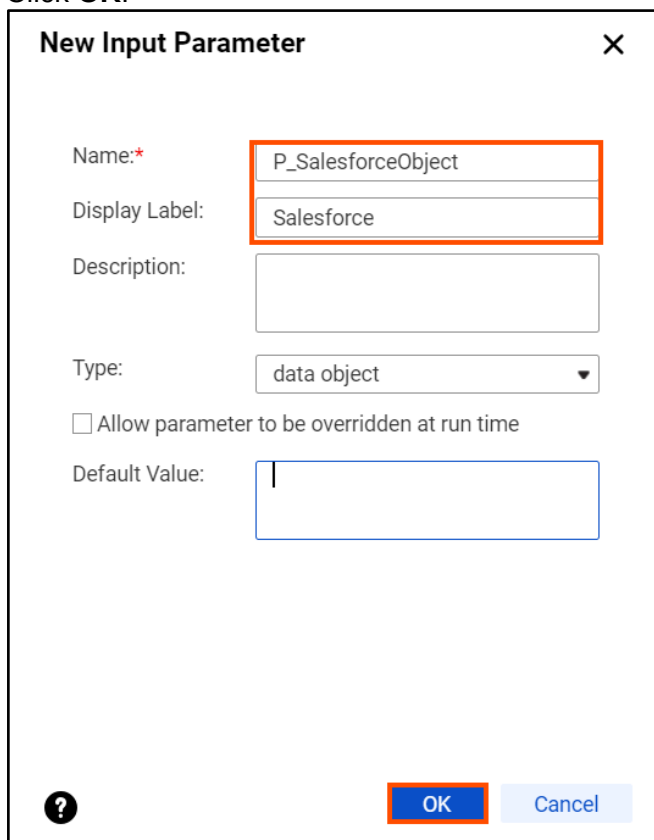
18. To create a new parameter, from the Parameter field, click **New Parameter**.



This screenshot is similar to the previous one, but the 'New Parameter...' button located next to the empty 'Parameter' field is highlighted with a red box.

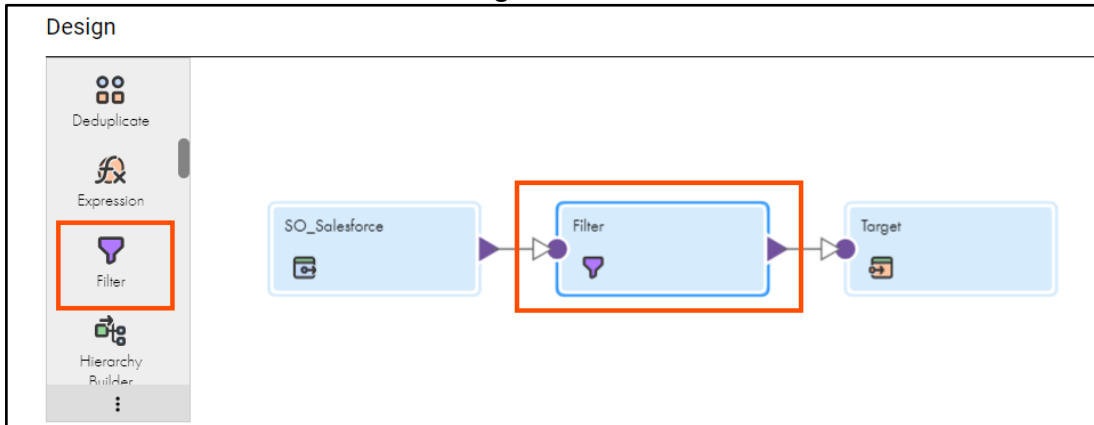
19. Enter Name as **P_SalesforceObject**, and the Display Label as **Salesforce**.

20. Click **OK**.



The 'New Input Parameter' dialog box is shown. The 'Name' field contains 'P_SalesforceObject' and the 'Display Label' field contains 'Salesforce'. Both fields are highlighted with a red box. The 'Type' is set to 'data object'. There is an unchecked checkbox for 'Allow parameter to be overridden at run time' and an empty 'Default Value' field. At the bottom, the 'OK' button is highlighted with a red box, next to a 'Cancel' button and a help icon.

21. From the list of available transformations, drag and drop a **Filter** transformation onto the link between **SO_Salesforce** and **Target**.



22. In the General section of the Filter properties, enter the Name as **FIL_RouteHighProbOpptys**.

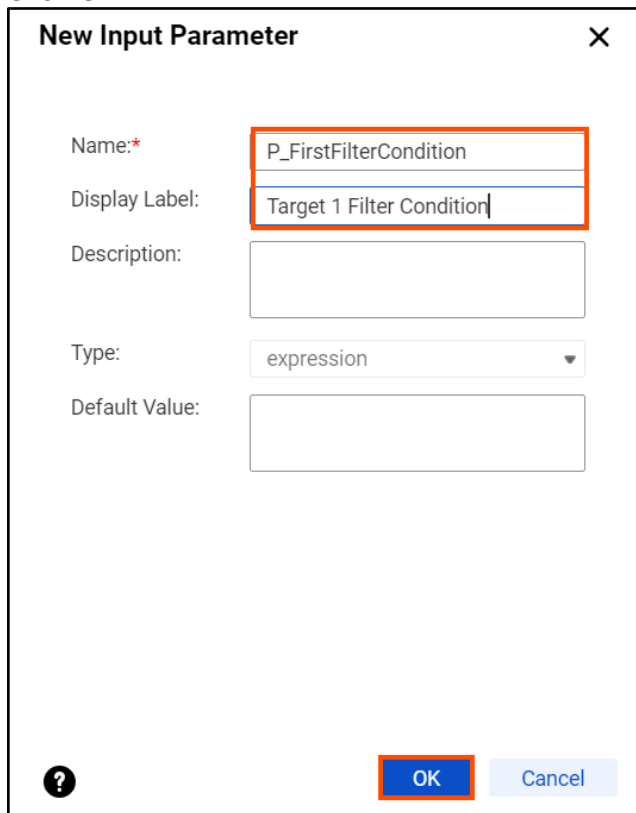
Properties		Preview	FIL_RouteHighProbOpptys
General	Name:*	FIL_RouteHighProbOpptys	
	Description:		
Incoming Fields			
Filter			

23. From the properties pane, click **Filter**.
24. From the Filter Condition drop-down, select **Completely Parameterized**.
25. To create a new filter parameter, click **New Parameter**.

Properties		Preview	FIL_RouteHighProbOpptys
General	Filter Condition:	Completely Parameterized ▼	
	Parameter:	New Parameter...	
Incoming Fields			
Filter			
Advanced			

26. Enter Name as **P_FirstFilterCondition**, and Display Label as **Target 1 Filter Condition**.

27. Click **OK**.



New Input Parameter [X]

Name:* P_FirstFilterCondition

Display Label: Target 1 Filter Condition

Description:

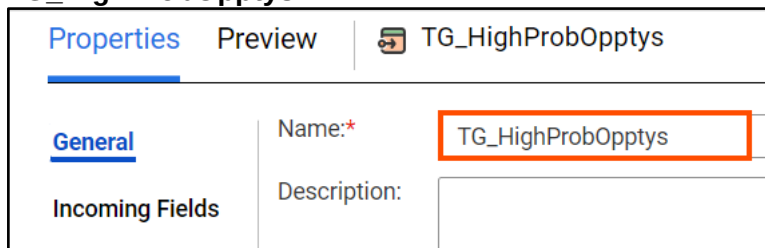
Type: expression

Default Value:

[?] [OK] [Cancel]

28. To configure the target, from the mapping canvas, click the **Target** transformation.

29. In the General section of the Target properties, enter the Name as **TG_HighProbOppty**.



Properties Preview TG_HighProbOppty

General

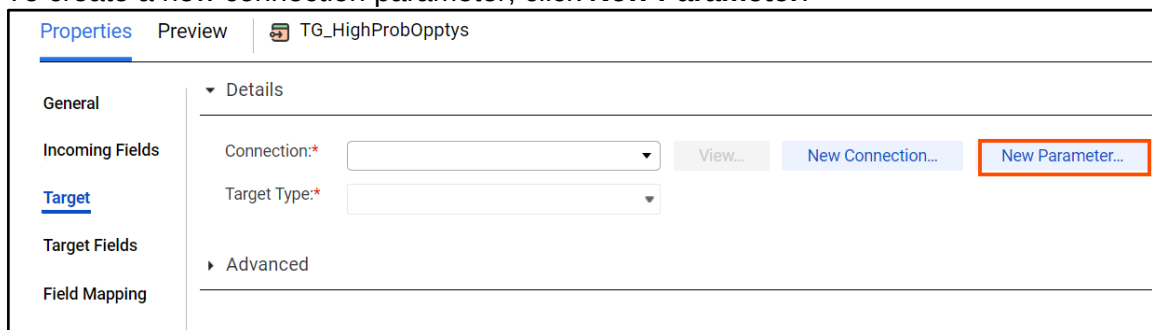
Name:* TG_HighProbOppty

Description:

Incoming Fields

30. From the properties pane, click **Target**.

31. To create a new connection parameter, click **New Parameter**.



Properties Preview TG_HighProbOppty

General

Target

Target Fields

Field Mapping

Details

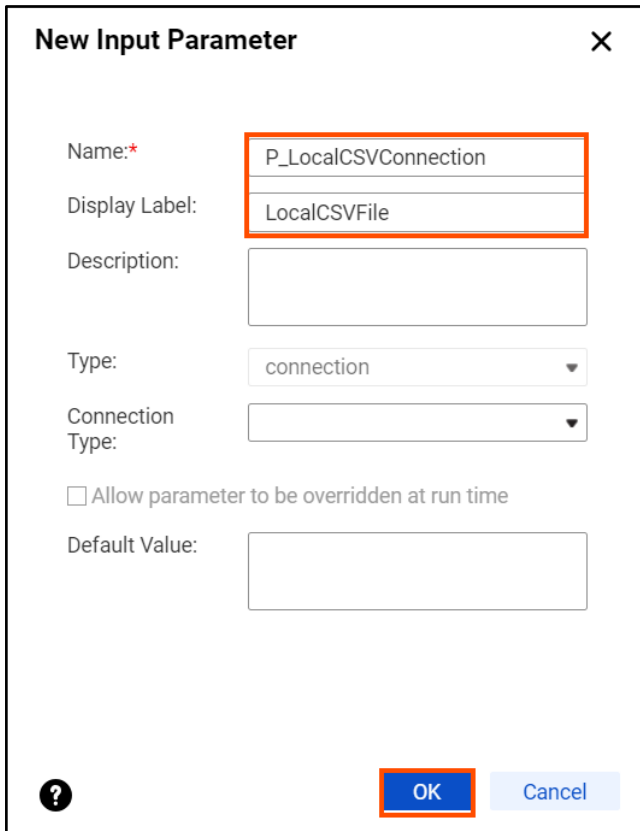
Connection:* [View...] [New Connection...] [New Parameter...]

Target Type:*

Advanced

32. Enter Name as **P_LocalCSVConnection**, and Display Label as **LocalCSVFile**.

33. Click **OK**.

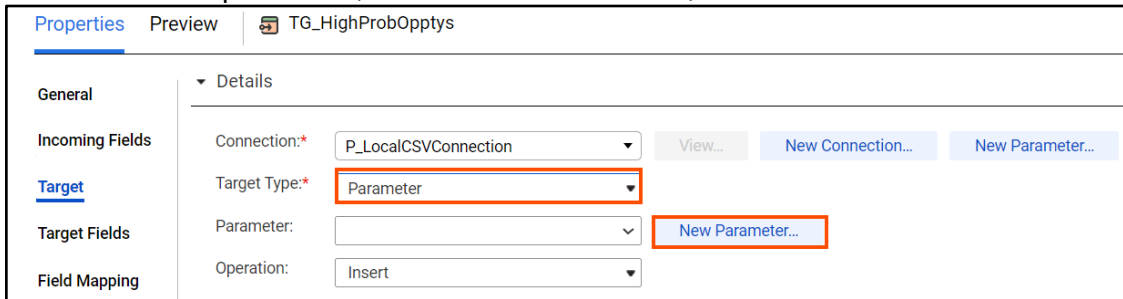


The 'New Input Parameter' dialog box is shown. It has a title bar with a close button (X). The fields are as follows:

- Name:** P_LocalCSVConnection (highlighted with a red box)
- Display Label:** LocalCSVFile (highlighted with a red box)
- Description:** (empty text box)
- Type:** connection (dropdown menu)
- Connection Type:** (empty dropdown menu)
- ☐ Allow parameter to be overridden at run time
- Default Value:** (empty text box)
- Buttons: ? (help), **OK** (highlighted with a red box), Cancel

34. From the Target Type drop-down, select **Parameter**.

35. To create a new parameter, from the Parameter field, click **New Parameter**.



The 'Properties' dialog box is shown, with the 'Target' tab selected. The 'Details' section contains the following fields:

- Connection:** P_LocalCSVConnection (dropdown menu)
- Target Type:** Parameter (dropdown menu, highlighted with a red box)
- Parameter:** (empty dropdown menu)
- Operation:** Insert (dropdown menu)

Buttons: View..., New Connection..., New Parameter... (highlighted with a red box), New Parameter... (highlighted with a red box)

36. Enter Name as **P_HighProb**, and the Display Label as **HighProbabilityOpportunities**.

37. Click **OK**.

New Input Parameter

×

Name:*

P_HighProb

Display Label:

HighProbabilityOpportunities

Description:

Type:

data object ▼

☐ Allow parameter to be overridden at run time

Default Value:

?

OK

Cancel

38. From the properties pane, click **Field Mapping**.

39. From the Field map options drop-down, select **Automatic**.

Properties

Preview

TG_HighProbOpptys

General

Field map options: Automatic ▼ **Note:** This option will automatically map any fields added later by name.

Incoming Fields

Incoming Fields: (0 of 0 mapped) Find

Target

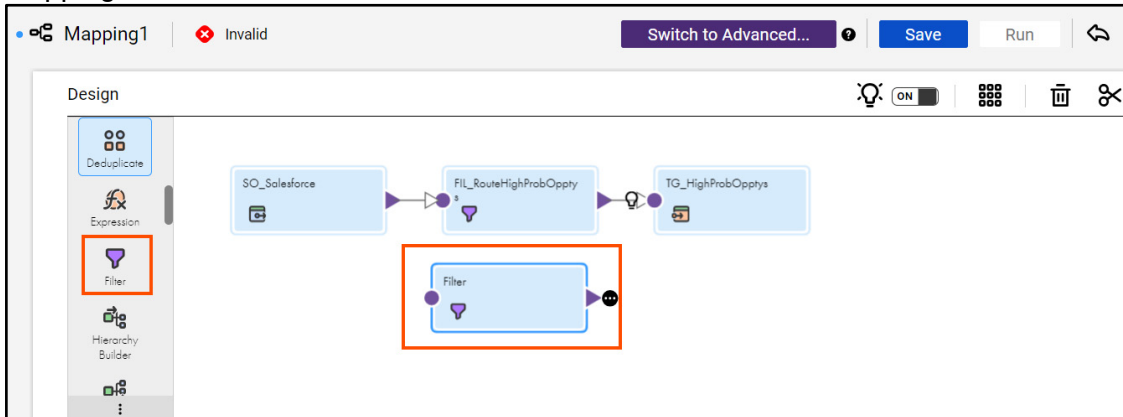
Target Fields: (0 of 0 mapped) Find

Field Mapping

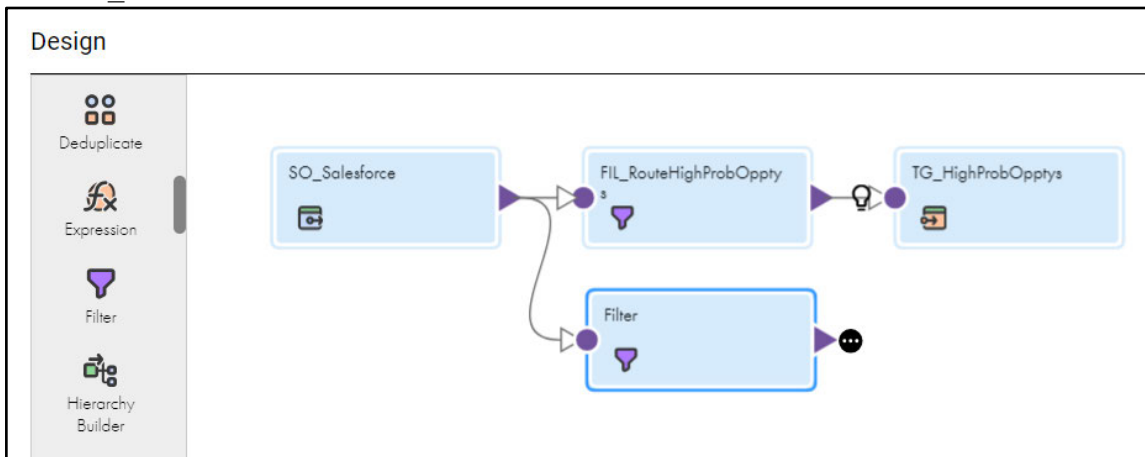
Field Name ↕ Mapped Field

There are no incoming fields. There are no target fields to show as the object has not been specified or you have chosen to

40. From the list of available transformations, drag and drop a **Filter** transformation onto the mapping canvas.



41. Link **SO_Salesforce** to the **Filter** transformation.



42. Select the **Filter** transformation from the mapping canvas.

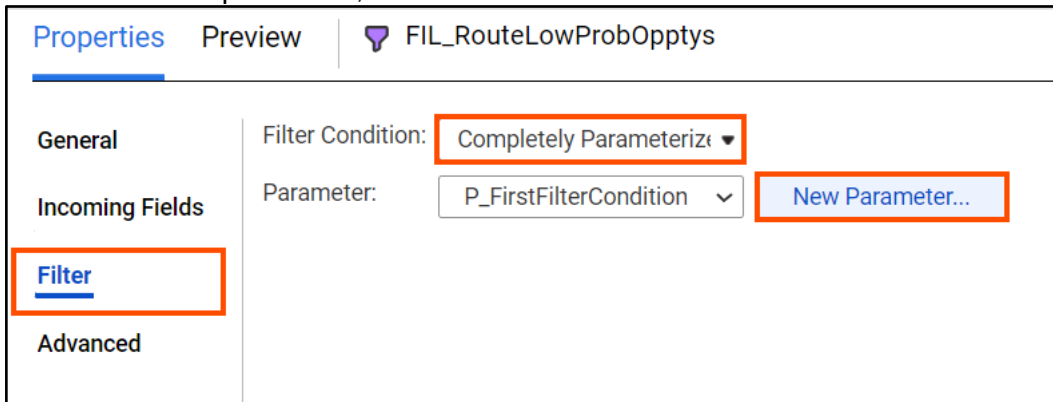
43. In the General section of the Filter properties, enter the Name as **FIL_RouteLowProbOppty**.

Properties		Preview	FIL_RouteLowProbOppty
General		Name:	FIL_RouteLowProbOppty
Incoming Fields		Description:	
Filter			

44. From the properties pane, click **Filter**.

45. From the Filter Condition drop-down, select **Completely Parameterized**.

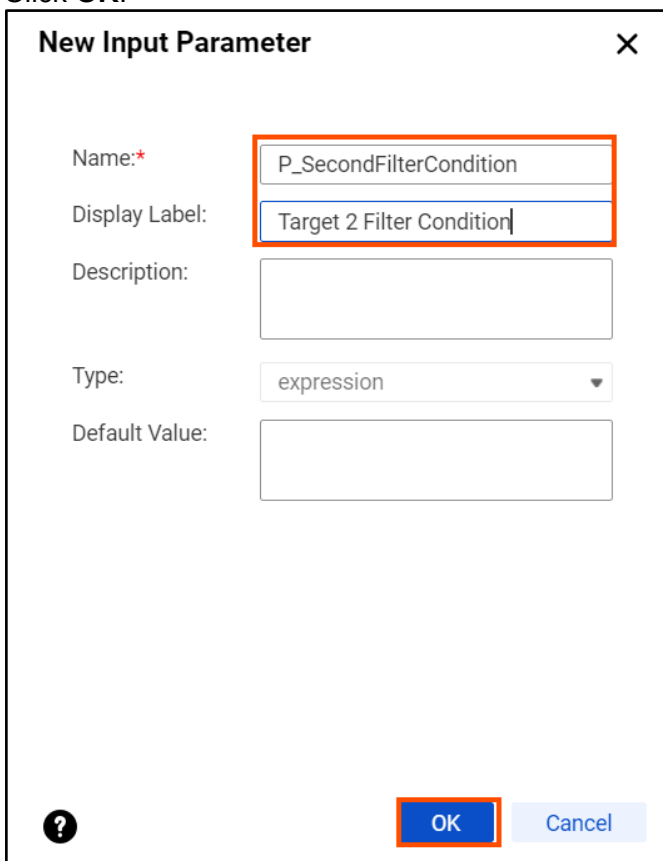
46. To create a new parameter, click **New Parameter**.



The screenshot shows the 'Properties' tab for a component named 'FIL_RouteLowProbOppty'. The 'Filter' tab is selected in the left-hand menu. In the main area, the 'Filter Condition' is set to 'Completely Parameterized' and the 'Parameter' is 'P_FirstFilterCondition'. A 'New Parameter...' button is highlighted in the top right corner.

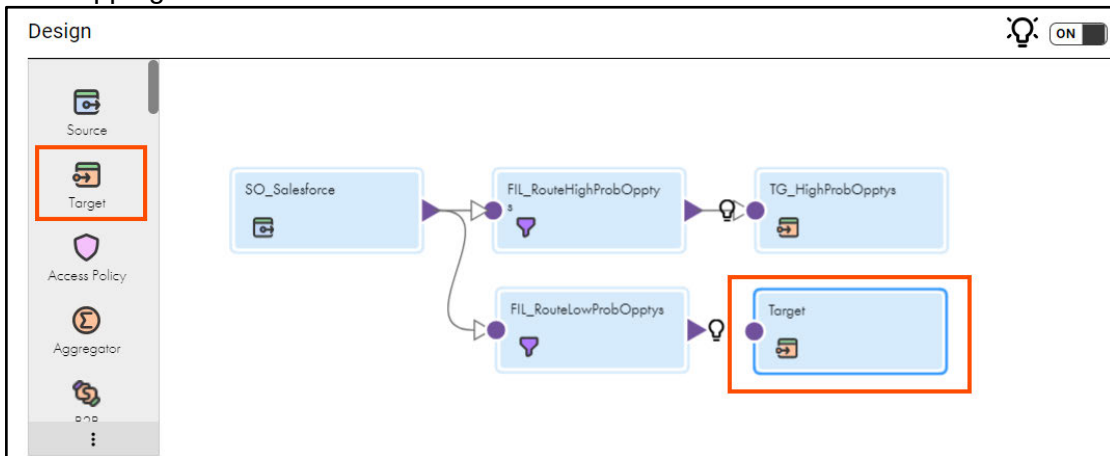
47. Enter Name as **P_SecondFilterCondition**, and Display Label as **Target 2 Filter Condition**.

48. Click **OK**.

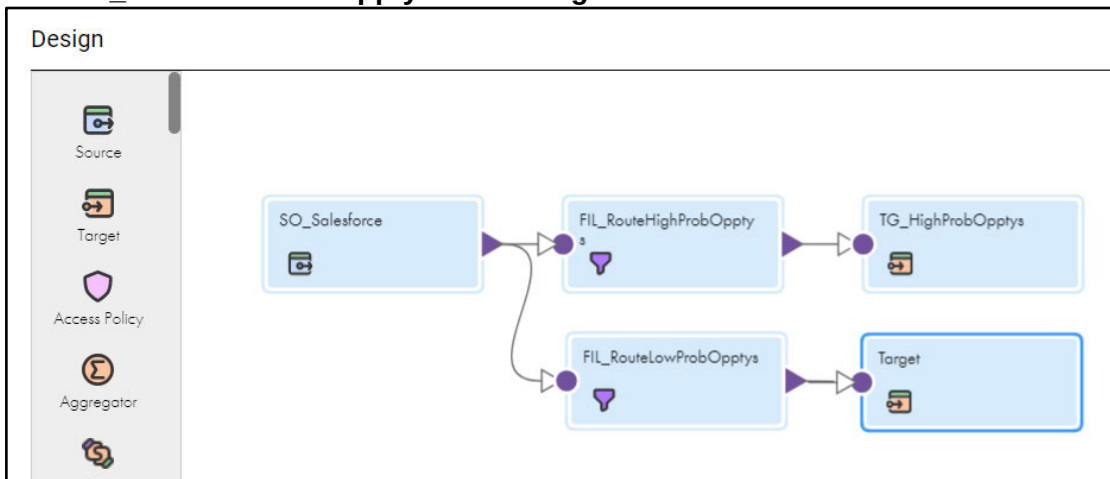


The screenshot shows the 'New Input Parameter' dialog box. The 'Name' field is 'P_SecondFilterCondition' and the 'Display Label' is 'Target 2 Filter Condition'. The 'Type' is set to 'expression'. The 'OK' button is highlighted in the bottom right corner.

49. From the list of available transformations, drag and drop a **Target** transformation onto the mapping canvas.



50. Link **FIL_RouteLowProbOppty** to the **Target** transformation.



51. Select the **Target** transformation from the mapping canvas.

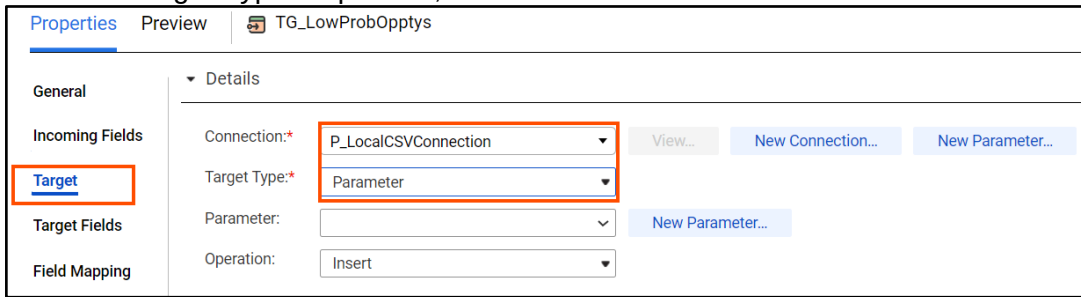
52. In the General section of Target properties, enter Name as **TG_LowProbOppty**.

Properties		Preview	TG_LowProbOppty
General			
Name:*		TG_LowProbOppty	
Description:			
Incoming Fields			
Target			

53. From the properties pane, click **Target**.

54. From the Connection drop-down, select **P_LocalCSVConnection**.

55. From the Target Type drop-down, select **Parameter**.



Properties Preview TG_LowProbOppty

General

Incoming Fields

Target

Target Fields

Field Mapping

Details

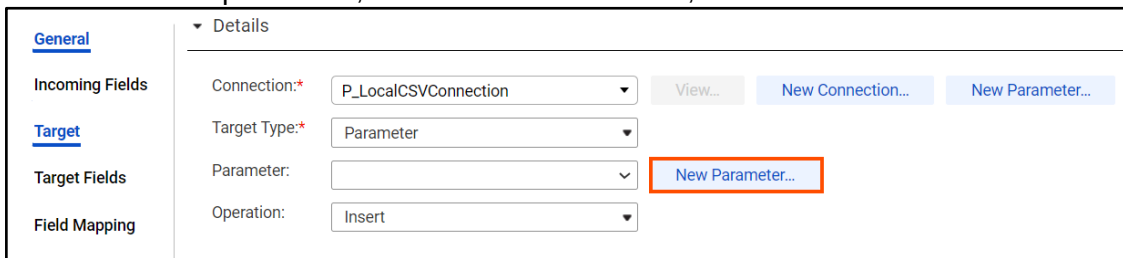
Connection:* P_LocalCSVConnection View... New Connection... New Parameter...

Target Type:* Parameter

Parameter: New Parameter...

Operation: Insert

56. To create a new parameter, from the Parameter field, click **New Parameter**.



General

Incoming Fields

Target

Target Fields

Field Mapping

Details

Connection:* P_LocalCSVConnection View... New Connection... New Parameter...

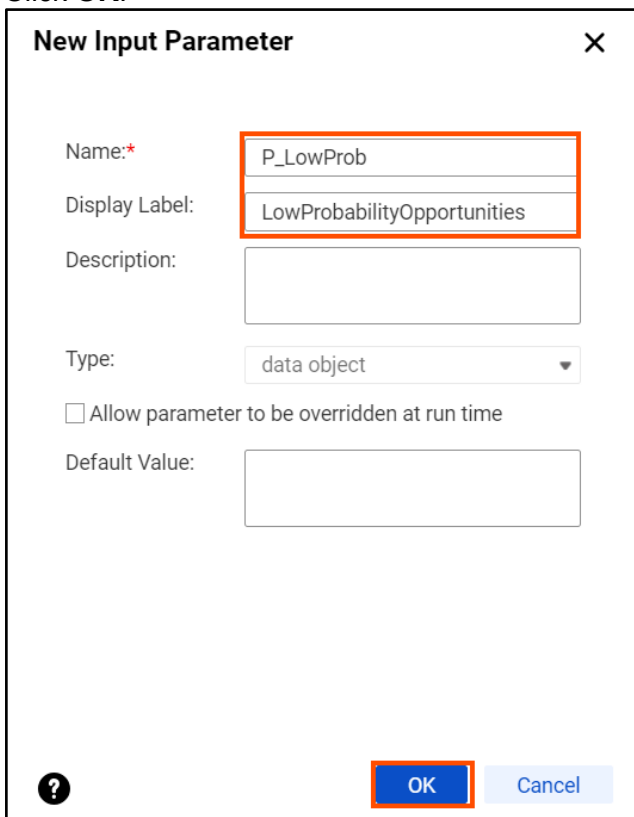
Target Type:* Parameter

Parameter: New Parameter...

Operation: Insert

57. Enter Name as **P_LowProb**, and the Display Label as **LowProbabilityOpportunities**.

58. Click **OK**.



New Input Parameter X

Name:* P_LowProb

Display Label: LowProbabilityOpportunities

Description:

Type: data object

☐ Allow parameter to be overridden at run time

Default Value:

? OK Cancel

59. From the properties pane, click **Field Mapping**.



60. From the Field map option drop-down, select **Automatic**.

The screenshot shows the 'Properties' dialog for a mapping named 'TG_LowProbOppty'. The 'Field map options' dropdown is set to 'Automatic'. The 'Field Mapping' tab is selected in the left sidebar. The 'Incoming Fields' and 'Target Fields' sections are empty, with messages indicating no fields are mapped.

61. Save and run the mapping.

62. From the Runtime Environment drop-down, select INFA-SERVER.

63. Click **Next**.

64. From the Salesforce (P_SalesforceConnection) Connection drop-down, select your **Salesforce** connection.

65. From the Salesforce Object field, click **Select**.

The screenshot shows the 'Run m_XX_MappingParameters' dialog with the 'Sources' tab selected. The 'Salesforce (P_SalesforceConnection) Connection' is set to 'XX_FirstName_SFDCDeveloper3 (Sale...'. The 'Source Type' is 'Single'. The 'Salesforce Object' field has a 'Select...' button highlighted. The 'Query Options' section is also visible.

66. From the list, select **Opportunity**.

Note: You can also use the search feature.

67. Click **Select**.

Select a source object, and then click Select. You can also search for a source object.

XX_FirstName_SFDCDeveloper (Salesforce)

Opportunity

Select	Name	Label	Description	Type
☒	Opportunity	Opportunity		
☐	OpportunityCompetitor	Opportunity: Competitor		
☐	OpportunityContactRole	Opportunity Contact Role		
☐	OpportunityFeed	Opportunity Feed		

Displaying 10 of 938 objects.

68. Click **Next**.

1 Definition 2 Sources 3 Targets 4 Input Parameters

Source Parameter Details

Salesforce (P_SalesforceConnection) Connection:*

Source Type:

Salesforce Object:

Query Options

Filter:

☐ Display technical names instead of labels

☐ Display source fields in alphabetical order

Data Preview

Opportunity (Total columns: 46)

Opportunity ID	Deleted	Account ID	Private	Name
006GB00001kEICNYA0	false	001GB00003AdSAzYAN	false	Edge Emergency Generator
006GB00001kEICXYA0	false	001GB00003AdSAzYAN	false	Edge Installation
006GB00001kEICVYA0	false	001GB00003AdSAzYAN	false	Edge SLA

69. From the LocalCSVFile (P_LocalCSVConnection) Connection drop-down, select your **target Flat File** connection.

70. To select the HighProbabilityOpportunities Object, click **Select** and choose the **HighProbabilityOpportunities.csv** file.



71. From the LowProbabilityOpportunities Object field, click **Select** and choose the **LowProbabilityOpportunities.csv** file.

① Definition ② Sources ③ Targets ④ Input Parameters

Target Parameter Details

LocalCSVFile (P_LocalCSVConnection) Connection:* FF_Target (Flat File)

HighProbabilityOpportunities Object: HighProbabilityOpportunities.csv

Operation:* Insert

☐ Display target fields in alphabetical order

LowProbabilityOpportunities Object: LowProbabilityOpportunities.csv

Operation:* Insert

☐ Display target fields in alphabetical order

72. Click **Next**.

73. To define Target 1 Filter Condition (P_FirstFilterCondition) Parameter Details, click **New**.

① Definition ② Sources ③ Targets ④ Input Parameters

Target 1 Filter Condition (P_FirstFilterCondition) Parameter Details* New...

Target 2 Filter Condition (P_SecondFilterCondition) Parameter Details* New...

74. Enter the filter condition as shown below and click **OK**.
Probability > 70

Data Filter

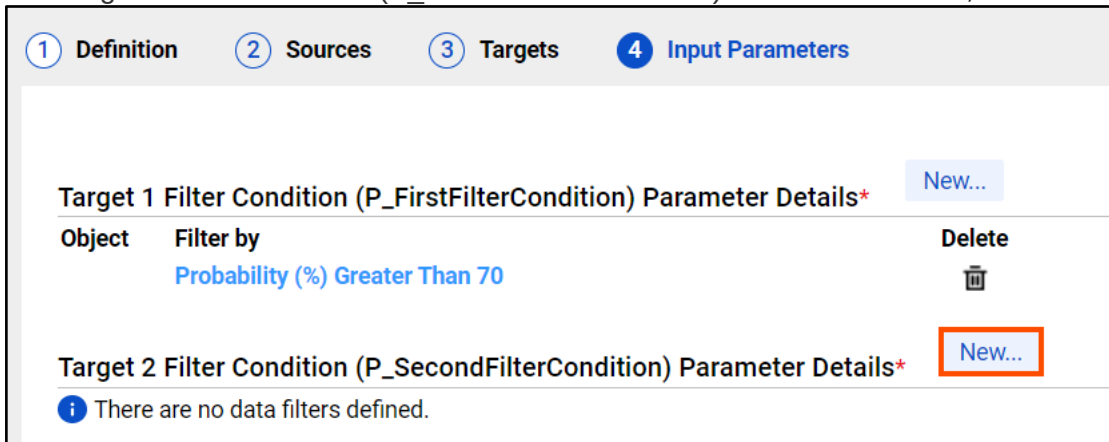
To define a data filter, select a field, select an operator, and then enter a filter value. Click OK to add the data filter to the task.

Filter by: Probability (%) Greater Than 70

Type: DOUBLE Example: 123456.78

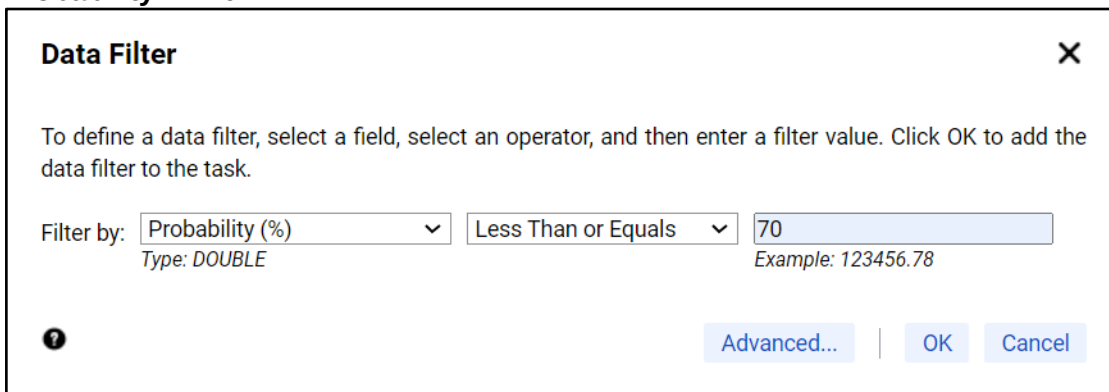
Advanced... OK Cancel

75. For Target 2 Filter Condition (P_SecondFilterCondition) Parameter Details, click **New**.



76. Enter the filter condition as shown below and click **OK**.

Probability <= 70



77. Click **Run**.

Monitor Status

78. To monitor the task status, navigate to the **My Jobs** page.

79. The task status changes to **Success**.

Note: You can refresh the job status if it does not change automatically.

80. Close the asset from the navigation pane.

Optional - Open the Dummy Mapping and View the Data

81. Navigate to your working folder and open the dummy mapping **m_XX_DUMMY_MAPPING**.

82. From the mapping canvas, select **Source** and go to the **Source** properties tab.

83. Change the Connection to your target Flat File connection. Click **Yes** in the Change Connection window.

84. Select the Object as your target – **HighProbabilityOpportunities.csv**, and preview the data.

Properties Preview  Source

General

Source

Fields

Partitions

Details

Connection:* FF_Tgt_Student_XX (Flat File) [View...](#) [New Connection...](#) [New Parameter...](#)

Source Type:* Single Object

Object:* **HighProbabilityOpportunities.csv** [Select...](#) [Formatting Options...](#) [Preview Data...](#)

Data Preview ✕

STAGENAME	AMOUNT	PROBABILITY	EXPECTEDREVENUE	TOTALOPPORTUNIT	CLOSEDATE	TYPE
Negotiation/Rev...	125000.0	90.0	112500.0		12/19/2023 00:...	Existing Cust
Closed Won	220000.0	100.0	220000.0		11/14/2023 00:...	New Custom
Closed Won	85000.0	100.0	85000.0		12/22/2023 00:...	New Custom
Proposal/Price ...	270000.0	75.0	202500.0		02/02/2024 00:...	Existing Cust
Closed Won	120000.0	100.0	120000.0		02/09/2024 00:...	Existing Cust
Closed Won	75000.0	100.0	75000.0		02/17/2024 00:...	New Custom
Closed Won	50000.0	100.0	50000.0		11/18/2023 00:...	New Custom
Closed Won	30000.0	100.0	30000.0		02/14/2024 00:...	Existing Cust
Negotiation/Rev...	270000.0	90.0	243000.0		12/15/2023 00:...	Existing Cust
Closed Won	50000.0	100.0	50000.0		12/09/2023 00:...	Existing Cust

[? Format](#) [Formatting Options...](#) [Done](#)

85. Similarly, observe the results of the target - **LowProbabilityOpportunities.csv**.

Properties Preview  Source

General

Source

Fields

Partitions

Details

Connection:* FF_Tgt_Student_XX (Flat File) [View...](#) [New Connection...](#) [New Parameter...](#)

Source Type:* Single Object

Object:* **LowProbabilityOpportunities.csv** [Select...](#) [Formatting Options...](#) [Preview Data...](#)

Data Preview
×

STAGENAME	AMOUNT	PROBABILITY	EXPECTEDREVENUE	TOTALOPPORTUNIT	CLOSEDATE	TYPE
Qualification	15000.0	10.0	1500.0		12/31/2023 00:...	New Custor
Id. Decision Ma...	15000.0	60.0	9000.0		11/09/2023 00:...	Existing Cu
Value Proposition	250000.0	50.0	125000.0		02/17/2024 00:...	Existing Cu
Prospecting	100000.0	10.0	10000.0		12/27/2023 00:...	
Value Proposition	80000.0	50.0	40000.0		11/12/2023 00:...	Existing Cu
Id. Decision Ma...	60000.0	60.0	36000.0		02/11/2024 00:...	
Perception Anal...	120000.0	70.0	84000.0		11/13/2023 00:...	Existing Cu
Needs Analysis	675000.0	20.0	135000.0		01/12/2024 00:...	Existing Cu
Id. Decision Ma...	35000.0	60.0	21000.0		02/23/2024 00:...	Existing Cu

?
Formatting Options...
Done

86. Save and close all the assets from the navigation pane.

This concludes the lab.

Module 9: Mapping Parameterization

Lab 9-2: Using In-Out Parameters for Incremental Data Loading

Overview:

In IICS, you can use an in-out parameter as a persistent task variable to manage an incremental data load.

In this lab, you will create a mapping with in-out parameter.

Objective:

- Use Input-Output parameters in a mapping

Scenario:

The Alaska outlet of NH suppliers is facing issues in maintaining sales data and wants to archive the older data. They also want to maintain the data based on the order Ids. So, they contact Ruby for help. Ruby discusses the issue with John, who suggests using the In-Out parameters in IICS for incremental data load.

In this lab, John uses the in-out parameters to define an orderid parameter in the mapping.

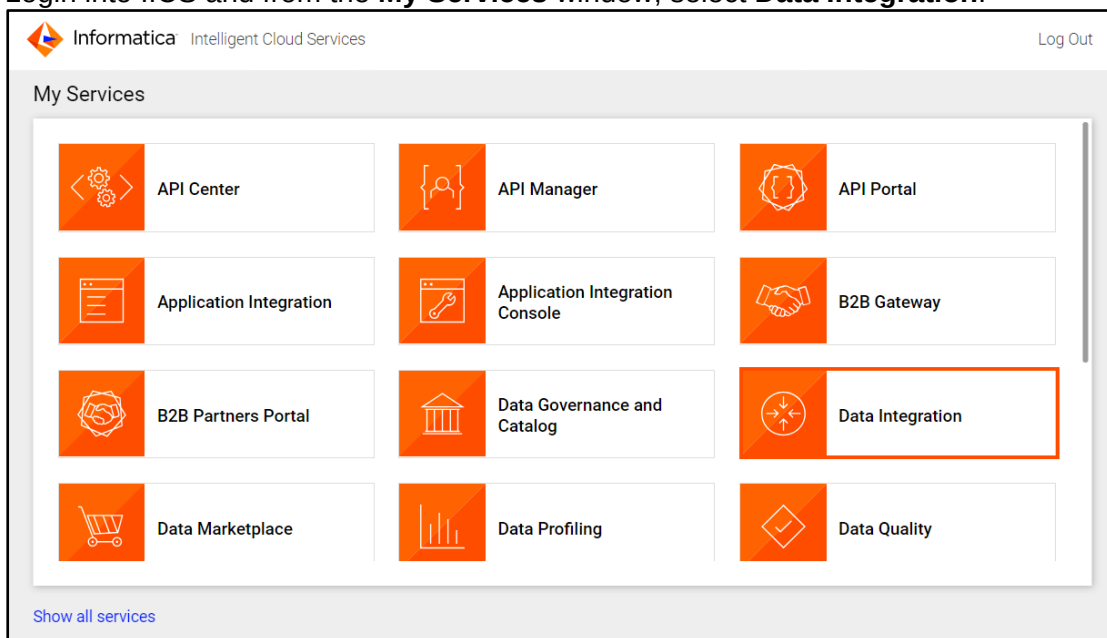
Duration:

15 minutes

Tasks

Create a Mapping

1. Login into IICS and from the **My Services** window, select **Data Integration**.



2. Create a new Mapping and name it **m_XX_InOutParameter**. Replace XX with your student number.

3. To create an input-output parameter, click .



4. From the In-Out Parameters section, click .

Parameters

×

Input Parameters

+

No Input Parameters.

In-Out Parameters

+

No In-Out Parameters.

5. Create the field as shown in the table below:

Name	Data Type	Precision	Default Value
p_orderid	integer	10	0

6. Click **OK**.

New In-Out Parameter

×

Name:*

p_orderid

Description:

Data Type:*

integer

Precision:*

10

Scale:

0

Default Value:

Retention Policy:*

On success or warning

Aggregation Type:*

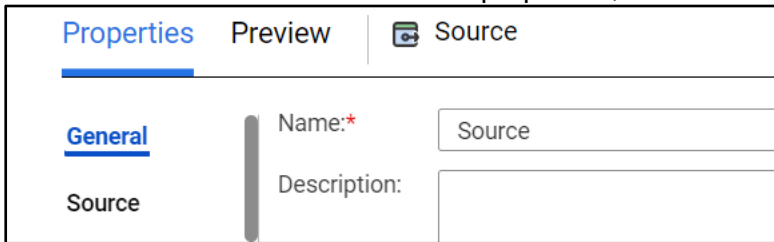
Max

?

OK

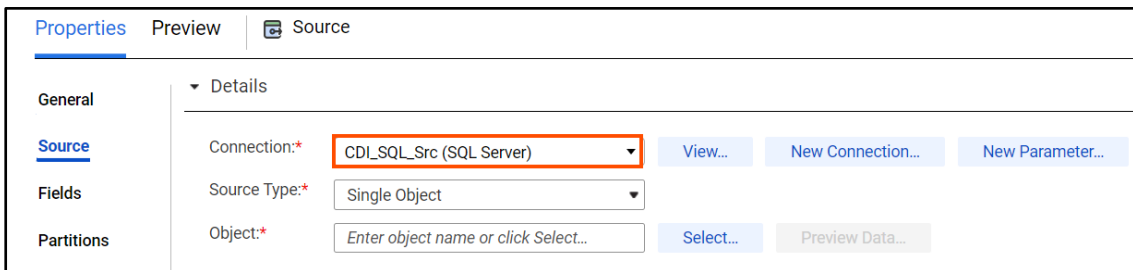
Cancel

7. Close the Parameters window.
8. To configure the source, from the mapping canvas, click the **Source** transformation.
9. In the General section of the Source properties, retain Name as **Source**.



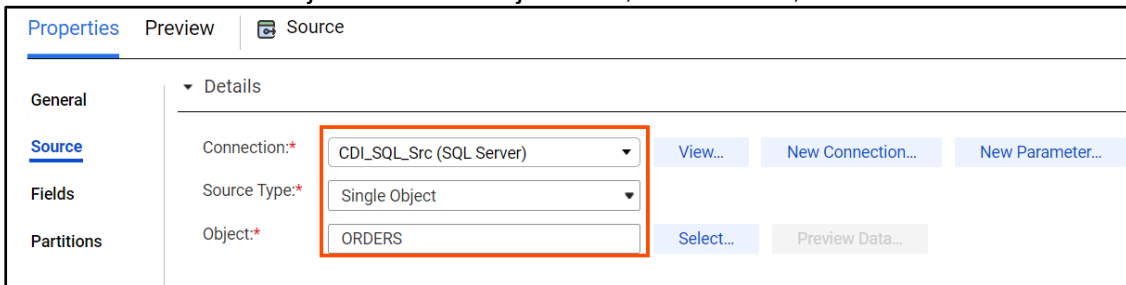
The screenshot shows the 'Source' transformation properties window. The 'General' tab is active. The 'Name' field is set to 'Source'. The 'Description' field is empty.

10. From the properties pane, click **Source**.
11. From the Connection drop-down, select **CDI_SQL_Src**.



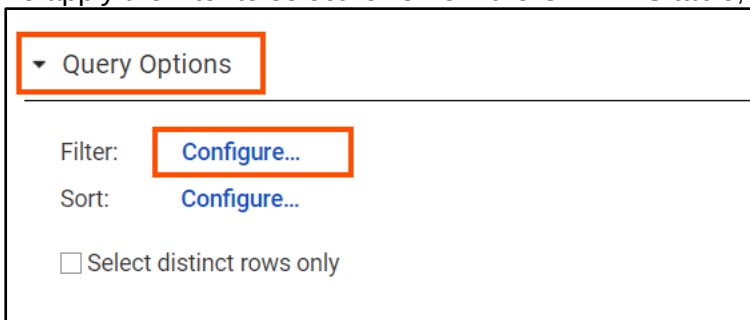
The screenshot shows the 'Source' transformation properties window. The 'Source' tab is active. The 'Connection' drop-down is set to 'CDI_SQL_Src (SQL Server)'. The 'Source Type' is set to 'Single Object'. The 'Object' field is empty.

12. To select the source object from the Object field, click **Select**, and choose ORDERS.



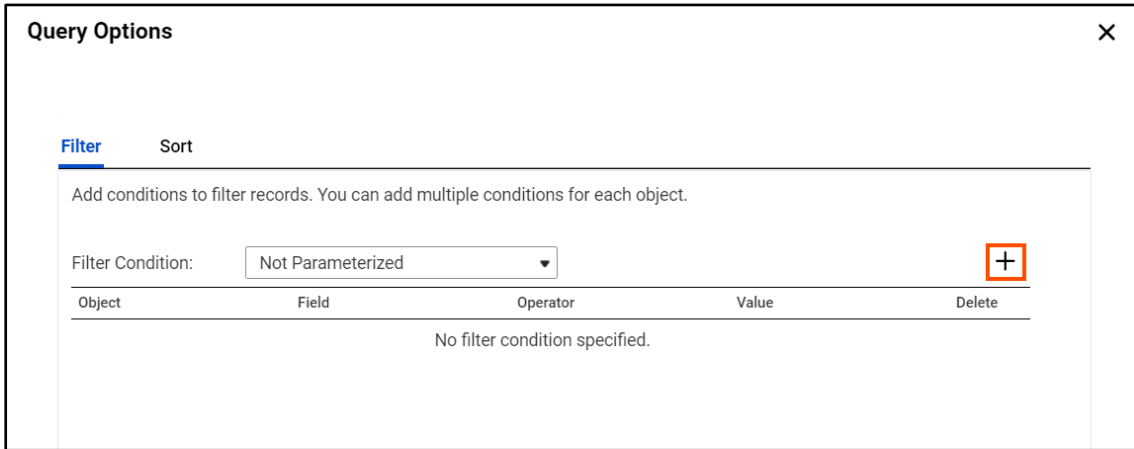
The screenshot shows the 'Source' transformation properties window. The 'Object' field is set to 'ORDERS'. The 'Select...' button is highlighted.

13. Scroll down and expand **Query Options**.
14. To apply the filter to select rows from the ORDERS table, click **Configure**.



The screenshot shows the 'Query Options' section. The 'Filter' button is highlighted. The 'Sort' button is also visible. The 'Select distinct rows only' checkbox is unchecked.

15. To add a filter condition, click .



Query Options [X]

Filter | Sort

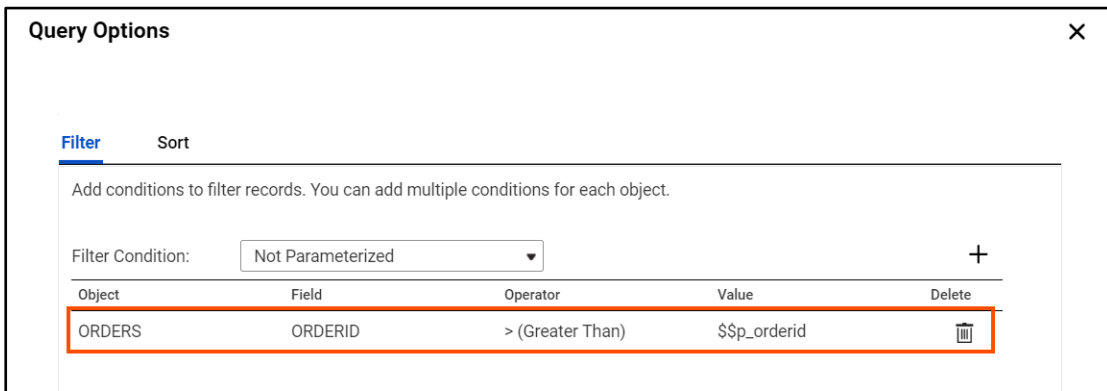
Add conditions to filter records. You can add multiple conditions for each object.

Filter Condition: Not Parameterized 

Object	Field	Operator	Value	Delete
No filter condition specified.				

16. Enter the details as shown in the table below:

Object	Field	Operator	Value
ORDERS	ORDERID	Greater Than (>)	\$\$p_orderid



Query Options [X]

Filter | Sort

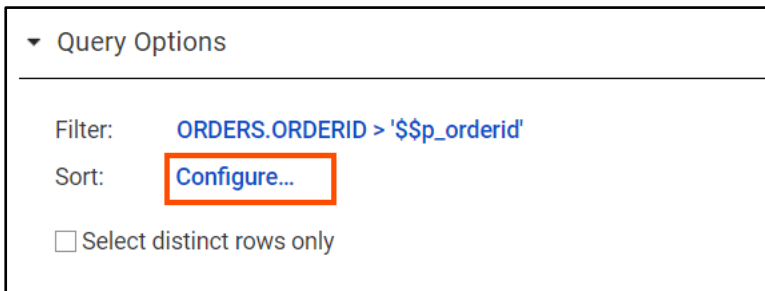
Add conditions to filter records. You can add multiple conditions for each object.

Filter Condition: Not Parameterized 

Object	Field	Operator	Value	Delete
ORDERS	ORDERID	> (Greater Than)	\$\$p_orderid	

17. Click **OK**.

18. To configure the sort condition, click **Configure**.



Query Options

Filter: ORDERS.ORDERID > '\$\$p_orderid'

Sort: Configure...

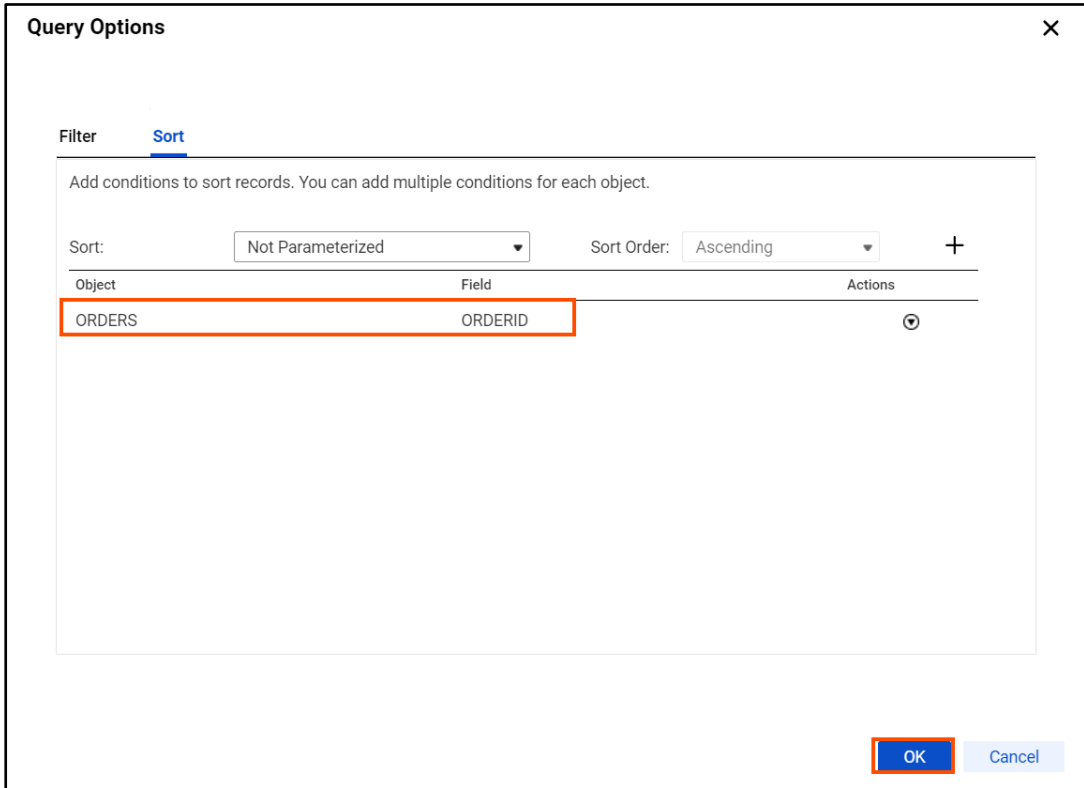
☐ Select distinct rows only

19. To add a sort condition, click .

20. Enter the details as shown in the table below:

Object	Field
ORDERS	ORDERID

21. Click **OK**.



Query Options [X]

Filter **Sort**

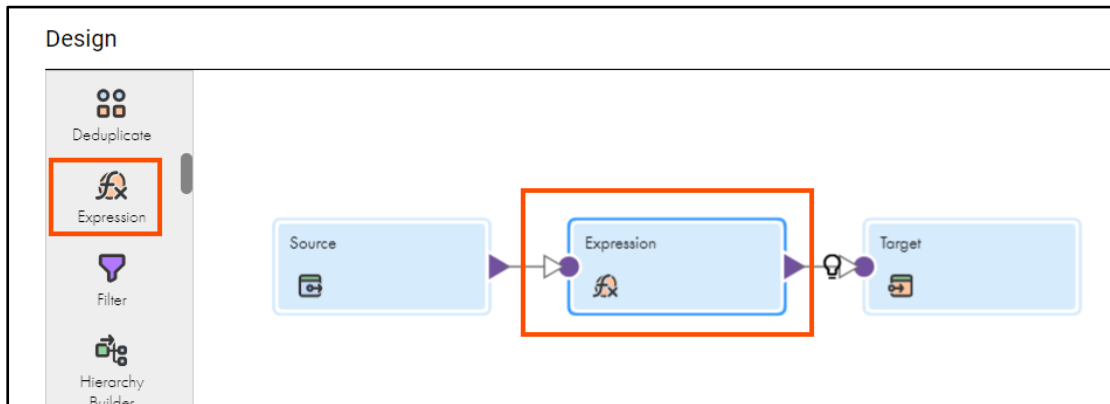
Add conditions to sort records. You can add multiple conditions for each object.

Sort: Not Parameterized Sort Order: Ascending +

Object	Field	Actions
ORDERS	ORDERID	⌵

OK Cancel

22. From the list of available transformations, drag and drop an **Expression** transformation on the link between **Source** and **Target**.



23. Select the **Expression** transformation from the mapping canvas.

24. From the properties pane, click **Expression**.

25. To add a new expression, click **+**.

26. Enter the details as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	P_ORDERID	integer	10	0

New Field

✕

Create new output field, variable field, input macro field or output macro field.

Field Type: Output Field

Name:* P_ORDERID

Type:* integer

Precision:* 10

Scale: 0

Default Value: ERROR('transformation error')

Description:

?
OK
Cancel

27. To configure the expression, click **Configure**.

Properties

Preview

Expression

General

Incoming Fields

Expression

Window

Advanced

Create simple expressions. You can also use expression macros to create complex expressions.

☐ Allow additional fields and expressions during task creation

Expressions

Field Name	Expression	Default Value	Type	Precision	Scale	Field Description
P_ORDERID	Configure...	ERROR('transformation error')	integer	10	0	

28. In the Expression field, copy and paste the following expression:
SetMaxVariable(\$\$p_orderid,ORDERID)

29. Validate the expression and click **OK**.

Field Expression: P_ORDERID(integer, 10, 0) ✕

Configure the expression by adding fields, functions, and variables. Use Ctrl + Space to see a list of autocomplete options.

Expression: Not Parameterized

✓ The expression is valid.

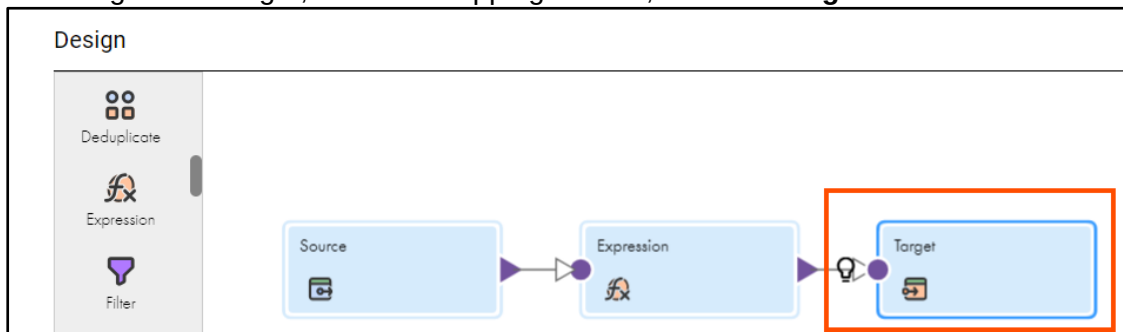
Fields	Expression
ORDERID	1 SetMaxVariable(\$\$p_orderid,ORDERID)
QUANTITY	
CUSTOMERID	
SHIPMENTTRACKINGNO	
ORDERSTATUS	
PRODUCTDESCRIPTION	
OUTLETID	

Select a field to see detailed information about it.

Validate

OK Cancel

30. To configure the target, from the mapping canvas, click the **Target** transformation.



31. From the properties pane, click **Target**.

32. From the Connection drop-down, select your **target Flat File** connection.

33. To select the target object from the Object field, click **Select**.

Properties Preview Target

General

Incoming Fields

Target

Target Fields

Field Mapping

Details

Connection:* FF_Target (Flat File) View... New Connection... New Parameter...

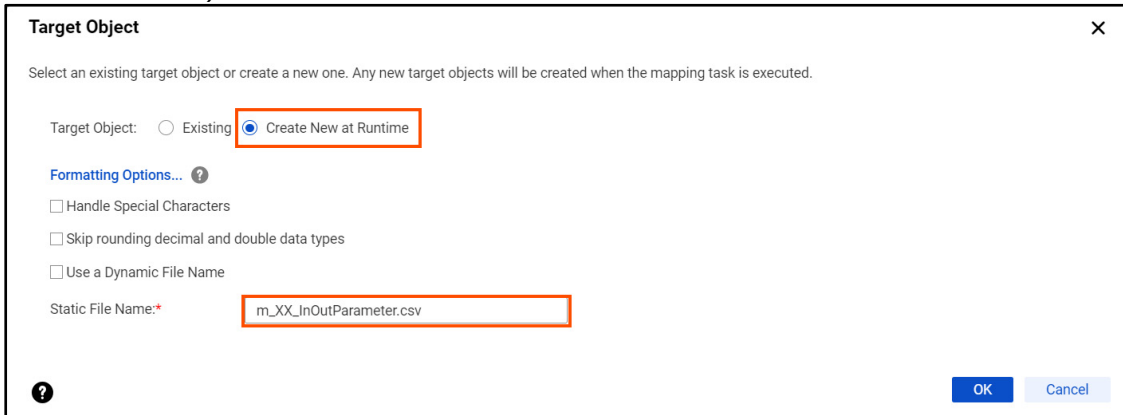
Target Type:* Single Object

Object:* Enter object name or click Select... Select... Formatting Options... Preview Data...

Operation: Insert

34. In the Target Object window, select **Create New at Runtime**.

35. In the Table Name field, enter **m_XX_InOutParameter.csv** (replace XX with your student number) and click **OK**.



Target Object [X]

Select an existing target object or create a new one. Any new target objects will be created when the mapping task is executed.

Target Object: ☐ Existing ☒ Create New at Runtime

[Formatting Options...](#) [?]

☐ Handle Special Characters

☐ Skip rounding decimal and double data types

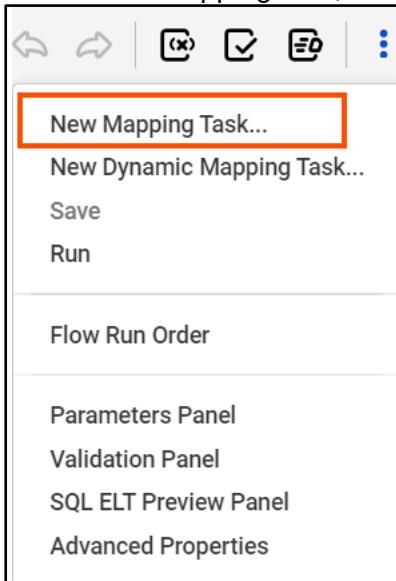
☐ Use a Dynamic File Name

Static File Name:*

[?] [OK] [Cancel]

36. **Save** the mapping.

37. To create a mapping task, click on the ellipsis icon and select **New Mapping Task**.



Navigation icons: [Back] [Forward] [Cancel] [Check] [Edit] [More]

- New Mapping Task...**
- New Dynamic Mapping Task...
- Save
- Run

Flow Run Order

Parameters Panel

Validation Panel

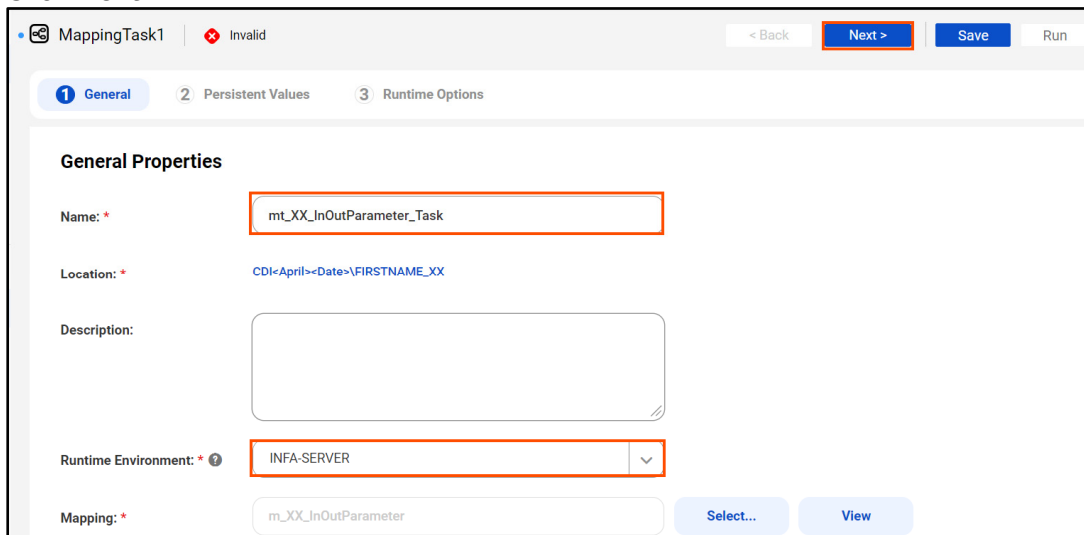
SQL ELT Preview Panel

Advanced Properties

38. In the Task Name field, enter **mt_XX_InOutParameter_Task**. Replace XX with your student number.

39. From the Runtime Environment drop-down, select **INFA-SERVER**.

40. Click **Next**.



MappingTask1 Invalid < Back **Next >** Save Run

1 General 2 Persistent Values 3 Runtime Options

General Properties

Name: * mt_XX_InOutParameter_Task

Location: * CDI<April>=<Date>\FIRSTNAME_XX

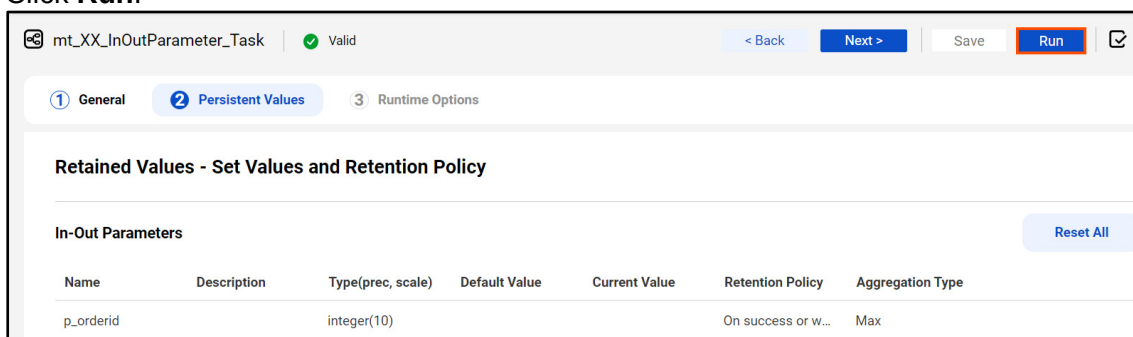
Description:

Runtime Environment: * INFA-SERVER

Mapping: * m_XX_InOutParameter Select... View

41. Retain the In-Out Parameters configuration and click **Save**.

42. Click **Run**.



mt_XX_InOutParameter_Task Valid < Back **Next >** Save **Run** ✓

1 General 2 **Persistent Values** 3 Runtime Options

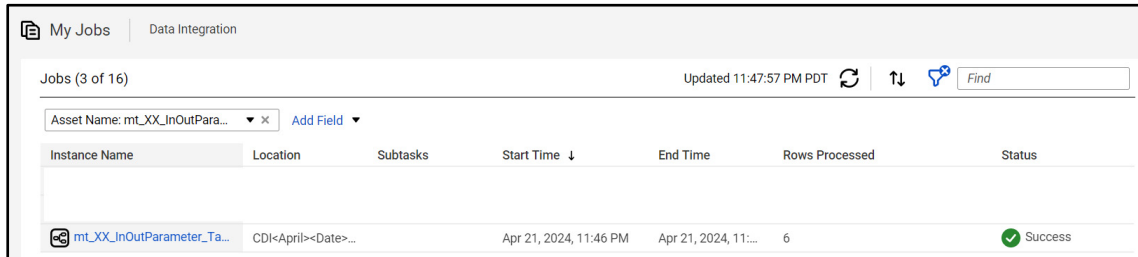
Retained Values - Set Values and Retention Policy

In-Out Parameters Reset All

Name	Description	Type(prec, scale)	Default Value	Current Value	Retention Policy	Aggregation Type
p_orderid		integer(10)			On success or w...	Max

Monitor Status

43. Monitor the job status and verify that 6 rows are processed by the mapping.



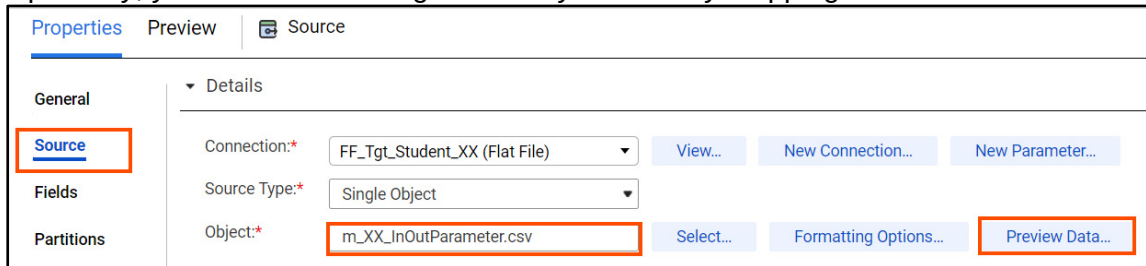
My Jobs | Data Integration

Jobs (3 of 16) Updated 11:47:57 PM PDT

Asset Name: mt_XX_InOutPara... Add Field

Instance Name	Location	Subtasks	Start Time ↓	End Time	Rows Processed	Status
mt_XX_InOutParameter-Ta...	CDI<April><Date>...		Apr 21, 2024, 11:46 PM	Apr 21, 2024, 11:...	6	Success

44. Optionally, you can view the target data in your dummy mapping.



Properties | Preview | Source

General

Source

Fields

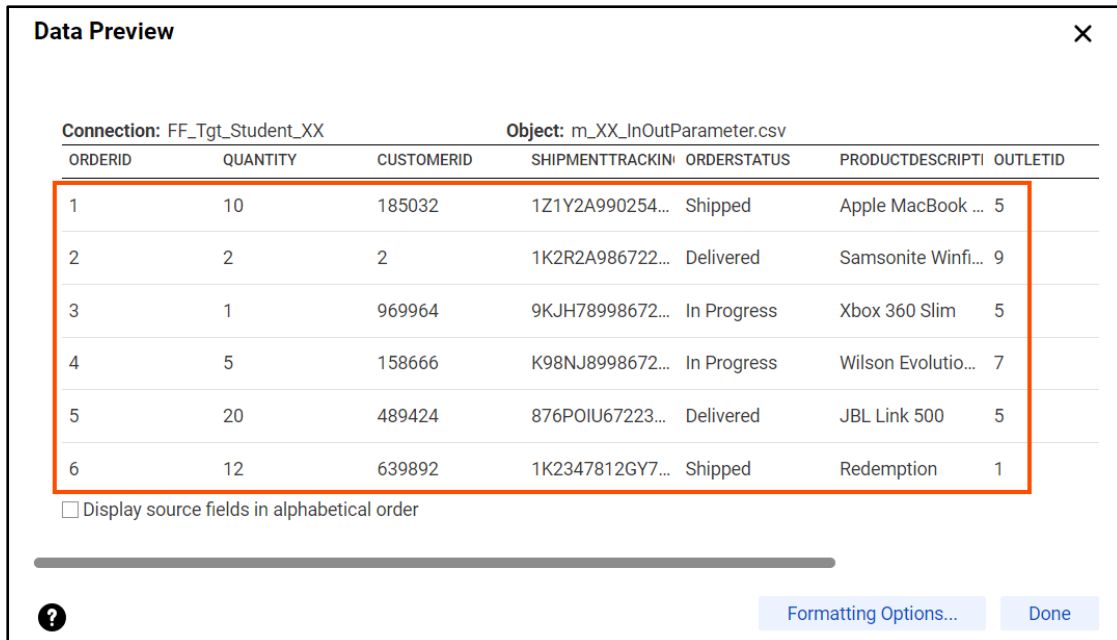
Partitions

Details

Connection:* FF_Tgt_Student_XX (Flat File) View... New Connection... New Parameter...

Source Type:* Single Object

Object:* m_XX_InOutParameter.csv Select... Formatting Options... Preview Data...



Data Preview

Connection: FF_Tgt_Student_XX Object: m_XX_InOutParameter.csv

ORDERID	QUANTITY	CUSTOMERID	SHIPMENTTRACKIN	ORDERSTATUS	PRODUCTDESCRIPTI	OUTLETID
1	10	185032	1Z1Y2A990254...	Shipped	Apple MacBook ...	5
2	2	2	1K2R2A986722...	Delivered	Samsonite Winfi...	9
3	1	969964	9KJH78998672...	In Progress	Xbox 360 Slim	5
4	5	158666	K98NJ8998672...	In Progress	Wilson Evolutio...	7
5	20	489424	876POIU67223...	Delivered	JBL Link 500	5
6	12	639892	1K2347812GY7...	Shipped	Redemption	1

☐ Display source fields in alphabetical order

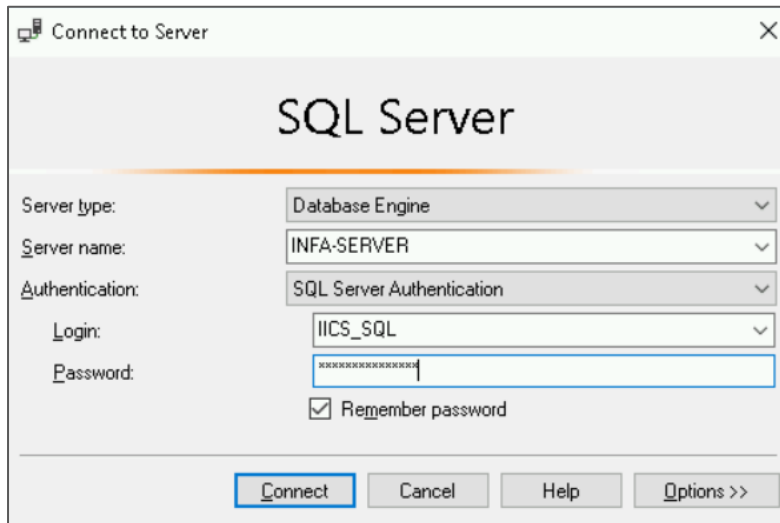
Formatting Options... Done

45. Save and close all the assets from the navigation pane.

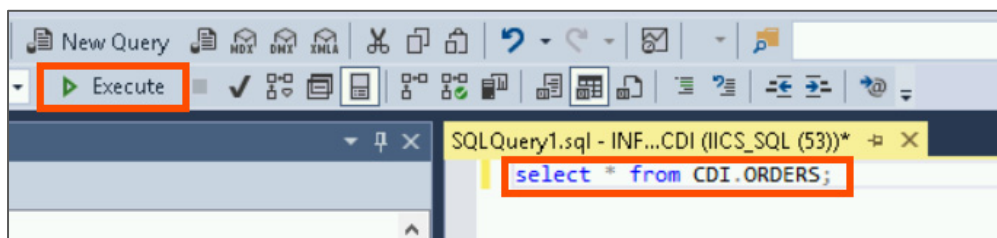
MANDATORY – Instructor Demonstration

This section has to be executed only by the instructors and NOT by the students (steps 46 to 54). Hence, all the students must wait till the instructor inserts a new record into the database table.

46. To verify the incremental load of orders data, from the Windows Start menu, open **SQL Server Management Studio**.
47. From the **Connections** section, enter the Login and Password as **IICS_SQL**.
48. Click **Connect**.



49. Expand the **Databases > CDI > Tables** and click on **CDI.ORDERS** table.
50. Click **New Query**.
51. To check the existing data in the **CDI.ORDERS** table, execute the below query.
select * from CDI.ORDERS;



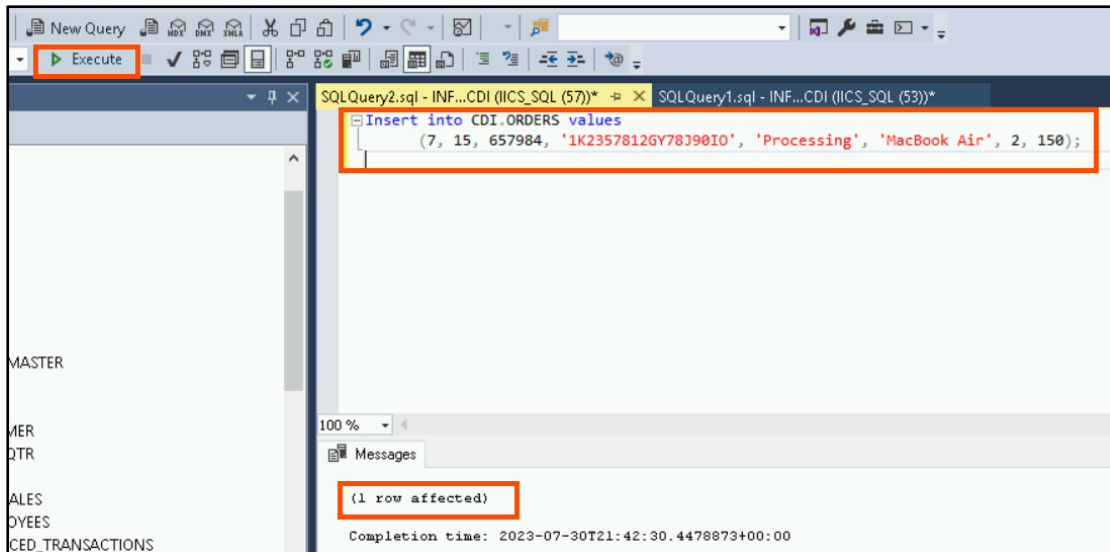
52. Add a new row and enter the below data.

ORDER ID	QUANTIT Y	CUSTOMER ID	SHIPMENT TRACKING NO	ORDER STATUS	PRODUCT DESCRIPTION	OUTLET ID	PRODUCT ID
7	15	657984	1K2357812GY78J90IO	Processing	MacBook Air	2	150

For Adding the new row, use the query shown below in a new query window:

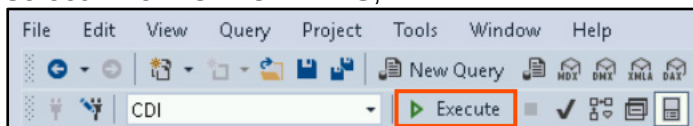
Insert into CDI.ORDERS values

(7, 15, 657984, '1K2357812GY78J90IO', 'Processing', 'MacBook Air', 2, 150);



53. Now, to check the newly inserted row, navigate back to the previous query window and re-run the command or enter the below query and click **Execute**.

select * from CDI.ORDERS;



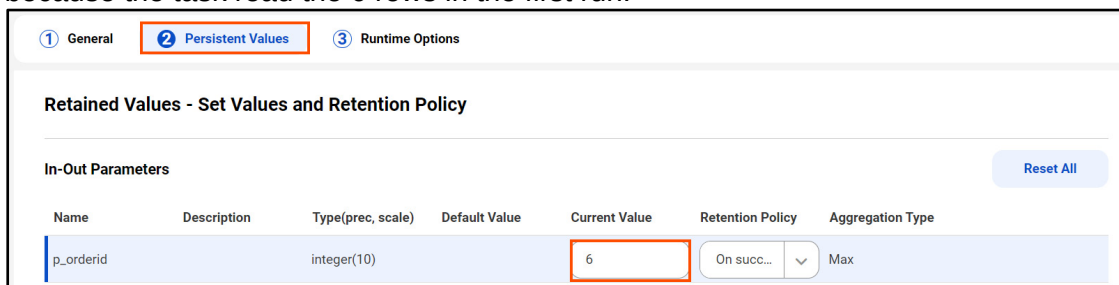
54. A new row is inserted in the table.

	ORDERID	QUANTITY	CUSTOMERID	SHIPMENTTRACKINGNO	ORDERSTATUS	PRODUCTDESCRIPTION	OUTLETID	PRODUCTID
1	1	10	185032	1Z1Y2A990254347812	Shipped	Apple MacBook Pro	5	109
2	2	2	2	1K2R2A986722347812	Delivered	Samsonite Winfield Luggage	9	140
3	3	1	969964	9KJH7899867234HJ8Y	In Progress	Xbox 360 Slim	5	136
4	4	5	158666	K98NJ899867234P009	In Progress	Wilson Evolution Game Basketball	7	127
5	5	20	489424	876P0IU672234712H5	Delivered	JBL Link 500	5	122
6	6	12	639892	1K2347812GY78J90I0	Shipped	Redemption	1	130
7	7	15	657984	1K2357812GY78J90I0	Processing	MacBook Air	2	150

Student Tasks (continued)

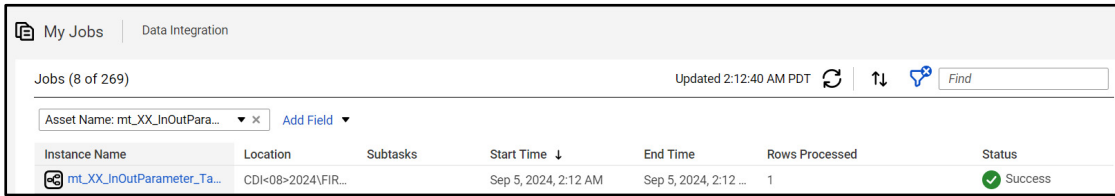
55. Navigate back to IICS, from your working directory, and edit the **mt_XX_InOutParameter_Task** asset.

56. Go to **Persistent Values** step and verify that the **Current Value** is set to **6**. This is because the task read the 6 rows in the first run.



57. Click **Save** and **Run** the task again.

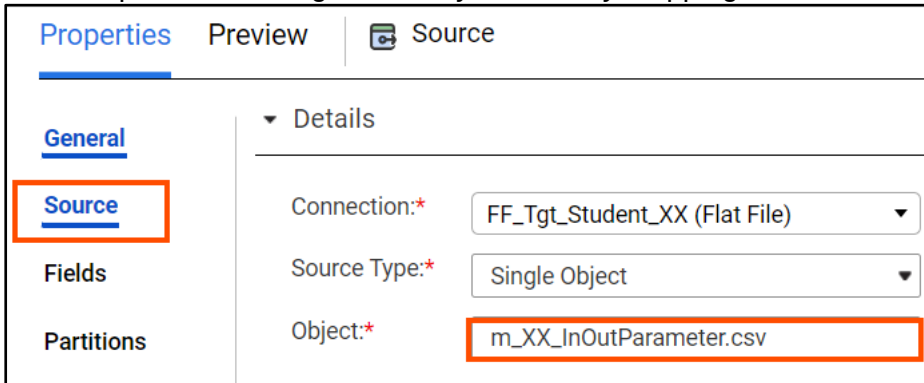
58. Go to the **My Jobs** page to verify the task status. Notice that only 1 row is processed in the second run of the task.



Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
mt_XX_InOutParameter_Ta...	CDI-08>2024\FIR...		Sep 5, 2024, 2:12 AM	Sep 5, 2024, 2:12 ...	1	Success

Optional

59. You can preview the target data in your dummy mapping.

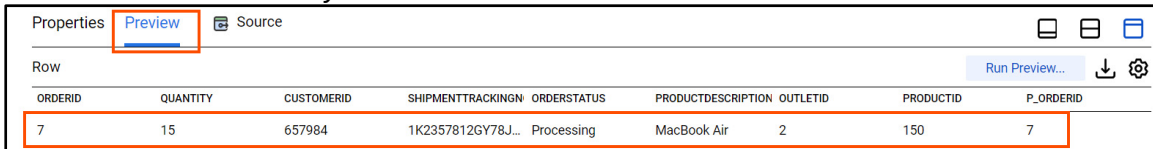


Properties | Preview | Source

General
Source
 Fields
 Partitions

Details
 Connection:* FF_Tgt_Student_XX (Flat File)
 Source Type:* Single Object
 Object:* m_XX_InOutParameter.csv

60. To view the updated target records, navigate to the Preview tab and click **Run Preview**. Observe that there is only 1 record.



ORDERID	QUANTITY	CUSTOMERID	SHIPMENTTRACKINGN	ORDERSTATUS	PRODUCTDESCRIPTION	OUTLETID	PRODUCTID	P_ORDERID
7	15	657984	1K2357812GY78J...	Processing	MacBook Air	2	150	7

61. Save and close all the assets.

This concludes the lab.

Module 10: Mapping Task

Lab 10-1: Creating a Mapping Task to Perform Basic History Tracking And Configure Advanced Settings

Overview:

The Mapping task allows you to process data based on the data flow logic defined in a Mapping. When you create a mapping task, you select the mapping for the task to use. On the Schedule page of the mapping task, you can define to run a mapping task manually or schedule it to run at a specific time or interval.

Objective:

- Create a Mapping to perform basic history tracking
- Create a Mapping Task
- Configure Email Notifications

Scenario:

Ruby wants to track the incoming data records and update the changes accordingly.

The process of updating or inserting records efficiently involves using the target table as a lookup to determine if a record represents a new or existing customer. This requires two instances of the target table in the mapping. New customer records are routed to a target object for INSERT operations, with a Sequence Generator providing surrogate keys. Existing customer records are routed to a target object for UPDATE operations, with any data changes captured. An Expression transformation outputs a timestamp for record currency. The Sequence Generator saves the last surrogate key value in the cloud repository for the next run, requiring a Mapping Task for this functionality.

Later, Ruby wants to monitor the job statuses of a few important jobs timely for 48 hours. Instead of verifying whether the job status is success or fail, Ruby will configure the email notifications property to get notified about the job statuses.

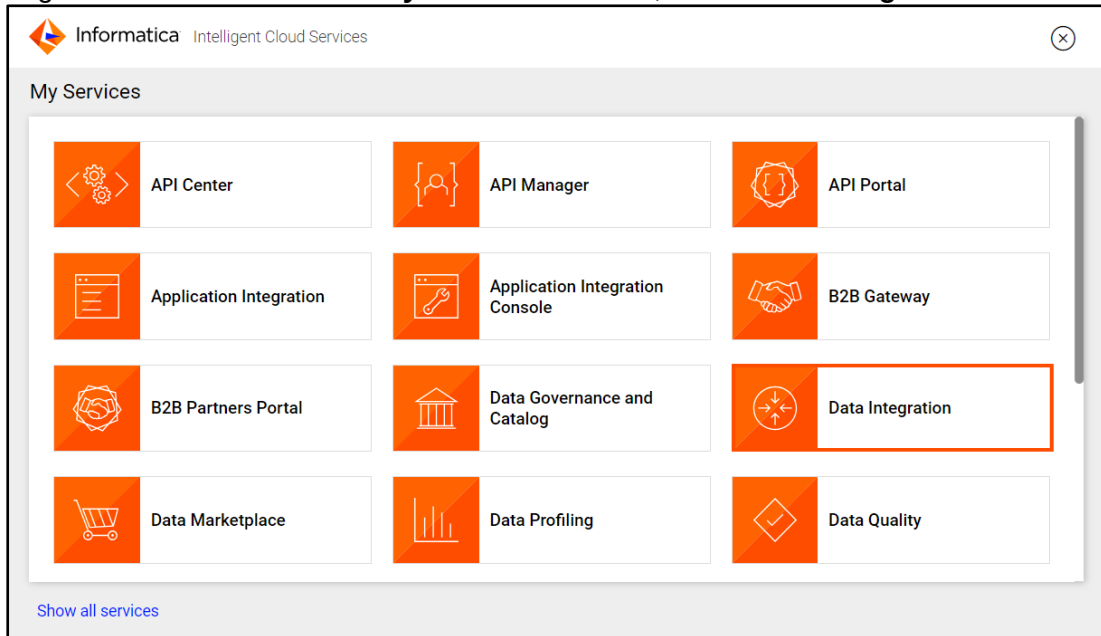
Duration:

40 minutes

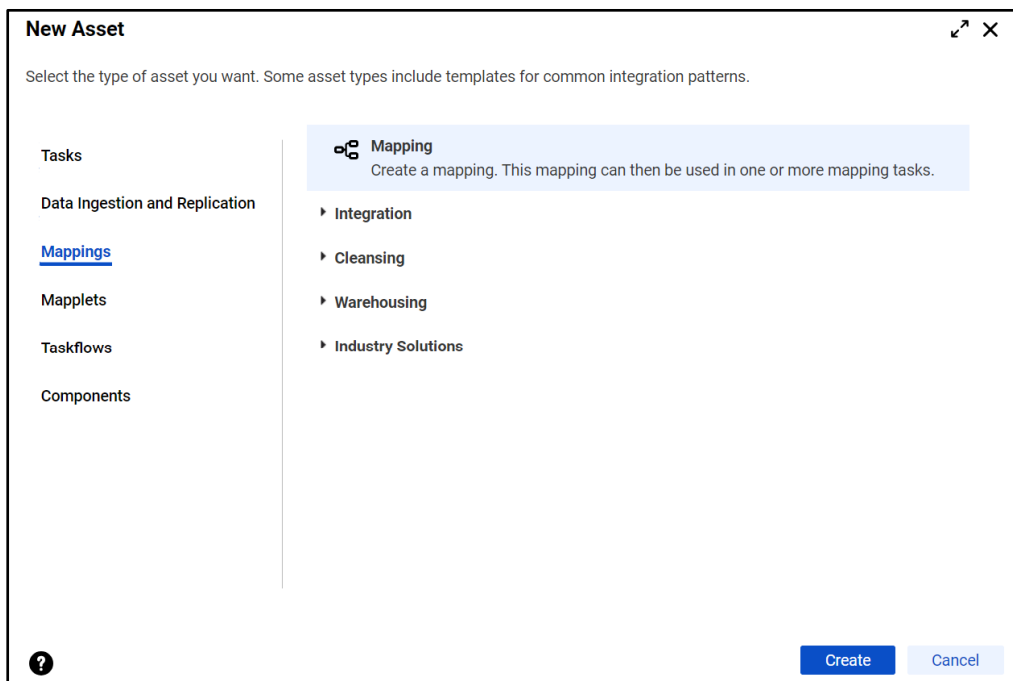
Tasks

Create a Mapping

1. Login into IICS and from the **My Services** window, select **Data Integration**.



2. To create a new asset, from the navigation pane, select **New**.
3. From the Mappings tab, select **Mapping** and click **Create**.



4. In the Name field of the mapping properties, enter **BASIC_HISTORY_TRACKING**.

- Verify that the asset Location is pointing to your working folder.

BASIC_HISTORY_TRACKING	
Name:*	BASIC_HISTORY_TRACKING
Location:*	CDI<April><Date>\FIRSTNAME_XX
Description:	

- To configure the source, from the mapping canvas, click the **Source** transformation.
- In the **General** section of Source properties, enter Name as **CUSTOMERS**.

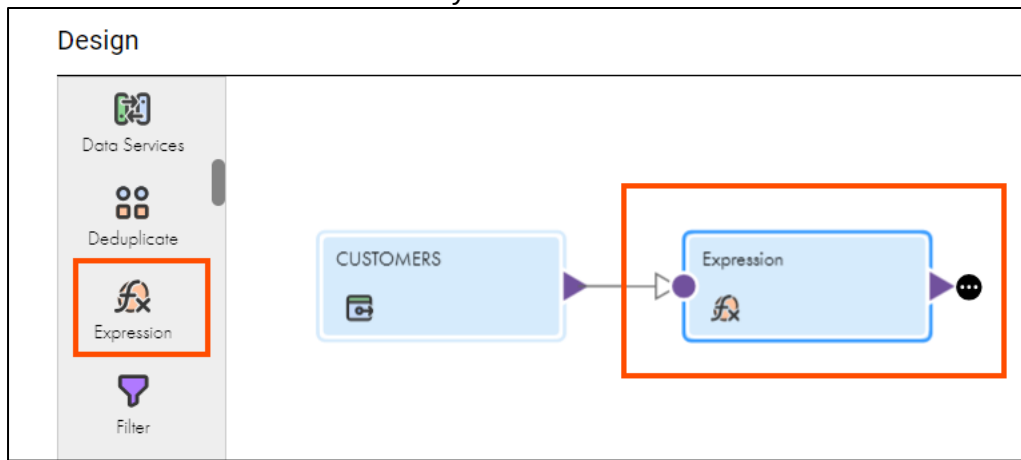
Properties	Preview	CUSTOMERS
General		
Source	Name:*	CUSTOMERS
	Description:	

- Delete** the link between Source and Target transformations in the mapping canvas.
- From the properties pane, click **Source**.
- In the Connection drop-down, select **FF_Source**.
- From the Object field, select **INITIAL_CUSTOMERS.CSV**.

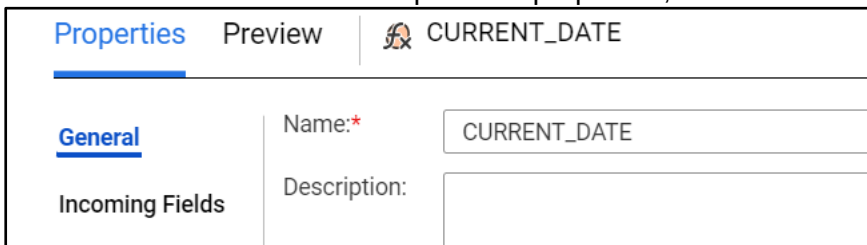
Properties	Preview	CUSTOMERS
General	Details	
Source	Connection: FF_Source (Flat File) View... New Connection... New Parameter... Source Type: Single Object Object: INITIAL_CUSTOMERS.CSV Select... Formatting Options... Preview Data...	
Fields		
Partitions		

Add Expression Transformation

12. From the list of available transformations, drag and drop the **Expression** transformation and link **CUSTOMERS** to the newly added transformation.



13. Select **Expression** transformation from the mapping canvas.
14. In the General section of the Expression properties, enter Name as **CURRENT_DATE**.



The screenshot shows the 'Properties' dialog box for the 'Expression' transformation. The 'General' tab is selected. The 'Name' field is set to 'CURRENT_DATE'. The 'Description' field is empty. The 'Incoming Fields' section is also empty.

15. From the properties pane, click **Expression**.
16. To add a new expression, click  .
17. Enter the details as shown in the table below and click **OK**.

Field Type	Name	Type	Precision	Scale
Output Field	CURRENT_DATE	date/time	29	9

New Field

✕

Create new output field, variable field, input macro field or output macro field.

Field Type:

Name:*

Type:*

Precision:*

Scale:

Default Value:

Description:

?

18. To configure the expression, click **Configure**.

Properties Preview CURRENT_DATE

General
Incoming Fields
Expression
Window
Advanced

Create simple expressions. You can also use expression macros to create complex expressions.

☐ Allow additional fields and expressions during task creation

Expressions

Field Name	Expression	Default Value	Type	Precision	Scale	Field Description
CURRENT_DATE	Configure...	ERROR('transformation error')	date/time	29	9	

19. In the Expression field, copy and paste the following expression:
SYSDATE

20. Click **Validate** and then click **OK**.

Field Expression: CURRENT_DATE(date/time, 29, 9) X

Configure the expression by adding fields, functions, and variables. Use Ctrl + Space to see a list of autocomplete options.

Expression: Not Parameterized

✓ The expression is valid.

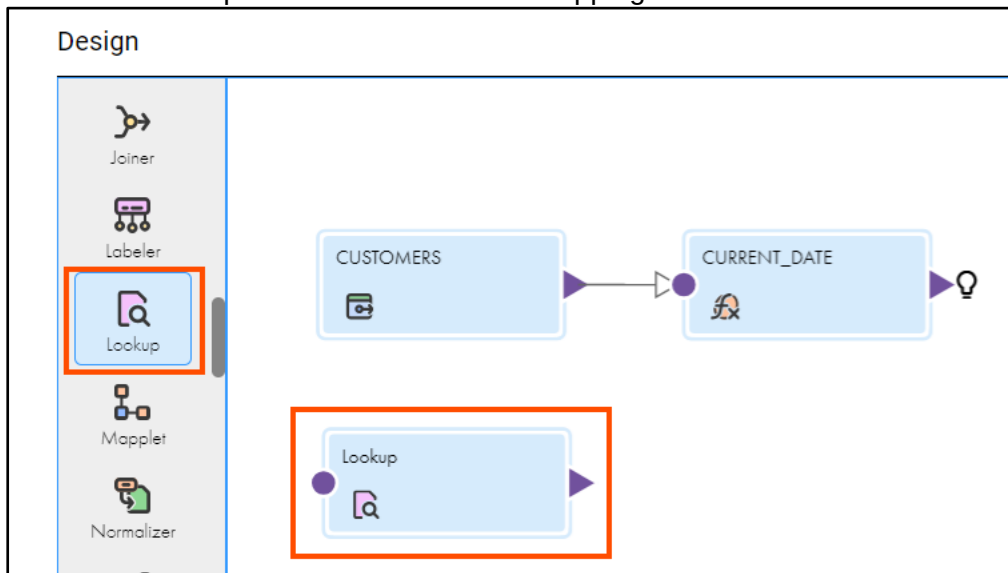
Fields	Expression
CUST_ID	1 SYSDATE
FIRST_NAME	
LAST_NAME	
ADDR1	
ADDR2	
CITY	
STATE	

Select a field to see detailed information about it.

Validate
OK Cancel

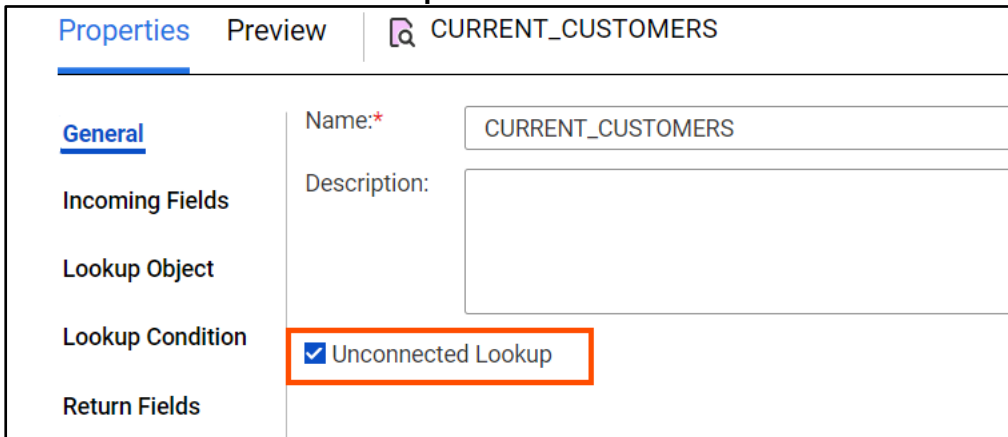
Add Lookup Transformation

21. From the list of available transformations, drag and drop a **Lookup** transformation on the mapping canvas.
22. Select the Lookup transformation on the mapping canvas.



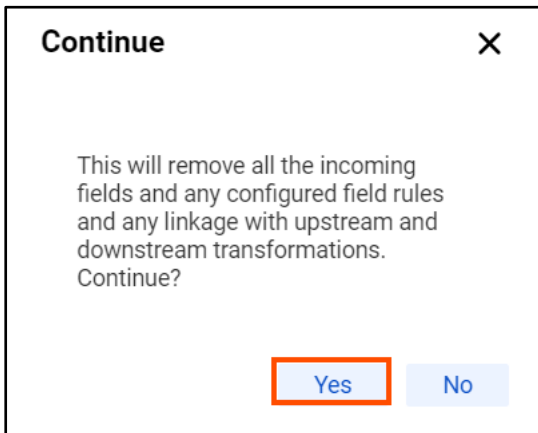
23. In the **General** section of Lookup properties, enter Name as **CURRENT_CUSTOMERS**.

24. Select the **Unconnected Lookup** checkbox



The screenshot shows the 'Properties' pane for the 'CURRENT_CUSTOMERS' object. The 'General' tab is selected. The 'Name' field is 'CURRENT_CUSTOMERS'. The 'Description' field is empty. The 'Lookup Condition' section has the 'Unconnected Lookup' checkbox checked, which is highlighted with a red rectangular box. Other options like 'Incoming Fields', 'Lookup Object', and 'Return Fields' are visible in the left sidebar.

25. In the **Continue** window, select **Yes**.



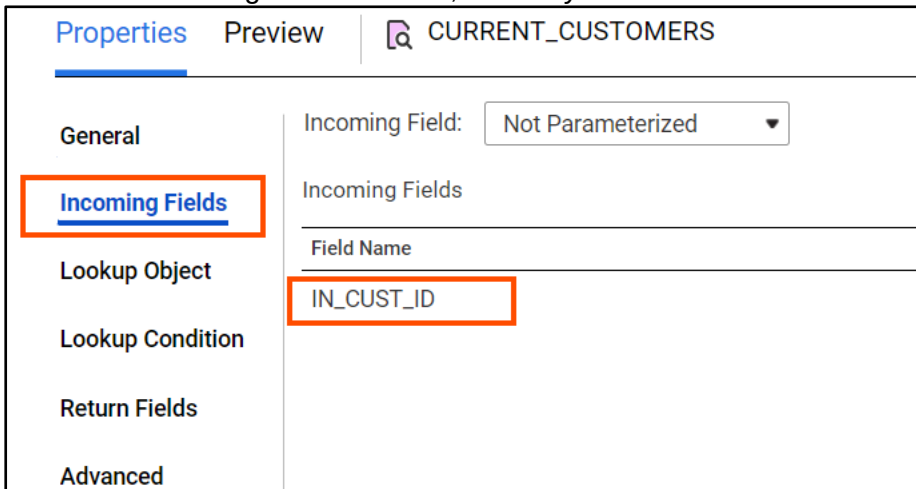
The screenshot shows a 'Continue' dialog box with a close button (X) in the top right. The text inside reads: 'This will remove all the incoming fields and any configured field rules and any linkage with upstream and downstream transformations. Continue?'. At the bottom, there are two buttons: 'Yes' and 'No'. The 'Yes' button is highlighted with a red rectangular box.

26. From the properties pane, click **Incoming Fields**.

27. To add a new incoming field, click .

28. In Field Name, enter **IN_CUST_ID**.

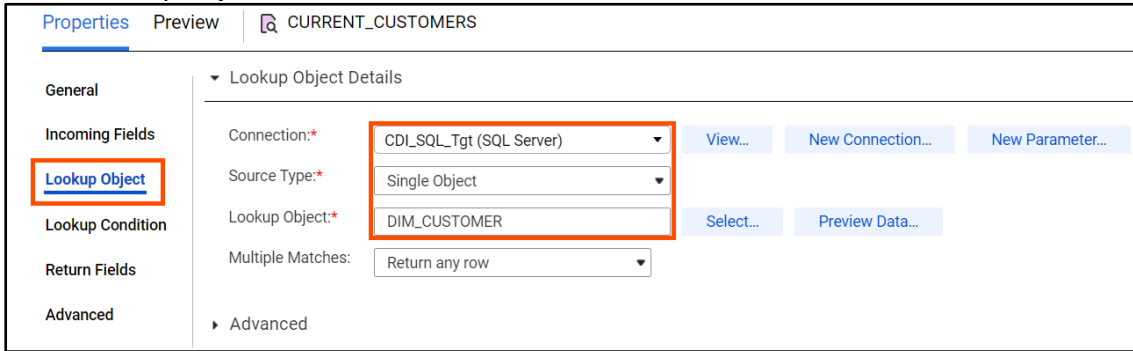
Note: After entering the field value, click anywhere outside the field to save the details.



The screenshot shows the 'Properties' pane for 'CURRENT_CUSTOMERS' with the 'Incoming Fields' tab selected. The 'Incoming Field' dropdown is set to 'Not Parameterized'. Below it, the 'Incoming Fields' section has a 'Field Name' input field containing 'IN_CUST_ID', which is highlighted with a red rectangular box. The left sidebar shows 'Incoming Fields' as the selected option.

29. From the properties pane, click **Lookup Object**.

30. From the Connection drop-down, select your target SQL connection **CDI_SQL_Tgt**.
31. Select lookup object as **DIM_CUSTOMER**.



Properties Preview CURRENT_CUSTOMERS

General

Lookup Object Details

Connection:* CDI_SQL_Tgt (SQL Server) View... New Connection... New Parameter...

Source Type:* Single Object

Lookup Object:* DIM_CUSTOMER Select... Preview Data...

Multiple Matches: Return any row

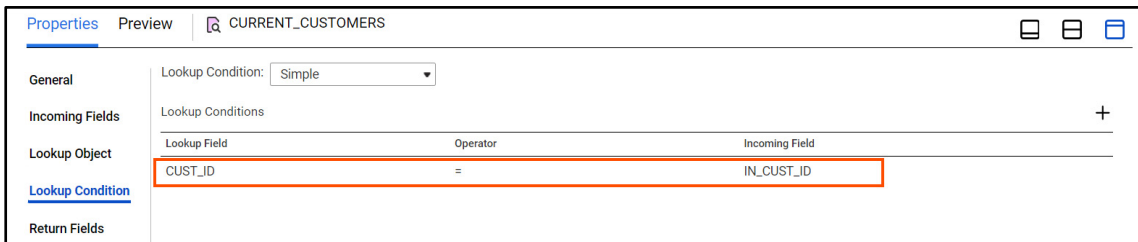
Advanced

32. From the properties pane, click **Lookup Condition**.

33. To add a new lookup condition, click **+**.

34. Set the lookup condition as shown in the table below:

Lookup Field	Operator	Incoming Field
CUST_ID	=	IN_CUST_ID



Properties Preview CURRENT_CUSTOMERS

General

Lookup Condition: Simple

Lookup Conditions

Lookup Field	Operator	Incoming Field
CUST_ID	=	IN_CUST_ID

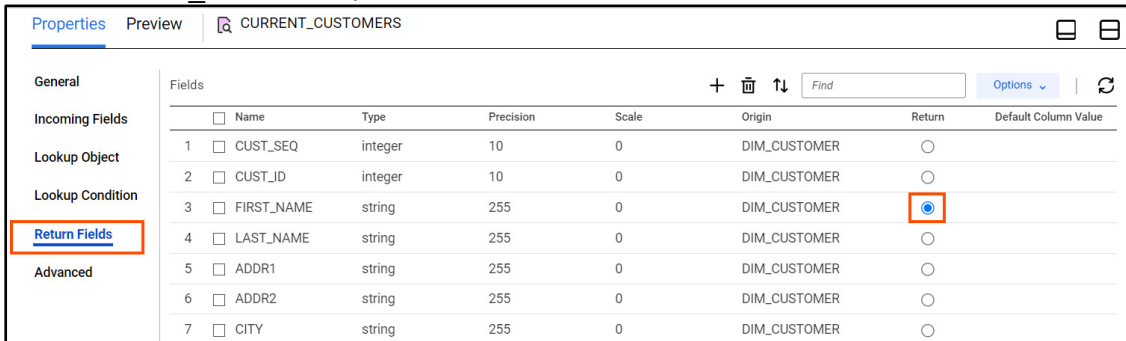
Lookup Object

Lookup Condition

Return Fields

35. From the properties pane, click **Return Fields**.

36. For the **FIRST_NAME** field, select **Return**.



Properties Preview CURRENT_CUSTOMERS

General

Fields

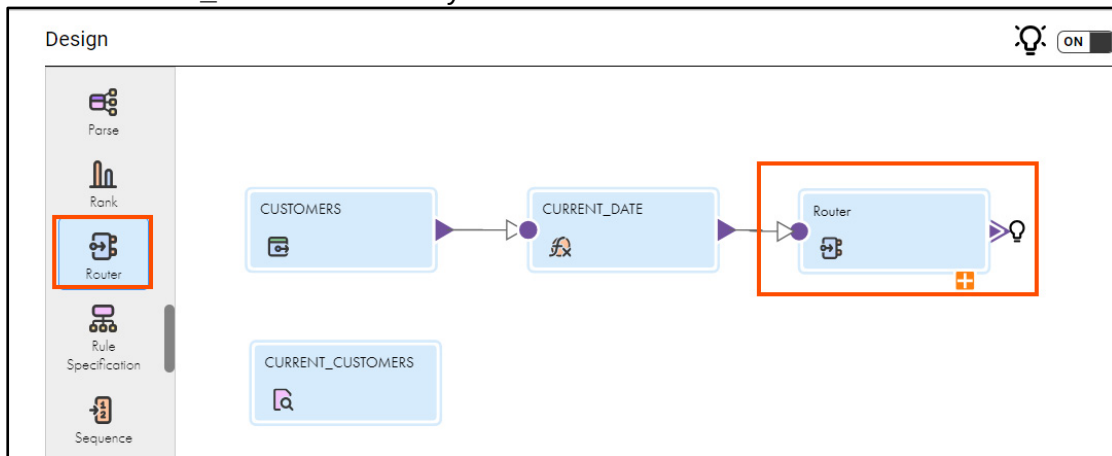
Name	Type	Precision	Scale	Origin	Return	Default Column Value
1 CUST_SEQ	integer	10	0	DIM_CUSTOMER	☐	
2 CUST_ID	integer	10	0	DIM_CUSTOMER	☐	
3 FIRST_NAME	string	255	0	DIM_CUSTOMER	☒	
4 LAST_NAME	string	255	0	DIM_CUSTOMER	☐	
5 ADDR1	string	255	0	DIM_CUSTOMER	☐	
6 ADDR2	string	255	0	DIM_CUSTOMER	☐	
7 CITY	string	255	0	DIM_CUSTOMER	☐	

Return Fields

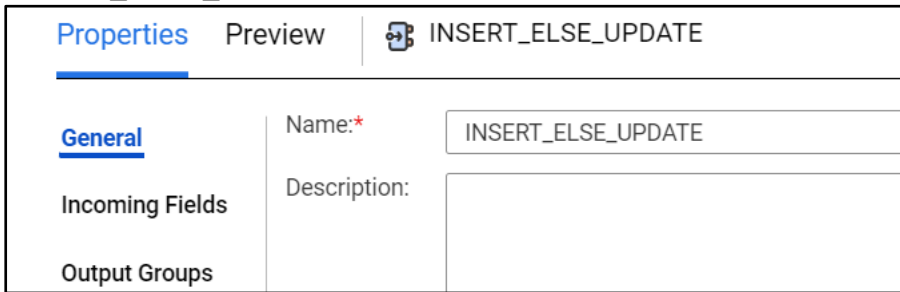
Advanced

Add Router Transformation

37. From the list of available transformations, drag and drop the **Router** transformation and link **CURRENT_DATE** to the newly added router transformation.

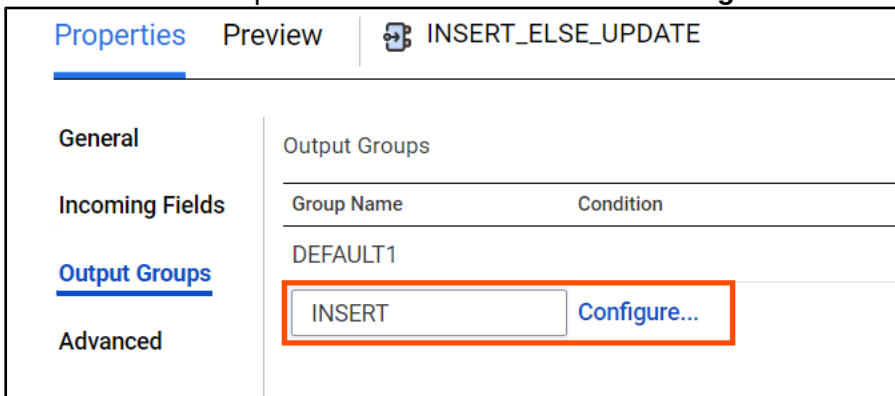


38. Select **Router** transformation from the mapping canvas.
39. In the **General** section of the Expression properties, enter Name as **INSERT_ELSE_UPDATE**.



The 'Properties' pane shows the 'General' section for the 'Router' transformation. The 'Name' field is set to 'INSERT_ELSE_UPDATE'. The 'Description' field is empty. The 'Incoming Fields' and 'Output Groups' sections are also visible.

40. From the properties pane, click **Output Groups**.
41. To add a new Output Group, click .
42. Enter the NewGroup Name as **INSERT** and click **Configure**.

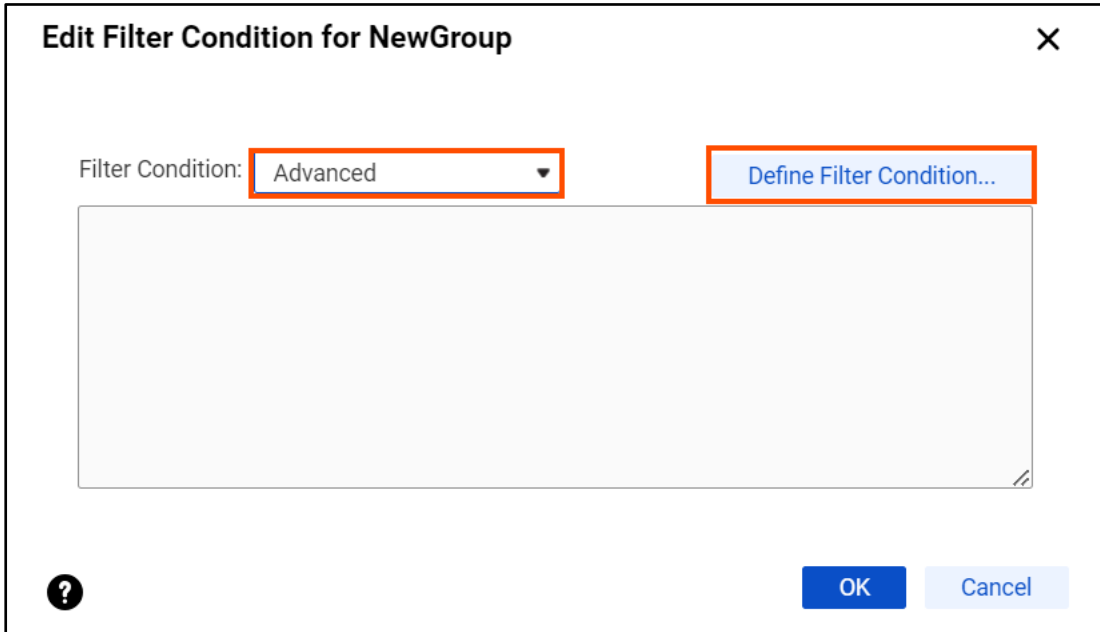


The 'Properties' pane shows the 'Output Groups' section for the 'Router' transformation. A table lists the output groups: 'DEFAULT1' and 'INSERT'. The 'INSERT' group is highlighted with a red box, and the 'Configure...' button next to it is also highlighted.

Group Name	Condition
DEFAULT1	
INSERT	Configure...

43. From the Filter Condition drop down, select **Advanced**.

44. Click on the **Define Filter Condition** button.



Edit Filter Condition for NewGroup

Filter Condition: Advanced Define Filter Condition...

? OK Cancel

45. In the Filter Condition field, enter the following expression:

ISNULL(:LKP.CURRENT_CUSTOMERS(CUST_ID))

46. Click **Validate** and then click **OK**.



Edit Filter Condition

✓ The expression is valid.

Fields Filter Condition Validate

CURRENT_DATE
CUST_ID
FIRST_NAME
LAST_NAME
ADDR1
ADDR2
CITY

ISNULL(:LKP.CURRENT_CUSTOMERS(CUST_ID))

Select a field to see detailed information about it.

OK Cancel

47. Click **OK**.

48. To add another Output Group, click +.

49. Enter the NewGroup Name as **UPDATE** and click **Configure**.

Properties

Preview

 INSERT_ELSE_UPDATE

General

Incoming Fields

Output Groups

Advanced

Output Groups

Group Name	Condition
DEFAULT1	
INSERT	ISNULL(:LKP.CURRENT_CUSTOMERS(CUST_ID))
UPDATE	Configure...

50. From the Filter Condition drop down, select **Advanced**.

51. Click on **Define Filter Condition**.

52. In the Filter Condition field, enter the following expression:

NOT ISNULL(:LKP.CURRENT_CUSTOMERS(CUST_ID))

Tip: You can copy the previous field's filter condition and paste it into this Update field. Edit the filter condition to include **NOT** at the beginning.

53. Click **Validate** and then click **OK**.

✕

Edit Filter Condition

✔ The expression is valid.

Fields

CURRENT_DATE
CUST_ID
FIRST_NAME
LAST_NAME
ADDR1
ADDR2
CITY

Filter Condition

NOT ISNULL(:LKP.CURRENT_CUSTOMERS(CUST_ID))

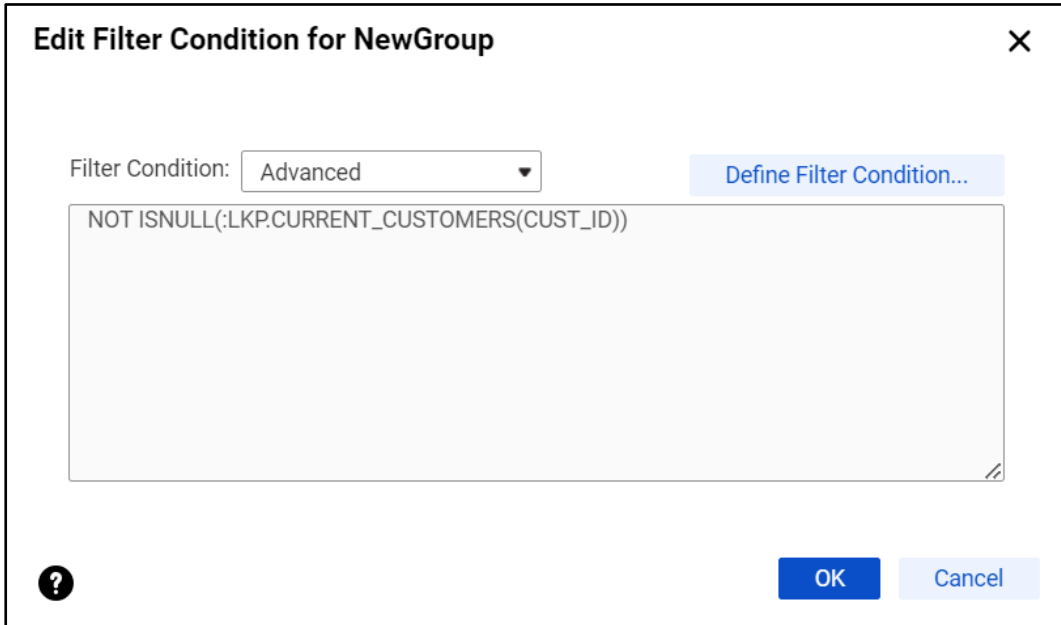
Validate

Select a field to see detailed information about it.

OK

Cancel

54. Click **OK**.



Edit Filter Condition for NewGroup [X]

Filter Condition: Advanced Define Filter Condition...

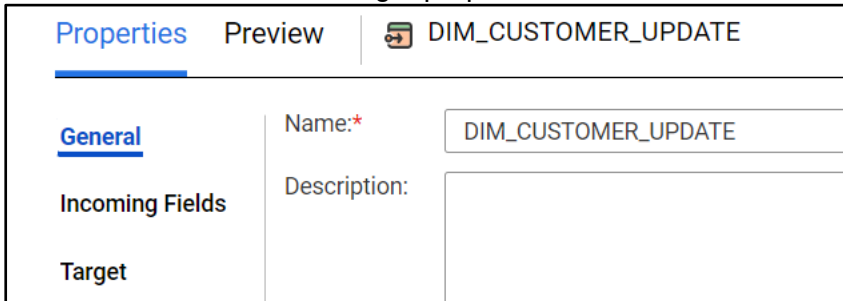
NOT ISNULL(:LKP.CURRENT_CUSTOMERS(CUST_ID))

[?] OK Cancel

Configure Target

53 Select **Target** transformation on the mapping canvas.

55. In the General section of Target properties, enter Name as **DIM_CUSTOMER_UPDATE**.



Properties **Preview** DIM_CUSTOMER_UPDATE

General

Name:* DIM_CUSTOMER_UPDATE

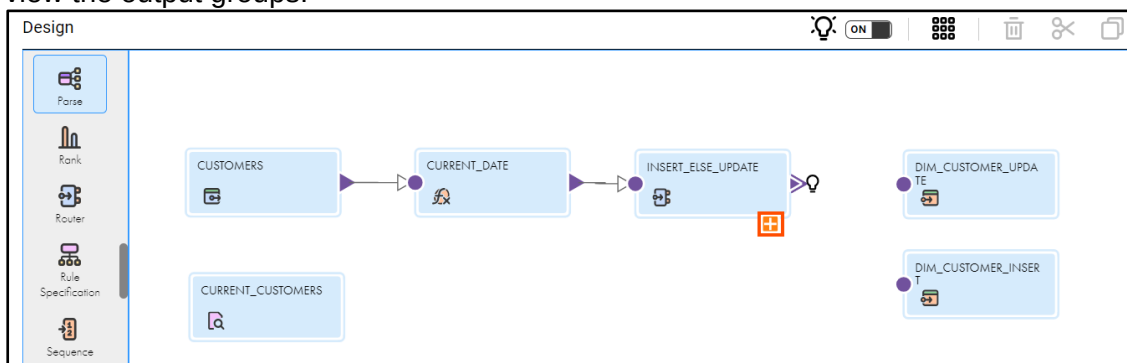
Description:

Incoming Fields

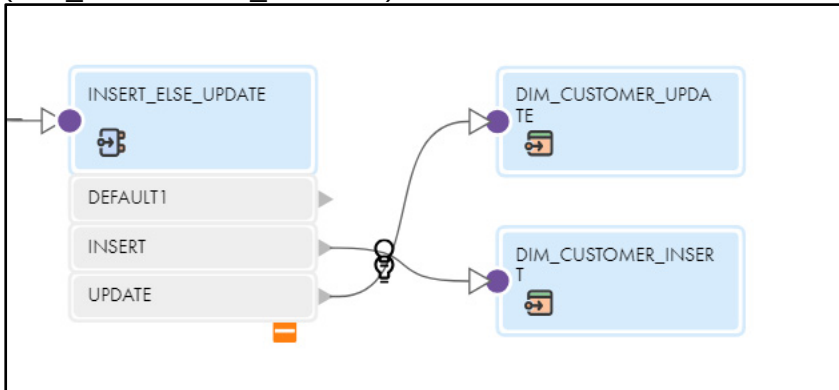
Target

56. Copy-paste the Target Object and name the copy **DIM_CUSTOMER_INSERT**.

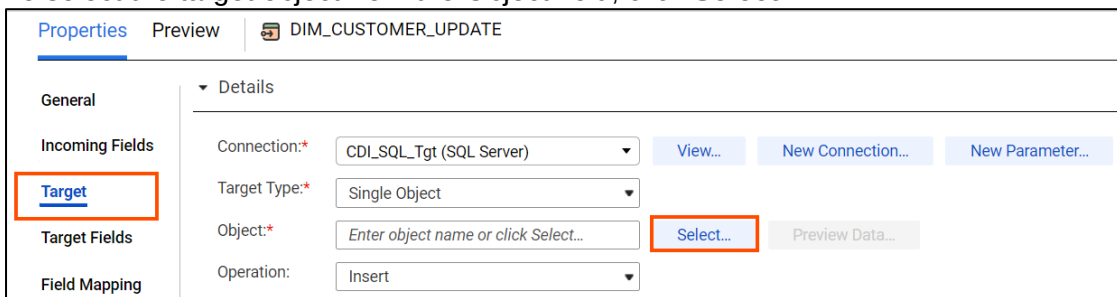
57. Click the  icon present at the bottom of the Router transformation to expand it and view the output groups.



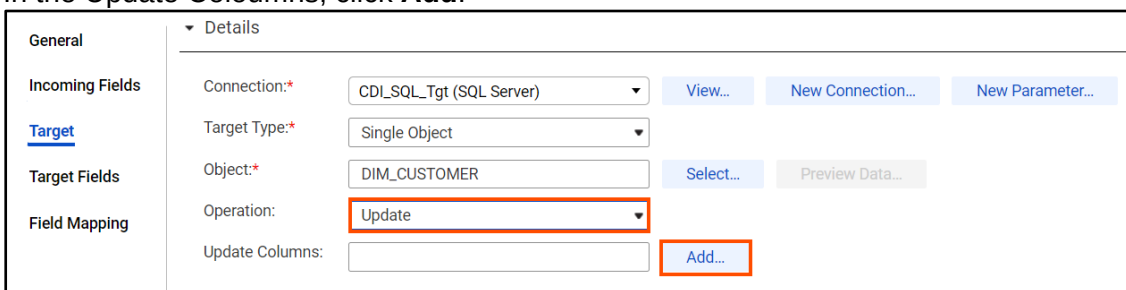
58. Link the **INSERT** output group of Router transformation to the INSERT target (DIM_CUSTOMER_INSERT).
59. Similarly, link the **UPDATE** output group of Router transformation to the UPDATE target (DIM_CUSTOMER_UPDATE) shown below:



60. From the properties pane of the DIM_CUSTOMER_UPDATE, click **Target**.
61. From the Connection drop-down, select **CDI_SQL_Tgt**.
62. To select the target object from the Object field, click **Select**.



63. From the list, select **DIM_CUSTOMER** and click **OK**.
64. From the Operation drop down, select **Update**.
65. In the Update Columns, click **Add**.



66. From the Target Columns, select **CUST_ID** and click .

67. Click **OK**.

Update Columns
✕

You can specify the update columns for the target object here. Select fields of the target object to be used as the update columns.

Target Columns

ADDR1
ADDR2
CITY
CUST_SEQ
EMAIL
FIRST_NAME
LAST_NAME

>
<
>>
<<

Update Columns

CUST_ID

OK
Cancel

68. Select **Field Mapping** from the properties pane.

69. Map the fields as shown below:

Tip: You can use the down arrow at Automap and choose **Exact Field Name** to map the fields, and manually map the **CURRENT_DATE** field to the **VALID_DATE** field.

Incoming Field	Target Fields
CUST_ID	CUST_ID
FIRST_NAME	FIRST-NAME
LAST_NAME	LAST_NAME
ADDR1	ADDR1
ADDR2	ADDR2
CITY	CITY
STATE	STATE
POSTAL_CODE	POSTAL_CODE
PHONE	PHONE
EMAIL	EMAIL
CURRENT_DATE	VALID_DATE

Properties Preview **DIM_CUSTOMER_UPDATE**

General

Incoming Fields: (11 of 11 mapped)

Field Name

CURRENT_DATE

Target

CUST_ID

Target Fields

FIRST_NAME

LAST_NAME

Field Mapping

ADDR1

ADDR2

CITY

STATE

POSTAL_CODE

PHONE

EMAIL

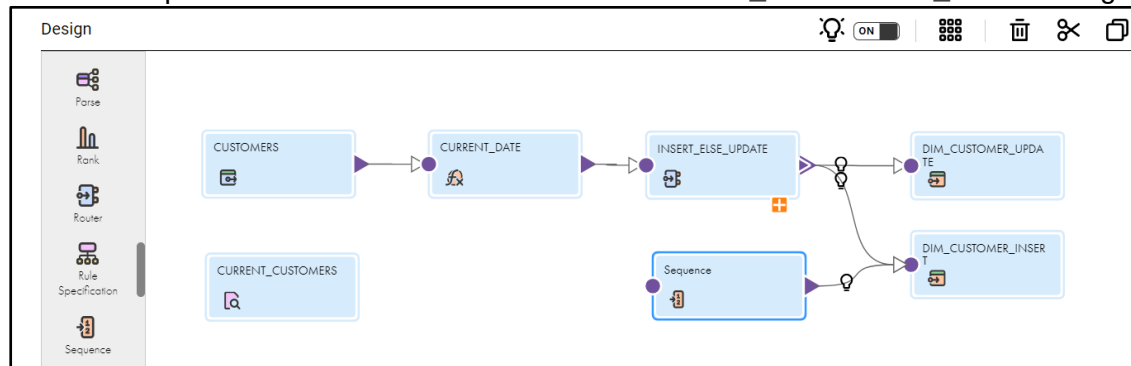
Target Fields: (11 of 12 mapped) **Automap**

Field Name	Mapped Field
CUST_SEQ	
CUST_ID	CUST_ID
FIRST_NAME	FIRST_NAME
LAST_NAME	LAST_NAME
ADDR1	ADDR1
ADDR2	ADDR2
CITY	CITY
STATE	STATE
POSTAL_CODE	POSTAL_CODE
PHONE	PHONE
EMAIL	EMAIL
VALID_DATE	CURRENT_DATE

Add a Sequence Generator transformation

70. From the list of available transformations, drag and drop the Sequence Generator transformation.

71. Link the Sequence Generator transformation with the **DM_CUSTOMER_INSERT** target.



72. Select the Sequence Generator transformation from the mapping canvas.

73. In the General section of its properties, enter Name as **CUST_SEQ**.

Properties Preview **CUST_SEQ**

General

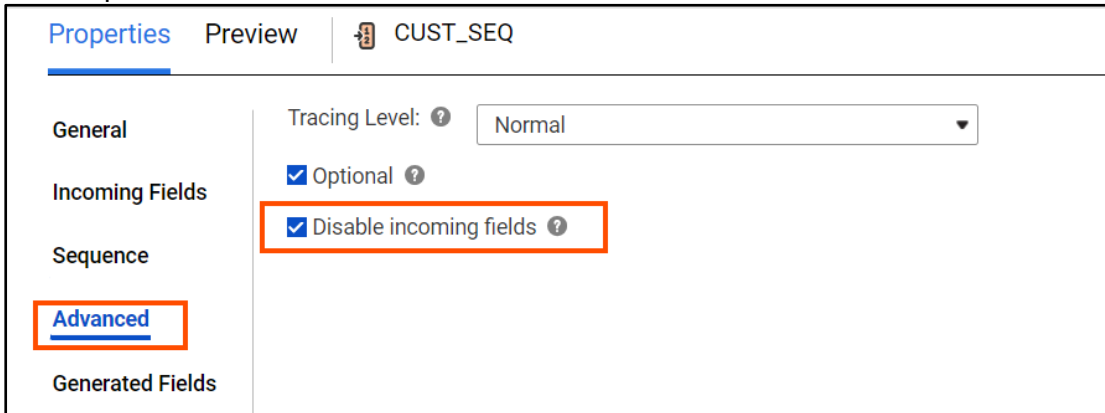
Name:*

Description:

Incoming Fields

Sequence

74. From the properties pane, select **Advanced** field and check the **Disable incoming fields** option.

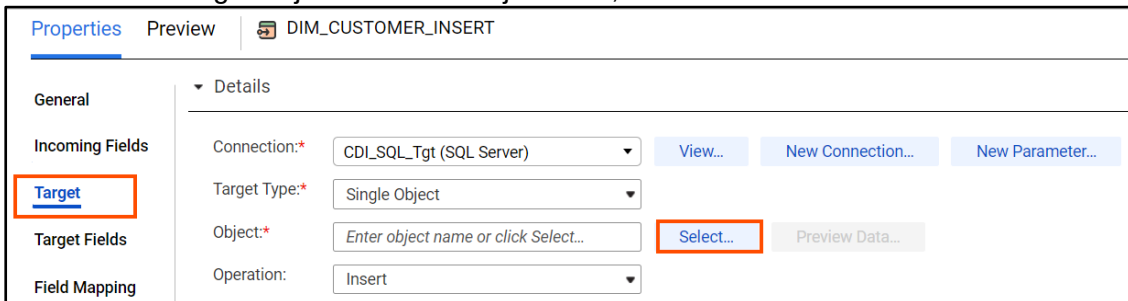


75. Now, select the target **DIM_CUSTOMER_INSERT**.

76. From the properties pane, click **Target**.

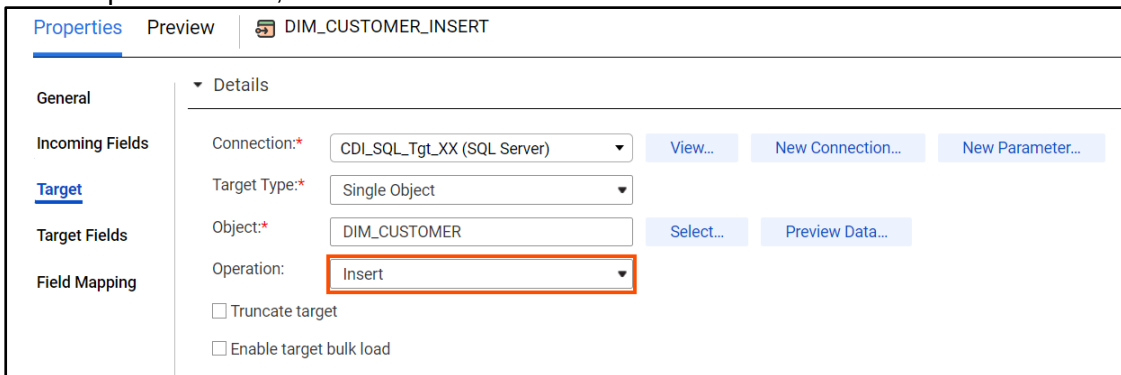
77. From the Connection drop-down, select **CDI_SQL_Tgt**.

78. To select the target object from the Object field, click **Select**.



79. From the list, select **DIM_CUSTOMER** and click **OK**.

80. In the Operation field, retain the field as **Insert**.



81. From the properties pane, click **Field Mapping**.

82. Map the following fields as shown below:

Incoming Field	Target Fields
NEXTVAL	CUST_SEQ
CUST_ID	CUST_ID
FIRST_NAME	FIRST_NAME

LAST_NAME	LAST_NAME
ADDR1	ADDR1
ADDR2	ADDR2
CITY	CITY
STATE	STATE
POSTAL_CODE	POSTAL_CODE
PHONE	PHONE
EMAIL	EMAIL
CURRENT_DATE	VALID_DATE

Properties Preview **DIM_CUSTOMER_INSERT**

General

Incoming Fields (12 of 13 mapped)

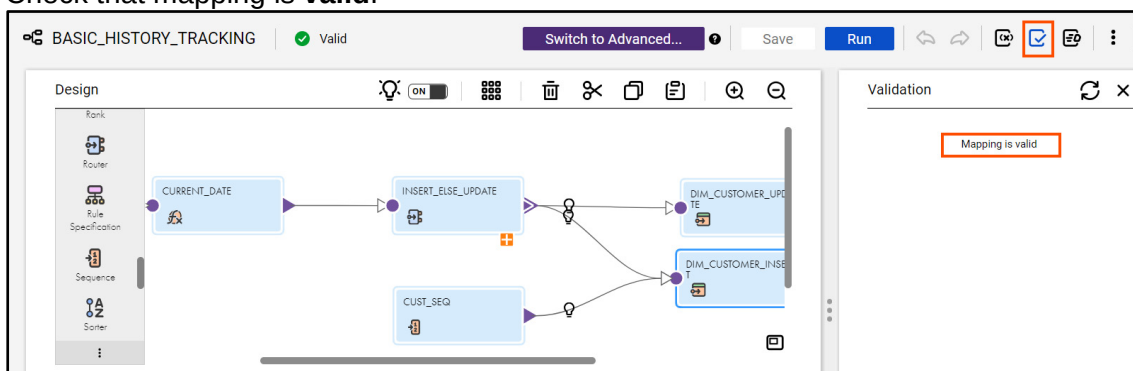
Field Name	Target
CURRENT_DATE	
CUST_ID	
FIRST_NAME	
LAST_NAME	
ADDR1	
ADDR2	
CITY	
STATE	
POSTAL_CODE	
PHONE	
EMAIL	
NEXTVAL	

Target Fields (12 of 12 mapped) **Automap**

Field Name	Mapped Field
CUST_SEQ	NEXTVAL
CUST_ID	CUST_ID
FIRST_NAME	FIRST_NAME
LAST_NAME	LAST_NAME
ADDR1	ADDR1
ADDR2	ADDR2
CITY	CITY
STATE	STATE
POSTAL_CODE	POSTAL_CODE
PHONE	PHONE
EMAIL	EMAIL
VALID_DATE	CURRENT_DATE

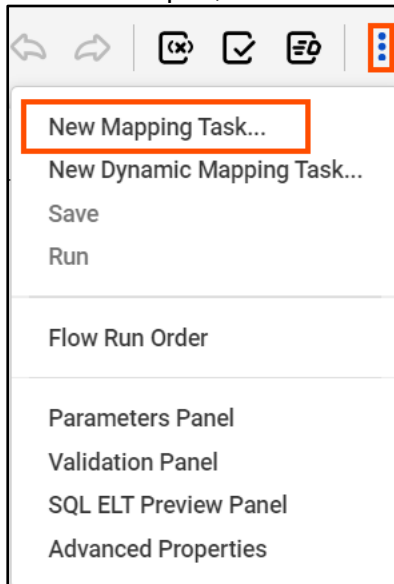
83. **Save** the mapping.

84. Check that mapping is **valid**.



Create a new Mapping Task

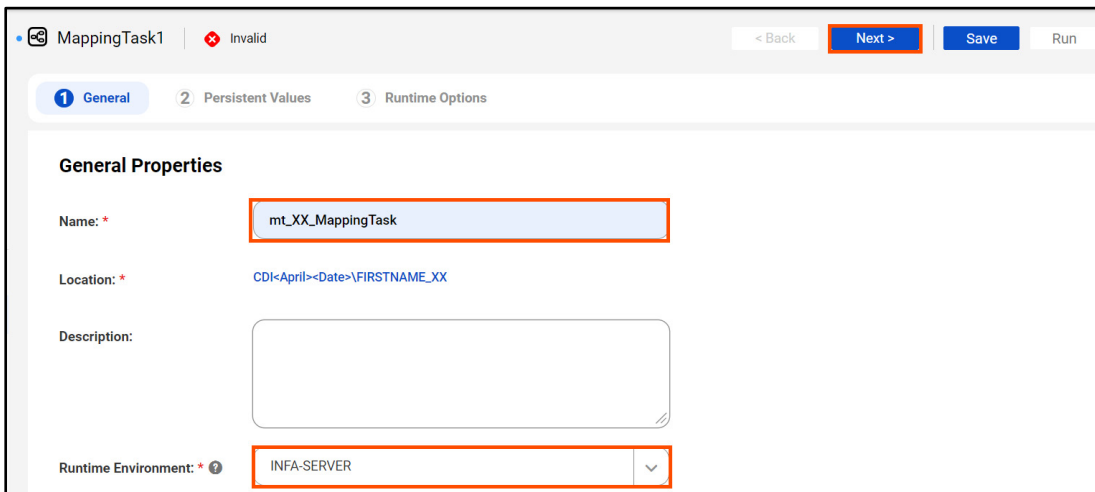
85. From the ellipse, select the **New Mapping Task**.



86. In the Task Name field, enter the Task Name as **mt_XX_MappingTask**.

87. From the Runtime Environment drop down, select **INFA-SERVER**.

88. Click **Next**.



MappingTask1 Invalid < Back **Next >** Save Run

1 General 2 Persistent Values 3 Runtime Options

General Properties

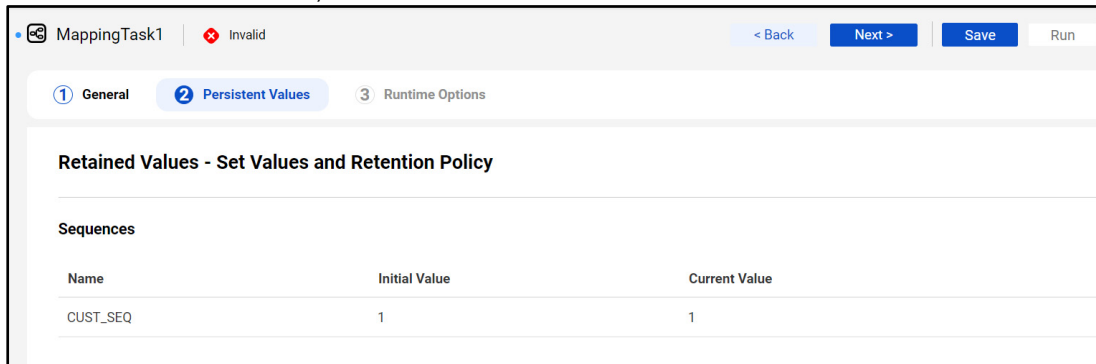
Name: * mt_XX_MappingTask

Location: * CDI<April><Date>\FIRSTNAME_XX

Description:

Runtime Environment: * INFA-SERVER

89. In the Persistent Values, retain the values and click **Next**.



MappingTask1 Invalid < Back Next > Save Run

1 General 2 Persistent Values 3 Runtime Options

Retained Values - Set Values and Retention Policy

Sequences

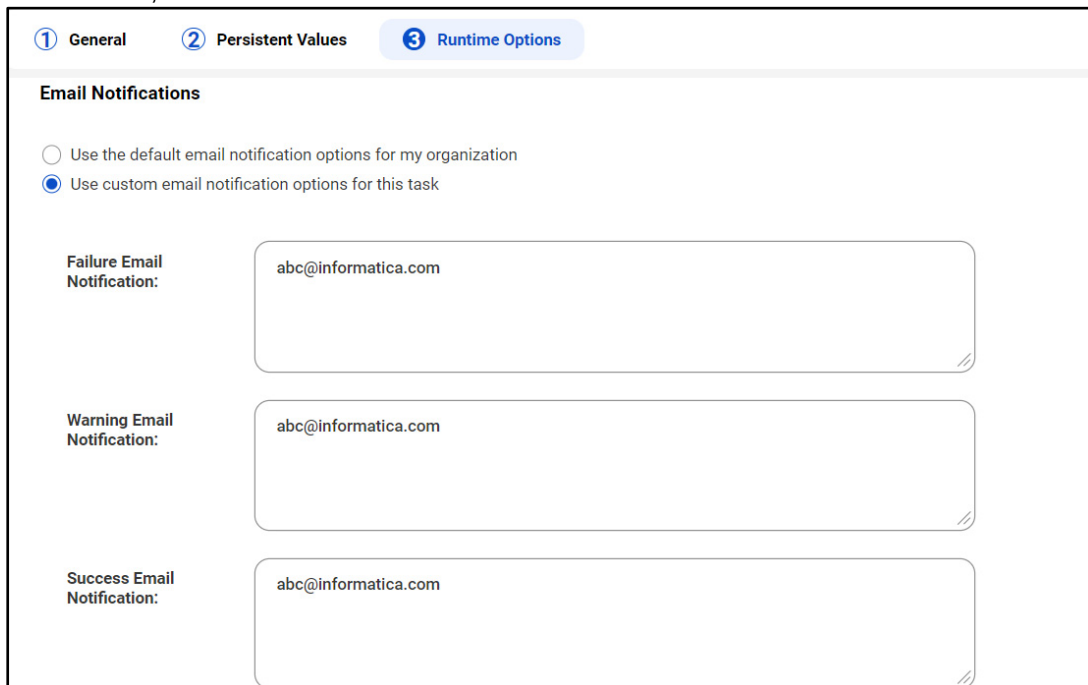
Name	Initial Value	Current Value
CUST_SEQ	1	1

Configure the Email Notifications

90. Scroll down to **Email Notification Options**, and select **Use custom email notification options for this task**.

Note: You can set it to receive emails if the mapping or Mapping task is successful, failed, or finished with some warnings.

91. Specify a valid and active email ID for the Failure Email Notification, Warning Email Notification, and Success Email Notification fields.



1 General 2 Persistent Values 3 Runtime Options

Email Notifications

☐ Use the default email notification options for my organization

☒ Use custom email notification options for this task

Failure Email Notification: abc@informatica.com

Warning Email Notification: abc@informatica.com

Success Email Notification: abc@informatica.com

92. Click **Save**. The mapping task will be saved.

93. **Run** the Mapping task.



mt_XX_MappingTask Valid < Back Next > Save Run

1 General 2 Persistent Values 3 Runtime Options

Monitor Status

94. Monitor the task status from the **My Jobs** page.

95. When the task completes, the status changes to **Success**.

My Jobs

Data Integration

Jobs (62)

Updated 11:07:58 PM PDT

Find

Instance Name	Location	Subtasks	Start Time ↓	End Time	Rows Processed	Status
mt_XX_MappingTask-2	CDI<April><Date>...		May 5, 2024, 11:07 PM	May 5, 2024, 11:0...	10	Success

96. Click on the **Instance Name** of the job.

My Jobs

Data Integration

Jobs (62)

Updated 11:07:58 PM PDT

Find

Instance Name	Location	Subtasks	Start Time ↓	End Time	Rows Processed	Status
mt_XX_MappingTask-2	CDI<April><Date>...		May 5, 2024, 11:07 PM	May 5, 2024, 11:0...	10	Success

97. Scroll down to the Individual Source/Target Results section and analyze the results. Confirm that all the records have been inserted and none of them are updated (Since we began with an empty target).

mt_XX_MappingTask-2

Job Properties

Task Name:

mt_XX_MappingTask

Instance ID:

2

Task Type:

Mapping Task

Started By:

cdi_training_01 through UI

Start Time:

May 5, 2024 11:07:23 PM

End Time:

May 5, 2024 11:07:36 PM

Duration:

13 seconds

Runtime Environment:

INFA-SERVER

Secure Agent:

INFA-SERVER

Results

Status:

Success

Success Rows:

10

Errors:

0

Session Log:

[Download Session Log](#)

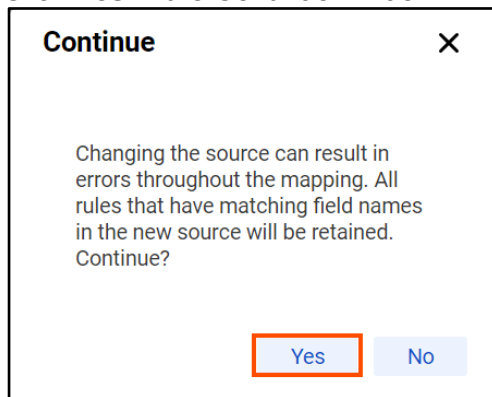
Individual Source/Target Results

Name	Success Rows	Affected Rows	Errors	Error Message
CUSTOMERS	10		0	
DIM_CUSTOMER_INSERT (DIM_CUSTOMER)	10	10	0	
DIM_CUSTOMER_UPDATE (DIM_CUSTOMER_1)	0	0	0	

98. Navigate to the BASIC_HISTORY_TRACKING mapping and select the source object **CUSTOMERS**.

99. From the properties section, go to the Source tab and change the source object to **UPDATE_CUSTOMERS.csv**.

100. Click **Yes** in the Continue window.



101. **Save** the mapping.

Note: Ensure that neither of the Target objects should have the **Truncate target** checkbox checked. Make sure that they are **UNCHECKED**.

102. Navigate to the mapping task (mt_XX_MappingTask) window.

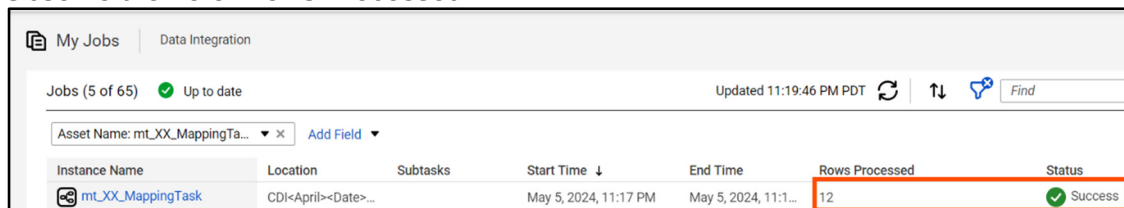
103. Run the mapping task.



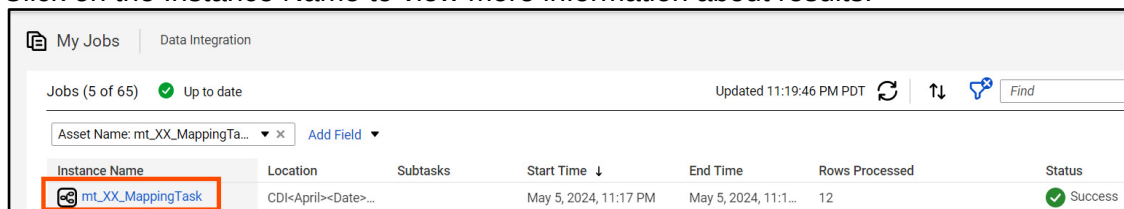
IMPORTANT: You must run the Mapping Task and not the mapping! If you run only the mapping, the Sequence Generator will start at "1" and the target database will reject all the records due to uniqueness constraint.

104. Navigate to the My Jobs page and wait till the job runs successfully.

105. Observe the no.of Rows Processed.



106. Click on the Instance Name to view more information about results.



107. Under the Individual Source/Target Results section, observe the result that most of the records have been updated, while two new customers have been inserted with the new surrogate keys.

Individual Source/Target Results				
Name	Success Rows	Affected Rows	Errors	Error Message
 CUSTOMERS	12		0	
 DIM_CUSTOMER_INSERT (DIM_CUSTOMER)	2	2	0	
 DIM_CUSTOMER_UPDATE (DIM_CUSTOMER_1)	12	12	0	

108. Close all the assets from the navigation pane.

109. Also, as configured, you will get emails on the registered email address with the task details. Verify your email inbox.

Mapping task "mt_XX_MappingTask" completed successfully. [Org Name: Informatica]



admin@informaticacloud.com

To 

Retention Policy Exchange online 3 Years delete (3 years)

Expires 5/6/2027

The Mapping Task **mt_XX_MappingTask** completed successfully.

Organization ID: 

Organization Name: Informatica

Start Time: May 6, 2024 11:46:36 AM (IST)

End Time: May 6, 2024 11:46:48 AM (IST)

Task Description:

Mapping name: BASIC_HISTORY_TRACKING

Source Connection: FF_Source

Source Object: INITIAL_CUSTOMERS.CSV

Target Connection: CDI_SQL_Tgt

Target Object: DIM_CUSTOMER

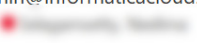
Target Connection: CDI_SQL_Tgt

Target Object: DIM_CUSTOMER

Result: **10 Success Rows, 0 Errors.**

You can access your Informatica Cloud account at <https://dm-us.informaticacloud.com/ma>

Mapping task "mt_XX_MappingTask" completed successfully. [Org Name: Informatica]

 admin@informaticacloud.com
To 

Retention Policy Exchange online 3 Years delete (3 years) Expires 5/6/2027

The Mapping Task **mt_XX_MappingTask** completed successfully.

Organization ID: 

Organization Name: Informatica

Start Time: May 6, 2024 11:47:41 AM (IST)

End Time: May 6, 2024 11:47:52 AM (IST)

Task Description:

Mapping name: BASIC_HISTORY_TRACKING

Source Connection: FF_Source

Source Object: UPDATE_CUSTOMERS.CSV

Target Connection: CDI_SQL_Tgt

Target Object: DIM_CUSTOMER

Target Connection: CDI_SQL_Tgt

Target Object: DIM_CUSTOMER

Result: **12 Success Rows, 0 Errors.**

You can access your Informatica Cloud account at <https://dm-us.informaticacloud.com/ma>

Thus, you have built a logic to perform the basic history tracking of the data records. This technique can be useful even if there is no traditional data warehouse involved.

This concludes the lab.

Module 11: Replication Task

Lab 11-1: Replicating Data to a Flat File

Overview:

A replication task replicates data from a source to a target.

Objective:

- Create a replication task to replicate data to a CSV file

Scenario:

Ruby is concerned that the updated data on the Salesforce Account object is not saved anywhere for backup purposes. So, John suggests creating a replication task to load Account objects in Salesforce to a flat file.

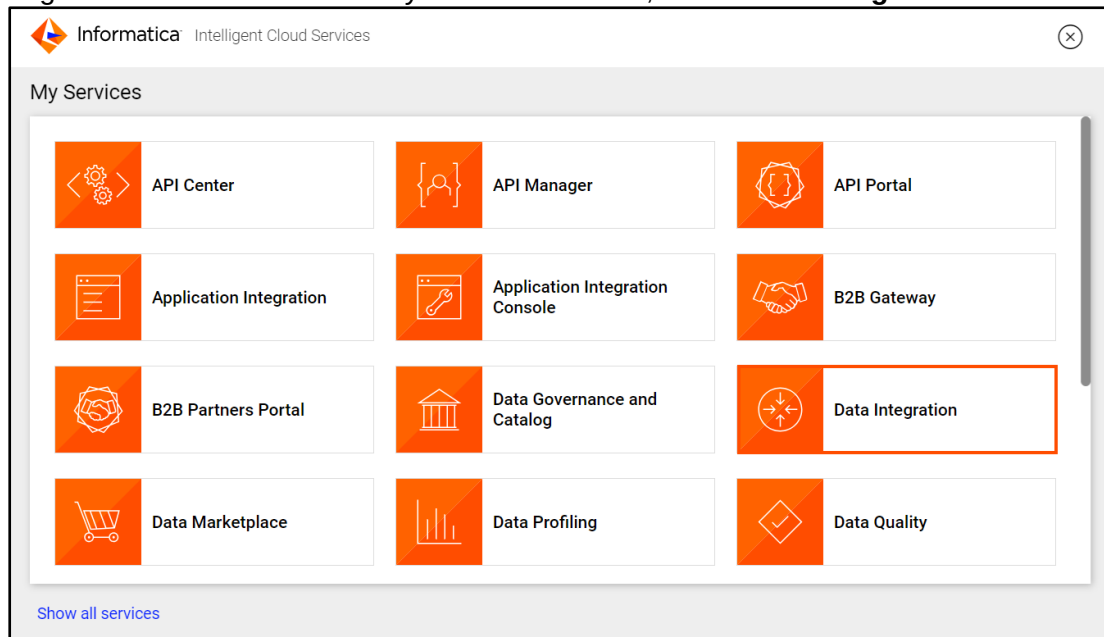
Duration:

10 minutes

Tasks

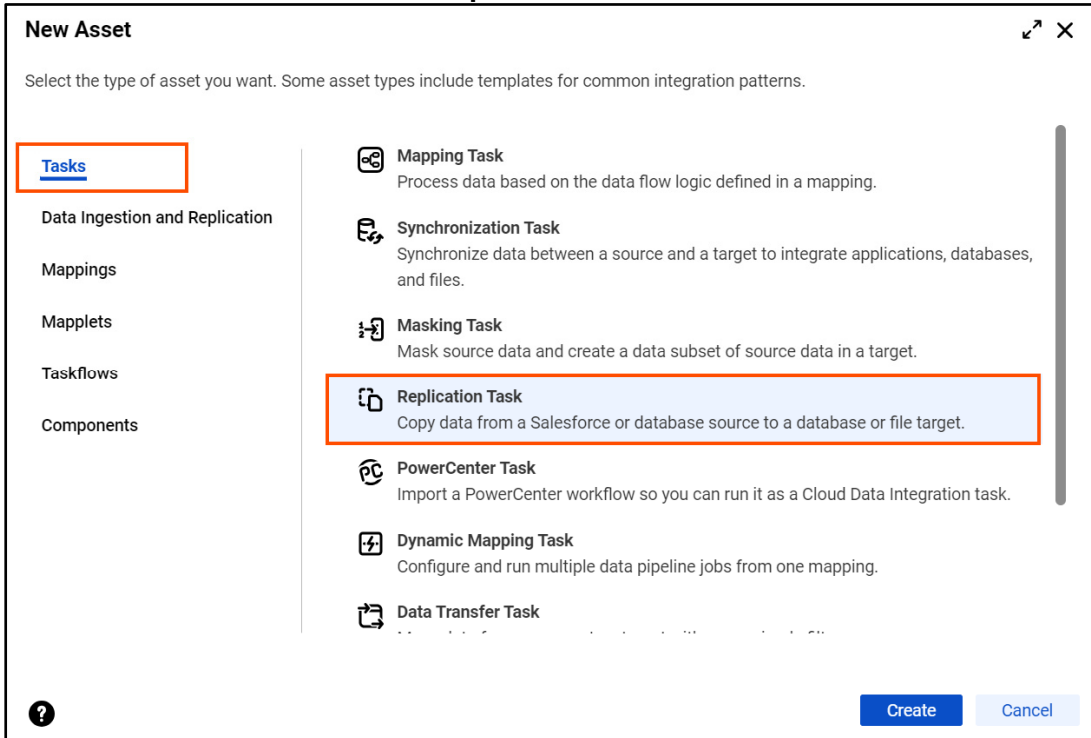
Create Replication Task

1. Login into IICS and from the My Services window, select **Data Integration**.



2. To create a new asset, from the navigation pane, select **New**.

3. In the New Asset window, select **Replication Task**, and click **Create**.



New Asset

Select the type of asset you want. Some asset types include templates for common integration patterns.

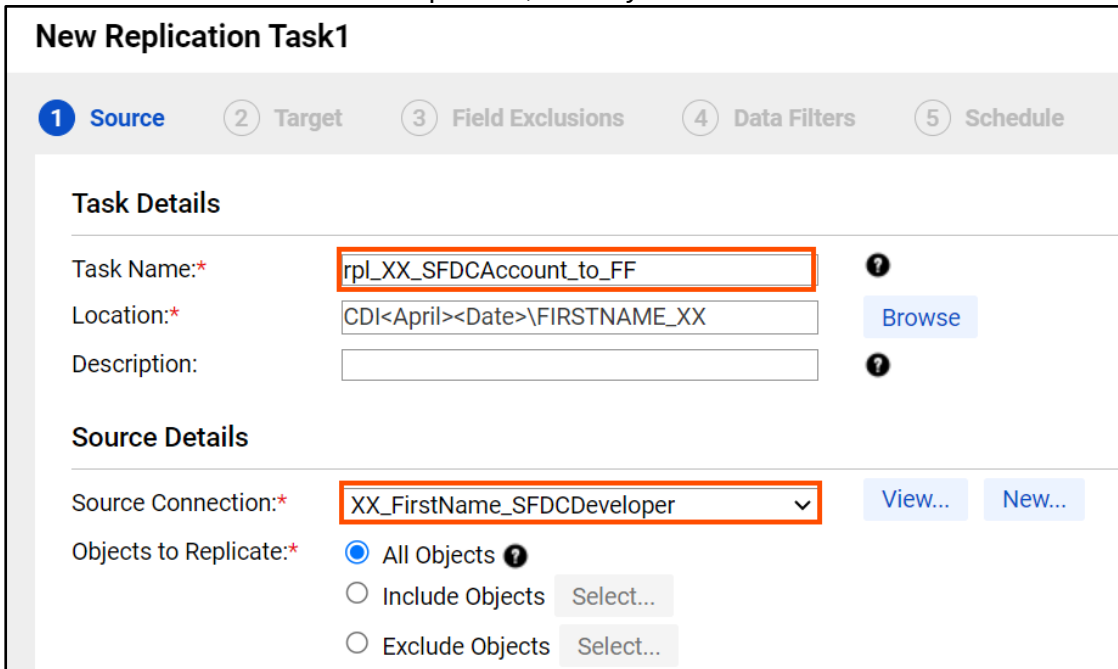
Tasks

- Data Ingestion and Replication
- Mappings
- Maplets
- Taskflows
- Components

- Mapping Task**
Process data based on the data flow logic defined in a mapping.
- Synchronization Task**
Synchronize data between a source and a target to integrate applications, databases, and files.
- Masking Task**
Mask source data and create a data subset of source data in a target.
- Replication Task**
Copy data from a Salesforce or database source to a database or file target.
- PowerCenter Task**
Import a PowerCenter workflow so you can run it as a Cloud Data Integration task.
- Dynamic Mapping Task**
Configure and run multiple data pipeline jobs from one mapping.
- Data Transfer Task**

Create **Cancel**

4. In the Task Name field, enter **rpl_XX_SFDCAccount_to_FF**. Replace XX with your student number.
5. Verify that the asset Location is pointing to your working folder.
6. From the Source Connection drop-down, select your **Salesforce** connection.



New Replication Task1

1 Source 2 Target 3 Field Exclusions 4 Data Filters 5 Schedule

Task Details

Task Name:* **rpl_XX_SFDCAccount_to_FF** ?

Location:* CDI<April><Date>\FIRSTNAME_XX Browse

Description: ?

Source Details

Source Connection:* **XX_FirstName_SFDCDeveloper** View... New...

Objects to Replicate:* ☒ All Objects ?

☐ Include Objects Select...

☐ Exclude Objects Select...

7. From the Objects to Replicate field, select **Include Objects**.

8. Click **Select**.

Source Details

Source Connection:*
XX_FirstName_SFDCDeveloper
View...
New...

Objects to Replicate:*
☐ All Objects ?
☒ Include Objects **Select...**
☐ Exclude Objects Select...

9. From the list, select **Account**.

10. Click **Select**.

Include Source Objects

Include a source object, and then click Select. You can also search for a source object.

Available Objects*
XX_FirstName_SFDCDeveloper3
 Name, Label, Description, Type
Search

Select	Name	Label	Description	Type
☐	AIInsightValue	AI Insight Value		
☐	AIRecordInsight	AI Record Insight		
☐	AcceptedEventRelation	Accepted Event Relation		
☒	Account	Account		

Included Objects*
Account

11. Click **Next**.

12. From the Connection drop-down, select your **target Flat File** connection.

13. Enter **SXX_** in the Target Prefix field.

1 Source
2 **Target**
3 Field Exclusions
4 Data Filters
5 Schedule

Target Details

Connection:*
FF_Target
View...
New...
?

Target Prefix:
SXX_
?

14. Click **Next**.

15. From the Field Exclusions section, click **Exclude Fields**.

1 Source
2 Target
3 **Field Exclusions**
4 Data Filters
5 Schedule

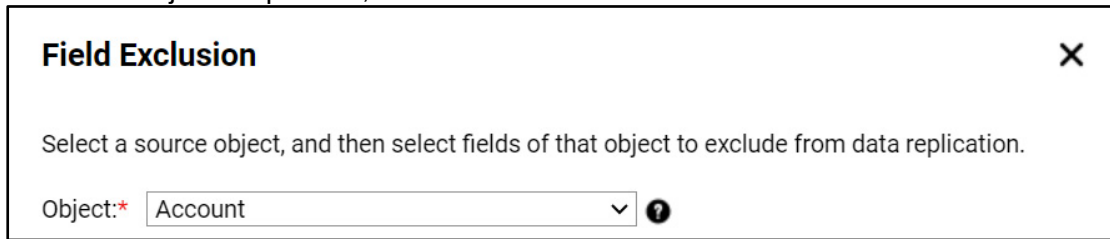
Field Exclusions

Exclude Fields...

There are no fields excluded for any object. All fields for all selected objects will be replicated.

The Field Exclusion window appears.

16. From the Object drop-down, select **Account**.



Field Exclusion ✕

Select a source object, and then select fields of that object to exclude from data replication.

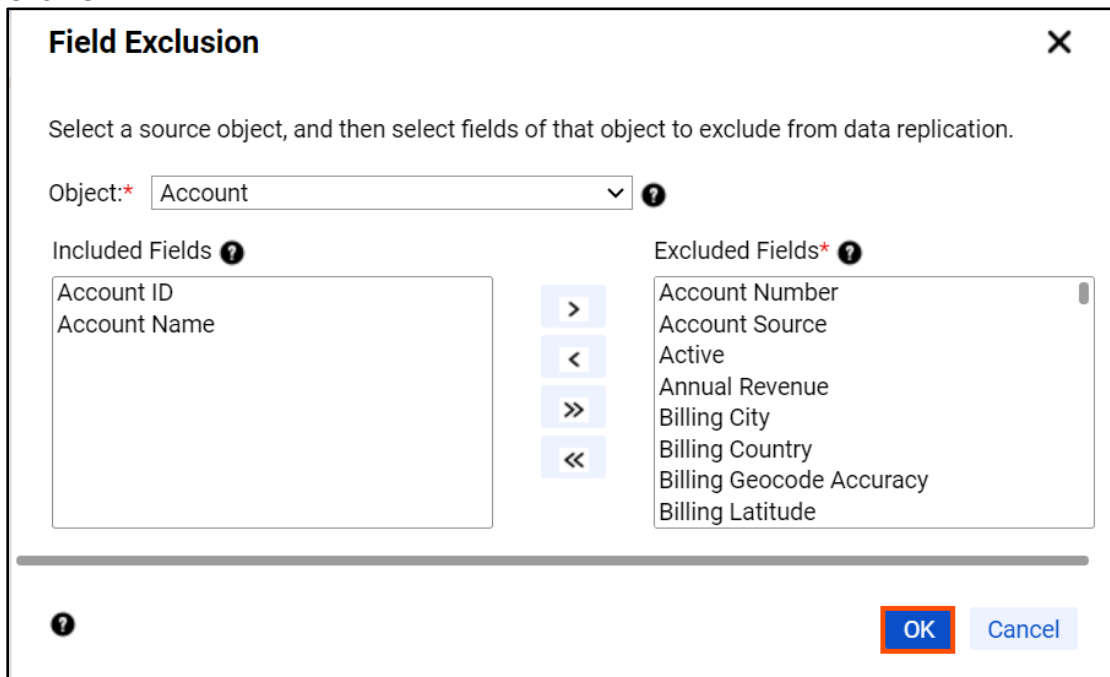
Object:* ?

17. In the Included Fields section, retain only the **Account Name** and **Account ID**.

Note: To retain the Account Name and Account ID, select the rest of the fields, and click



18. Click **OK**.



Field Exclusion ✕

Select a source object, and then select fields of that object to exclude from data replication.

Object:* ?

Included Fields ?

Account ID
Account Name

>

<

>>

<<

Excluded Fields* ?

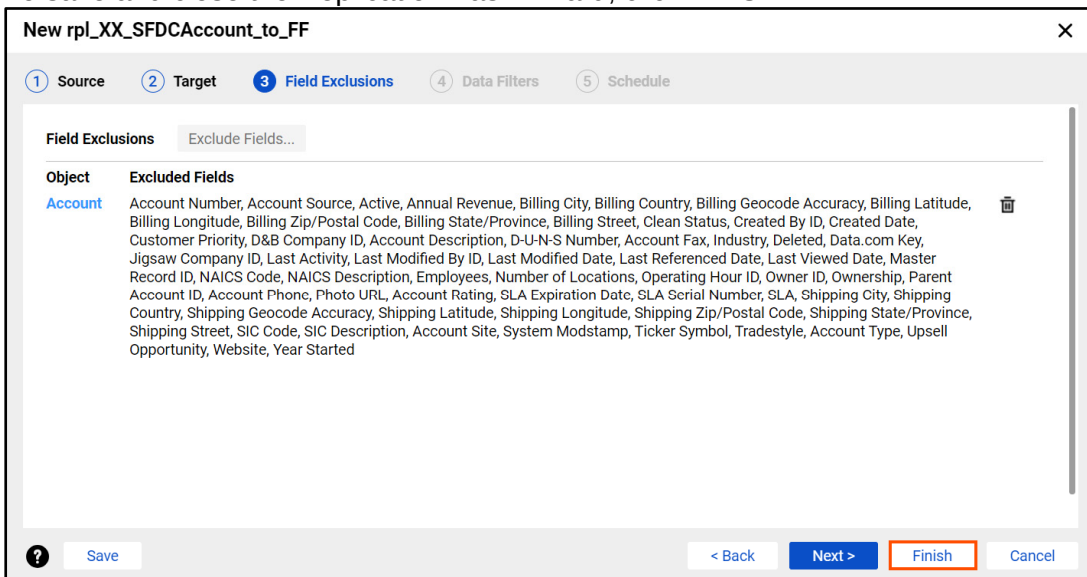
Account Number
Account Source
Active
Annual Revenue
Billing City
Billing Country
Billing Geocode Accuracy
Billing Latitude

?

OK

Cancel

19. To save and close the Replication Task wizard, click **Finish**.



New rpl_XX_SFDCAccount_to_FF

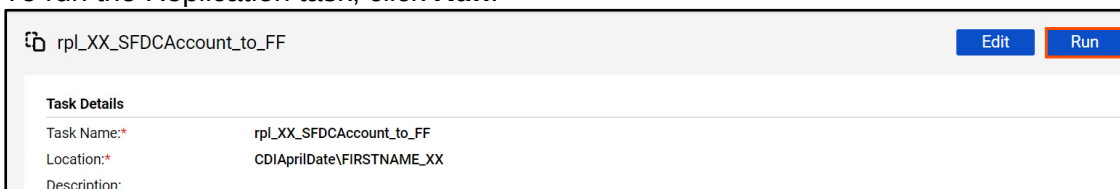
1 Source 2 Target 3 **Field Exclusions** 4 Data Filters 5 Schedule

Field Exclusions Exclude Fields...

Object	Excluded Fields
Account	Account Number, Account Source, Active, Annual Revenue, Billing City, Billing Country, Billing Geocode Accuracy, Billing Latitude, Billing Longitude, Billing Zip/Postal Code, Billing State/Province, Billing Street, Clean Status, Created By ID, Created Date, Customer Priority, D&B Company ID, Account Description, D-U-N-S Number, Account Fax, Industry, Deleted, Data.com Key, Jigsaw Company ID, Last Activity, Last Modified By ID, Last Modified Date, Last Referenced Date, Last Viewed Date, Master Record ID, NAICS Code, NAICS Description, Employees, Number of Locations, Operating Hour ID, Owner ID, Ownership, Parent Account ID, Account Phone, Photo URL, Account Rating, SLA Expiration Date, SLA Serial Number, SLA, Shipping City, Shipping Country, Shipping Geocode Accuracy, Shipping Latitude, Shipping Longitude, Shipping Zip/Postal Code, Shipping State/Province, Shipping Street, SIC Code, SIC Description, Account Site, System Modstamp, Ticker Symbol, Tradestyle, Account Type, Upsell Opportunity, Website, Year Started

? Save < Back Next > **Finish** Cancel

20. To run the Replication task, click **Run**.



rpl_XX_SFDCAccount_to_FF Edit **Run**

Task Details

Task Name:*	rpl_XX_SFDCAccount_to_FF
Location:*	CDI\AprilDate\FIRSTNAME_XX
Description:	

Monitor Task

21. To monitor the task, from the navigation pane, click **My Jobs**.
22. When the task is completed, the status changes to **Success**.

Jobs (1 of 21)		Updated 4:21:32 AM PDT    Find				
Asset Name: rpl_XX_SFDCAcco...  Add Field 						
Instance Name	Location	Subtasks	Start Time ↓	End Time	Rows Processed	Status
 rpl_XX_SFDCAccount_...	CDI<April><Da...	1 task	Apr 22, 2024, 4:21 ...	Apr 22, 2024, ...	13	 Success

Note: The number of processed rows can change depending on the data in the Salesforce Account object.

23. Close the asset from the navigation pane.
24. Optionally, you can view the target data in your dummy mapping.

[Properties](#)
[Preview](#)
[Source](#)

[General](#)

[Source](#)

Fields

Partitions

Details

Connection:*
FF_Tgt_Student_XX (Flat File)
View...
New Connection...
New Parameter...

Source Type:*
Single Object

Object:*
S01_ACCOUNT.csv
Select...
Formatting Options...
Preview Data...

Data Preview

×

Connection: FF_Tgt_Student_XX		Object: S01_ACCOUNT.csv	
ID	NAME		
001GB00003AdSB9YAN	GenePoint		
001GB00003AdSB7YAN	United Oil & Gas, UK		
001GB00003AdSB8YAN	United Oil & Gas, Singapore		
001GB00003AdSAzYAN	Edge Communications		
001GB00003AdSB0YAN	Burlington Textiles Corp of America		
001GB00003AdSB1YAN	Pyramid Construction Inc.		
001GB00003AdSB2YAN	Dickenson plc		
001GB00003AdSB3YAN	Grand Hotels & Resorts Ltd		
001GB00003AdSB5YAN	Express Logistics and Transport		
001GB00003AdSB6YAN	University of Arizona		

☐ Display source fields in alphabetical order


[Formatting Options...](#)
[Done](#)

This concludes the lab.

Module 12: Masking Task

Lab 12-1: Creating a Masking Task

Overview:

A masking task masks source data and creates a data subset of the source data in the target.

In this lab, you will mask the account phone number using a masking task.

Objective:

- Create a masking task to mask the phone number

Scenario:

Ruby wants to use the customer data for development and testing purposes. However, as customer information is sensitive data, John suggests using the Masking task to mask customer phone number details.

Duration:

10 minutes

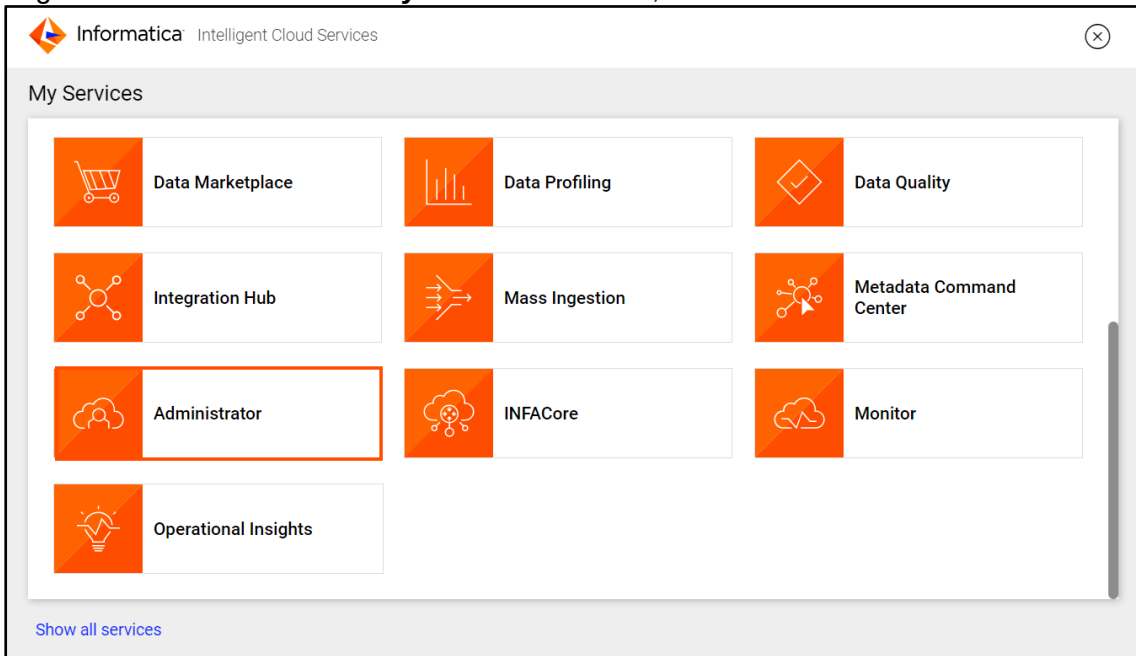
Tasks

Create Masking Task

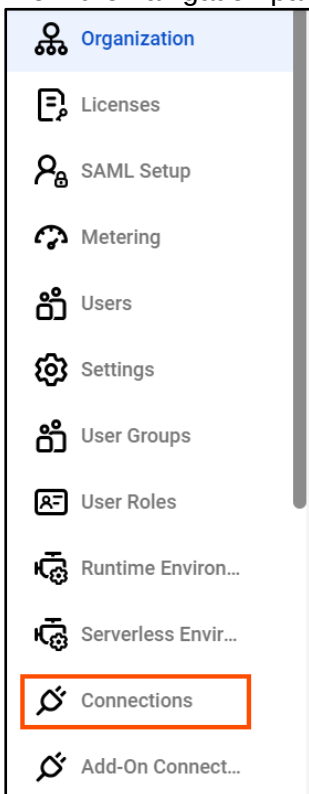
1. The following screenshot shows the data of the **Outlets.csv** file used in creating the Synchronization Task lab. Note the phone numbers for some of the accounts.

A	B	C	D	E	F	G	H	I
Outlet_ID	Outlet_Name	Phone_Number	Street	City	State	ZipCode	Country	Employees
1	NH D'needs	754-3010	11 West Park Road	Pasadena	CA	91001	USA	4
2	NH Mart	281-6000	6620 W. Broad St.	Rampart	AK	99767	USA	4
3	NH Groceries	522-8175	7227 W Harry St	Winslow	AZ	86047	USA	4
4	NH Lifestyle	864-8000	2145 Hamilton Ave	Salida	CO	81201	USA	5
5	NH Digital	314-3600	6601 Hawkinsville Rd	Milford	DE	19963	USA	5
6	NH Trends	782-9000	95 N Moorland Rd	Miami	FL	33101	USA	4
7	NH Everyday	454-3071	114 Division Ave N	Bloomington	IN	47401	USA	4
8	NH Digiworld	547-2518	651 Holiday Dr	Pineville	LA	71359	USA	5
9	NH Supplies	612-6700	32 Old Slip	Grand Rapids	MI	49501	USA	5

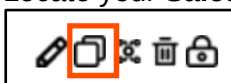
2. Login into IICS and from the **My Services** window, select **Administrator**.



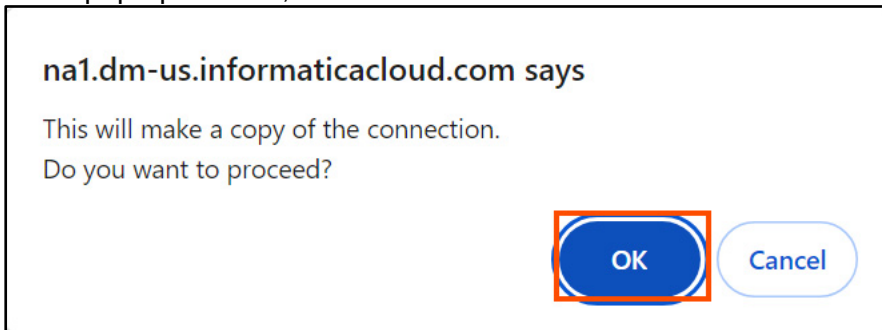
3. From the navigation pane, select **Connections**.



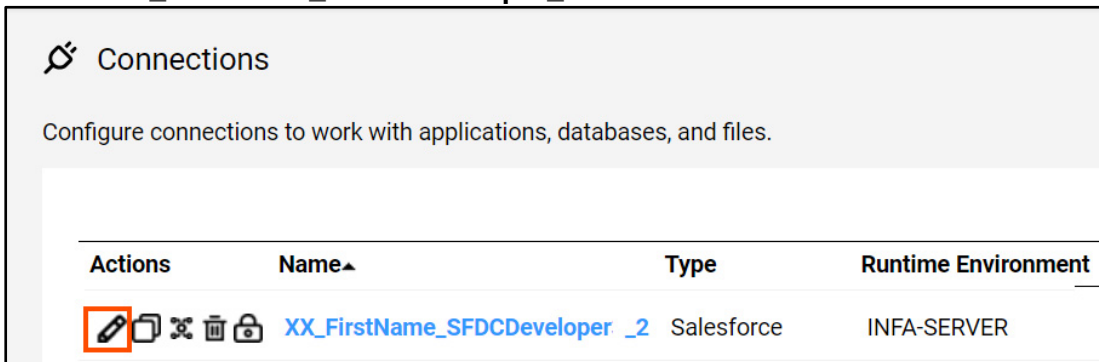
4. Locate your **Salesforce** connection and click on  to make a copy of the connection.



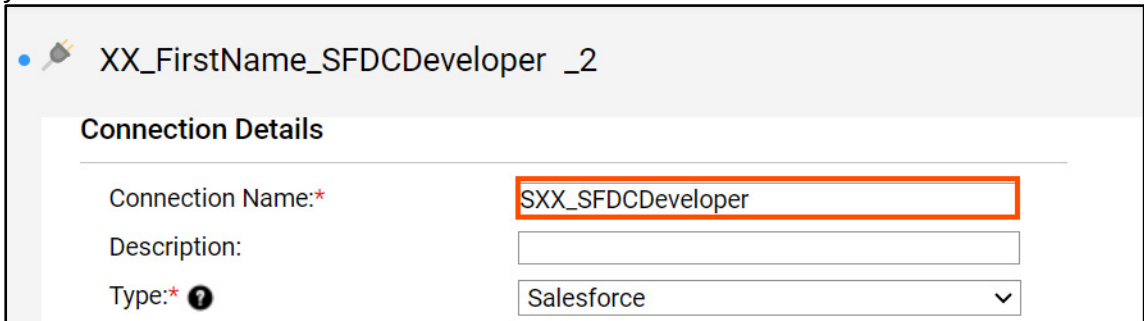
- In the pop-up window, click **OK**.



- Edit the **XX_FirstName_SFDCDeveloper_2** connection.



- Update the connection name to **SXX_SFDCDeveloper**. Make sure to replace XX with your student ID/number.

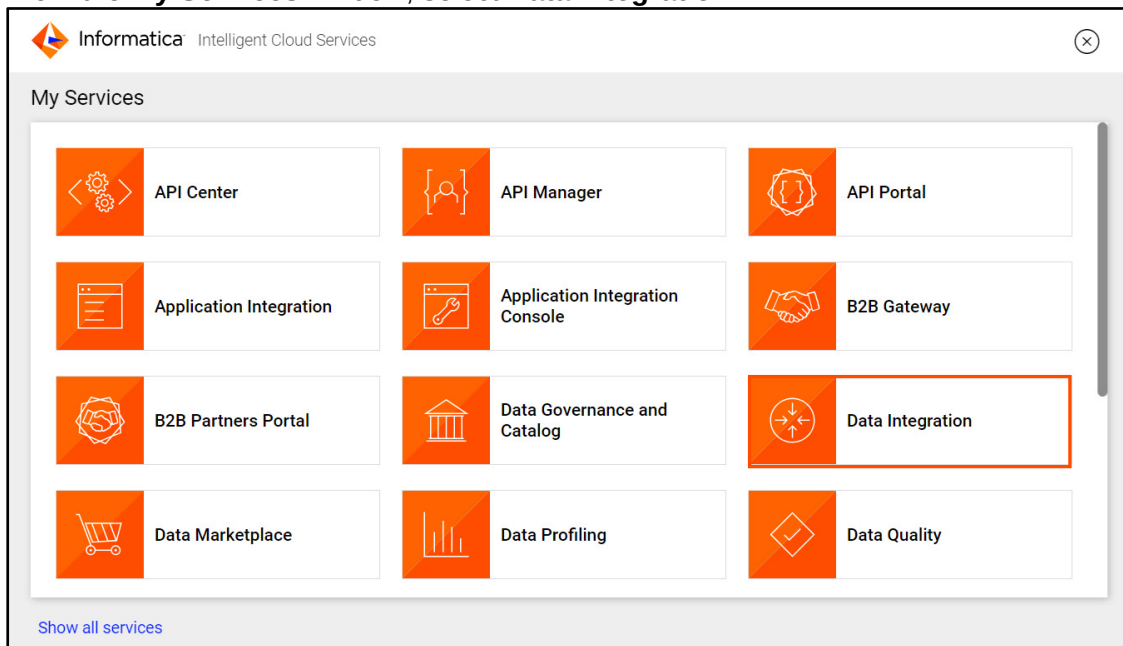


Note: When you create a Masking Task, the connection name cannot start with a number. So, you must update the connection name to use it in the Masking task.

- Test** the connection and **save** the changes. Make sure that the connection is valid. If you encounter any issues while testing the connection, for example: “**..username, password, or security token is not valid..**”, reset your security token and specify the valid details.
- To switch between the available services, from the toolbar, select the drop-down next to **Administrator**.

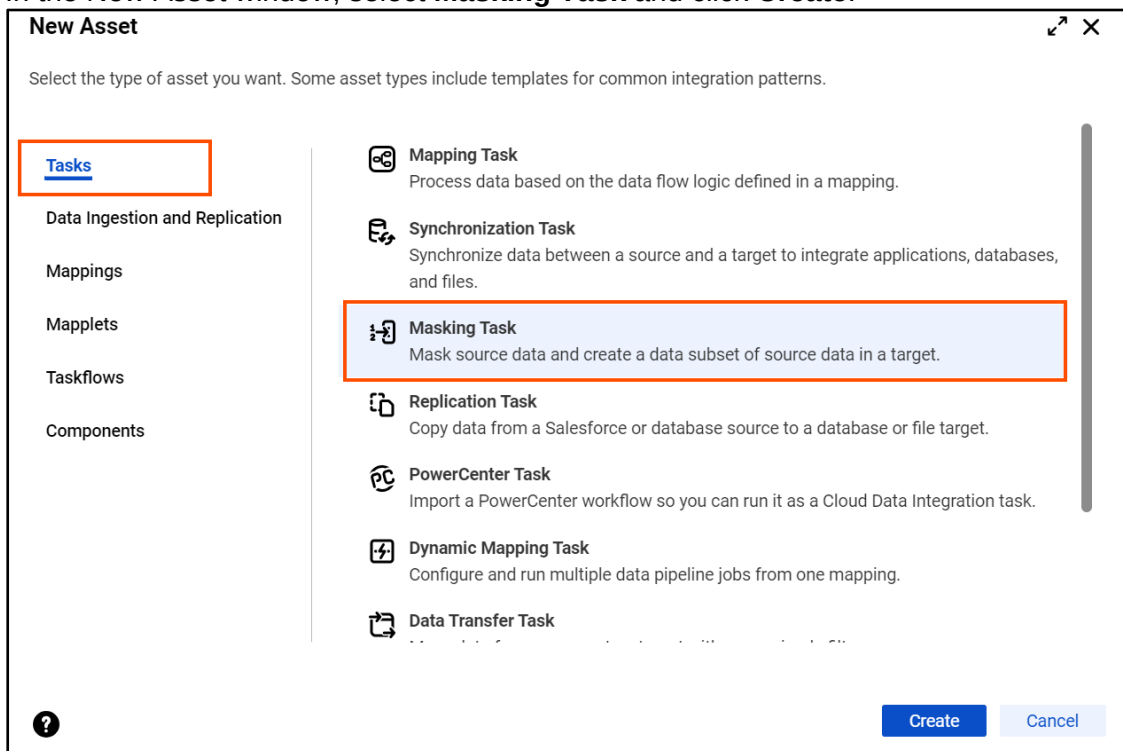


10. From the **My Services** window, select **Data Integration**.



11. To create a new asset, from the navigation pane, select **New**.

12. In the New Asset window, select **Masking Task** and click **Create**.



13. In the Task Name field, enter **mask_XX_PhoneNumber_Masking**. Replace XX with your student number.

14. Verify that the asset Location is pointing to your working folder.

15. Click **Next**.



16. From the Source Connection drop-down, select your duplicate **Salesforce** connection **SXX_SFDCDeveloper**.
17. From the Source Object drop-down, select **Account**.

New mask_XX_PhoneNumber_Masking

① Definition ② **Source** ③ Target ④ Data Filters ⑤ Masking ⑥ Schedule

Source Details

Connection:* SXX_SFDCDeveloper View... New...

Source Type:* ☒ Single ☐ Multiple

Source Object:* Account Select...

☒ Display technical names instead of labels
☐ Display source fields in alphabetical order

18. Click **Next**.
19. Verify that the **Same as source?** field is selected.

① Definition ② Source ③ **Target** ④ Data Filters ⑤ Masking ⑥ Schedule

Target Details

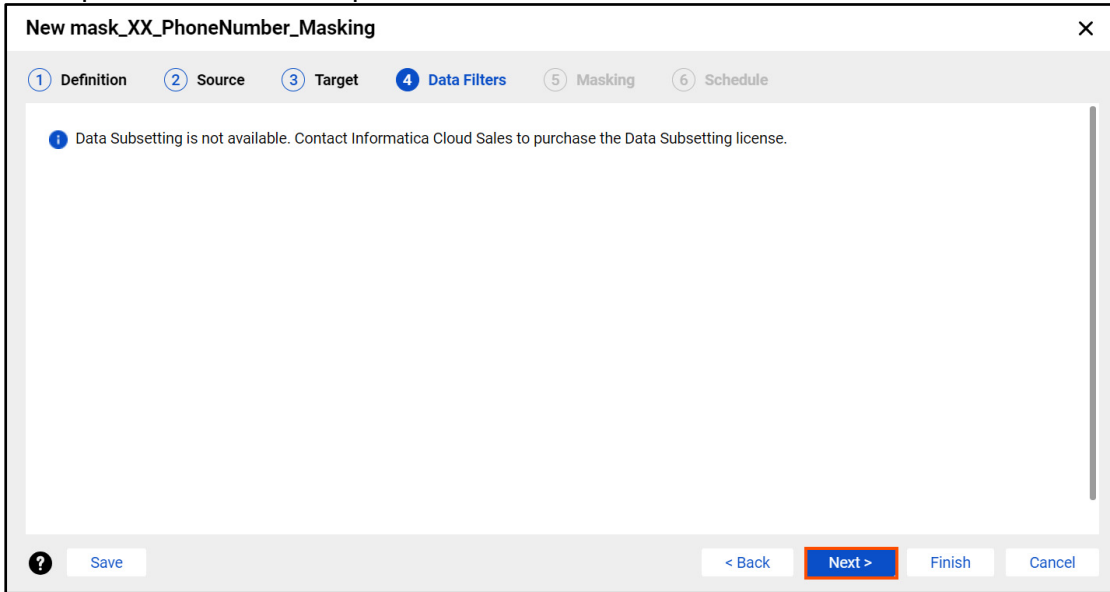
☒ Same as source? ⓘ

Connection:* SXX_SFDCDeveloper1 View... New... ⓘ

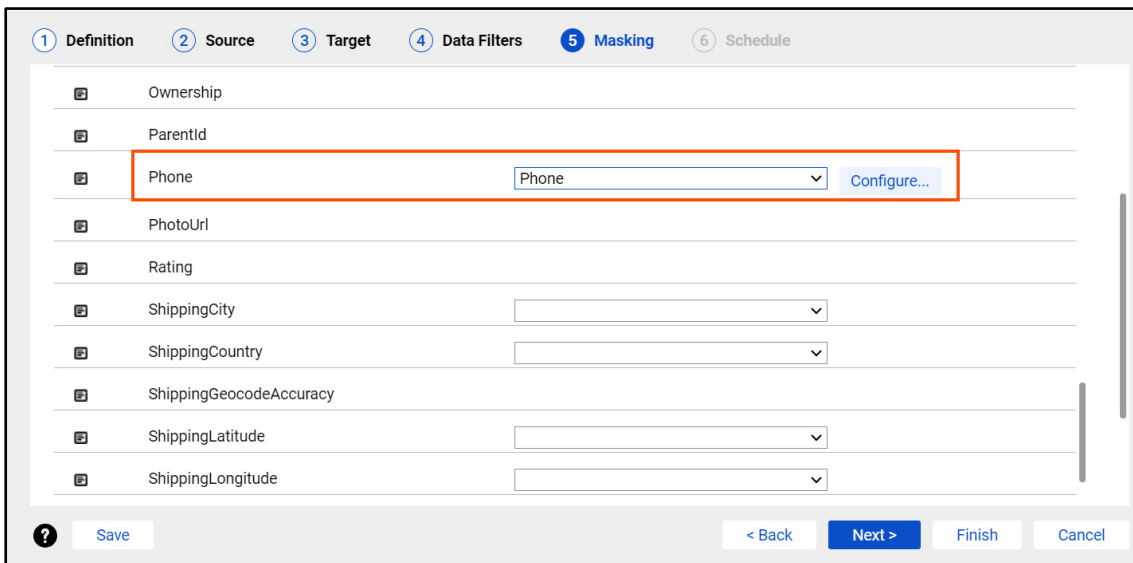
Task Operation:* Update ⓘ

20. Click **Next**.

21. To skip the Data Filters step, click **Next**.



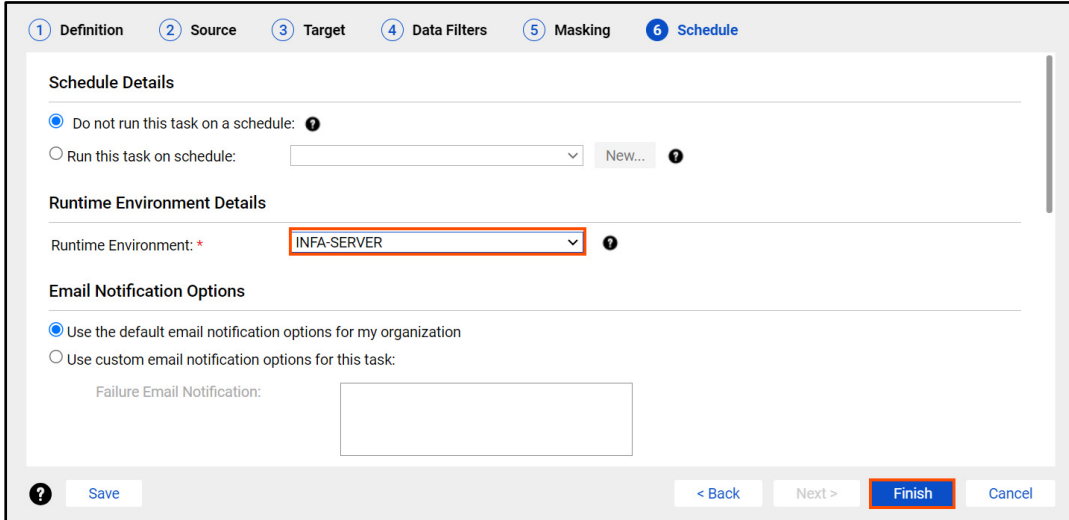
22. In the Define Masking Rules section, from the Phone field drop-down, select **Phone**.
Note: To mask all the fields of the Account object, you can use the Default Assignment option. For this lab, we will just mask the Phone field.



23. Click **Next**.

24. From the Runtime Environment drop-down, select **INFA-SERVER**.

25. To save and close the Masking Task wizard, click **Finish**.



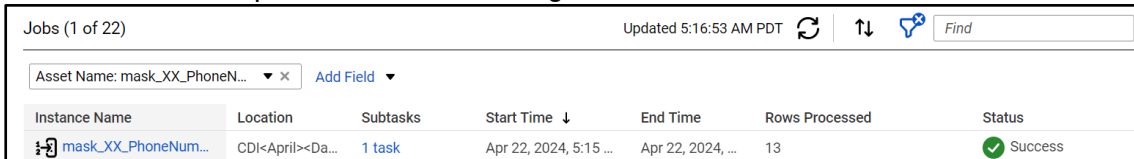
26. To run the Masking task, click **Run**.



Monitor the Status

27. To monitor the task, from the navigation pane, click **My Jobs**.

28. When the task completes, the status changes to **Success**.



Instance Name	Location	Subtasks	Start Time ↓	End Time	Rows Processed	Status
mask_XX_PhoneNum...	CDI<April><Da...	1 task	Apr 22, 2024, 5:15 ...	Apr 22, 2024, ...	13	Success

Note: The number of processed rows can change depending on the data in the Salesforce Account object.

29. Close the asset from the navigation pane.

Examine Results

30. Log in to your Salesforce Developer account using your credentials.

Note: You can use the below mentioned link to login into Salesforce:

<https://login.salesforce.com/>

31. On the Salesforce homepage, from the available tabs, select **Accounts**.



32. Observe the phone number for accounts noted earlier.

33. Verify that the phone numbers have been masked.

This concludes the lab.

Module 12: Masking Task

Lab 12-2: Using Data Masking Transformation in a Mapping

Overview:

Data Masking transformation changes the sensitive production data into realistic test data for non-production environments. The Data Masking transformation modifies source data based on masking rules that you configure for each column.

Objective:

- Configure a mapping using Data Masking transformation
- Mask the Surname, DOB, and State field values in the source data

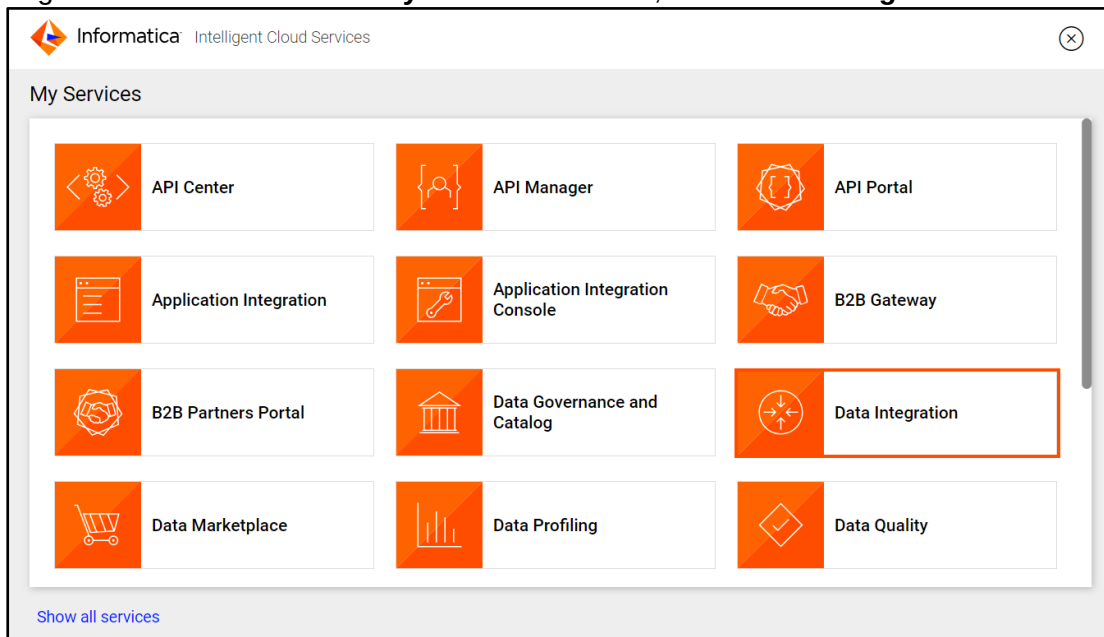
Duration:

10 minutes

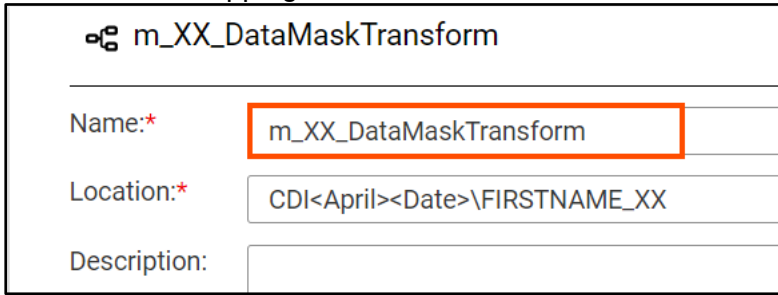
Tasks

Create Mapping

1. Login into IICS and from the **My Services** window, select **Data Integration**.



2. Create a new mapping and name it **m_XX_DataMaskTransform**.



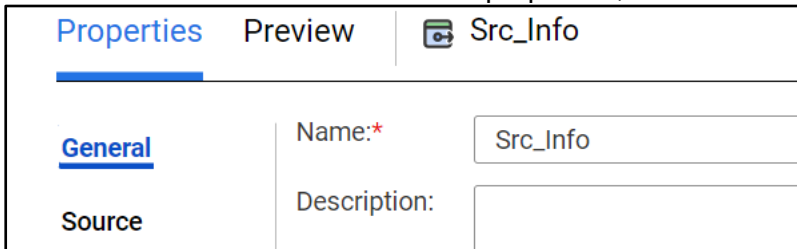
m_XX_DataMaskTransform

Name:* **m_XX_DataMaskTransform**

Location:* CDI<April><Date>\FIRSTNAME_XX

Description:

3. To configure the source, from the mapping canvas, click the **Source** transformation.
4. In the **General** section of the Source properties, enter the Name as **Src_Info**.



Properties Preview **Src_Info**

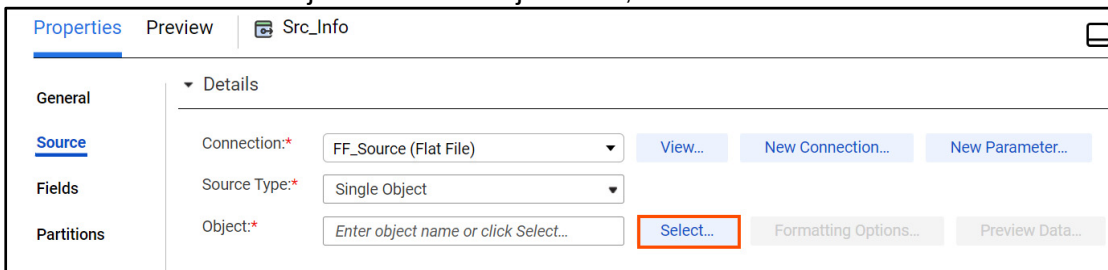
General

Name:* **Src_Info**

Description:

Source

5. From the properties pane, click **Source**.
6. From the Connection drop-down, select **FF_Source**.
7. To select the source object from the Object field, click **Select**.



Properties Preview **Src_Info**

General

Source

Fields

Partitions

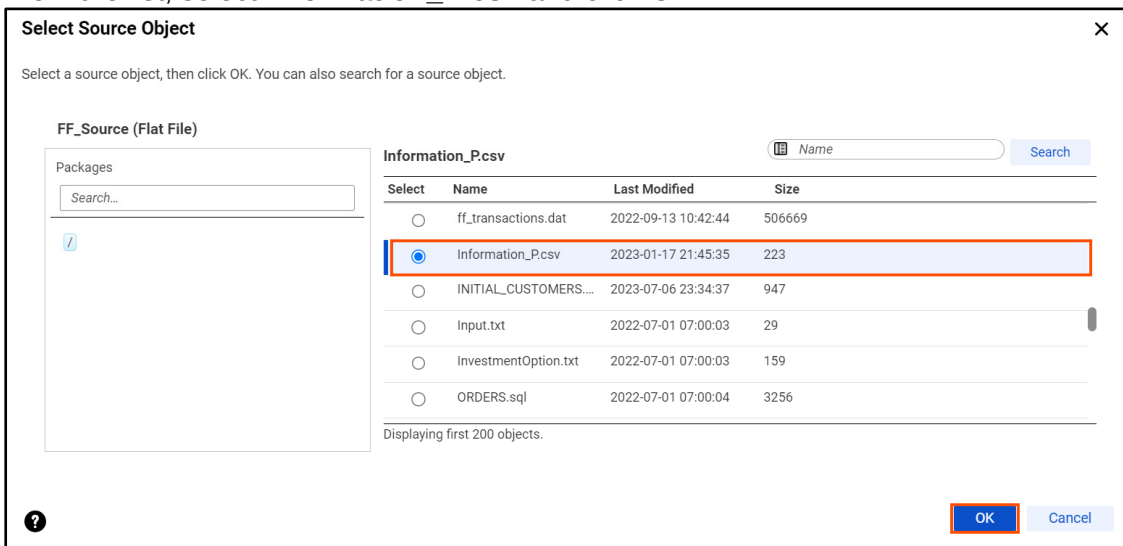
Details

Connection:* **FF_Source (Flat File)** View... New Connection... New Parameter...

Source Type:* Single Object

Object:* Enter object name or click Select... **Select...** Formatting Options... Preview Data...

8. From the list, select **Information_P.csv** and click **OK**.



Select Source Object

Select a source object, then click OK. You can also search for a source object.

FF_Source (Flat File)

Packages

Search...

Information_P.csv

Name Search

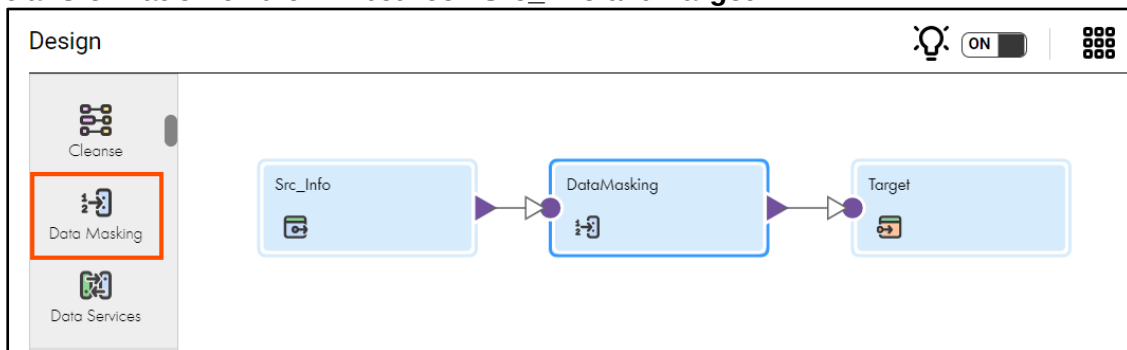
Select	Name	Last Modified	Size
☐	ff_transactions.dat	2022-09-13 10:42:44	506669
☒	Information_P.csv	2023-01-17 21:45:35	223
☐	INITIAL_CUSTOMERS...	2023-07-06 23:34:37	947
☐	Input.txt	2022-07-01 07:00:03	29
☐	InvestmentOption.txt	2022-07-01 07:00:03	159
☐	ORDERS.sql	2022-07-01 07:00:04	3256

Displaying first 200 objects.

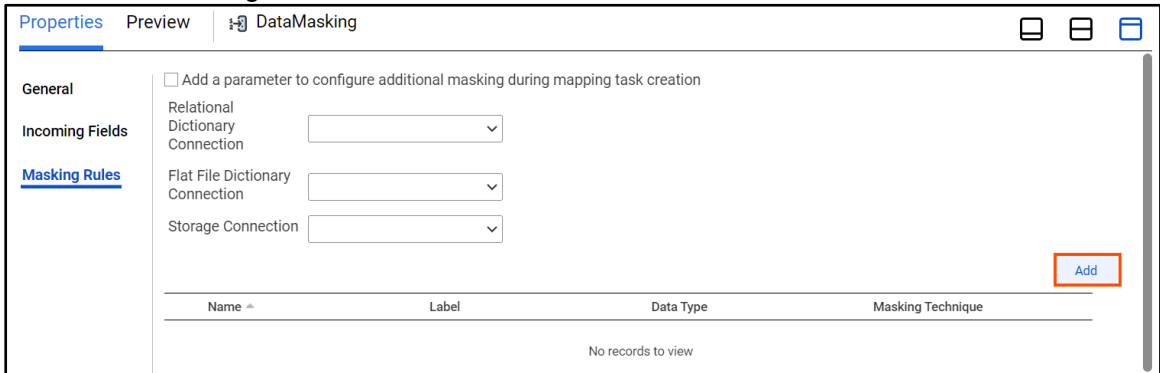
OK Cancel

Add Data Masking Transformation

- From the list of available transformations, drag and drop the **Data Masking transformation** on the link between **Src_Info** and **Target**.

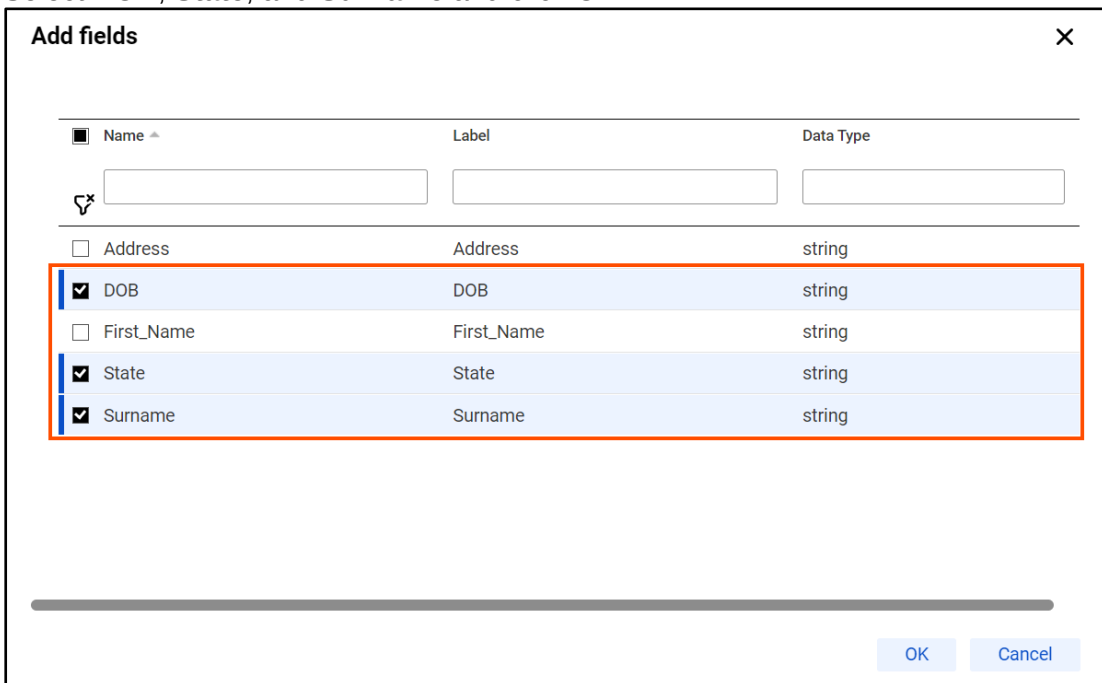


- Select the **DataMasking** transformation from the mapping canvas.
- From the properties pane, click **Masking Rules**.
- To add the Masking rules, click **Add**.



The Properties pane for the DataMasking transformation is shown. The 'Masking Rules' tab is selected. The 'General' section has a checkbox 'Add a parameter to configure additional masking during mapping task creation'. The 'Incoming Fields' section has dropdowns for 'Relational Dictionary Connection', 'Flat File Dictionary Connection', and 'Storage Connection'. The 'Masking Rules' section has an 'Add' button highlighted. Below is a table with columns: Name, Label, Data Type, and Masking Technique. The table is currently empty, showing 'No records to view'.

- Select **DOB**, **State**, and **Surname** and click **OK**.



The 'Add fields' dialog box is shown. It has a table with columns: Name, Label, and Data Type. The following fields are selected (checked):

Name	Label	Data Type
☒ Address	Address	string
☒ DOB	DOB	string
☐ First_Name	First_Name	string
☒ State	State	string
☒ Surname	Surname	string

At the bottom, there are 'OK' and 'Cancel' buttons.

Properties Preview DataMasking

General

☐ Add a parameter to configure additional masking during mapping task creation

Incoming Fields

Relational Dictionary Connection

Masking Rules

Flat File Dictionary Connection

Storage Connection

Name	Label	Data Type	Masking Technique	
1	DOB	DOB	string	Configure...
2	State	State	string	Configure...
3	Surname	Surname	string	Configure...

14. To configure the DOB, click **Configure**.

15. From the Configure Masking Rules drop down, select the **Random** option

Configure Masking Rules

Masking Technique:

Configure Masking Rules ✕

Masking Technique: Random

Use Mask Format: ☒ Yes ☐ No

Mask Format: R

Filter Source: ☐ Yes ☒ No

Source Filter Type:

Source Filter Chars:

Filter Target: ☐ Yes ☒ No

Target Filter Type:

Target Filter Chars:

OK
Cancel

16. Review all the other default settings and click **OK**.

17. Similarly, configure the **State** and **Surname** fields, by selecting **Substitution state** and **Substitution last name** respectively.

Properties
Preview
DataMasking

General

Incoming Fields

Masking Rules

☐ Add a parameter to configure additional masking during mapping task creation

Relational Dictionary Connection

Flat File Dictionary Connection

Storage Connection

Add

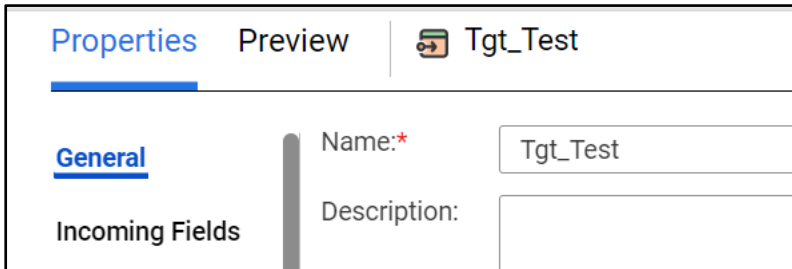
	Name	Label	Data Type	Masking Technique
1	DOB	DOB	string	Random
2	State	State	string	Substitution state
3	Surname	Surname	string	Substitution last name

Configure Target Transformation

18. Link the **Data Masking** transformation with the **Target** transformation if it is not linked.

19. Select the **Target** transformation from the mapping canvas.

20. In the General section of Target properties, enter Name as **Tgt_Test**.



The screenshot shows the 'Target Properties' dialog box with the 'Properties' tab selected. Under the 'General' section, the 'Name' field is populated with 'Tgt_Test'. The 'Description' field is empty. The 'Incoming Fields' section is also visible but empty.

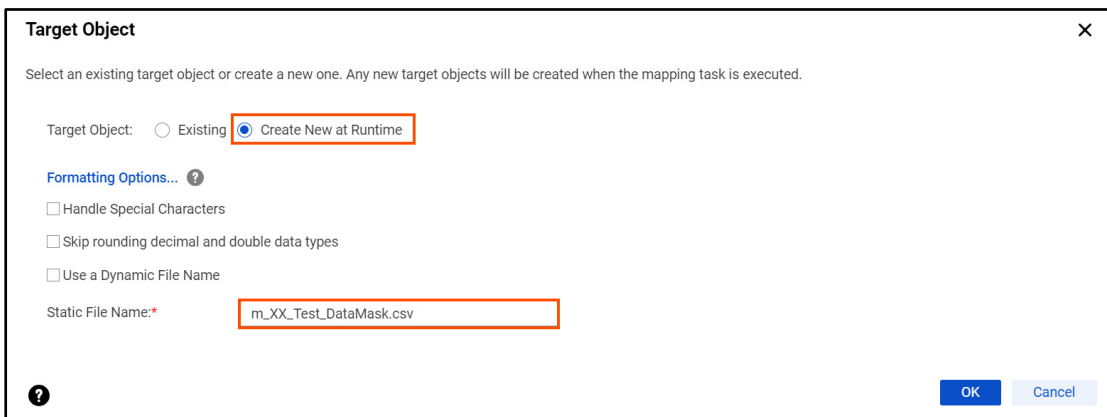
21. From the properties pane, select **Target**.

22. From the Connection drop-down, select your **target Flat File** connection.

23. To choose an object, click **Select**.

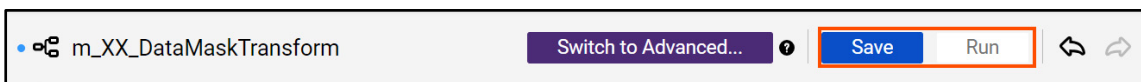
24. In the Objects fields, select **Create New at Runtime**.

25. In the File Name, enter the name as **m_XX_Test_DataMask.csv** and click **OK**.



The screenshot shows the 'Target Object' dialog box. The 'Target Object' section has 'Create New at Runtime' selected. Under 'Formatting Options...', there are three unchecked checkboxes: 'Handle Special Characters', 'Skip rounding decimal and double data types', and 'Use a Dynamic File Name'. The 'Static File Name' field is set to 'm_XX_Test_DataMask.csv'. The 'OK' button is highlighted.

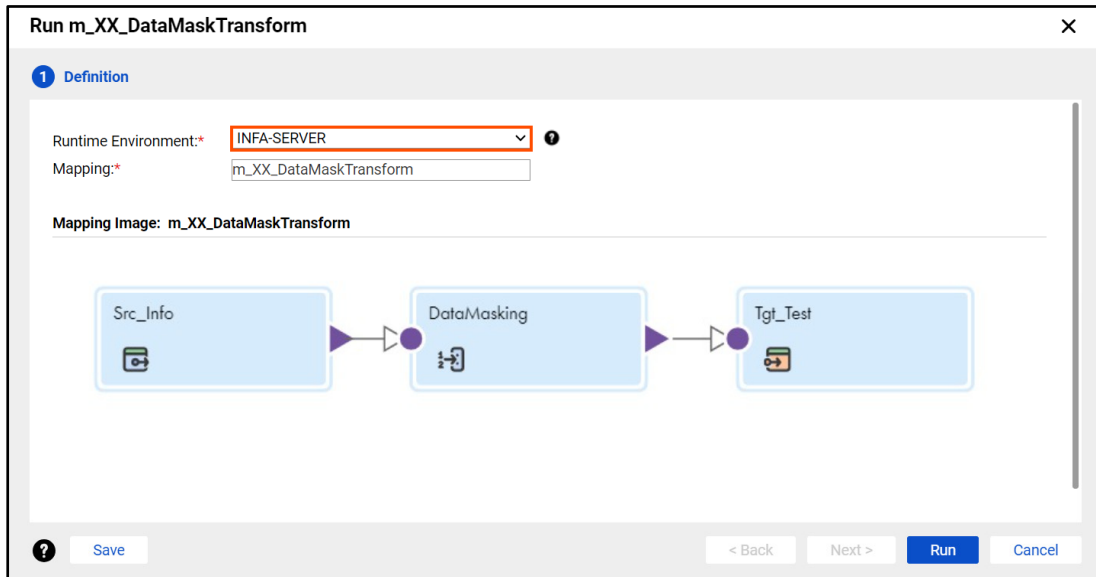
26. Save and run the mapping.



The screenshot shows the mapping toolbar for 'm_XX_DataMaskTransform'. The 'Save' button is highlighted with a red box. Other buttons include 'Switch to Advanced...', 'Run', and a 'Refresh' button.

27. From the Runtime Environment drop-down, select **INFA-SERVER**.

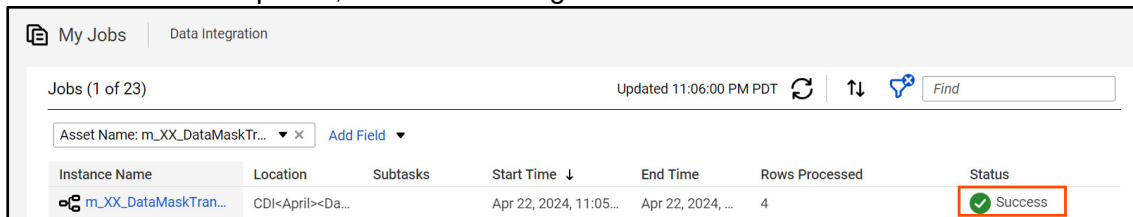
28. Click **Run**.



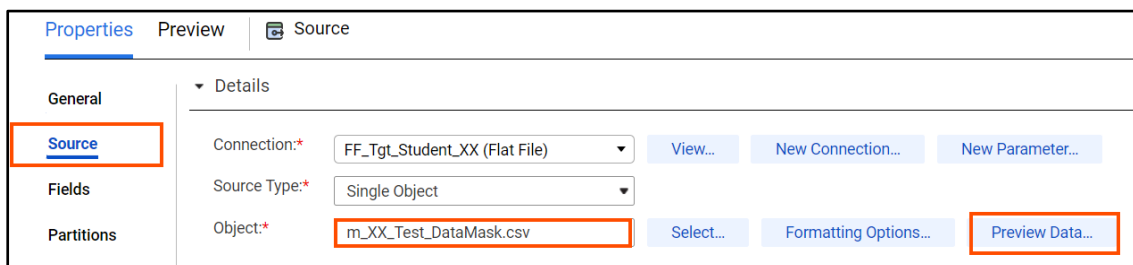
Monitor Status

29. Monitor the task status from the **My Jobs** page.

30. When the task completes, the status changes to **Success**.



31. Optionally, you can view results using your dummy mapping.



Data Preview

Connection: FF_Tgt_Student_XX

Object: m_XX_Test_DataMask.csv

out_First_Name	out_Surname	out_DOB	out_Address	out_State
Uma	Millard	]]5';K"	24, Atkins Avenue	Arkansas
John	Brandewie	-Ch&X!!L<	49, Wheeler Road	North Carolina
Keiko	Donohve	zT nCE,Clb	13, Draker Drive	Indiana
Isadora	Runzler	=")Cp:(e	20, Fountain Ce...	Arkansas

☐ Display source fields in alphabetical order

?

Formatting Options...

Done

32. Save and close all the assets from the navigation pane.

This concludes the lab.

Module 13: Taskflows

Lab 13-1: Creating a Linear Taskflow

Overview:

A Taskflow enables you to add multiple data integration tasks and run them in a specific sequence or in parallel.

In this lab, you will create Linear Taskflow to group multiple data integration tasks and run them in the order that is specified.

Objective:

- Create a Linear taskflow
- Configure email notifications for Linear taskflow

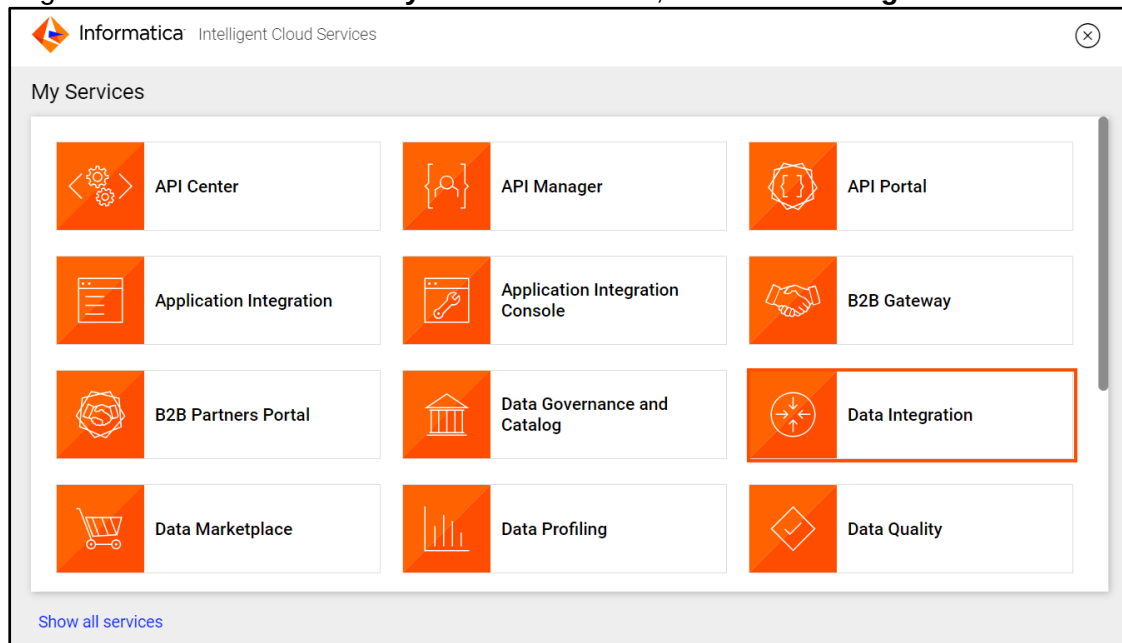
Duration:

20 minutes

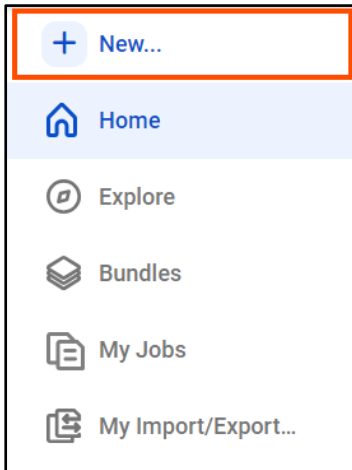
Tasks

Create Taskflow

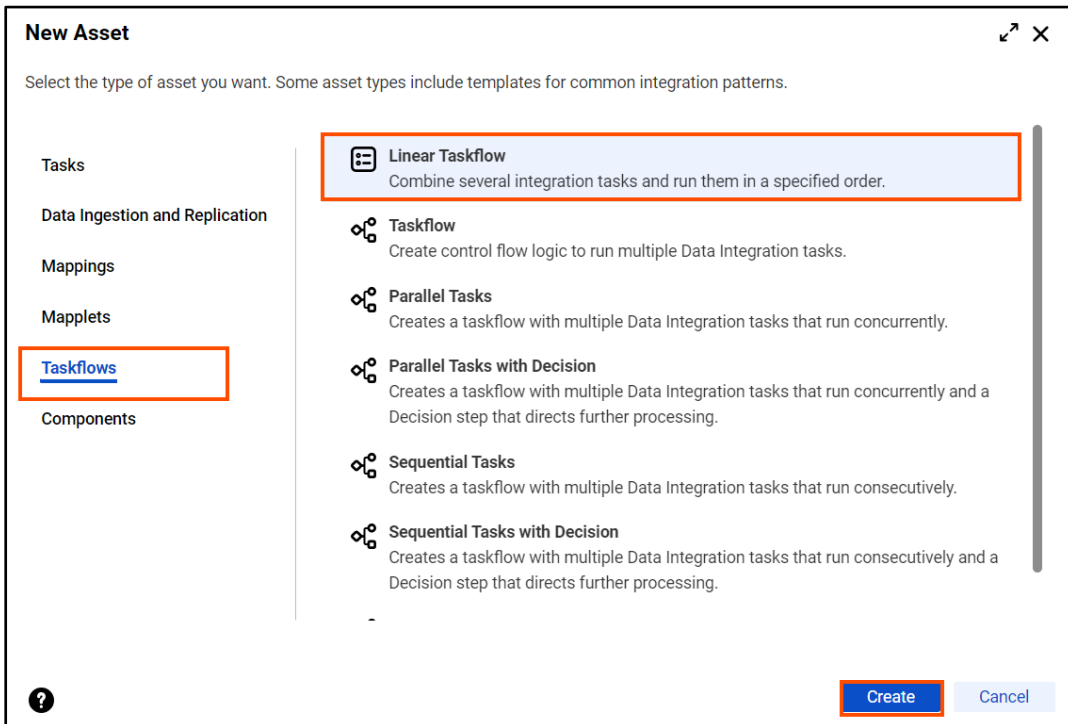
1. Login into IICS and from the **My Services** window, select **Data Integration**.



2. From the navigation pane, select **New**.



3. From the New Asset window, click the **Taskflows** tab.
4. Select **Linear Taskflow** and click **Create**.



5. In the Name field, enter **tf_XX_load_Payment_Taskflow**.

•  **Linear Taskflow1**

Linear Taskflow Details

Linear Taskflow Name:* ?

Location:* [Browse](#)

Description: ?

6. Verify that the asset Location is pointing to your working folder.

7. From the **List of Tasks** tab, click on **Add Task**.

Linear Taskflow Details

Linear Taskflow Name:* ?

Location:* [Browse](#)

Description: ?

Schedule: ☒ Do not run this task on a schedule ?
☐ Run this task on schedule: [New...](#) ?

List of Tasks [Add Task...](#) [Refresh](#)

There are no tasks associated with this linear taskflow. Click Add Task to add tasks to the linear taskflow.

8. Go to your work folder. You will add the following pre-created mapping tasks from this folder:

- syn_XX_OutletsLoad
- syn_XX_Employee
- syn_XX_UPSERT_CUSTOMERS
- mask_XX_PhoneNumber_Masking

9. Select the task **syn_XX_OutletsLoad** and click **Select**.

10. Click on **Add Task** again and go to your working folder.

11. Select the task **syn_XX_Employee** and click **Select**.

12. Similarly, add other tasks – **syn_XX_UPSERT_CUSTOMERS** and **mask_XX_PhoneNumber_Masking**.

List of Tasks Add Task... Refresh					
Sequence	Name	Type	Stop on Error	Stop on Warning	Delete
☐	CDI<April><Date>\FIRSTNAME_XX\syn_XX_OutletsLoad	Synchronization	☐	☐	
☐	CDI<April><Date>\FIRSTNAME_XX\syn_XX_Employee	Synchronization	☐	☐	
☐	CDI<April><Date>\FIRSTNAME_XX\syn_XX_UPSERT_CUSTOMERS	Synchronization	☐	☐	
☐	CDI<April><Date>\FIRSTNAME_XX\mask_XX_PhoneNumber_Masking	Masking	☐	☐	

13. You can see the **Sequence** of the Mapping Task as 1,2, 3, and 4 respectively.

List of Tasks Add Task... Refresh					
Sequence	Name	Type	Stop on Error	Stop on Warning	Delete
1	CDI<April><Date>\FIRSTNAME_XX\syn_XX_OutletsLoad	Synchronization	☐	☐	
2	CDI<April><Date>\FIRSTNAME_XX\syn_XX_Employee	Synchronization	☐	☐	
3	CDI<April><Date>\FIRSTNAME_XX\syn_XX_UPSERT_CUSTOMERS	Synchronization	☐	☐	
4	CDI<April><Date>\FIRSTNAME_XX\mask_XX_PhoneNumber_Masking	Masking	☐	☐	

Note: You can change the Sequence number of the Mapping Tasks to any other order of your preference.

14. Check the **Stop on Error** checkboxes for all the 4 tasks.

15. Also, check the **Stop on Warning** checkboxes for all the tasks.

List of Tasks Add Task... Refresh					
Sequence	Name	Type	Stop on Error	Stop on Warning	Delete
1	CDI<April><Date>\FIRSTNAME_XX\syn_XX_OutletsLoad	Synchronization	☒	☒	
2	CDI<April><Date>\FIRSTNAME_XX\syn_XX_Employee	Synchronization	☒	☒	
3	CDI<April><Date>\FIRSTNAME_XX\syn_XX_UPSERT_CUSTOMERS	Synchronization	☒	☒	
4	CDI<April><Date>\FIRSTNAME_XX\mask_XX_PhoneNumber_Masking	Masking	☒	☒	

16. From the Email Notification Options, select the **Use custom email notification options for this task** option.

17. Enter your valid email address in the **Failure Email Notification**, **Warning Email Notification**, and **Success Email Notification**.

Email Notification Options

☐ Use the default email notification options for my organization

☒ Use custom email notification options for this task:

Failure Email Notification:

Warning Email Notification:

Success Email Notification:

18. Save and run the Linear Taskflow.

• tf_XX_load_Payment_Taskflow Save Run

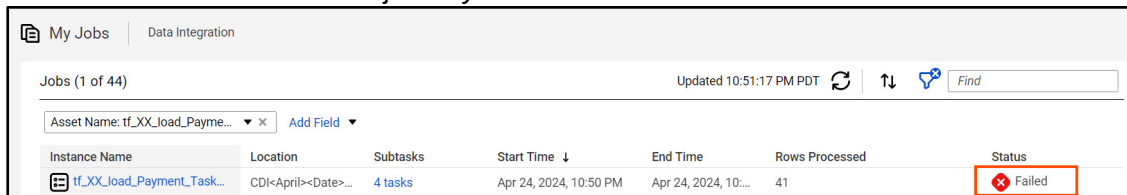
Linear Taskflow Details

Linear Taskflow Name: * tf_XX_load_Payment_Taskflow

Monitor the Status

19. To monitor the task, from the navigation pane, click **My Jobs**.

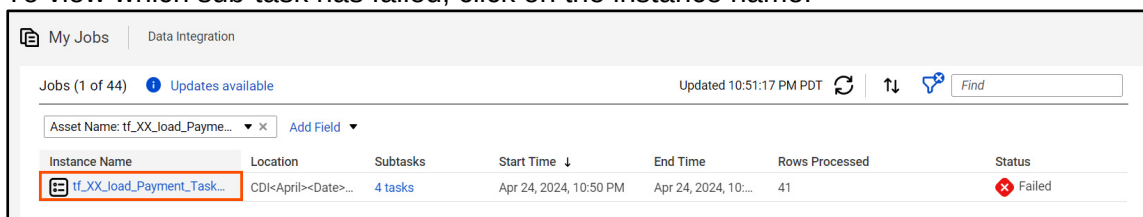
20. It may take 1 to 2 minutes for the task to complete. When the task run is complete, you will see that the status of the job says **Failed**.



Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
tf_XX_load_Payment_Task...	CDI<April><Date>...	4 tasks	Apr 24, 2024, 10:50 PM	Apr 24, 2024, 10:...	41	Failed

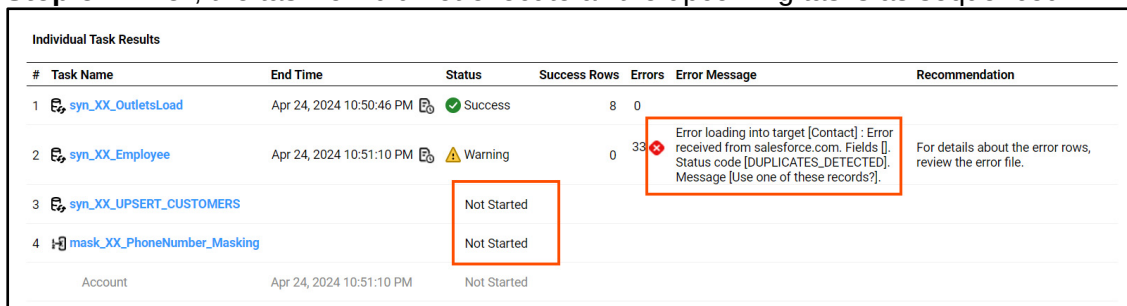
Note: You can refresh the page if the status does not change automatically.

21. To view which sub-task has failed, click on the instance name.



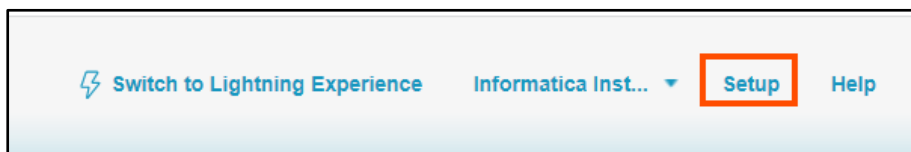
Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
tf_XX_load_Payment_Task...	CDI<April><Date>...	4 tasks	Apr 24, 2024, 10:50 PM	Apr 24, 2024, 10:...	41	Failed

22. Scroll down and observe that the error has occurred because of loading duplicate records into the salesforce object **Contact** (as you have previously run the task and loaded data into the Contact object). Also, note that, as you selected the checkbox – **Stop on Error**, the taskflow did not execute all the upcoming tasks as sequenced.

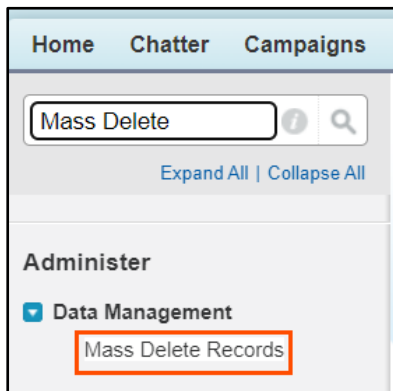


#	Task Name	End Time	Status	Success Rows	Errors	Error Message	Recommendation
1	syn_XX_OutletsLoad	Apr 24, 2024 10:50:46 PM	Success	8	0		
2	syn_XX_Employee	Apr 24, 2024 10:51:10 PM	Warning	0	33	Error loading into target [Contact] : Error received from salesforce.com. Fields []. Status code [DUPLICATES_DETECTED]. Message [Use one of these records?].	For details about the error rows, review the error file.
3	syn_XX_UPSERT_CUSTOMERS		Not Started				
4	mask_XX_PhoneNumber_Masking		Not Started				
	Account	Apr 24, 2024 10:51:10 PM	Not Started				

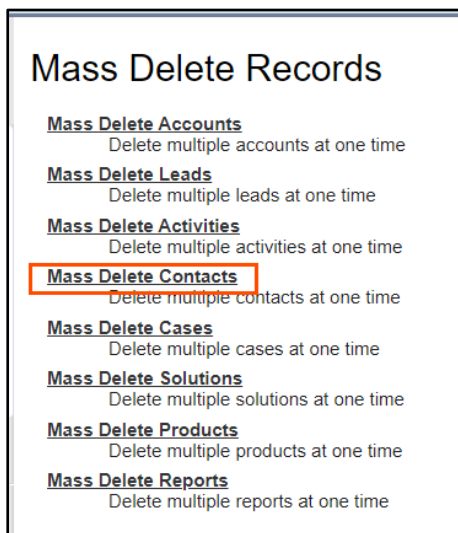
23. To troubleshoot the issue, you need to clear all the contacts records from Salesforce. Login into your Salesforce account and go to the **Setup** page.



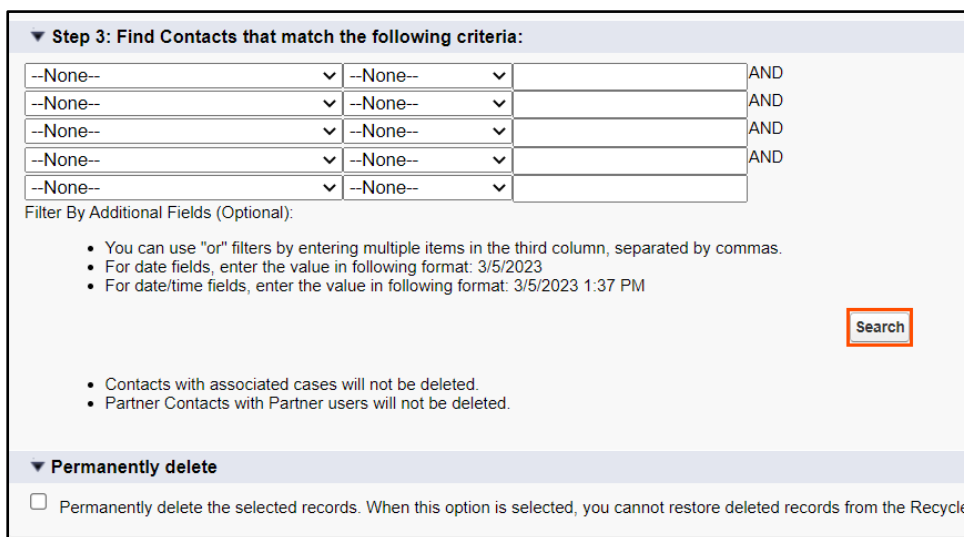
24. From the Search box on your left, enter Mass Delete and click the **Mass Delete Records** section.



25. Click on the **Mass Delete Contacts** link.



26. Scroll down and click **Search**.



▼ Step 3: Find Contacts that match the following criteria:

--None--	▼	--None--	▼		AND
--None--	▼	--None--	▼		AND
--None--	▼	--None--	▼		AND
--None--	▼	--None--	▼		AND
--None--	▼	--None--	▼		

Filter By Additional Fields (Optional):

- You can use "or" filters by entering multiple items in the third column, separated by commas.
- For date fields, enter the value in following format: 3/5/2023
- For date/time fields, enter the value in following format: 3/5/2023 1:37 PM

Search

▼ Permanently delete

☐ Permanently delete the selected records. When this option is selected, you cannot restore deleted records from the Recycle

27. Now you will see all the records that belong to Contacts. Select all the records including the permanent delete checkbox and click **Delete**.

- Contacts with associated cases will not be deleted.
- Partner Contacts with Partner users will not be deleted.

▼ Permanently delete

☒ Permanently delete the selected records. When this option is selected, you cannot restore deleted records from the Recycle Bin. Please

Delete

☒	Name	Account Name	Title	Phone	Email
☒	Bailey, Patrick	NH Trends		319-812-6957	patrick.bailey@nhsupplies.com
☒	Baker, Nancy	NH Supplies		229-336-5117	nancy.baker@nhsupplies.com
☒	Brown, Donna	NH Everyday		212-434-7910	donna.brown@nhsupplies.com
☒	Brown, Lillian	NH D'needs		239-812-9041	lillian.brown@nhsupplies.com
☒	Butler, Brenda	NH Supplies		480-814-8284	brenda.butler@nhsupplies.com
☒	Collins, Carl	NH Digiworld		479-740-7633	carl.collins@nhsupplies.com
☒	Cooper, Daniel	NH Digital		701-510-8162	daniel.cooper@nhsupplies.com

28. Back in Data Integration service window, restart the job and view the results.

Job Properties
 Task Name: [tf_XX_load_Payment_Taskflow](#)
 Instance ID: 2
 Task Type: Linear Taskflow

Results
 Status: ✖ Failed
 Success Rows: 8
 Errors: 33

My Jobs
Data Integration

Jobs (3 of 49) Updated 11:48:33 PM PDT Find

Asset Name: [tf_XX_load_Payme...](#) Add Field

Instance Name	Location	Subtasks	Start Time ↓	End Time	Rows Processed	Status
tf_XX_load_Payment_Task...	CDI<April><Date>...	4 tasks	Apr 24, 2024, 11:47 PM	Apr 24, 2024, 11:...	91	✔ Success

29. To view the status of the subtask, click on **4 tasks**.

30. Verify that all the four tasks are executed successfully.

My Jobs
Data Integration > tf_XX_load_Payment_Taskflow-3

Jobs (4) i Updates available Updated 11:49:26 PM PDT Find

Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
mask_XX_PhoneNumber_M...	CDI<April><Date>...	1 task	Apr 24, 2024, 11:47 PM	Apr 24, 2024, 11:...	38	✔ Success
syn_XX_UPSERT_CUSTOM...	CDI<April><Date>...		Apr 24, 2024, 11:47 PM	Apr 24, 2024, 11:...	12	✔ Success
syn_XX_Employee-5	CDI<April><Date>...		Apr 24, 2024, 11:47 PM	Apr 24, 2024, 11:...	33	✔ Success
syn_XX_OutletsLoad-6	CDI<April><Date>...		Apr 24, 2024, 11:47 PM	Apr 24, 2024, 11:...	8	✔ Success

31. Open your email inbox and check the task notification emails. It may take 2 to 3 minutes for the email notifications to initiate.

32. Close the assets from the navigation pane.

This concludes the lab.

Module 13: Taskflows

Lab 13-2: Creating a Taskflow with Conditional Logic

Overview:

A Taskflow enables you to add multiple data integration tasks and run them in a specific sequence or in parallel.

Use a taskflow to control the execution sequence of a data transfer task, dynamic mapping task, mapping task, PowerCenter task, or synchronization task based on the output of the previous task.

Objective:

- Create a taskflow to run the Task.
- Get an email notification.
- Use conditional logic and take decision based on output of the task.

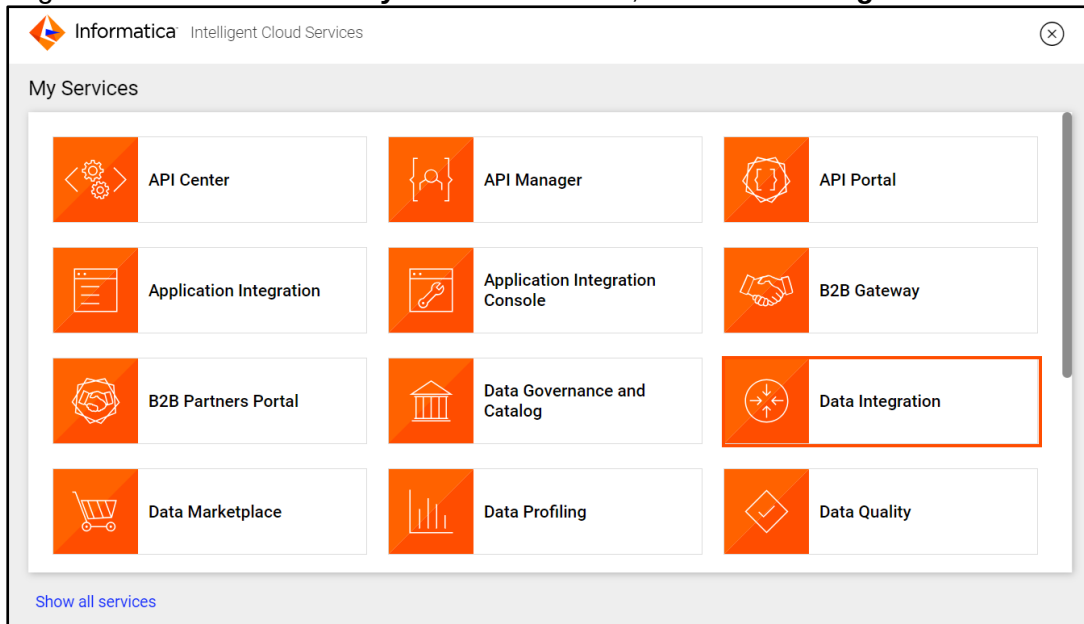
Duration:

40 minutes

Tasks

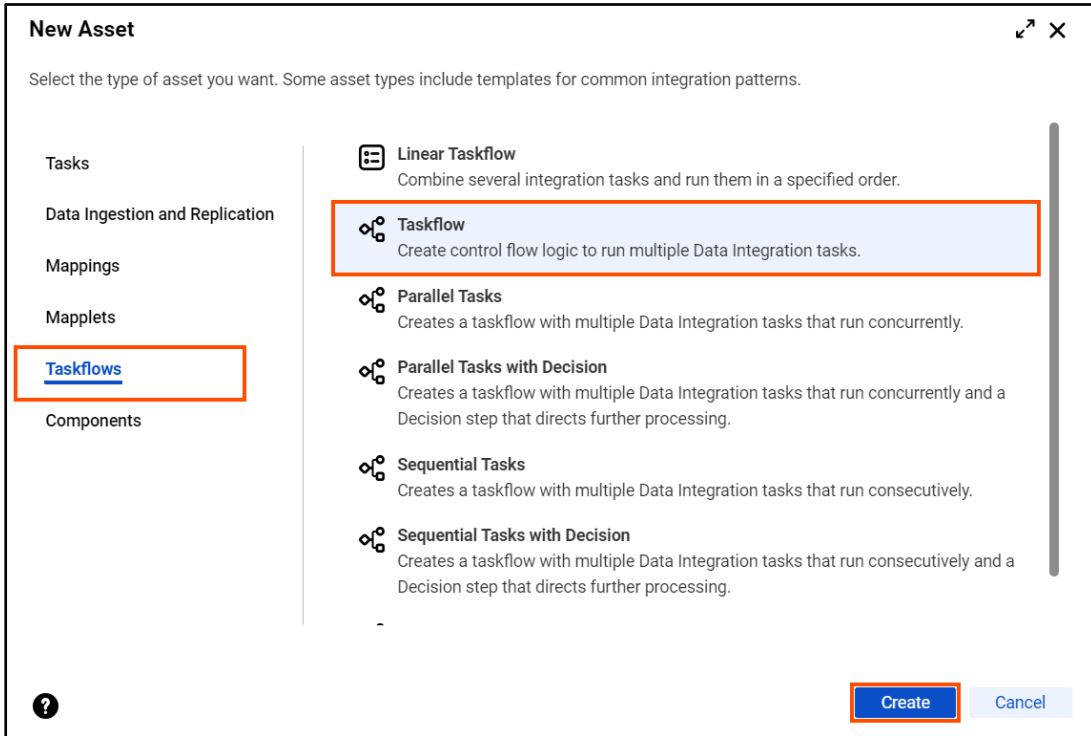
Create Taskflow

1. Login to IICS and from the **My Services** window, select **Data Integration**.



2. From the navigation pane, select **New**.
3. From the New Asset window, click the **Taskflows** tab.

4. Select **Taskflow** and click **Create**.



New Asset

Select the type of asset you want. Some asset types include templates for common integration patterns.

Tasks

Data Ingestion and Replication

Mappings

Maplets

Taskflows

Components

Linear Taskflow
Combine several integration tasks and run them in a specified order.

Taskflow
Create control flow logic to run multiple Data Integration tasks.

Parallel Tasks
Creates a taskflow with multiple Data Integration tasks that run concurrently.

Parallel Tasks with Decision
Creates a taskflow with multiple Data Integration tasks that run concurrently and a Decision step that directs further processing.

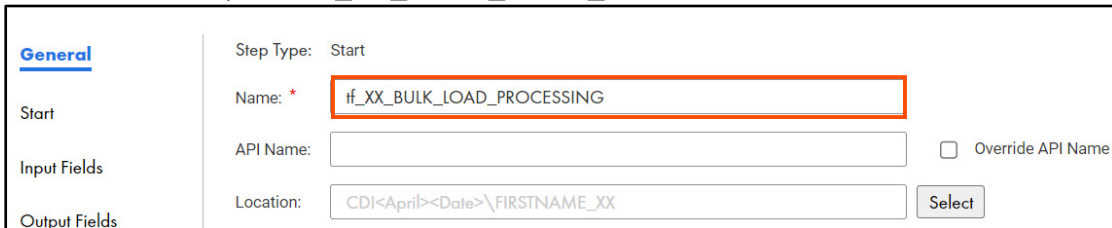
Sequential Tasks
Creates a taskflow with multiple Data Integration tasks that run consecutively.

Sequential Tasks with Decision
Creates a taskflow with multiple Data Integration tasks that run consecutively and a Decision step that directs further processing.

Create **Cancel**

5. From the Taskflow properties pane, select **General**.

6. In the Name field, enter **tf_XX_BULK_LOAD_PROCESSING**.



General

Step Type: Start

Name: * **tf_XX_BULK_LOAD_PROCESSING**

API Name: ☐ Override API Name

Location: **Select**

Start

Input Fields

Output Fields

7. From the properties pane, navigate to **Temp Fields**.

8. To add new Temp Field, click .



tf_XX_BULK_LOAD_PROCESSING Properties

General

Start

Input Fields

Output Fields

Temp Fields

Temp Fields (0)

Name	Type	Description	Initial Value
No data to display			



9. Enter the details as shown in table below.

Name	ROW_ERRORS
Type	Integer
Description	Total row level load errors in the first three tasks
Initial Value	0 (click the edit icon and enter the value)

10. To add another Temp Field, click .

11. Enter the details as shown in table below.

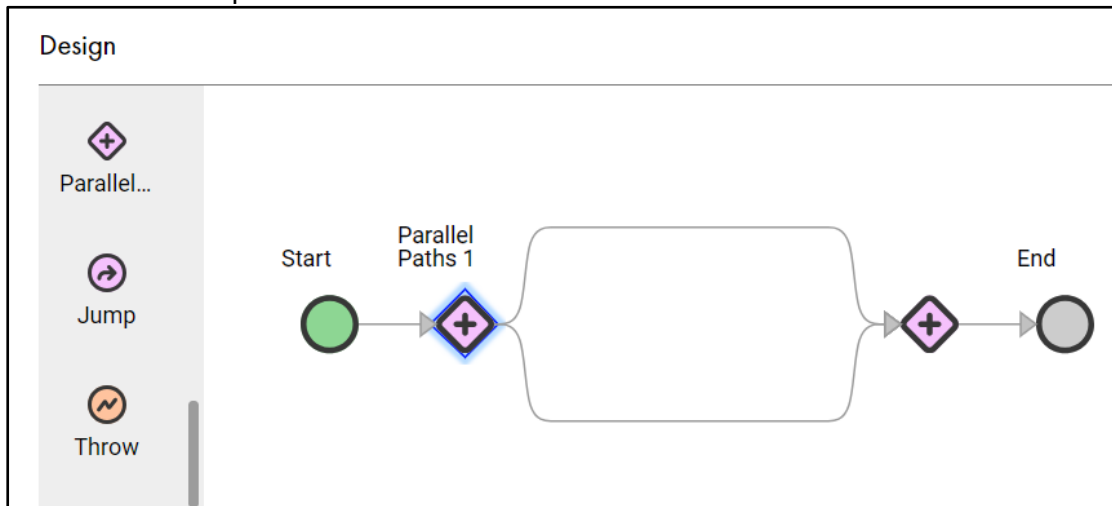
Name	INITIAL_TASK_STATUS
Type	Integer
Description	Will be zero if the first three tasks succeed
Initial Value	0

12. Similarly add another Temp field with the following configurations:

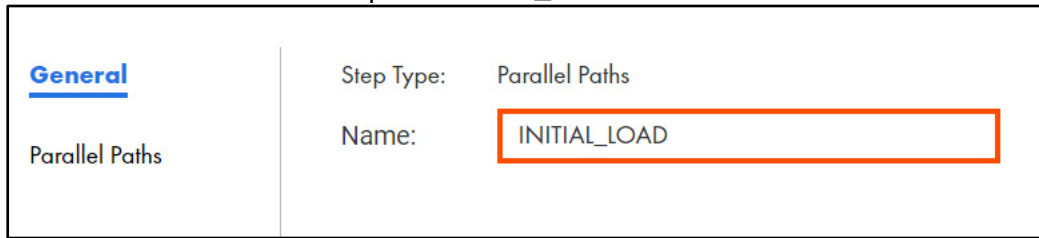
Name	INITIAL_LOAD_SUCCESS
Type	Integer
Description	Will be zero if both ROW_ERRORS and INITIAL_TASK_STATUS are zero
Initial Value	0

if_XX_BULK_LOAD_PROCESSING Properties				
General	Temp Fields (3)			
Start				
Input Fields				
Output Fields				
<u>Temp Fields</u>				
	Name	Type	Description	Initial Value
	ROW_ERRORS	Integer	Total row level load errors in the ...	0
	INITIAL_TASK_STATUS	Integer	Will be zero if the first three task...	0
	INITIAL_LOAD_SUCCESS	Integer	Will be zero if both ROW_ERROR...	0

13. From the Design pane, drag and drop the **Parallel Paths** step onto the link between Start and End steps.

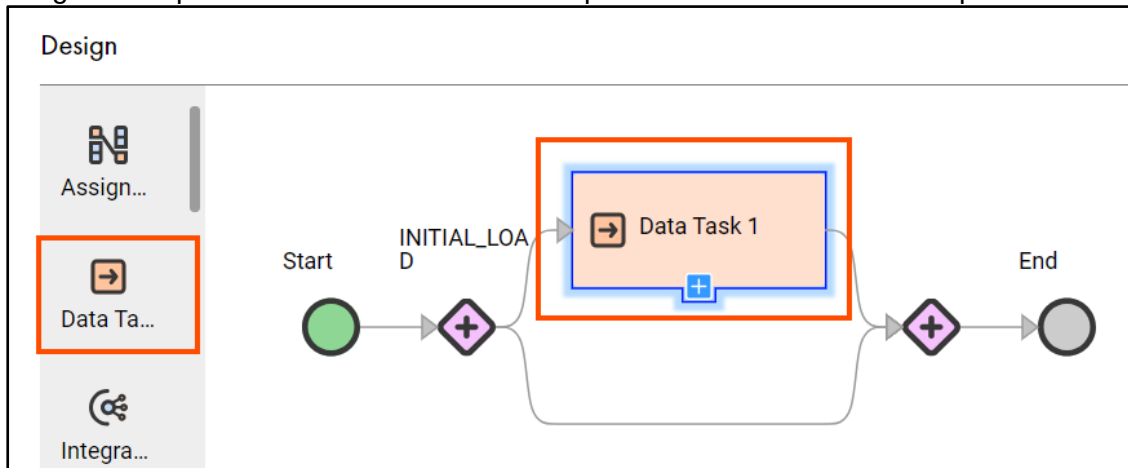


14. Name the Parallel Paths step as **INITIAL_LOAD**.



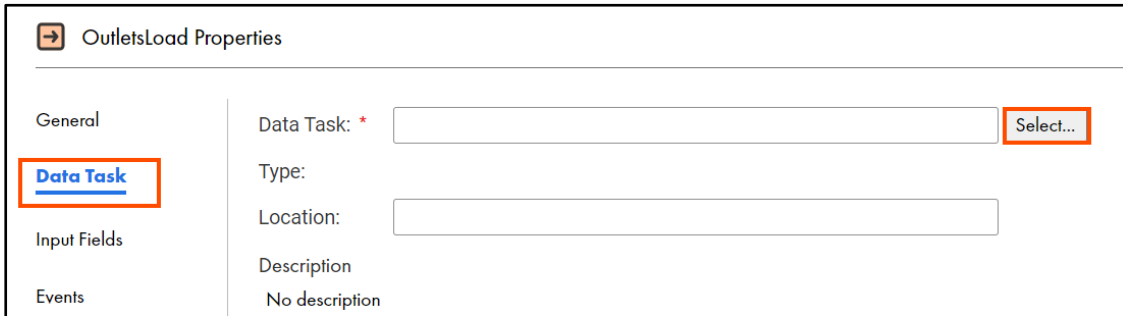
The screenshot shows the 'General' properties section for the 'Parallel Paths' step. The 'Step Type' is 'Parallel Paths'. The 'Name' field is highlighted with a red border and contains the text 'INITIAL_LOAD'.

15. Drag and drop the Data task onto to the first path of the Parallel Paths step.



16. In the General properties section, enter **OutletsLoad** as the name of the Data task.

17. With the Data task being selected, navigate to the **Data Task** properties tab. To choose the task, click **Select**.



OutletsLoad Properties

General

Data Task

Input Fields

Events

Data Task: * **Select...**

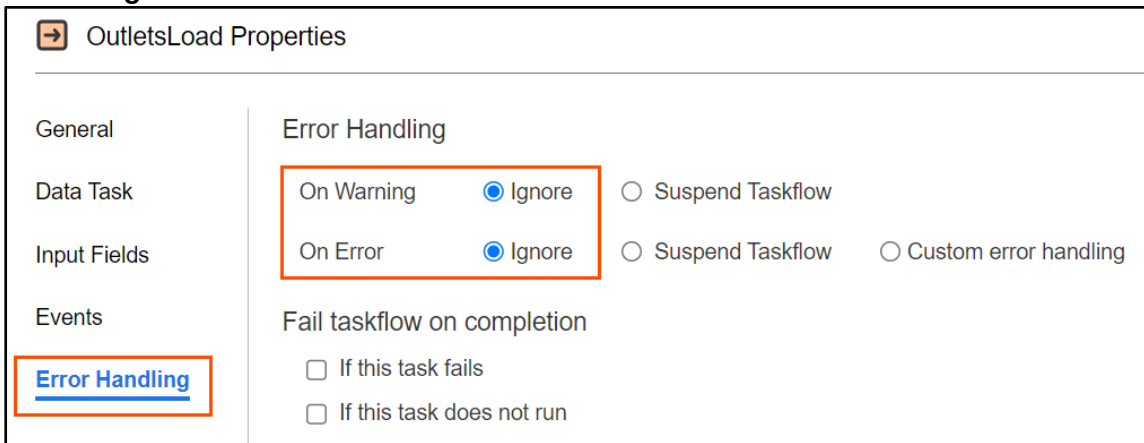
Type:

Location:

Description

No description

18. Navigate to your working folder and select the **syn_XX_OutletsLoad** task.
19. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.



OutletsLoad Properties

General

Data Task

Input Fields

Events

Error Handling

On Warning ☒ Ignore ☐ Suspend Taskflow

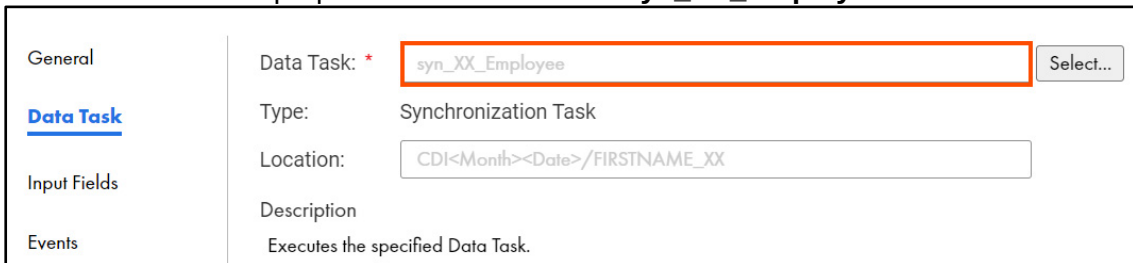
On Error ☒ Ignore ☐ Suspend Taskflow ☐ Custom error handling

Fail taskflow on completion

☐ If this task fails

☐ If this task does not run

20. Drag and drop another data task onto the 2nd Parallel path and name it as **Employee**.
21. Go to the Data Task properties tab and add the **syn_XX_Employee** task.



General

Data Task

Input Fields

Events

Data Task: * **Select...**

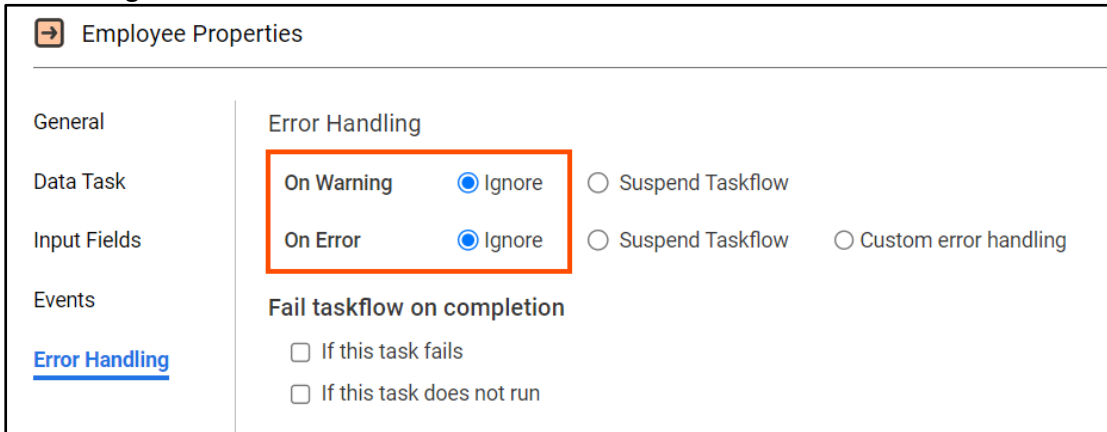
Type: Synchronization Task

Location:

Description

Executes the specified Data Task.

22. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.



The screenshot shows the 'Employee Properties' dialog box with the 'Error Handling' tab selected. The 'On Warning' and 'On Error' options are both set to 'Ignore', which is highlighted by a red box. The 'Fail taskflow on completion' section is also visible.

Property	Value
On Warning	Ignore
On Error	Ignore
Suspend Taskflow	Not selected
Custom error handling	Not selected

23. Select the Parallel Paths step from the canvas and navigate to the Parallel Paths tab. You can view 2 default paths. Click the plus icon to add the 3rd path to it.

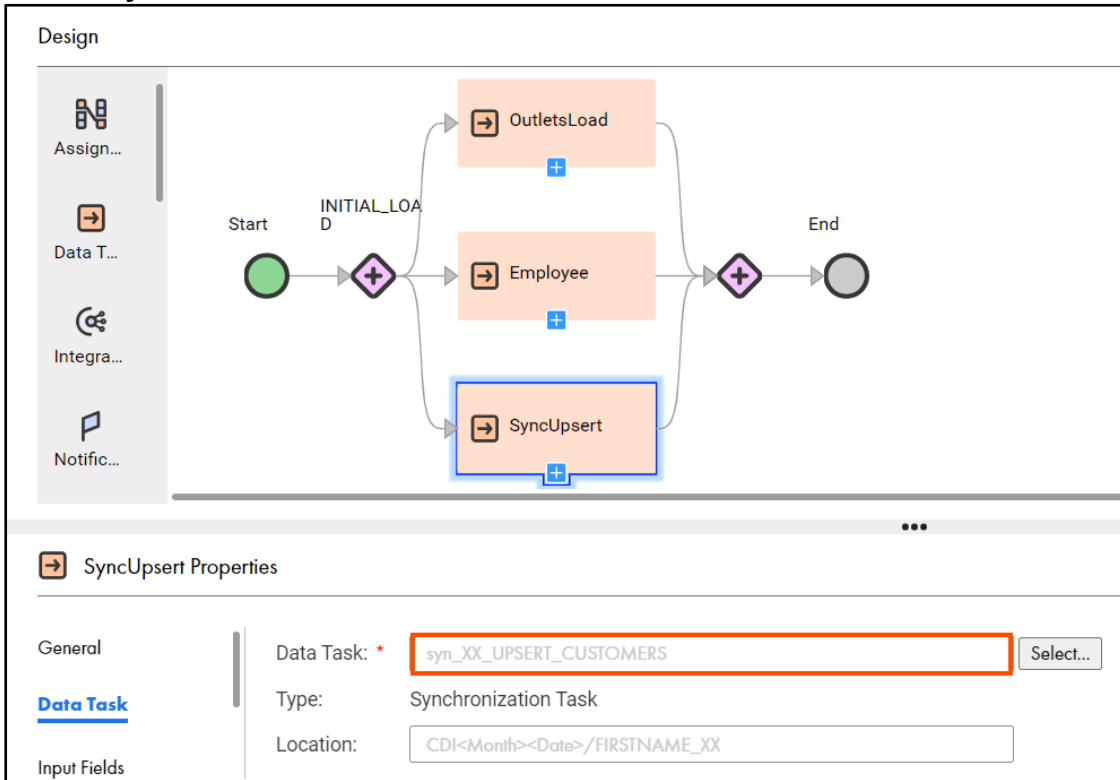


The screenshot shows the 'INITIAL_LOAD Properties' dialog box with the 'Parallel Paths' tab selected. There are three paths listed: Path 0, Path 1, and Path 2. Path 2 is highlighted with a red box, and a plus icon is visible in the top right corner of the table area.

Path	Description	Status
Path 0	Path starting with Data Task (OutletsLoad)	✓
Path 1	Path starting with Data Task (Employee)	✓ ✓
Path 2	New Path	✓

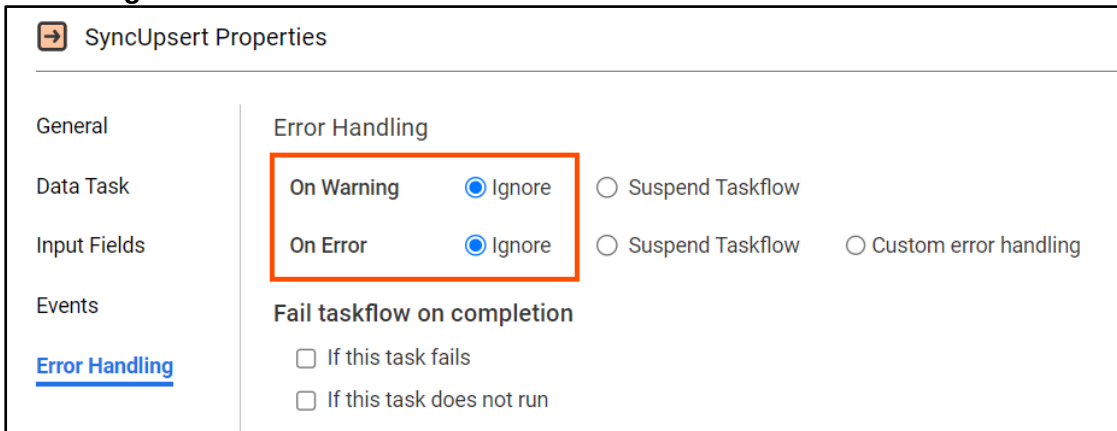
24. Drag and drop another Data task step onto the 3rd parallel path and name it as **SyncUpsert**.

25. Navigate to the Data Task properties tab and from your working directory, select and add the **syn_XX_UPSERT_CUSTOMERS** task.



The screenshot shows the Informatica Designer interface. The top pane displays a taskflow diagram with a 'Start' node, a split connector, three parallel tasks ('OutletsLoad', 'Employee', and 'SyncUpsert'), a join connector, and an 'End' node. The 'SyncUpsert' task is highlighted with a blue border. The bottom pane shows the 'SyncUpsert Properties' dialog. The 'Data Task' field is set to 'syn_XX_UPSERT_CUSTOMERS' and is highlighted with a red border. The 'Type' is 'Synchronization Task' and the 'Location' is 'CDI<Month><Date>/FIRSTNAME_XX'.

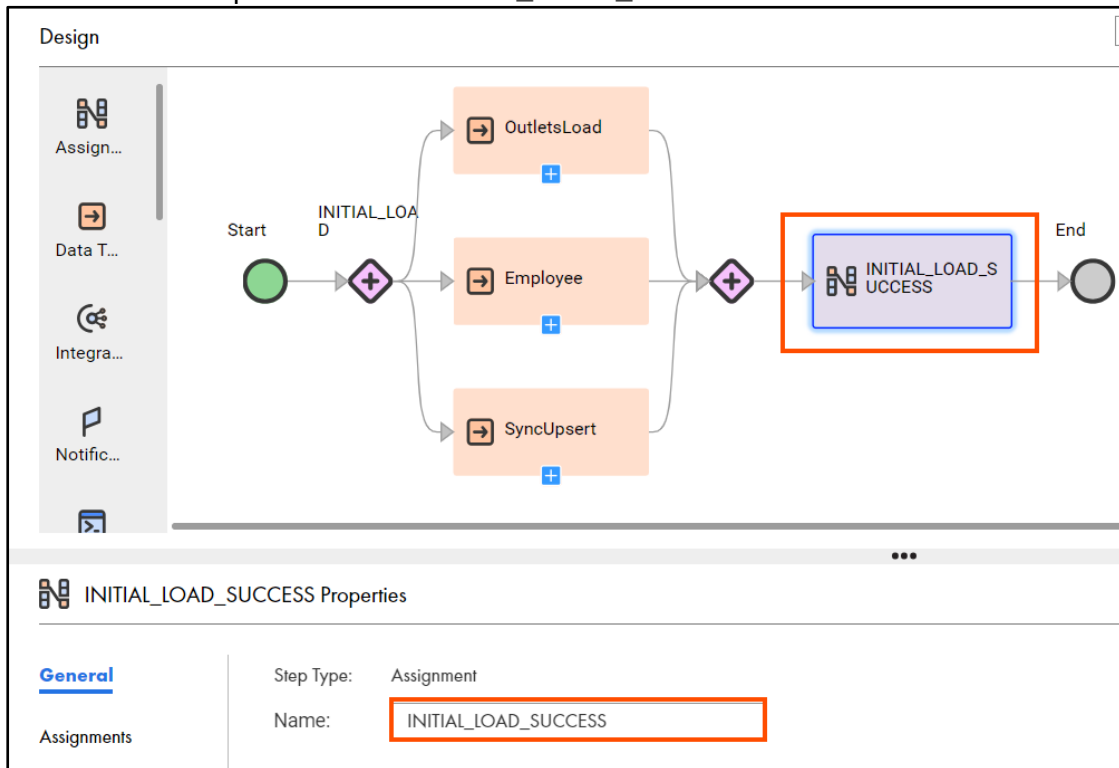
26. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.



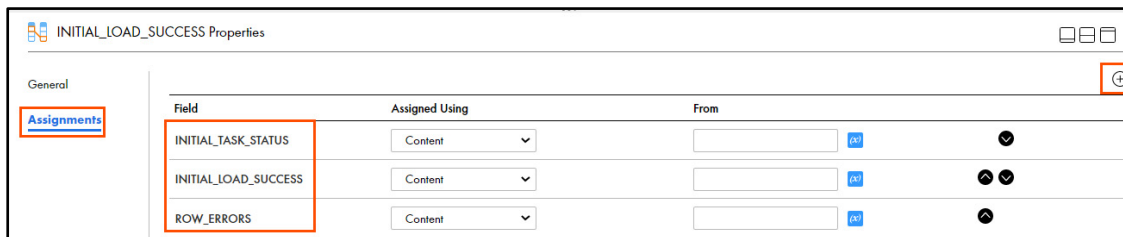
The screenshot shows the 'SyncUpsert Properties' dialog with the 'Error Handling' tab selected. The 'On Warning' and 'On Error' options are both set to 'Ignore', which are highlighted with a red border. The 'Fail taskflow on completion' section has two unchecked checkboxes: 'If this task fails' and 'If this task does not run'.

27. Save your work.

28. Drag and drop an Assignment step from the Design panel and place it right after the Parallel Paths step. Name it as **INITIAL_LOAD_SUCCESS**.



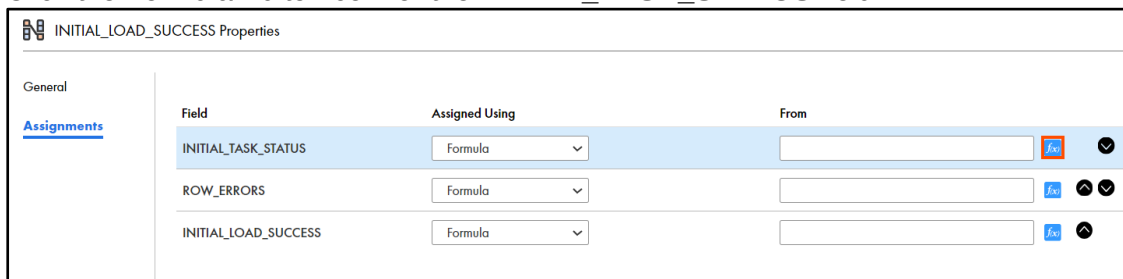
29. Go to the Assignments properties tab and using the plus icon, add **INITIAL_TASK_STATUS**, **ROW_ERRORS**, and **INITIAL_LOAD_SUCCESS** fields.



Field	Assigned Using	From
INITIAL_TASK_STATUS	Content	[+] ✓
INITIAL_LOAD_SUCCESS	Content	[+] ✓ ✓
ROW_ERRORS	Content	[+] ✓

30. Change the **Assigned Using** column property for all the three fields to **Formula**.

31. Click the Formula Editor icon for the **INITIAL_TASK_STATUS** field.



Field	Assigned Using	From
INITIAL_TASK_STATUS	Formula	[fx] ✓
ROW_ERRORS	Formula	[fx] ✓ ✓
INITIAL_LOAD_SUCCESS	Formula	[fx] ✓

32. Set the Expression as shown below:

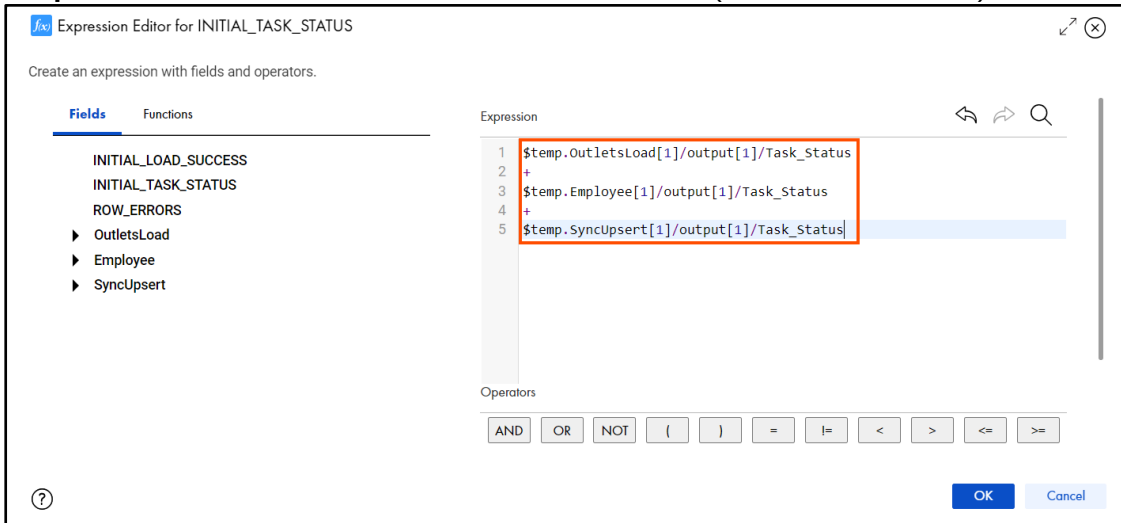
\$temp.OutletsLoad[1]/output[1]/Task_Status
+

\$temp.Employee[1]/output[1]/Task_Status

+

\$temp.SyncUpsert[1]/output[1]/Task_Status

Tip: To configure the expression, from the Fields section, expand each of the Object's **Output Parameters** section and add the **Task Status** (From all the 3 tasks).



33. Click **OK**.

Note: Using the up and down arrows next to the fields, keep the **INITIAL_LOAD_SUCCESS** field to the last.



34. Similarly, for the ROW_ERRORS field, go to the expression editor and set the following expression using the Output Parameters:

\$temp.OutletsLoad[1]/output[1]/Failed_Target_Rows

+

\$temp.OutletsLoad[1]/output[1]/TotalTransErrors

+

\$temp.Employee[1]/output[1]/Failed_Target_Rows

+

\$temp.Employee[1]/output[1]/TotalTransErrors

+

\$temp.SyncUpsert[1]/output[1]/Failed_Target_Rows

+

\$temp.SyncUpsert[1]/output[1]/TotalTransErrors



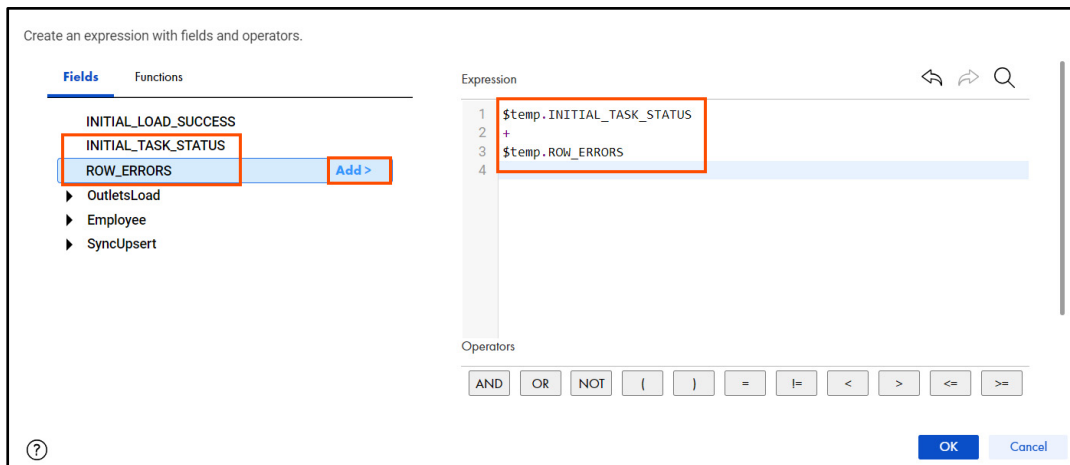
35. Click **OK**.

36. Similarly, for **INITIAL_LOAD_SUCCESS**, set the following expression:

\$temp.INITIAL_TASK_STATUS

+

\$temp.ROW_ERRORS

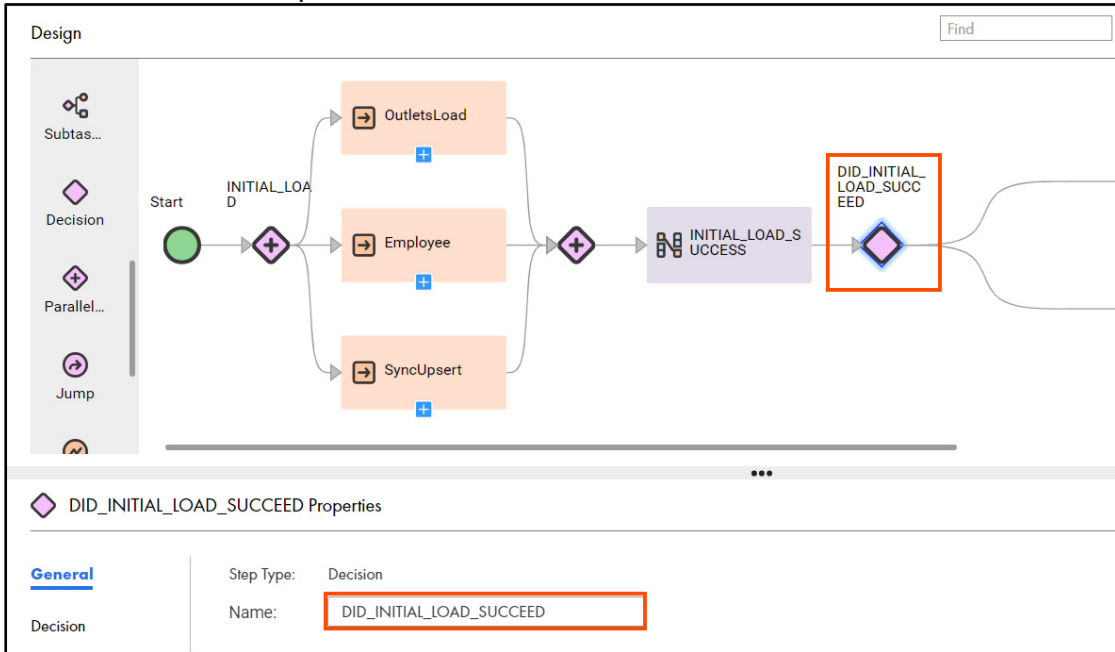


37. Click **OK**.

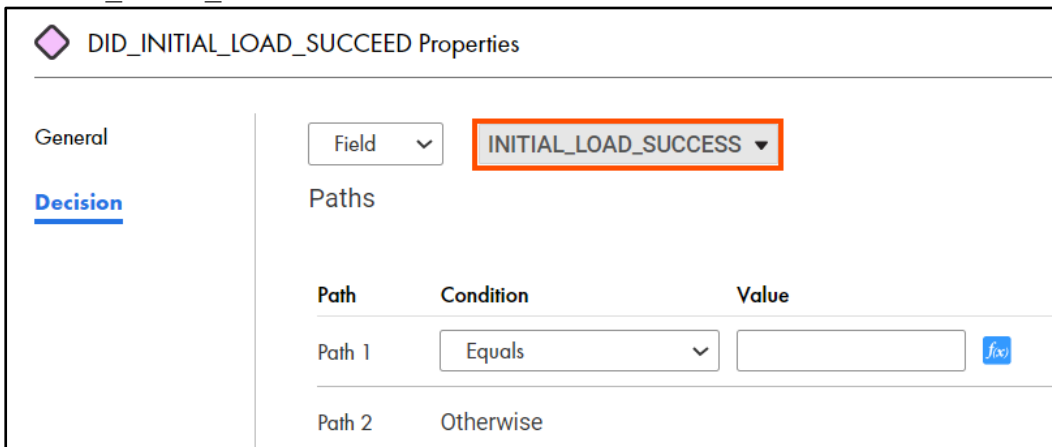
38. Save your work.

39. Drag and drop a Decision step onto the canvas and place it next to the Assignment step.

40. Name the Decision step as **DID_INITIAL_LOAD_SUCCEED**.

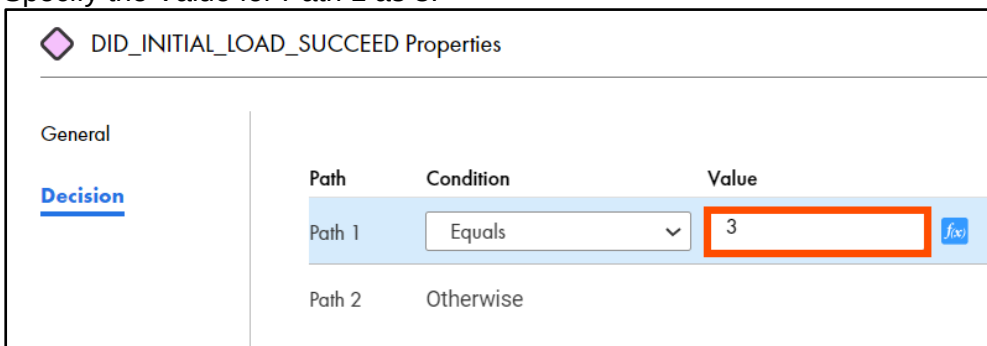


41. Navigate to the Decision properties tab and select the decision field as **INITIAL_LOAD_SUCCESS**.



The screenshot shows the **DID_INITIAL_LOAD_SUCCEED Properties** dialog box. The **Decision** tab is selected. The **Field** dropdown is set to **INITIAL_LOAD_SUCCESS**. The **Paths** section shows two paths: Path 1 with the condition **Equals** and Path 2 with the condition **Otherwise**.

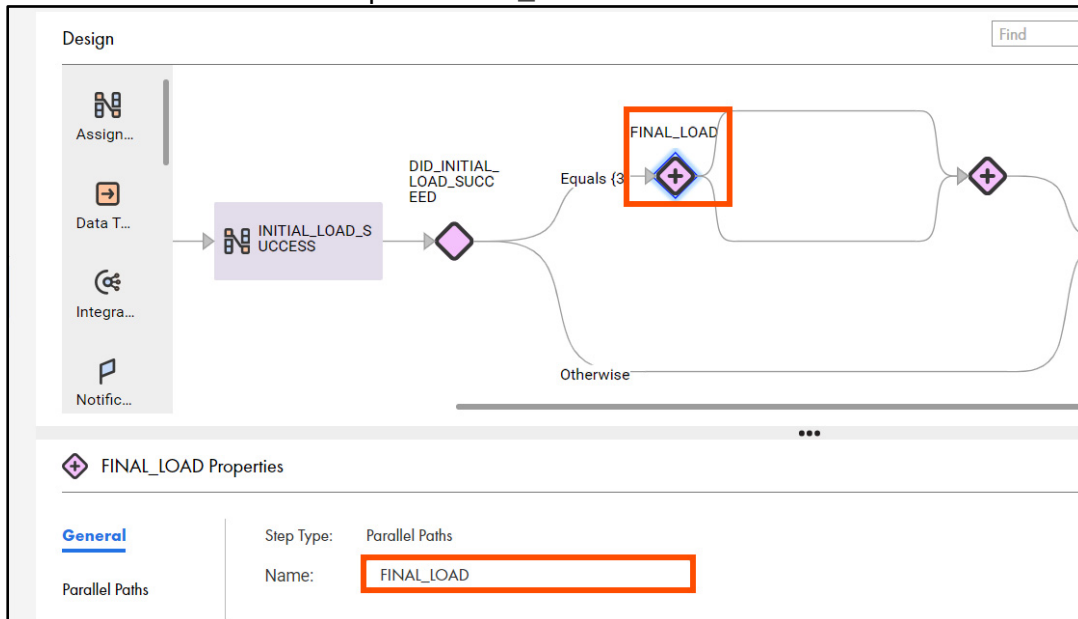
42. Specify the Value for Path 1 as **3**.



The screenshot shows the **DID_INITIAL_LOAD_SUCCEED Properties** dialog box. The **Decision** tab is selected. The **Field** dropdown is set to **INITIAL_LOAD_SUCCESS**. The **Paths** section shows two paths: Path 1 with the condition **Equals** and the value **3** (highlighted with a red box), and Path 2 with the condition **Otherwise**.

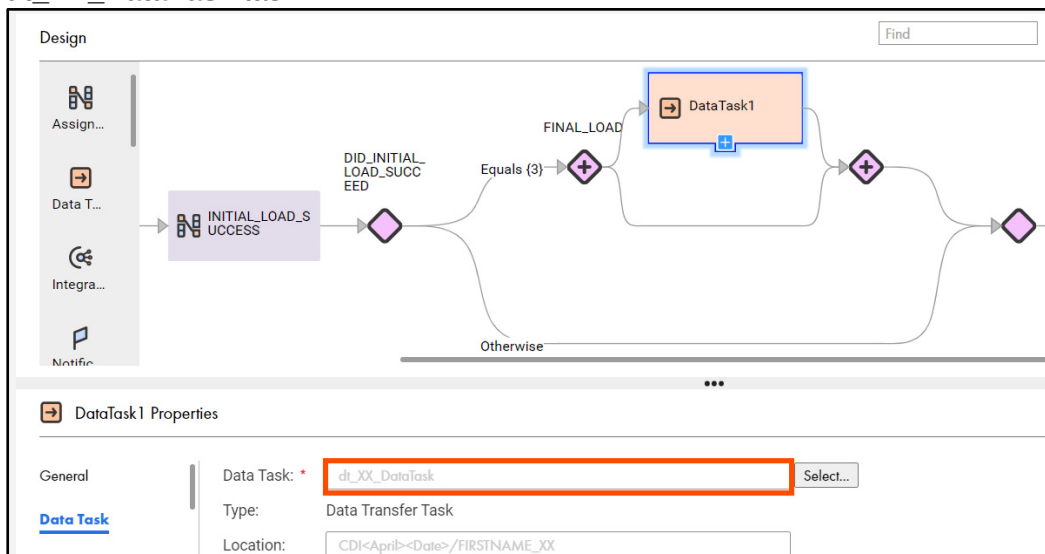
43. In path 1 of decision step (value = 3 path), drag and drop a Parallel Paths step.

44. Name the Parallel Paths step as **FINAL_LOAD**.



45. Save your work. To the first parallel path, add a Data task step and retain the default name.

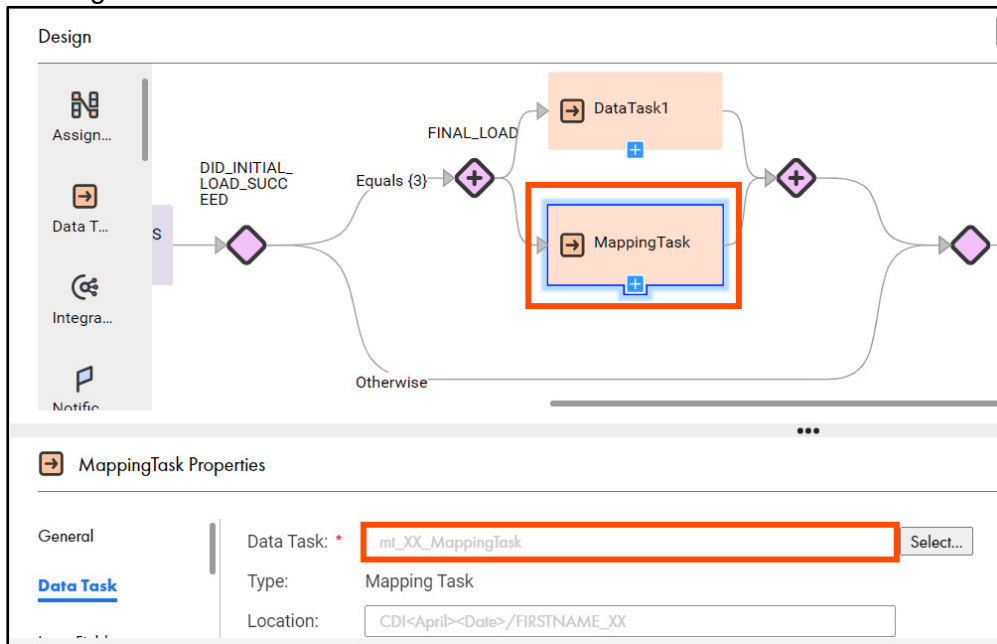
46. From the Data Task properties tab, navigate to your working folder and select the **dt_XX_DataTask** task.



47. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.

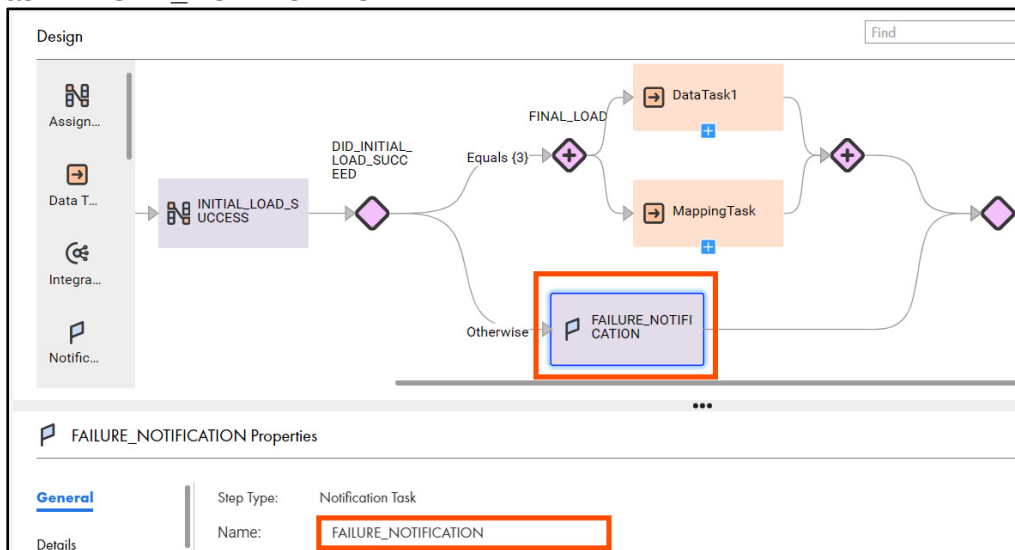
48. Onto the second parallel path, add another data task step and name it as **MappingTask**.

49. From its Data Task properties tab, select the **mt_XX_MappingTask** task from your working folder.



50. From the properties pane, select **Error Handling** and set both **On Warning** and **On Error** to **Ignore**.

51. Onto the Otherwise path of Decision step, drag and drop a Notification step and name it as **FAILURE_NOTIFICATION**.



52. Go to the **Details** properties tab of the Notification step.

53. Provide a valid email Id for the Email To property field and Email Subject as **PROBLEM WITH BULK LOAD PROCESSING**.

FAILURE_NOTIFICATION Properties

General

Notification Method * ☒ Email ☐ In-app notification ☐ Trigger a process ☐ Webhook

Email To * (X)

Email Cc (X)

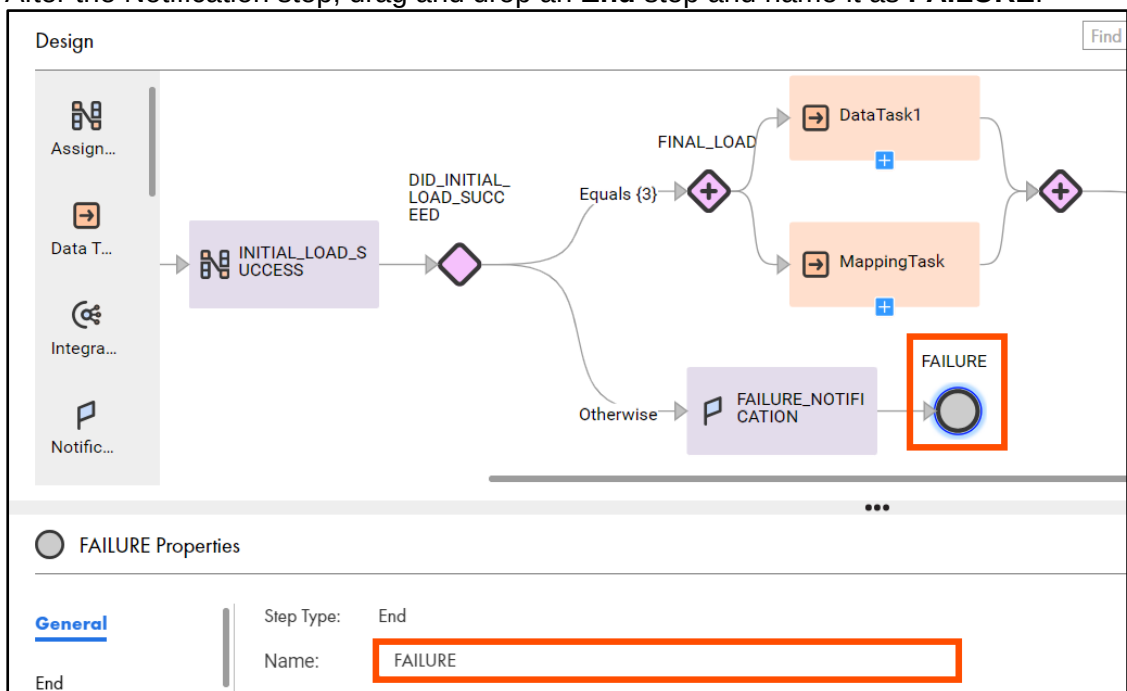
Email Bcc (X)

Email Subject (X)

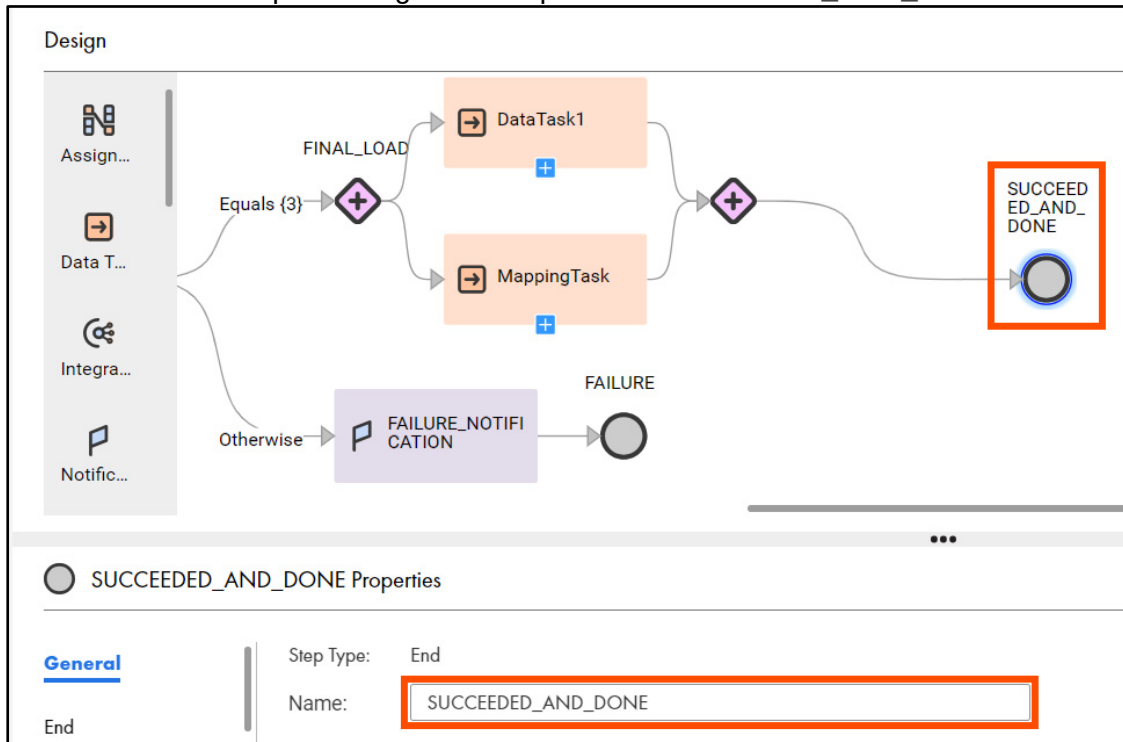
Email Content Type * ☐ HTML ☒ Plain Text

54. Save your work at regular intervals.

55. After the Notification step, drag and drop an **End** step and name it as **FAILURE**.



56. Rename the End step following Decision path to **SUCCEEDED_AND_DONE**.



57. **Save** the Taskflow.

58. Publish the Taskflow by clicking on the **Publish** option next to Save.



59. Run the Taskflow.



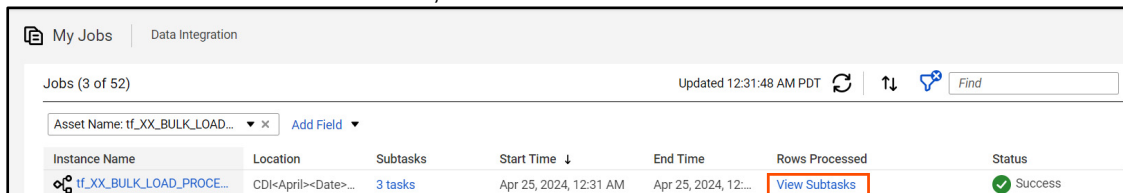
Monitor Status

60. To monitor the task, from the navigation pane, click **My Jobs**.

61. When the task is completed, the status changes to **Success**.

Note: You can refresh the page if the status does not change automatically.

62. To view the status of the subtask, click **View Subtasks**.



Instance Name	Location	Subtasks	Start Time ↓	End Time	Rows Processed	Status
tf_XX_BULK_LOAD_PROCE...	CDI<April><Date>...	3 tasks	Apr 25, 2024, 12:31 AM	Apr 25, 2024, 12:...	View Subtasks	Success

63. Observe that you see a warning status for 1 task, and it is expected. This is the same duplication error that was observed in the previous lab exercise. In this case, you will

receive an email alert informing about the task failure.

My Jobs | Data Integration > tf_XX_BULK_LOAD_PROCESSING1-968051535334707200

Jobs (3) Updated 12:32:32 AM PDT

Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
syn_XX_OutletsLoad-9	CDI<April><Date>...		Apr 25, 2024, 12:31 AM	Apr 25, 2024, 12:...	8	Success
syn_XX_Employee-8	CDI<April><Date>...		Apr 25, 2024, 12:31 AM	Apr 25, 2024, 12:...	33	Warning
syn_XX_UPSERT_CUSTOM...	CDI<April><Date>...		Apr 25, 2024, 12:31 AM	Apr 25, 2024, 12:...	12	Success

PROBLEM WITH BULK LOAD PROCESSING

Retention: Exchange online 3 Years delete (3 years) Expires: Sun 4/25/2027 1:01 PM

admin@informaticacloud.com
To: [REDACTED]

64. To go back to the previous window, click **Data Integration** next to My Jobs.

My Jobs | **Data Integration** > tf_XX_BULK_LOAD_PROCESSING1-972024372462460928

Jobs (3) Updated 12:11:03 AM PDT

Instance Name	Location	Subtasks	Start Time	End Time	Rows Processed

65. Click on the instance name.

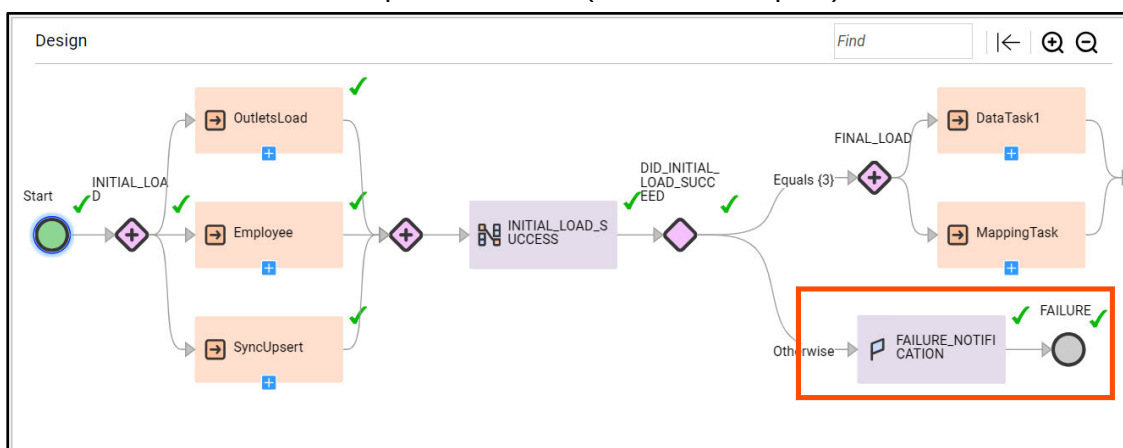
My Jobs | Data Integration

Jobs (3 of 52)

Asset Name: tf_XX_BULK_LOAD... Add Field

Instance Name	Location	Subtasks	Start Time
tf_XX_BULK_LOAD_PROCE...	CDI<April><Date>...	3 tasks	Apr 25, 2024, 12:31 AM

66. Observe the executed path. It has taken the failure path since one of the associated subtasks has failed due to duplicate records (as seen in step 58).



Note: If you want to run your taskflow in the success path1, delete the existing records (as mentioned in the previous lab exercise) in the target of the failed subtask and re-run the taskflow.

This concludes the lab.

Module 14: IICS REST APIs

Lab 14-1: Creating a Mapping Using a REST V2 Connector

Overview:

REST is a web standards-based architecture that uses HTTP Protocol for data communication. These commonly used APIs allow you to create web-based applications.

In this lab, you will use the REST V2 Connector to connect to OpenWeather API and to get the current weather information.

Objective:

- Create a REST connection using the REST V2 connector
- Get the JSON message and write it to a Flat File

Scenario:

John receives a request to implement weather information for requested location. He uses a REST V2 Connector to connect to the OpenWeather API. He uses the unique API key to fetch the current weather information for the city specified in the request message. After the information is received from the API in the JSON format, he creates a target file at runtime to write the output.

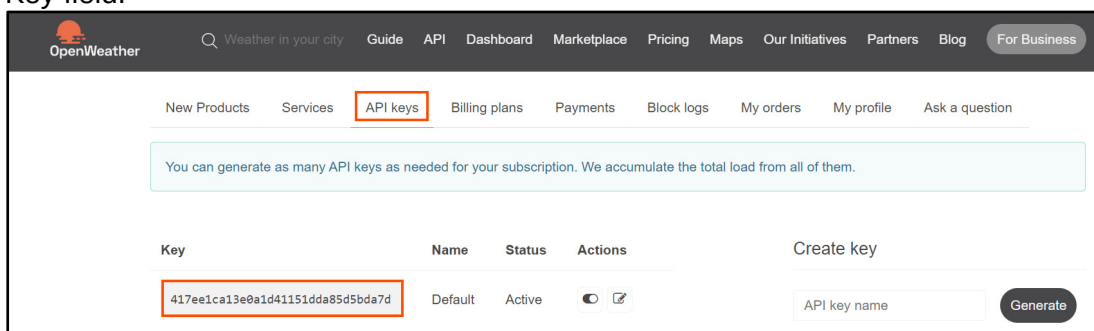
Duration:

30 minutes

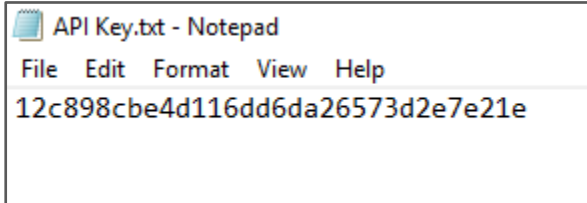
Tasks

Sign up to Access OpenWeatherAPI

1. Open a web browser and enter the following URL in a new tab:
https://home.openweathermap.org/users/sign_up
Note: You can bookmark this link for future use.
2. Create a new account in OpenWeather using the link provided.
Note: After you create an account, you can access your unique API key.
3. To access your unique API Key, select the **API Keys** tab, and copy the API key from the Key field.

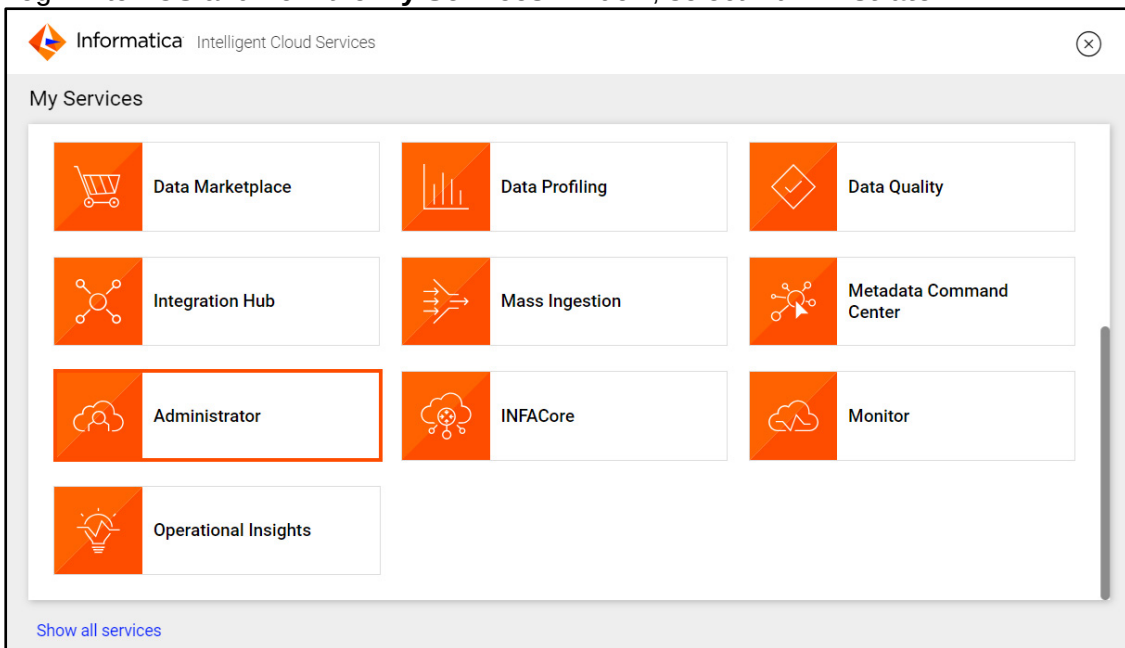


4. Paste the API key in a notepad and save the notepad file on your system.

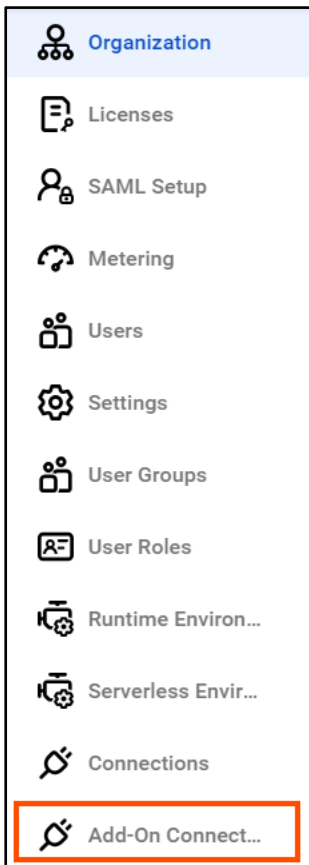


Enable REST V2 Connector

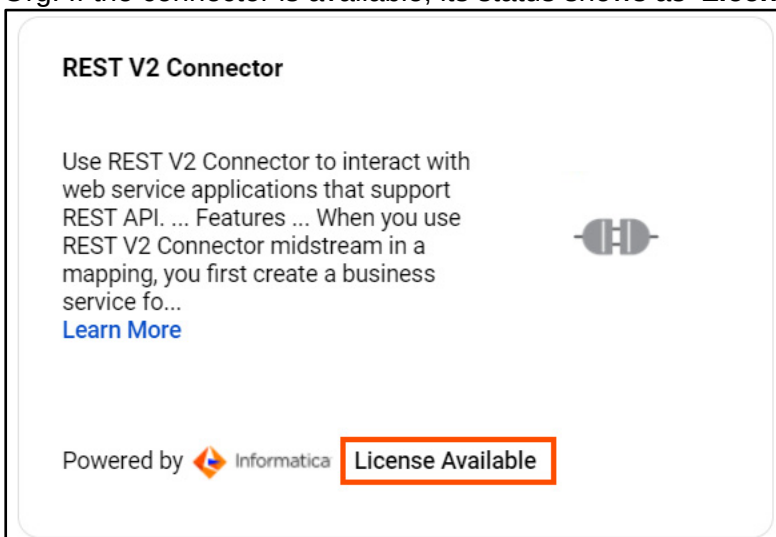
5. Login into IICS and from the **My Services** window, select **Administrator**.



6. From the navigation pane, select **Add-On Connectors**.



7. Locate the **REST V2 Connector** and check if the REST V2 connector is available for the Org. If the connector is available, its status shows as '**License Available**'.

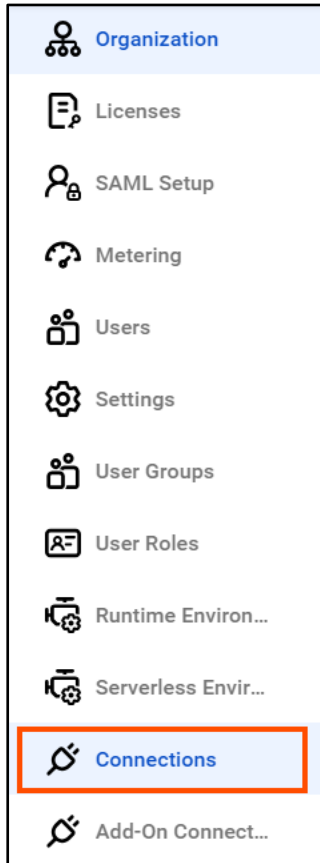


Note: If the connector is already available for the org, you can skip the steps to enable the connector.

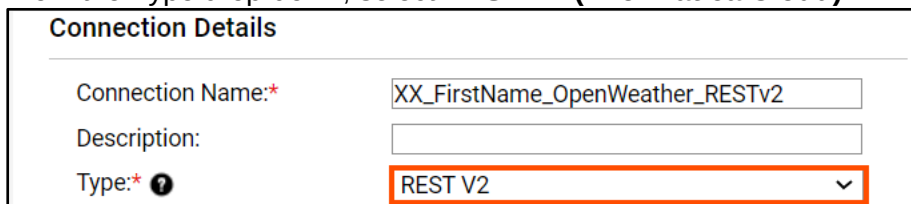
8. If the REST V2 Connector is not available, click **Start Free Trial** to enable the connector.

Create a REST Connection

9. From the navigation pane, select **Connections**.



10. Create a new connection.
11. In the Name field, enter **XX_FirstName_OpenWeather_RESTv2**.
12. From the Type drop-down, select **REST V2 (Informatica Cloud)**.



The screenshot shows the 'Connection Details' form with the following fields:

- Connection Name*: XX_FirstName_OpenWeather_RESTv2
- Description: (empty field)
- Type*: **REST V2** (selected in the dropdown menu, highlighted with a red box)

13. From the **Runtime Environment** drop-down, select **INFA-SERVER**.
14. From the Authentication drop-down, select **Standard**.
15. Verify that the Authentication Type is set to **NONE**.
16. In the Swagger File Path field, enter the following path of the json file:
C:\IICSLabFiles\SrcFiles\Swagger_api_openweathermap_org_072248.json

17. Verify that from the **Advanced Settings** Proxy Type is selected as **Platform Proxy**.

Connection Details

Connection Name:* XX_FirstName_OpenWeather_RESTv2

Description:

Type:* ? REST V2

Runtime Environment:* ? INFA-SERVER

Authentication:* ? Standard

Authentication Type: ? **NONE**

Auth User ID: ?

Auth Password: ?

OAuth Consumer Key: ?

OAuth Consumer Secret: ?

OAuth Token: ?

OAuth Token Secret: ?

Swagger File Path:* ? **C:\IICSLabFiles\SrcFiles\Swagger_api_openv**

Advanced Settings

TrustStore File Path: ?

TrustStore Password: ?

KeyStore File Path: ?

KeyStore Password: ?

Proxy Type: ? **Platform Proxy**

Proxv Configuration: <host><port>

18. Test the connection.




 The test for this connection was successful.

Connection Details

Connection Name:* XX_FirstName_OpenWeather_RESTv2

Description:

Type:* ? REST V2

Runtime Environment:* ? INFA-SERVER

Let's configure the REST V2 connection properties to interact with web service applications built on REST architecture.

Before you begin

Before you get started, be sure to complete the prerequisites.

19. Save the connection.




 The test for this connection was successful.

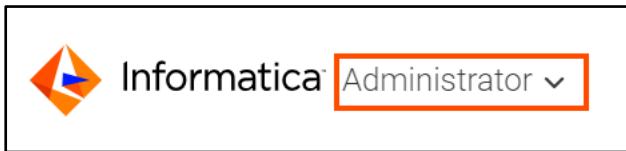
Connection Details

Connection Name:* XX_FirstName_OpenWeather_RESTv2

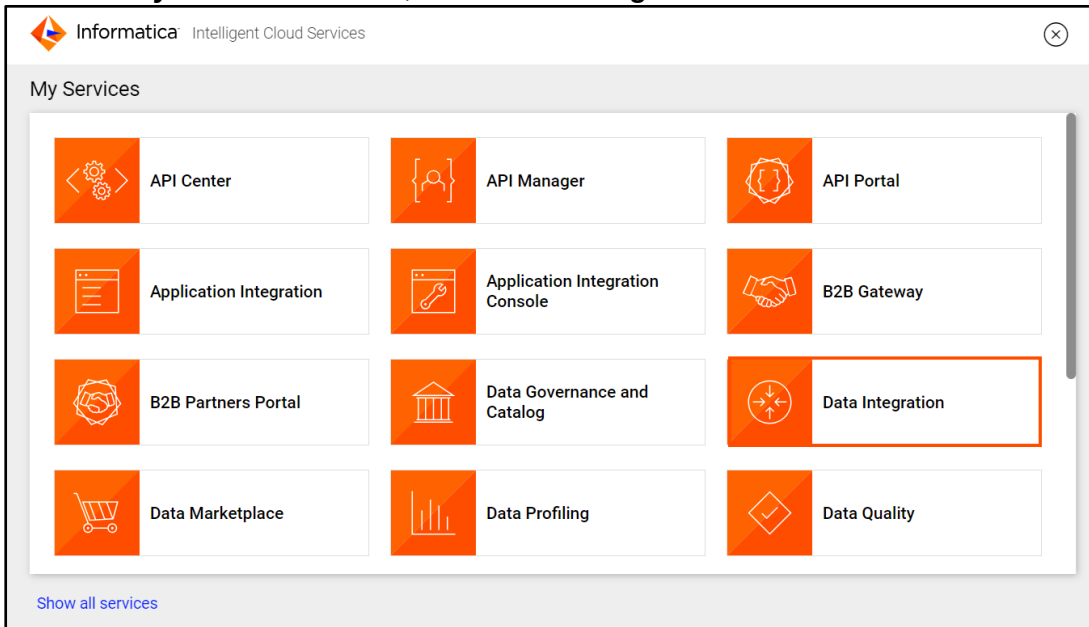
Let's configure the REST V2 connection properties to interact with web service applications built on REST architecture.

Create a Mapping

20. To switch between the available services, from the toolbar, select the current service **Administrator**.



21. From the **My Services** window, select **Data Integration**.



22. Create a new Mapping.

23. In the Name field, enter **m_XX_REST_GetWeather_By_City**.

24. Verify that the asset Location is pointing to your working directory.

m_XX_REST_GetWeather_By_City	
Name:*	m_XX_REST_GetWeather_By_City
Location:*	CDI<April><Date>\FIRSTNAME_XX

25. To configure the source, from the mapping canvas, click the **Source** transformation.

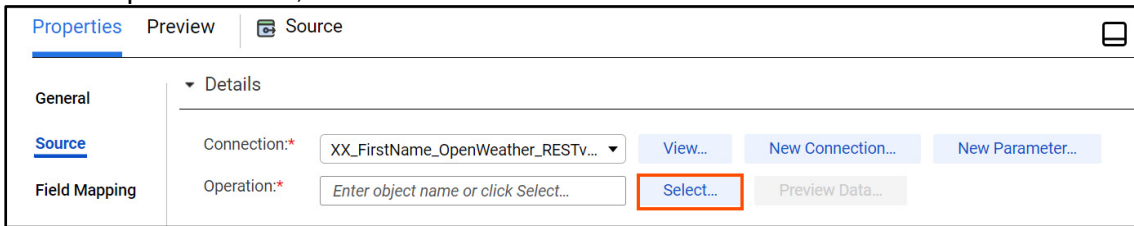
26. In the **General** section of the Source properties, retain the Name as **Source**.

Properties Preview Source	
General	
Source	Name: * Source Description:

27. From the properties pane, click **Source**.

28. From the Connection drop-down, select **XX_FirstName_OpenWeather_RESTv2**.

29. For the Operation field, click **Select**.



Properties | Preview | Source

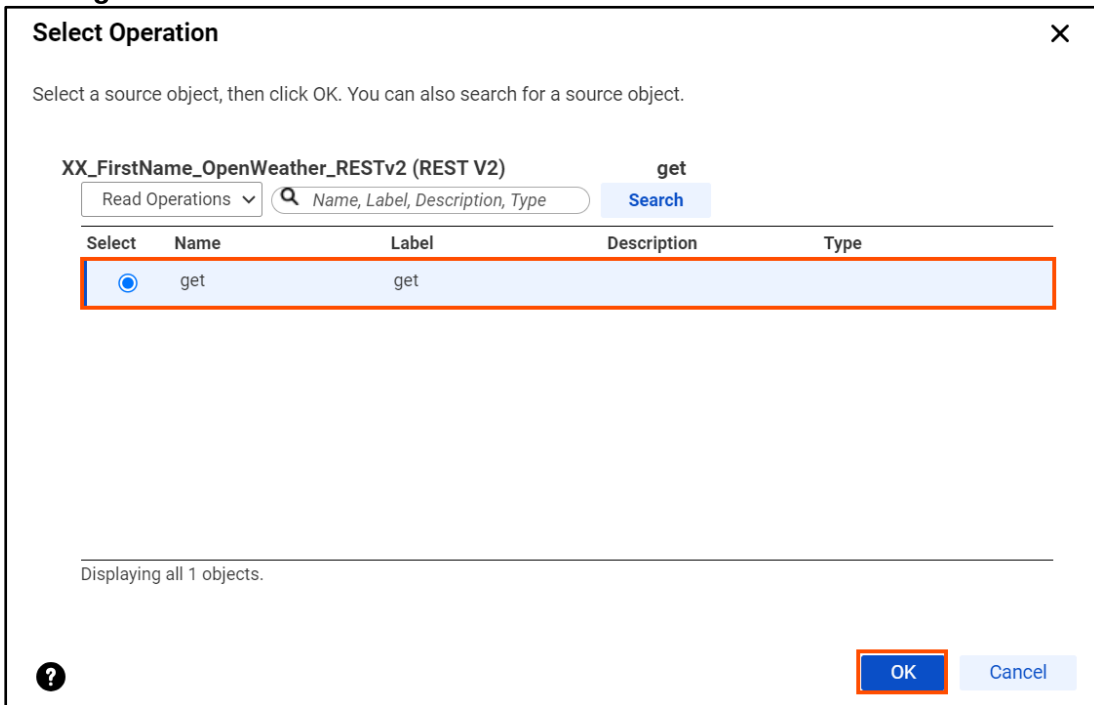
General | Source | Field Mapping

Details

Connection:* XX_FirstName_OpenWeather_RESTv... View... New Connection... New Parameter...

Operation:* Enter object name or click Select... **Select...** Preview Data...

30. Select **get** and click **OK**.



Select Operation

Select a source object, then click OK. You can also search for a source object.

XX_FirstName_OpenWeather_RESTv2 (REST V2) get

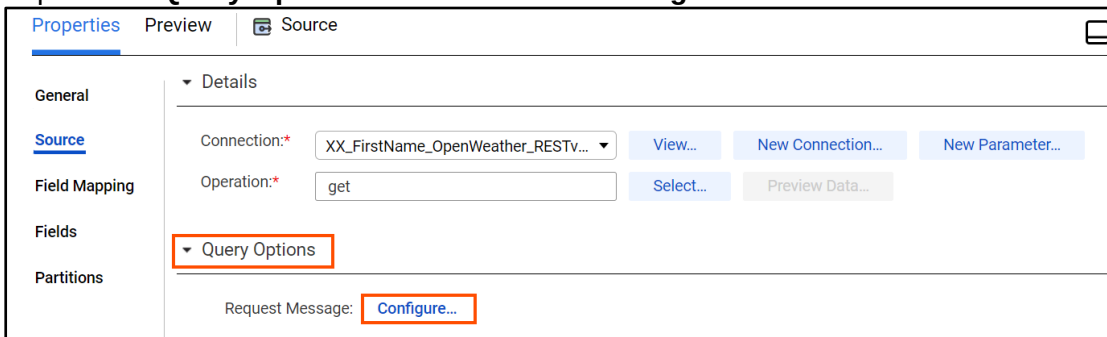
Read Operations Search Name, Label, Description, Type

Select	Name	Label	Description	Type
☒	get	get		

Displaying all 1 objects.

OK Cancel

31. Expand the **Query Options** section and click **Configure**.



Properties | Preview | Source

General | Source | Field Mapping | Fields | Partitions

Details

Connection:* XX_FirstName_OpenWeather_RESTv... View... New Connection... New Parameter...

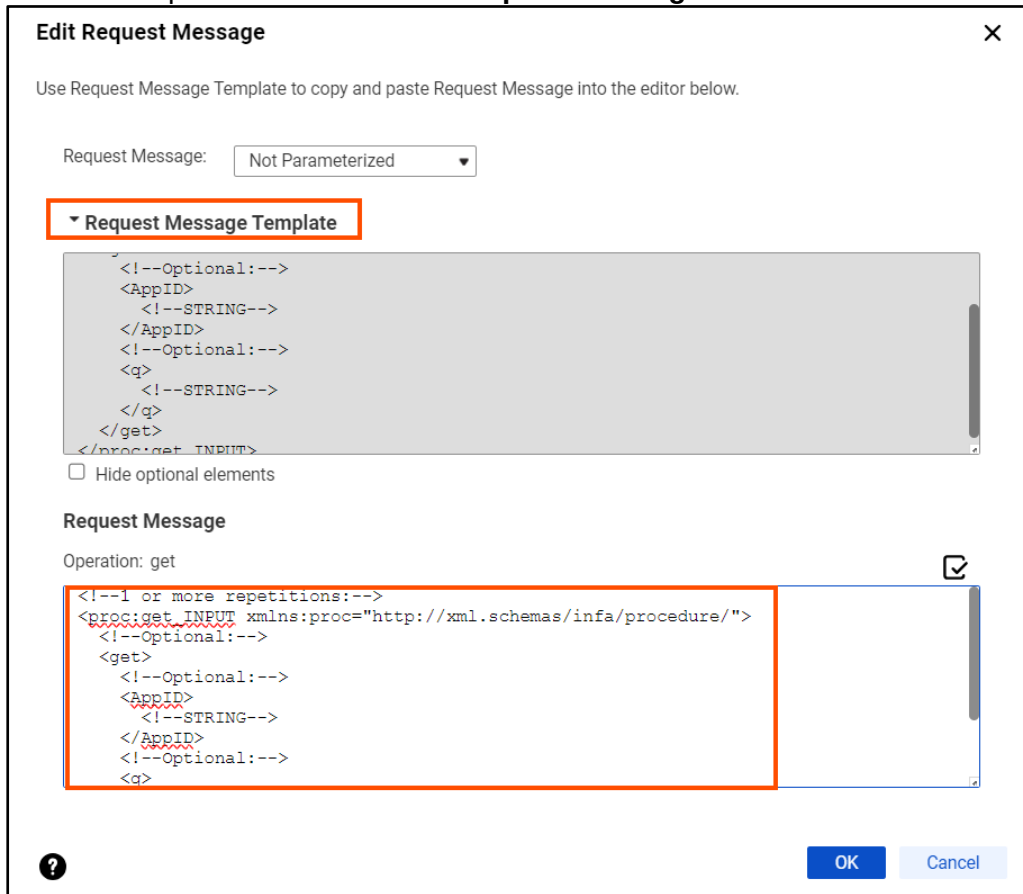
Operation:* get Select... Preview Data...

Query Options

Request Message: **Configure...**

32. Expand the **Request Message Template** section and copy all the contents from the field.

33. Paste the copied contents into the **Request Message** field.



Edit Request Message [X]

Use Request Message Template to copy and paste Request Message into the editor below.

Request Message: Not Parameterized

Request Message Template

```
<!--Optional:-->
<AppID>
  <!--STRING-->
</AppID>
<!--Optional:-->
<q>
  <!--STRING-->
</q>
</get>
</proc:get INPUT>
```

☐ Hide optional elements

Request Message

Operation: get

```
<!--1 or more repetitions:-->
<proc:get INPUT xmlns:proc="http://xml.schemas/infra/procedure/">
  <!--Optional:-->
  <get>
    <!--Optional:-->
    <AppID>
      <!--STRING-->
    </AppID>
    <!--Optional:-->
    <q>
```

[?] [OK] [Cancel]

34. In the Request Message field, in between **<AppID> </AppID>**, enter the unique API Key saved earlier.

35. In the Request Message field, in between **<q> </q>**, enter **London**.

Example:

<AppID>

12c898cbe4d116dd6da26573d2e7e21e

</AppID>

<q>

London

</q>

36. To validate the message, click  and click **OK**.

Request Message

Operation: get

```
<!--Optional:-->
<AppID>
12c898cbe4d116dd6da26573d2e7e21e
</AppID>
<!--Optional:-->
<q>
London
</q>
</get>
</proc:get_INPUT>
```


OK
Cancel

37. From the properties pane, click **Field Mapping**.

38. From the Element Name section, select **root** and select **Map all descendants**.

Properties | Preview | Source

General

Source

Field Mapping

Fields

Partitions

Select elements of Response to be mapped to Output Fields. Output groups and keys

Response Structure (0 of 35 mapped)

Show Fields: All Find

Element Name	Cardinality
☒ root	1-1
☐	0-1
☐	0-1
☐	0-1

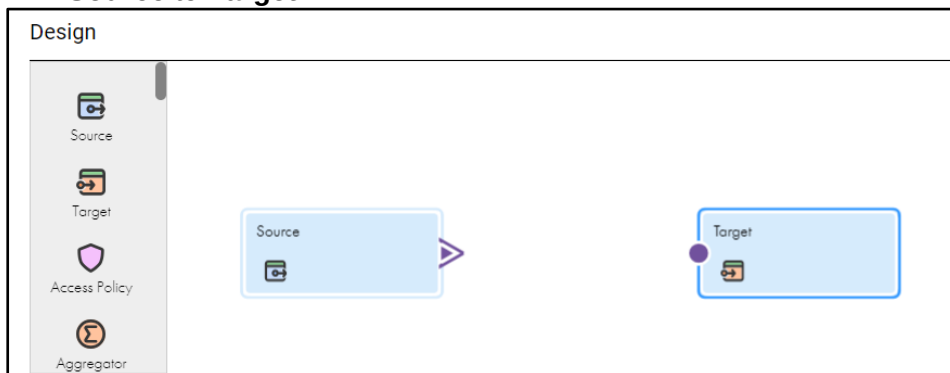
Map all descendants

Map immediate children

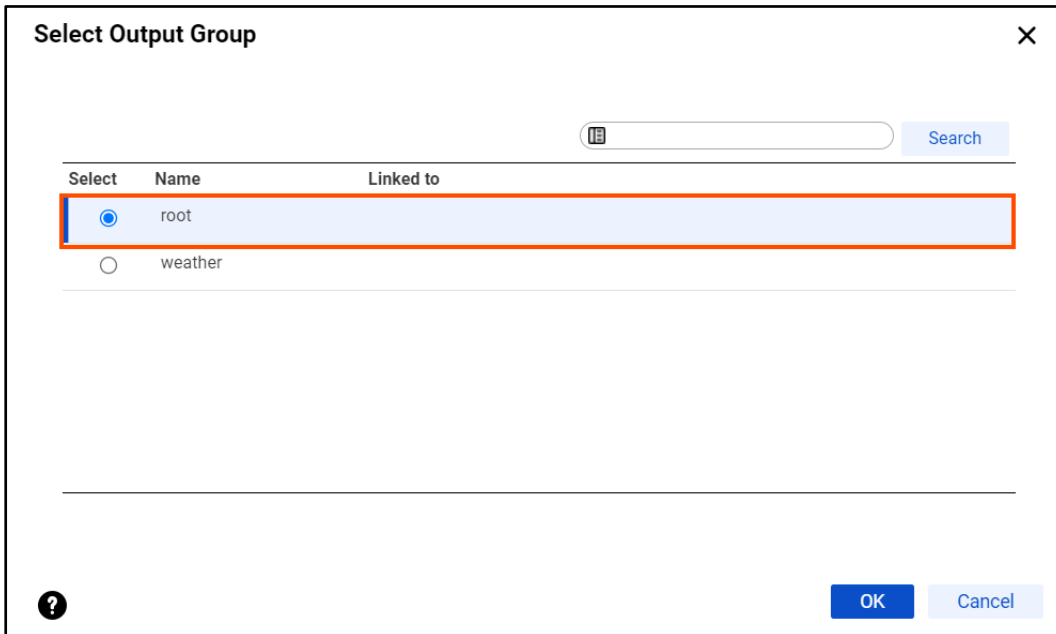
Unmap all descendants

Unmap immediate children

39. Link **Source** to **Target**.



40. Select **root** and click **OK**.

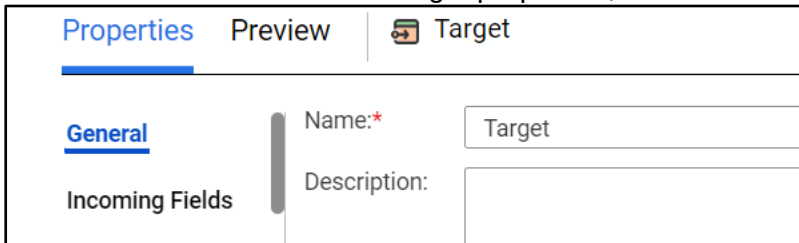


The 'Select Output Group' dialog box shows a table with two rows. The first row, labeled 'root', is selected and highlighted with a red border. The second row, labeled 'weather', is not selected. The 'Select' column has radio buttons next to each row. The 'Name' column contains the text 'root' and 'weather'. The 'Linked to' column is empty. At the bottom right, there are 'OK' and 'Cancel' buttons. A help icon (?) is at the bottom left.

Select	Name	Linked to
☒	root	
☐	weather	

41. To configure the target, from the mapping canvas, click the **Target** transformation.

42. In the **General** section of the Target properties, retain the Name as **Target**.



The 'Target' properties dialog box has three tabs: 'Properties', 'Preview', and 'Target'. The 'Properties' tab is active. Under the 'General' section, the 'Name' field is set to 'Target' and the 'Description' field is empty. The 'Incoming Fields' section is also empty.

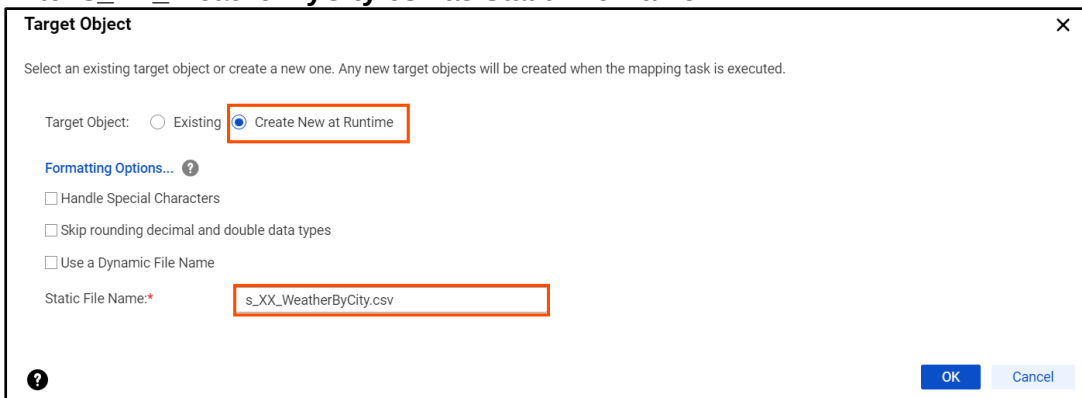
43. From the properties pane, click **Target**.

44. From the Connection drop-down, select your **target Flat File** connection.

45. To select the target object, from the Object field, click **Select**.

46. In the Target Object window, select **Create New at Runtime**.

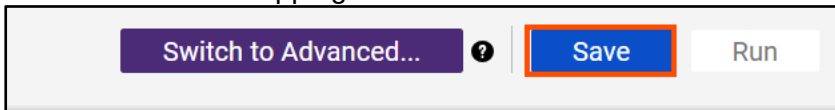
47. Enter **s_XX_WeatherByCity.csv** as Static File Name.



The 'Target Object' dialog box has a title bar and a close button. It contains a message: 'Select an existing target object or create a new one. Any new target objects will be created when the mapping task is executed.' Below this, there are two radio buttons: 'Existing' and 'Create New at Runtime'. The 'Create New at Runtime' radio button is selected and highlighted with a red border. Under the 'Formatting Options...' section, there are three checkboxes: 'Handle Special Characters', 'Skip rounding decimal and double data types', and 'Use a Dynamic File Name'. All three are unchecked. At the bottom, there is a 'Static File Name:' label and a text field containing 's_XX_WeatherByCity.csv', which is also highlighted with a red border. At the bottom right, there are 'OK' and 'Cancel' buttons. A help icon (?) is at the bottom left.

48. Click **OK**.

49. Save and run the mapping.



50. From the Runtime Environment drop-down, select INFA-SERVER.

51. Click **Run**.

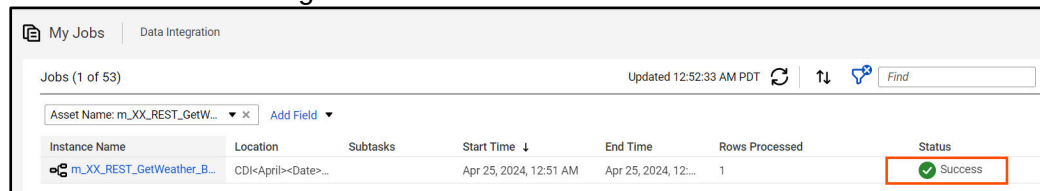
Monitor Status

52. To monitor the mapping status, from the navigation pane, click **My Jobs**.

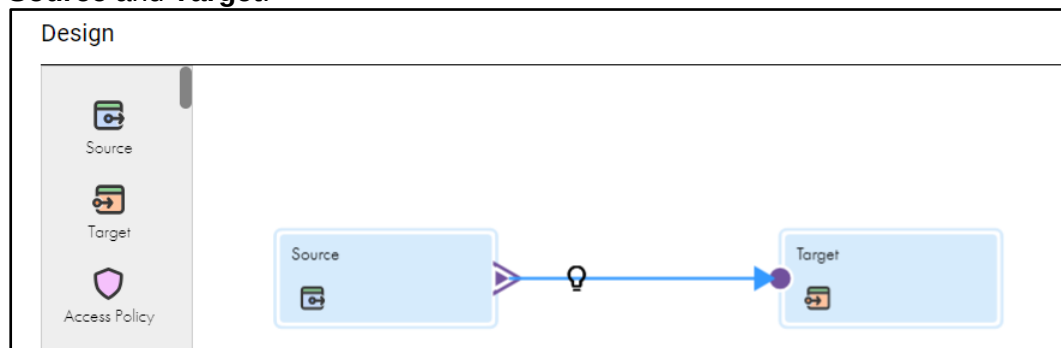
53. When the task completes, the status changes to **Success**.

Note:

- i. If the task completes with 0 processed rows, click on the Instance Name, and check the error message.

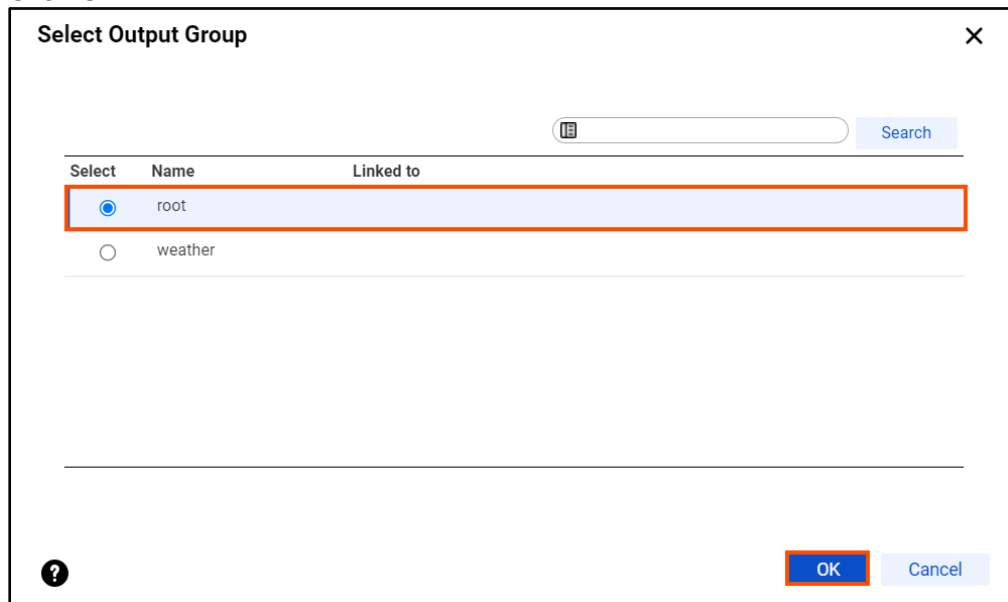


- ii. If you see the following error message, perform the below mentioned steps:
[ERROR] log4j:ERROR A "org.apache.log4j.ConsoleAppender" object is not assignable to a "org.apache.log4j.Appender" variable,
then, open the REST_GetWeather_By_City mapping and delete the link between **Source** and **Target**.



- iii. Again, link Source to Target.
- iv. In the **Select Output Group** window, select **root**.

v. Click **OK**.



The dialog box titled "Select Output Group" has a close button (X) in the top right corner. Below the title bar, there is a search bar with a magnifying glass icon and a "Search" button. Below the search bar is a table with three columns: "Select", "Name", and "Linked to". The table contains two rows: "root" and "weather". The "root" row is selected, indicated by a blue circle in the "Select" column and a blue highlight. The "OK" button is highlighted with a red border.

Select	Name	Linked to
☒	root	
☐	weather	

At the bottom left is a help icon (?). At the bottom right are "OK" and "Cancel" buttons.

vi. Save and run the mapping again.

54. Close the asset from the navigation pane.

55. Optionally, you can preview the target data records using your dummy mapping.

This concludes the lab.

Module 14: IICS REST APIs

Lab 14-2: Using Web Services Transformation in a Mapping

Overview:

A web service is a system software that enables machine-to-machine interaction over a network. It makes applications platform-and-technology independent.

In this lab, you will create a business service. You will also create a mapping that calls the business service in a mapping using web service transformation.

Objective:

- Create a Business Service
- Use Web Services transformation in a mapping

Scenario:

Ruby wants to change the format of billing for NH Retail outlets. She wants to convert the price information from numbers to words. So, John suggests using a business service to convert numbers to words.

In this lab, John creates a business service that connects to the NumberToDollars web service. He also creates a mapping in IICS and calls the business service in the mapping using the web service transformation. The transformation converts a list of numbers received as input into words to generate a flat file output.

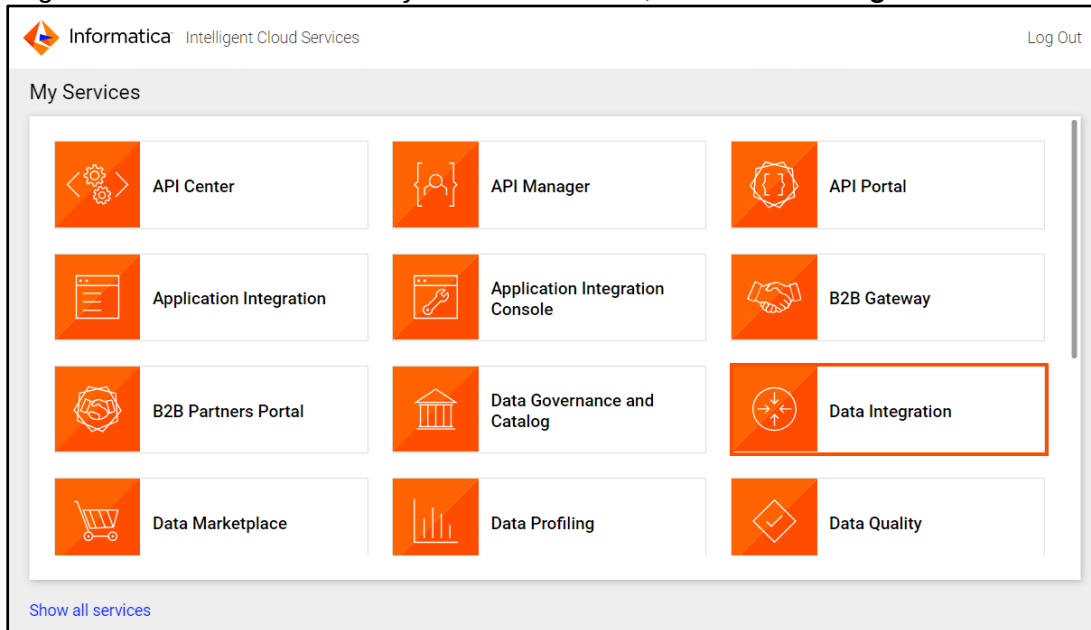
Duration:

25 minutes

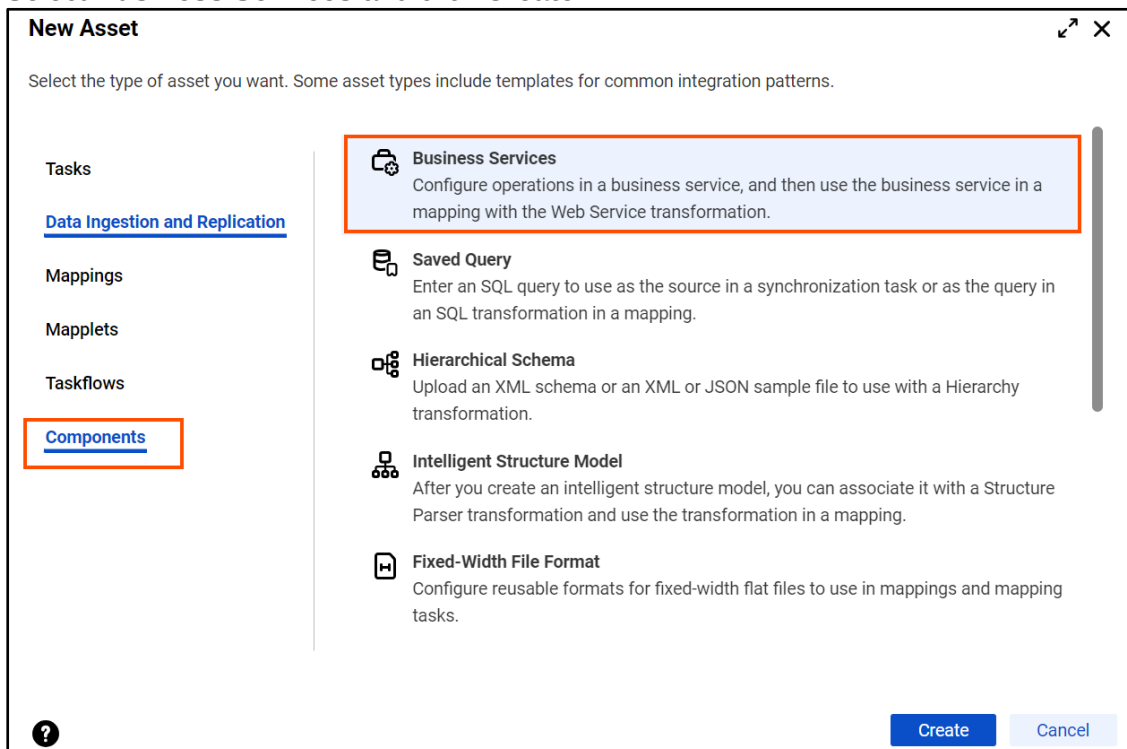
Tasks

Create Business Service

1. Login into IICS and from the My Services window, select **Data Integration**.

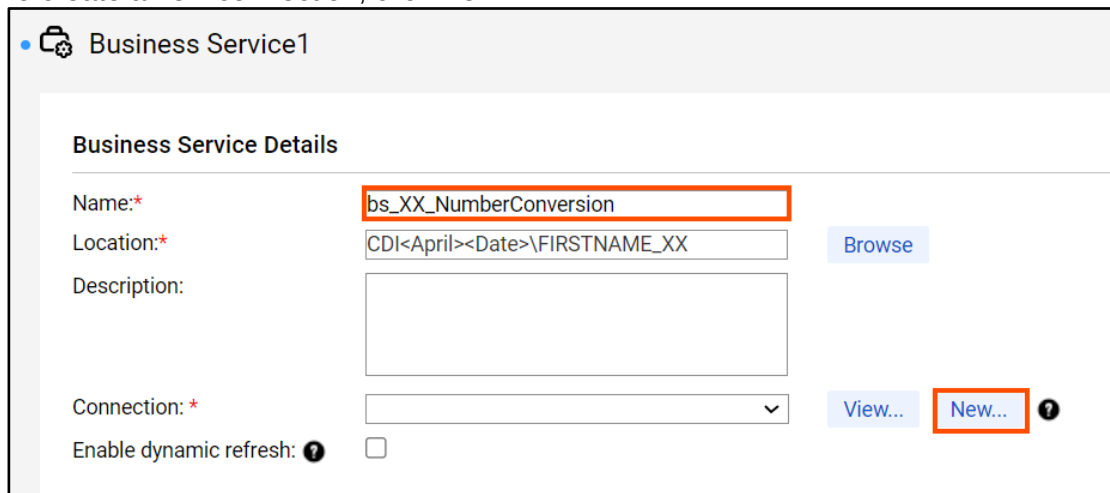


2. From the navigation pane, select **New**.
3. From the New Asset window, click the **Components** tab.
4. Select **Business Services** and click **Create**.



5. In the Name field, enter **bs_XX_NumberConversion**.

6. To create a new connection, click **New**.



Business Service1

Business Service Details

Name:*

Location:* [Browse](#)

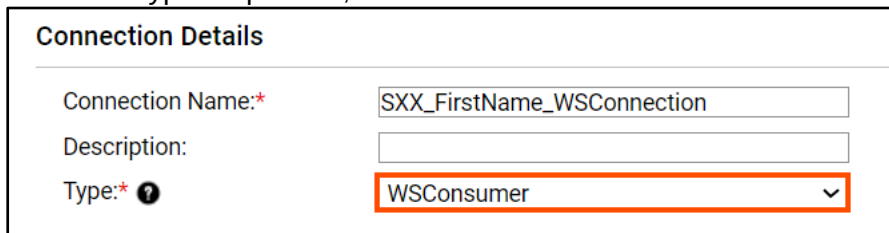
Description:

Connection: * [View...](#) [New...](#) ⓘ

Enable dynamic refresh: ⓘ ☐

7. In the Name field, enter **SXX_FirstName_WSConnection**.

8. From the Type drop-down, select **WSConsumer**.



Connection Details

Connection Name:*

Description:

Type:* ⓘ

9. From the Runtime Environment drop-down, select **INFA-SERVER**.

10. From the Authentication drop-down, select **Other Authentication**.

11. In the WSDL URL field, enter the following URL:

<https://www.dataaccess.com/webservicesserver/numberconversion.wso?WSDL>

12. In the Endpoint URL field, enter the following URL:

<https://www.dataaccess.com/webservicesserver/numberconversion.wso>

13. Click **OK**.



WSConsumer Connection Properties

Runtime Environment:* ⓘ

Authentication:* ⓘ

Other Authentication Connection Properties

WSDL URL:*

Endpoint URL:*

HTTP Username:

HTTP Password:

The following table describes the Web Service Consumer connection properties:

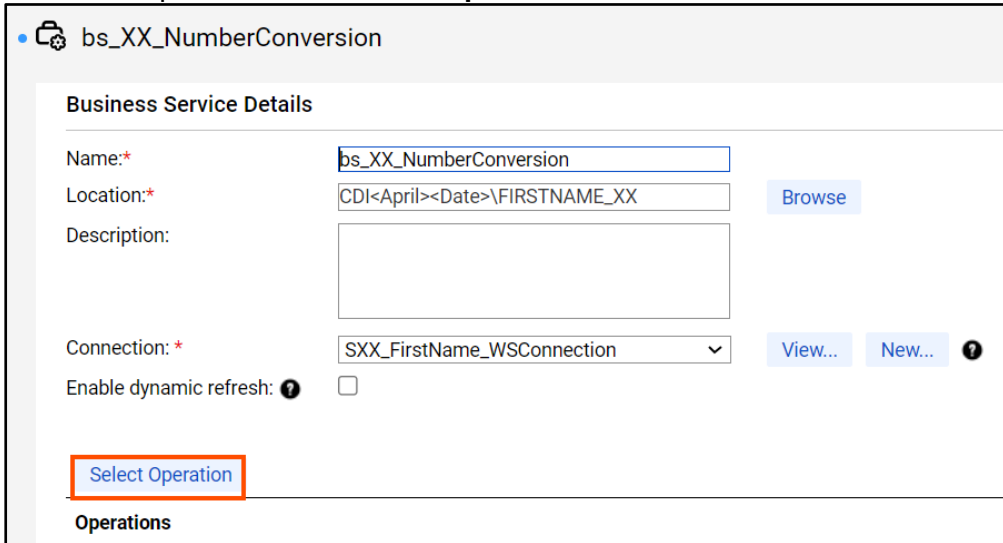
Property	Description
Connection Name	Enter a unique name for the connection.
Description	Provide a relevant description for the connection.

ⓘ [OK](#) [Cancel](#)

14. From the Connection drop-down, select **SXX_FirstName_WSConnection**.

Note: Skip this step if the created connection is already selected.

15. To add an operation, click **Select Operation**.



bs_XX_NumberConversion

Business Service Details

Name:* bs_XX_NumberConversion

Location:* CDI<April><Date>\FIRSTNAME_XX [Browse](#)

Description:

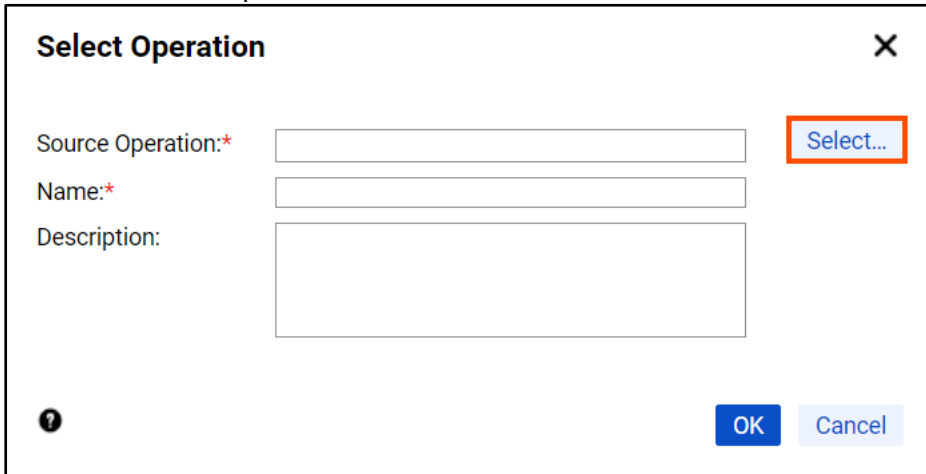
Connection: * SXX_FirstName_WSCConnection [View...](#) [New...](#) ⓘ

Enable dynamic refresh: ⓘ ☐

[Select Operation](#)

Operations

16. To select source operation, click **Select**.



Select Operation ✕

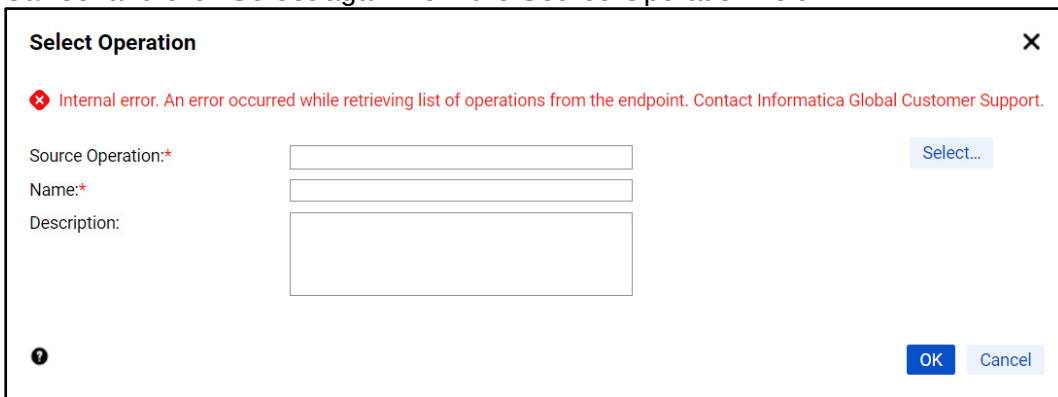
Source Operation:* [Select...](#)

Name:*

Description:

ⓘ [OK](#) [Cancel](#)

Note: If you get the Internal error message as shown in the screenshot below, click **Cancel** and click **Select** again from the Source Operation field.



Select Operation ✕

✖ Internal error. An error occurred while retrieving list of operations from the endpoint. Contact Informatica Global Customer Support.

Source Operation:* [Select...](#)

Name:*

Description:

ⓘ [OK](#) [Cancel](#)

17. From the list, select **NumberToDollars**.

18. Click **Select**.

Source Operation

NumberToDollars

Search

Select	Name
☐	NumberToWords
☒	NumberToDollars

Displaying all 2 objects.

?

Select

Cancel

Note: This action redirects you to the Select Operation window.

19. Click **OK**.

Select Operation

Source Operation:*

NumberToDollars

Select...

Name:*

NumberToDollars

Description:

?

OK

Cancel

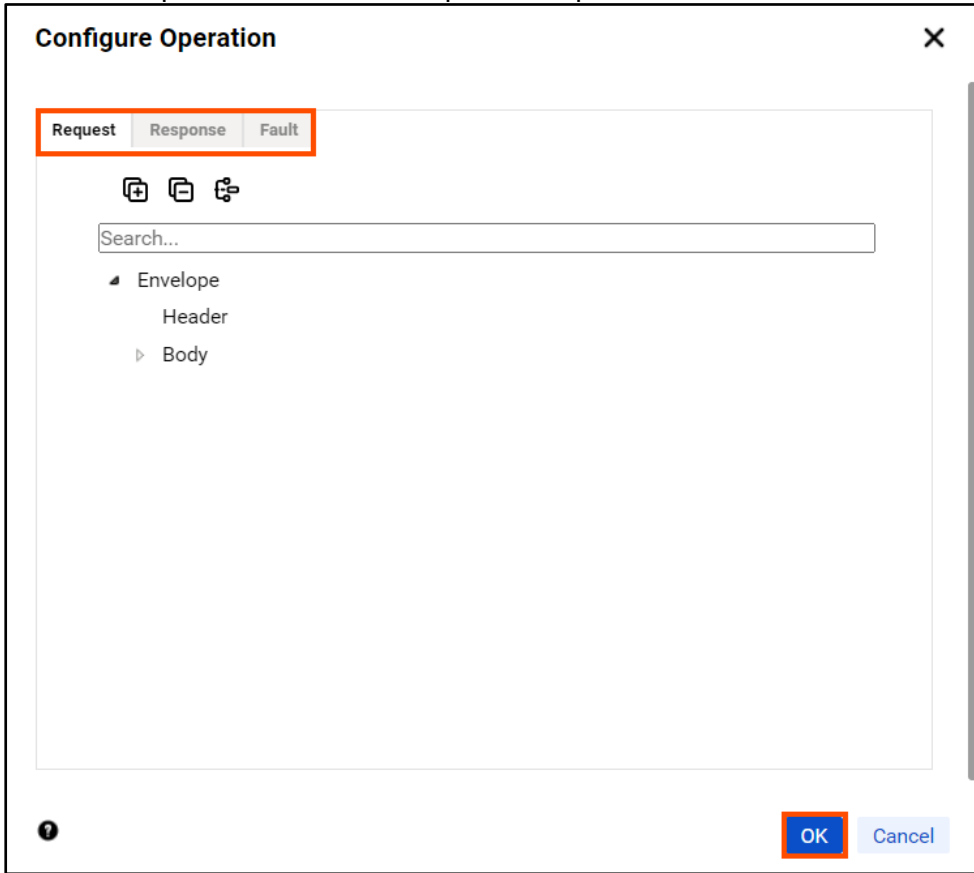
20. Click **Configure** to set the request, response, and fault parameters.

Select Operation

Operations

Delete	Name	Origin Name	Description
	NumberToDollars	NumberToDollars	Configure...

21. View all the parameters in the Request, Response, and Fault tabs and click **OK**.



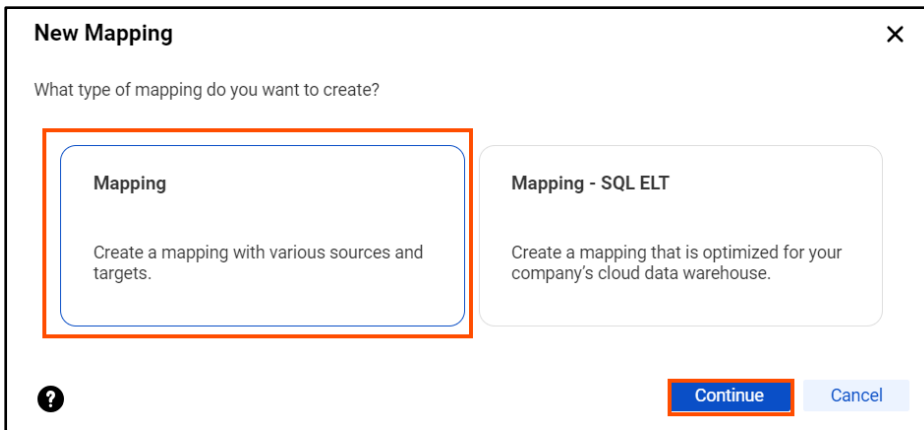
The **Configure Operation** dialog box is shown with the **Request** tab selected. The **Request** tab is highlighted with a red box. Below the tabs, there are three icons: a plus sign, a document, and a refresh. A search bar is present with the text "Search...". Below the search bar, there is a tree view showing the structure of the request: **Envelope** (expanded), **Header**, and **Body**. At the bottom right, the **OK** button is highlighted with a red box, and the **Cancel** button is also visible.

22. Save the business service.

Create a Mapping

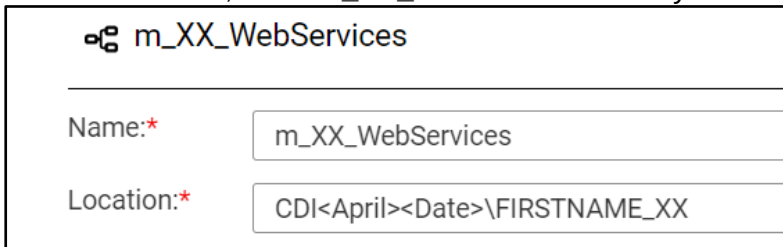
23. Create a new Mapping.

Note: If you get What type of mapping do you want to create? Select **Mapping** and click **Continue**.



The **New Mapping** dialog box is shown with the question "What type of mapping do you want to create?". There are two options: **Mapping** and **Mapping - SQL ELT**. The **Mapping** option is highlighted with a red box. Below the **Mapping** option, there is a description: "Create a mapping with various sources and targets." Below the **Mapping - SQL ELT** option, there is a description: "Create a mapping that is optimized for your company's cloud data warehouse." At the bottom right, the **Continue** button is highlighted with a red box, and the **Cancel** button is also visible.

24. In the Name field, enter **m_XX_WebServices**. Verify the asset location.



m_XX_WebServices

Name:* m_XX_WebServices

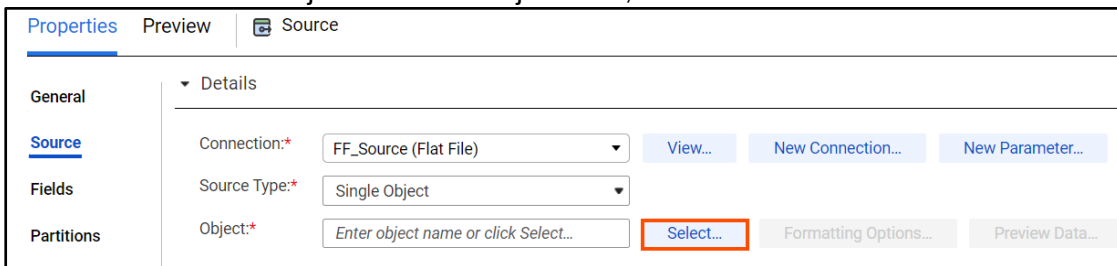
Location:* CDI<April><Date>\FIRSTNAME_XX

25. To configure the source, from the mapping canvas, click the **Source** transformation.

26. From the properties pane, click **Source**.

27. From the Connection drop-down, select **FF_Source**.

28. To select the source object from the Object field, click **Select**.



Properties Preview **Source**

General

Source

Fields

Partitions

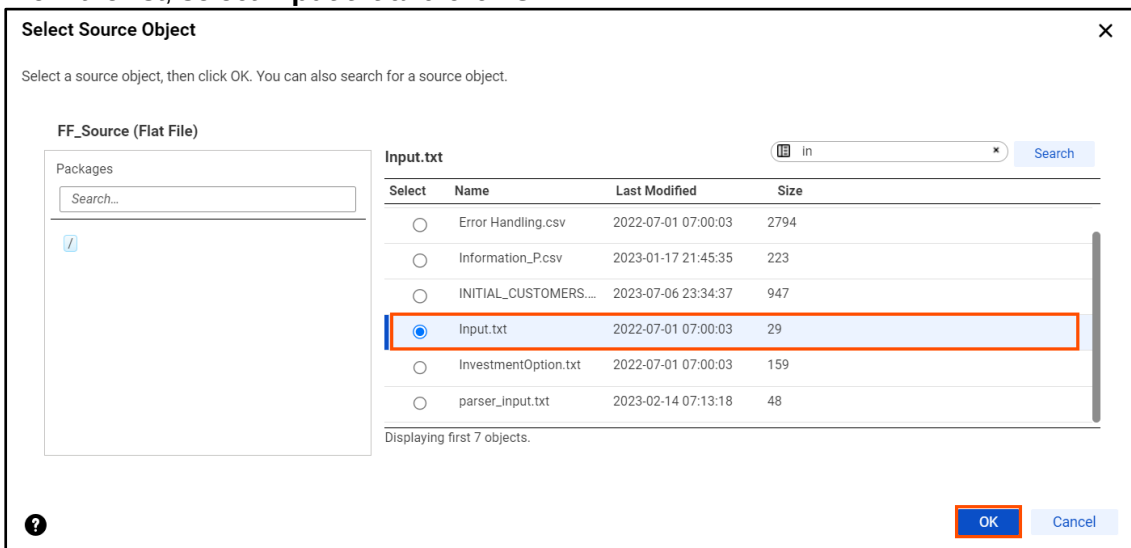
Details

Connection:* FF_Source (Flat File) View... New Connection... New Parameter...

Source Type:* Single Object

Object:* Enter object name or click Select... **Select...** Formatting Options... Preview Data...

29. From the list, select **Input.txt** and click **OK**.



Select Source Object

Select a source object, then click OK. You can also search for a source object.

FF_Source (Flat File)

Packages

Search...

Input.txt

Select Name Last Modified Size

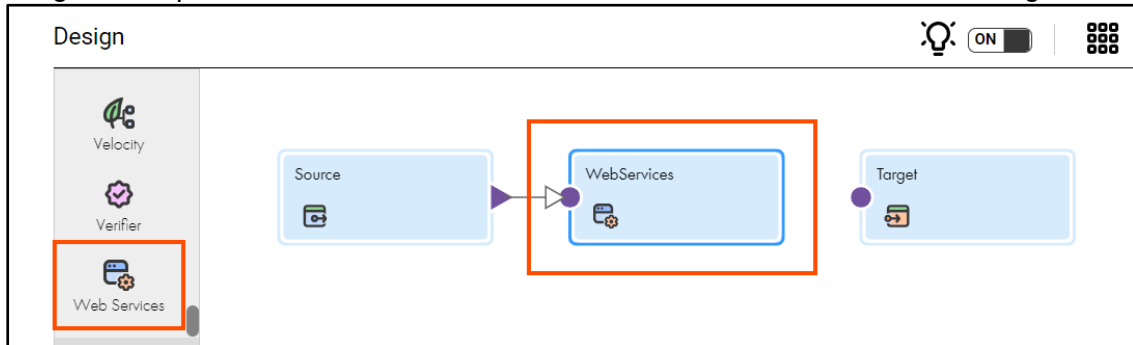
☐	Error Handling.csv	2022-07-01 07:00:03	2794
☐	Information_P.csv	2023-01-17 21:45:35	223
☐	INITIAL_CUSTOMERS....	2023-07-06 23:34:37	947
☒	Input.txt	2022-07-01 07:00:03	29
☐	InvestmentOption.txt	2022-07-01 07:00:03	159
☐	parser_input.txt	2023-02-14 07:13:18	48

Displaying first 7 objects.

OK Cancel

Add Web Services Transformation

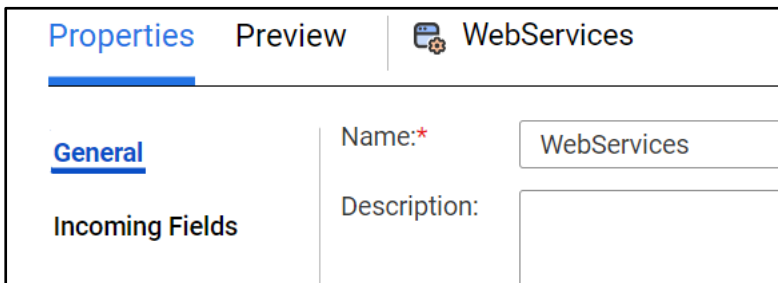
30. Drag and drop a **Web Services** transformation between the source and the target.



Note: The Source transformation gets linked to the Web Service transformation.

31. Select the **WebServices** transformation on the mapping canvas.

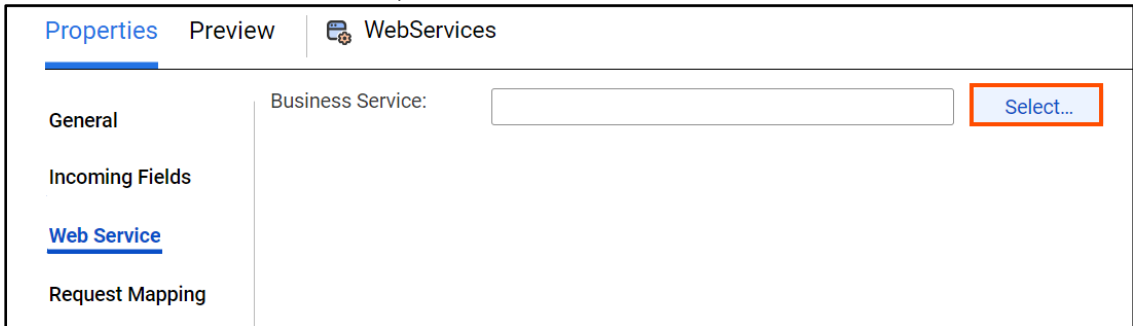
32. In the **General** section of the WebServices properties, retain the Name as **WebServices**.



The 'WebServices' properties dialog box is shown with the 'General' tab selected. The 'Name' field is set to 'WebServices'.

33. From the properties pane, click **Web Service**.

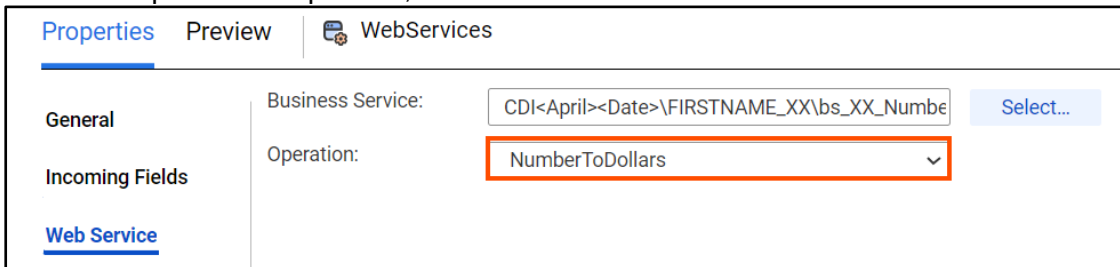
34. To select the business service, click **Select**.



The 'WebServices' properties dialog box is shown with the 'Web Service' tab selected. The 'Business Service' field is empty, and the 'Select...' button is highlighted with an orange box.

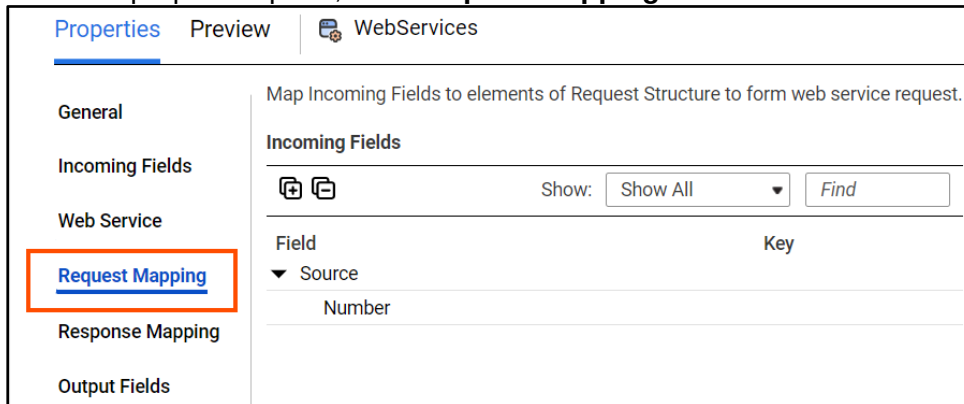
35. Navigate to your working directory and select the **bs_XX_NumberConversion** business service.

36. From the Operation drop-down, select **NumberToDollars**.



The 'WebServices' properties dialog box is shown with the 'Web Service' tab selected. The 'Business Service' field contains the text 'CDI<April><Date>\FIRSTNAME_XX\bs_XX_Numbe', and the 'Operation' dropdown is set to 'NumberToDollars' (highlighted with an orange box).

37. From the properties pane, click **Request Mapping**.



Properties Preview WebServices

General

Incoming Fields

Web Service

Request Mapping

Response Mapping

Output Fields

Map Incoming Fields to elements of Request Structure to form web service request.

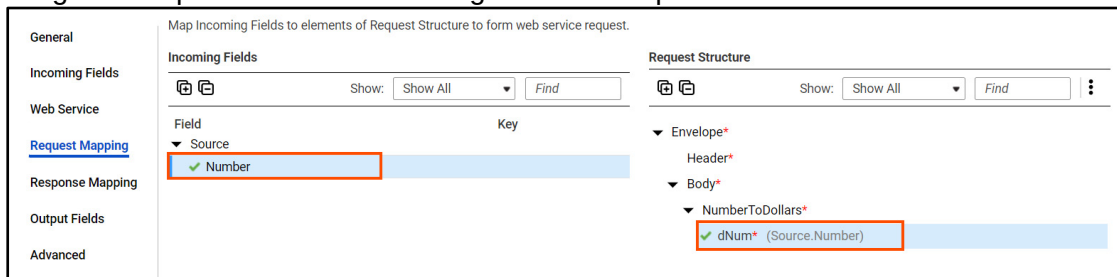
Incoming Fields

Show: Show All Find

Field	Key
Source	
Number	

38. From the Request Structure section, drill-down to **Body > NumberToDollars > dNum**.

39. Drag and drop the **Number** incoming field and drop it on **dNum**.



General

Incoming Fields

Web Service

Request Mapping

Response Mapping

Output Fields

Advanced

Map Incoming Fields to elements of Request Structure to form web service request.

Incoming Fields

Show: Show All Find

Field	Key
Source	
Number	

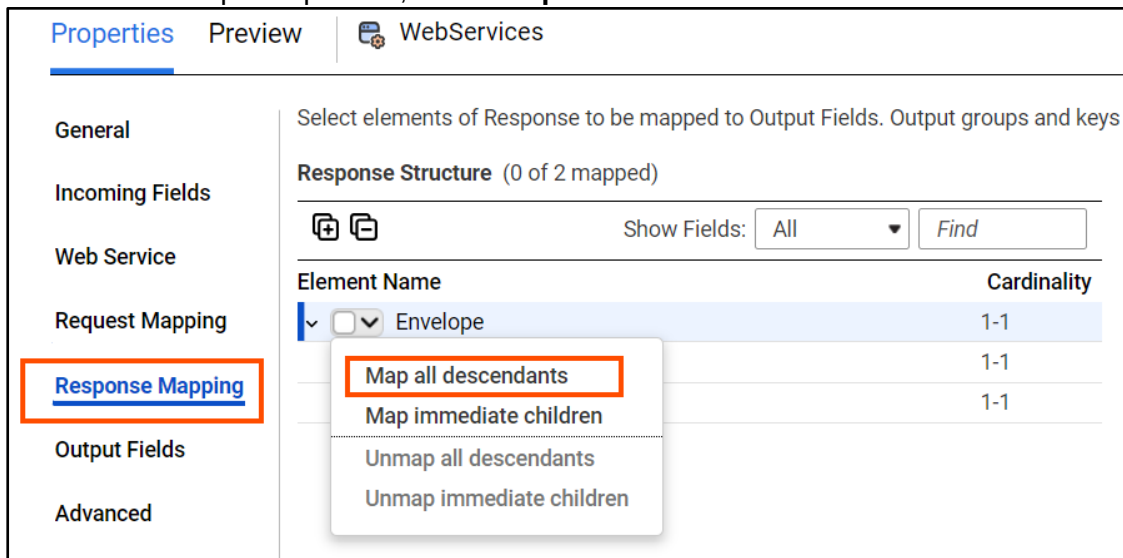
Request Structure

Show: Show All Find

- Envelope*
 - Header*
 - Body*
 - NumberToDollars*
 - dNum* (Source.Number)**

40. From the properties pane, click **Response Mapping**.

41. From the Envelope drop-down, select **Map all descendants**.



Properties Preview WebServices

General

Incoming Fields

Web Service

Request Mapping

Response Mapping

Output Fields

Advanced

Select elements of Response to be mapped to Output Fields. Output groups and keys

Response Structure (0 of 2 mapped)

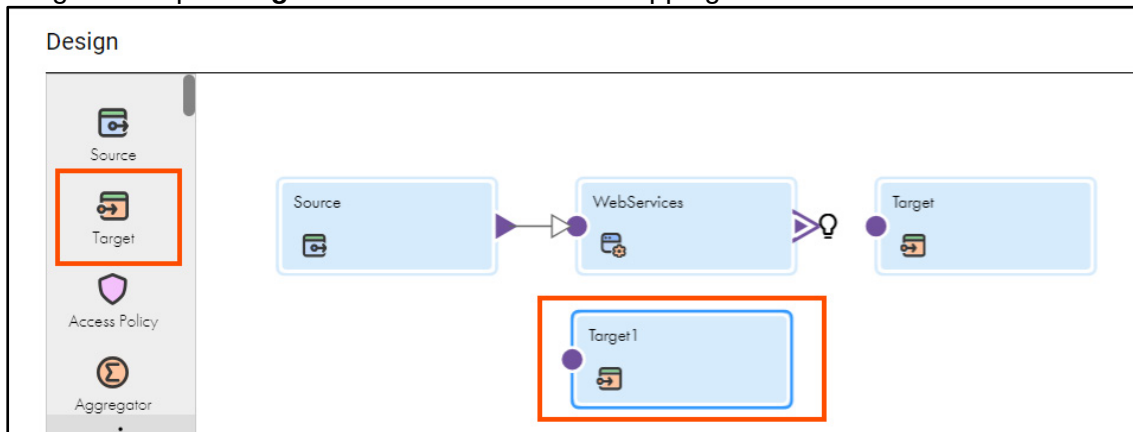
Show Fields: All Find

Element Name	Cardinality
Envelope	1-1
	1-1
	1-1

- Map all descendants
- Map immediate children
- Unmap all descendants
- Unmap immediate children

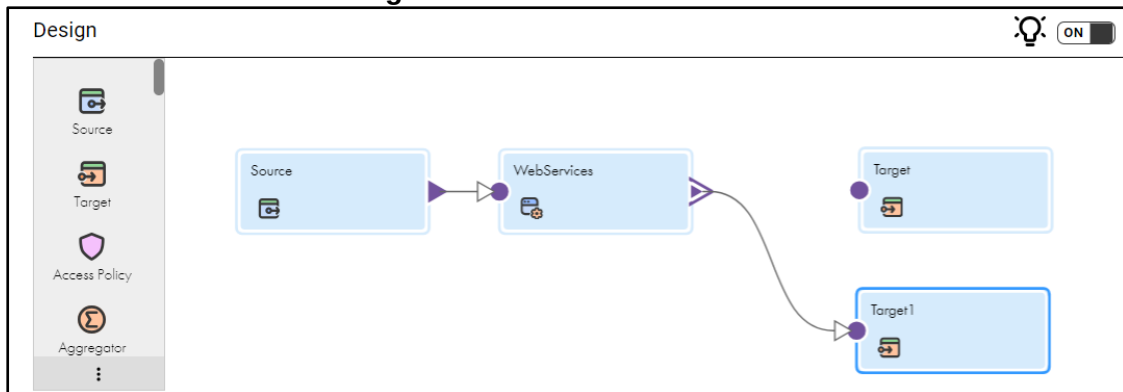
Add a Target Transformation

42. Drag and drop a **Target** transformation on the mapping canvas.



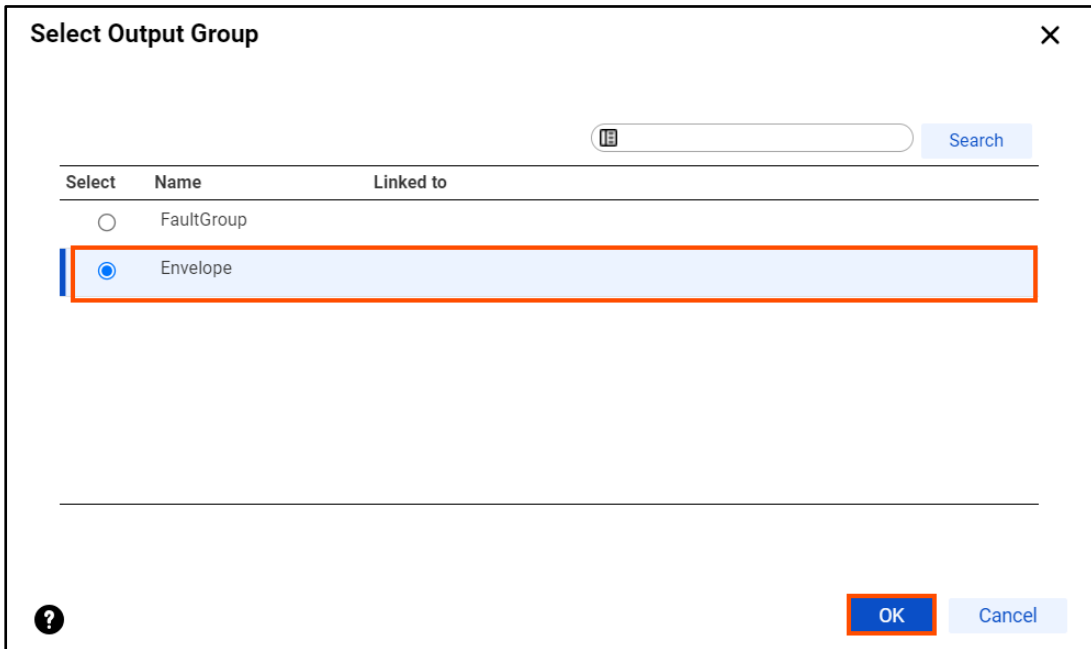
43. Select the added Target transformation (**Target 1**) on the mapping canvas.

44. Link the **WebServices** to **Target1**.



45. From the list, select **Envelope**.

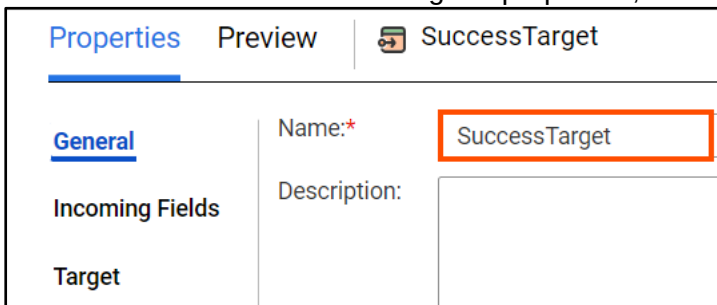
46. Click **OK**.



The dialog box titled "Select Output Group" has a search bar at the top with a magnifying glass icon and a "Search" button. Below the search bar is a table with three columns: "Select", "Name", and "Linked to". The table contains two rows: "FaultGroup" and "Envelope". The "Envelope" row is highlighted with a blue background and an orange border. At the bottom right of the dialog are "OK" and "Cancel" buttons. A question mark icon is located at the bottom left.

Select	Name	Linked to
☐	FaultGroup	
☒	Envelope	

47. In the **General** section of the Target 1 properties, enter Name as **SuccessTarget**.

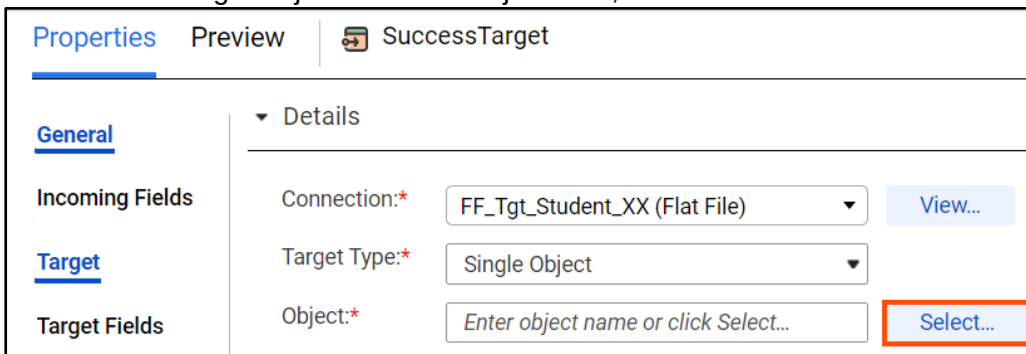


The "Properties" dialog box for "SuccessTarget" has two tabs: "Properties" and "Preview". The "Properties" tab is active. It has three sections: "General", "Incoming Fields", and "Target". The "General" section is expanded, showing "Name:*" with the value "SuccessTarget" and "Description:" with an empty text area. The "Incoming Fields" and "Target" sections are collapsed.

48. From the properties pane, click **Target**.

49. From the Connection drop-down, select your **target Flat File** connection.

50. To select the target object from the Object field, click **Select**.



The "Properties" dialog box for "SuccessTarget" has two tabs: "Properties" and "Preview". The "Properties" tab is active. It has three sections: "General", "Incoming Fields", and "Target". The "Target" section is expanded, showing "Details". The "Details" section has three fields: "Connection:*" with a dropdown menu showing "FF_Tgt_Student_XX (Flat File)", "Target Type:*" with a dropdown menu showing "Single Object", and "Object:*" with a text area containing "Enter object name or click Select...". A "Select..." button is highlighted with an orange border. There are also "View..." and "Select..." buttons.

51. In the Target Object window, select **Create New at Runtime**.

52. Enter **SuccessResponse.csv** as Static File Name.

53. Click **OK**.

Target Object X

Select an existing target object or create a new one. Any new target objects will be created when the mapping task is executed.

Target Object: ☐ Existing ☒ Create New at Runtime

[Formatting Options...](#) ?

☐ Handle Special Characters

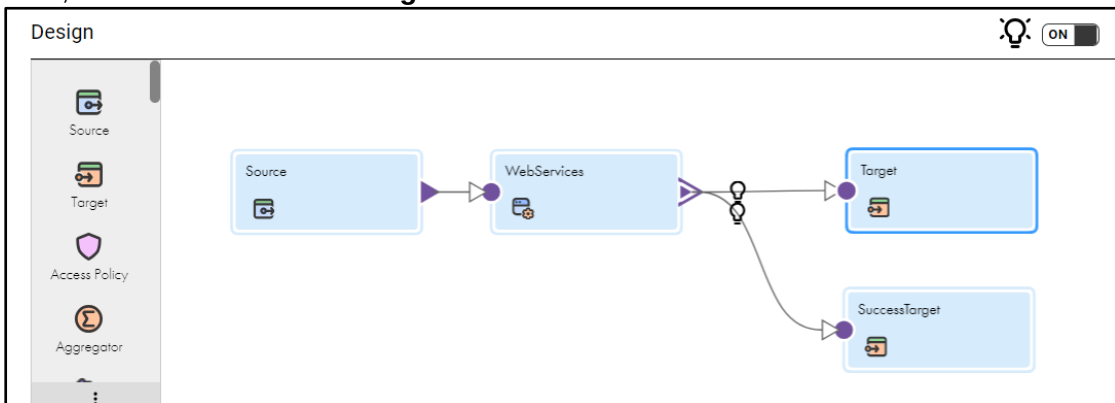
☐ Skip rounding decimal and double data types

☐ Use a Dynamic File Name

Static File Name:*

? **OK** Cancel

54. Now, link **WebServices** to **Target**.



55. From the list, select **FaultGroup** and click **OK**.

Select Output Group X

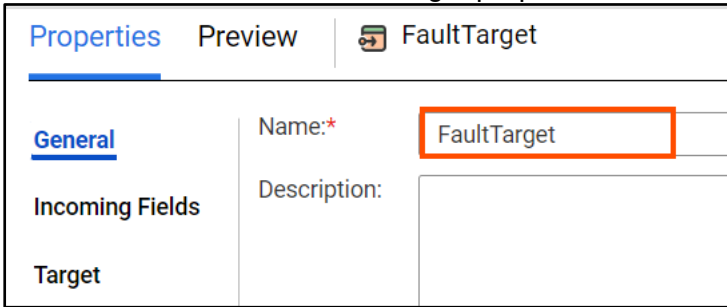
Search

Select	Name	Linked to
☒	FaultGroup	
☐	Envelope	SuccessTarget

? **OK** Cancel

56. Select the Target transformation on the mapping canvas.

57. In the **General** section of the Target properties, enter the Name as **FaultTarget**.

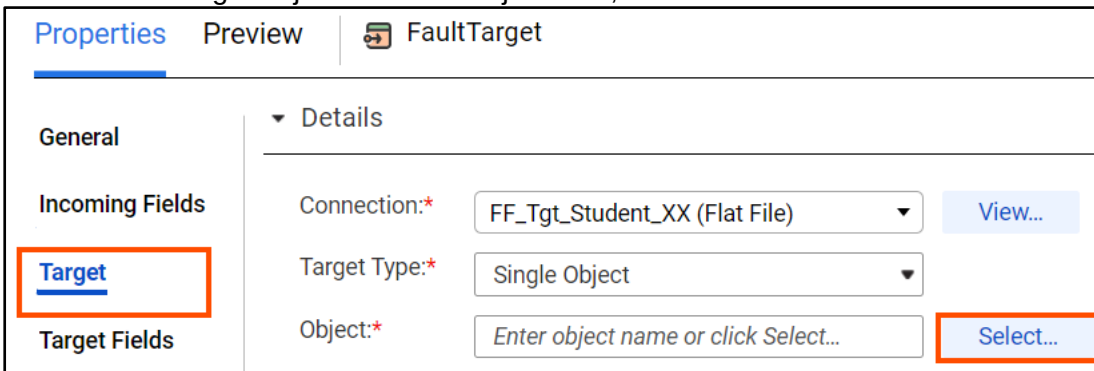


The screenshot shows the 'Properties' pane for a target named 'FaultTarget'. The 'General' tab is selected. The 'Name' field is set to 'FaultTarget' and is highlighted with a red box. The 'Description' field is empty. The 'Incoming Fields' and 'Target' sections are also visible.

58. From the properties pane, click **Target**.

59. From the Connection drop-down, select your **target Flat File** connection.

60. To select the target object from the Object field, click **Select**.

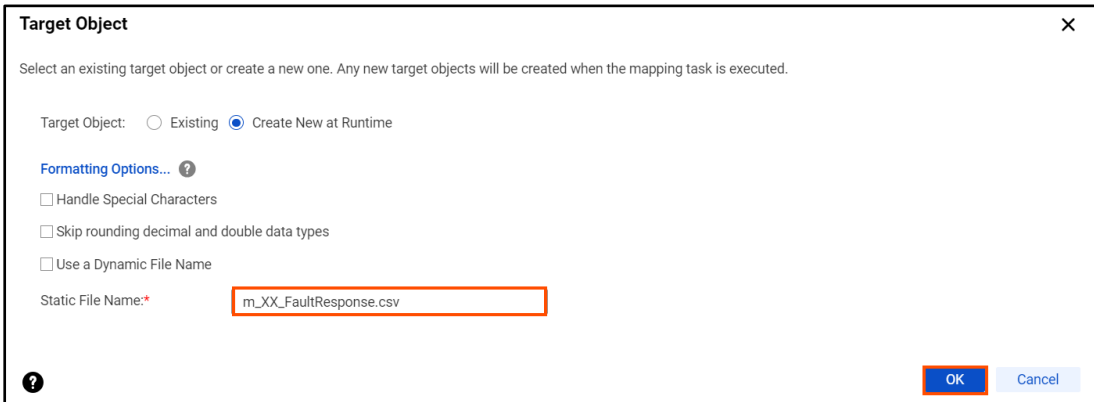


The screenshot shows the 'Properties' pane for a target named 'FaultTarget'. The 'Target' tab is selected and highlighted with a red box. The 'Details' section shows the 'Connection' set to 'FF_Tgt_Student_XX (Flat File)' and the 'Target Type' set to 'Single Object'. The 'Object' field is empty, and the 'Select...' button is highlighted with a red box.

61. In the Target Object window, select **Create New at Runtime**.

62. Enter **m_XX_FaultResponse.csv** as Static File Name.

63. Click **OK**.



The screenshot shows the 'Target Object' dialog. The 'Create New at Runtime' radio button is selected. The 'Static File Name' field is set to 'm_XX_FaultResponse.csv' and is highlighted with a red box. The 'OK' button is highlighted with a red box.

64. Save and run the mapping.



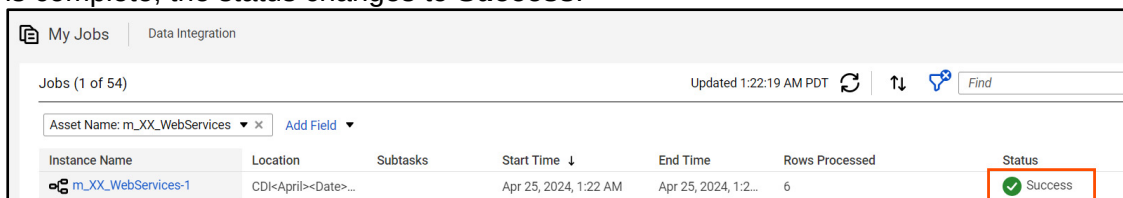
The screenshot shows the mapping toolbar with the 'Save' and 'Run' buttons highlighted with a red box.

65. From the Runtime Environment drop-down, select INFA-SERVER.

66. Click **Run**.

Monitor Status

67. To monitor the mapping status, from the navigation pane, click **My Jobs**. When the task is complete, the status changes to **Success**.



My Jobs		Data Integration				
Jobs (1 of 54)		Updated 1:22:19 AM PDT				
Asset Name: m_XX_WebServices		Add Field				
Instance Name	Location	Subtasks	Start Time ↓	End Time	Rows Processed	Status
m_XX_WebServices-1	CDI<April><Date>...		Apr 25, 2024, 1:22 AM	Apr 25, 2024, 1:2...	6	Success

68. Close the asset from the navigation pane.

69. Optionally, you can preview data in the dummy mapping.

This concludes the lab.

Module 14: IICS REST APIs

Lab 14-3: Running a Mapping Task Using REST API

Overview:

An API Connector uses the REST API to access programs and activities in Informatica Cloud and perform various tasks like logging into IICS, running a task, and much more.

In this lab, you will use the REST API to log in to your Org and run tasks.

Objective:

- Use a REST client application to call the login resource and obtain a session ID
- Start a Mapping task
- Log out of the Informatica Cloud API session

Scenario:

John demonstrates to Ruby how she can use REST APIs to log in and run tasks in IICS. He uses the Postman client to log in to the Org and obtain a session ID. He will use the session ID to run a mapping task using the REST API and then log out from the Org.

Duration:

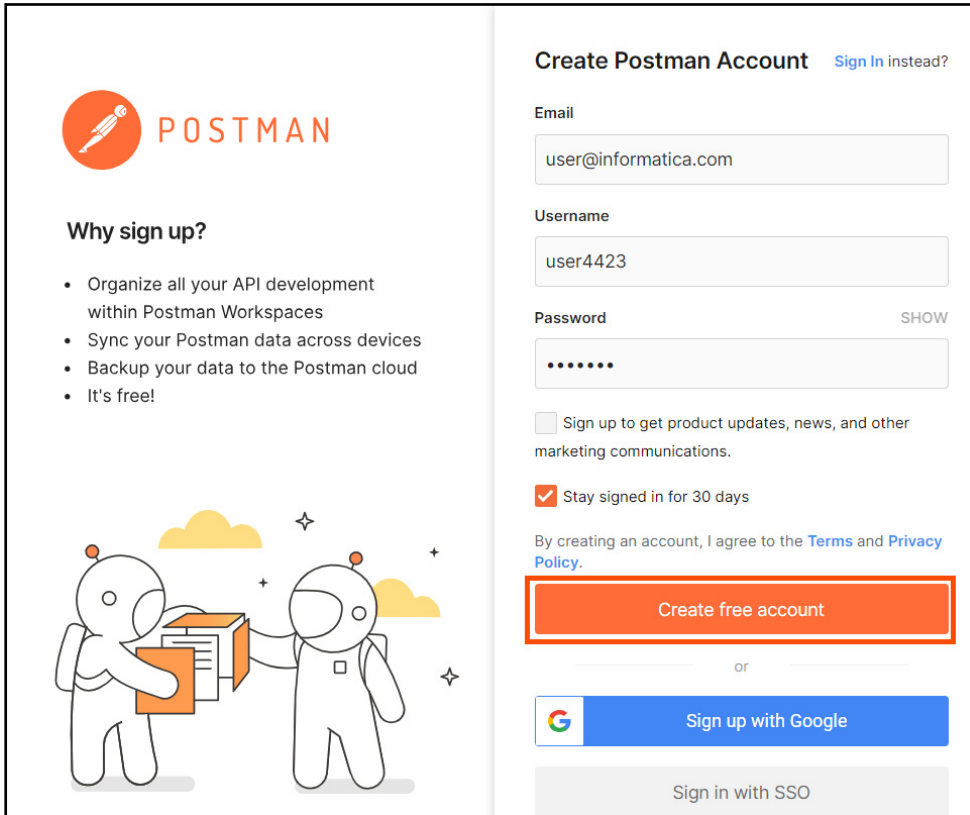
25 minutes

Tasks

Create a Postman Account

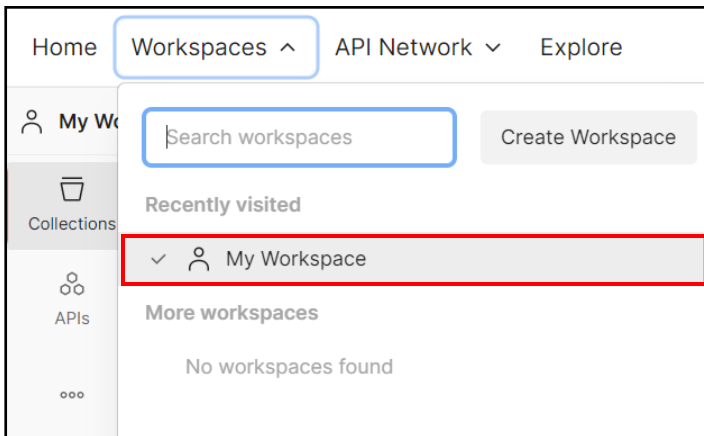
1. Open a web browser and enter the following link to open Postman:
<https://identity.getpostman.com/>
2. Click **Create Account**.

3. Provide a valid email Id and fill in all the required details. Click **Create free account**.



The image shows the Postman account creation page. On the left, there's a section titled "Why sign up?" with a list of benefits: "Organize all your API development within Postman Workspaces", "Sync your Postman data across devices", "Backup your data to the Postman cloud", and "It's free!". Below this is an illustration of two astronauts. On the right, the "Create Postman Account" form is displayed. It includes fields for "Email" (filled with "user@informatica.com"), "Username" (filled with "user4423"), and "Password" (masked with dots). There are checkboxes for "Sign up to get product updates, news, and other marketing communications." and "Stay signed in for 30 days". A link to "Terms and Privacy Policy" is present. The "Create free account" button is highlighted with a red border. Below it, there are options to "Sign up with Google" and "Sign in with SSO".

4. You will get an account confirmation email to the registered email id. Confirm your postman account.
5. Now you will be logged into Postman. If not, login into Postman using your credentials.
6. From the Workspace drop-down, select **My Workspace**.



The image shows the Postman workspace selection interface. At the top, there are navigation tabs: "Home", "Workspaces" (selected), "API Network", and "Explore". Below the tabs, there's a sidebar with icons for "My Workspace", "Collections", "APIs", and a menu icon. The main area shows a search bar for "Search workspaces" and a "Create Workspace" button. Under "Recently visited", the "My Workspace" option is highlighted with a red border. Below this, there's a section for "More workspaces" which currently shows "No workspaces found".

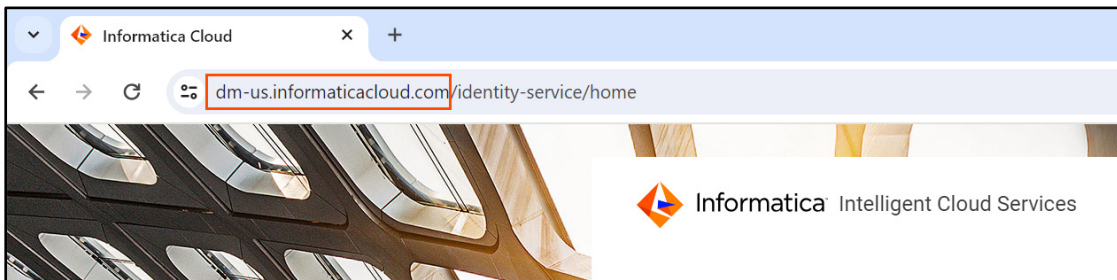
Send a Login Request

7. Click on a new tab in the workspace section and from the drop-down, select **POST**.
8. Enter the URL in the request URL field in the following format:
https://<Informatica URL>/ma/api/v2/user/login

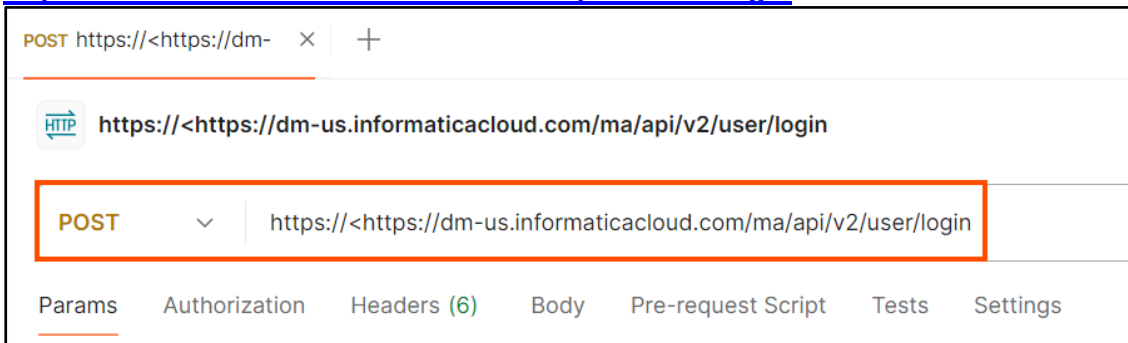
Note: The Informatica URL can change based on the POD (Point of Deployment) region of your IICS Org. Your POD region is based on the location of your Informatica Intelligent Cloud Services data center. Use one of the following POD regions

- For North America, use **us**
- For Europe, use **em**
- For Asia, use **ap**

You can identify the Informatica URL from the IICS login page, as shown in the image below:



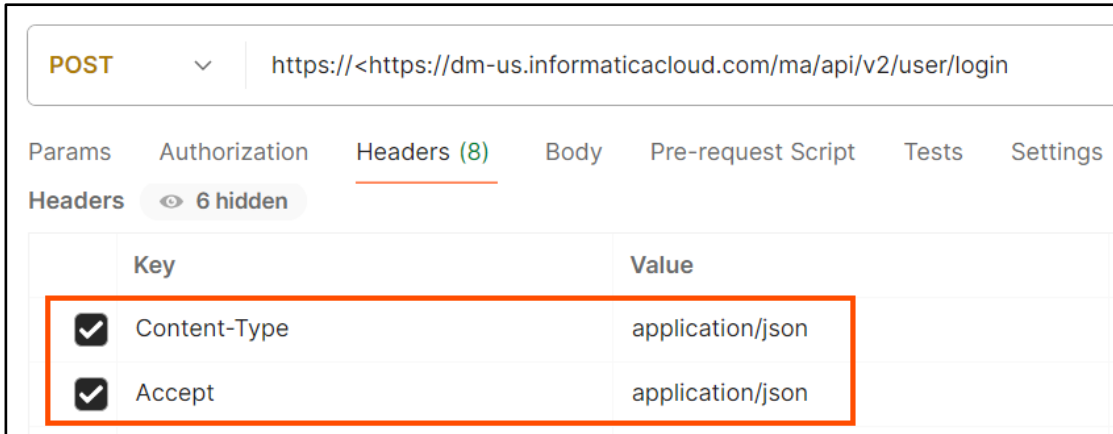
9. For the lab environment provided by the instructor, enter the following URL:
<https://dm-us.informaticacloud.com/ma/api/v2/user/login>



Note: Make sure to remove extra space from the URL after pasting the URL into the request field.

10. Select the **Headers** tab.
11. Enter Key and Value, as shown in the table below:

Key	Value
Content-Type	application/json
Accept	application/json



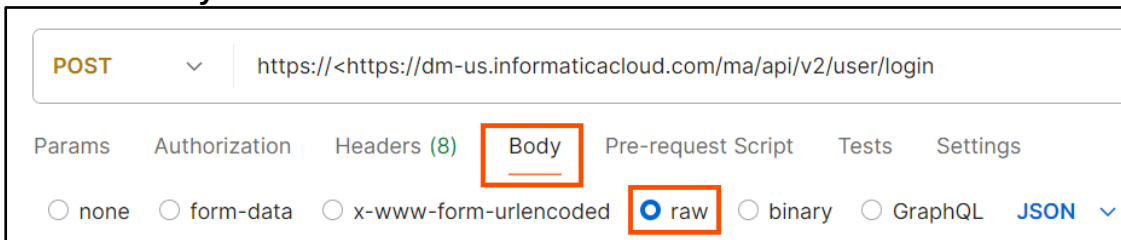
POST ▼ <https://<https://dm-us.informaticacloud.com/ma/api/v2/user/login>

Params Authorization Headers (8) Body Pre-request Script Tests Settings

Headers 6 hidden

Key	Value
☒ Content-Type	application/json
☒ Accept	application/json

12. Select the **Body** tab and select **raw**.



POST ▼ <https://<https://dm-us.informaticacloud.com/ma/api/v2/user/login>

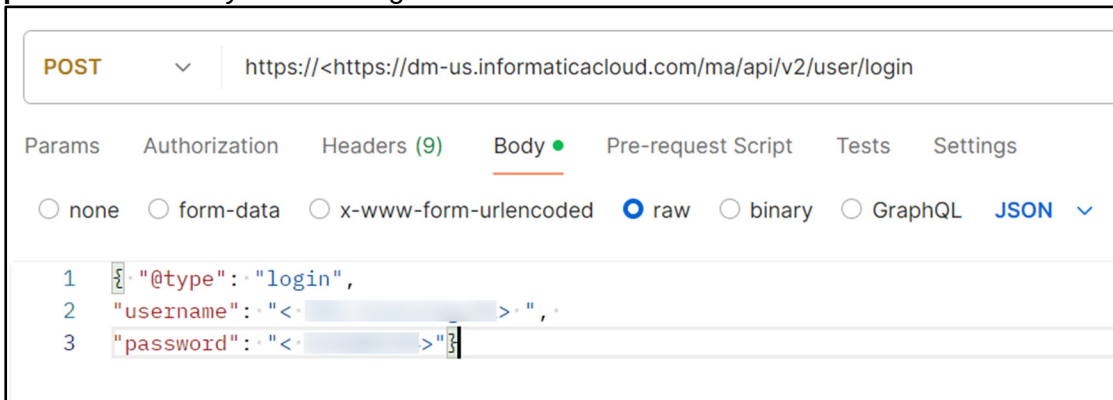
Params Authorization Headers (8) Body Pre-request Script Tests Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼

13. Enter the following syntax:

```
{ "@type": "login",
  "username": "< Informatica Cloud username> ",
  "password": "< Informatica Cloud password>"}
```

14. In the syntax, replace < Informatica Cloud username> and < Informatica Cloud password> with your IICS Org credentials.



POST ▼ <https://<https://dm-us.informaticacloud.com/ma/api/v2/user/login>

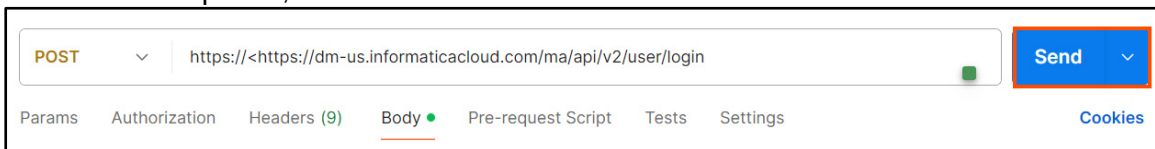
Params Authorization Headers (9) Body ● Pre-request Script Tests Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼

```
1 { "@type": "login",
2   "username": "< > ",
3   "password": "< >" }
```

Note: For this lab, we have masked the username and password in the screenshot.

15. To view the response, click **Send**.



POST ▼ <https://<https://dm-us.informaticacloud.com/ma/api/v2/user/login>

Params Authorization Headers (9) Body ● Pre-request Script Tests Settings

Send ▼ Cookies

16. The status of the response is **200 OK**.

17. From the response, note down the values for **ServerUrl** and **icSessionId** in a notepad.



```

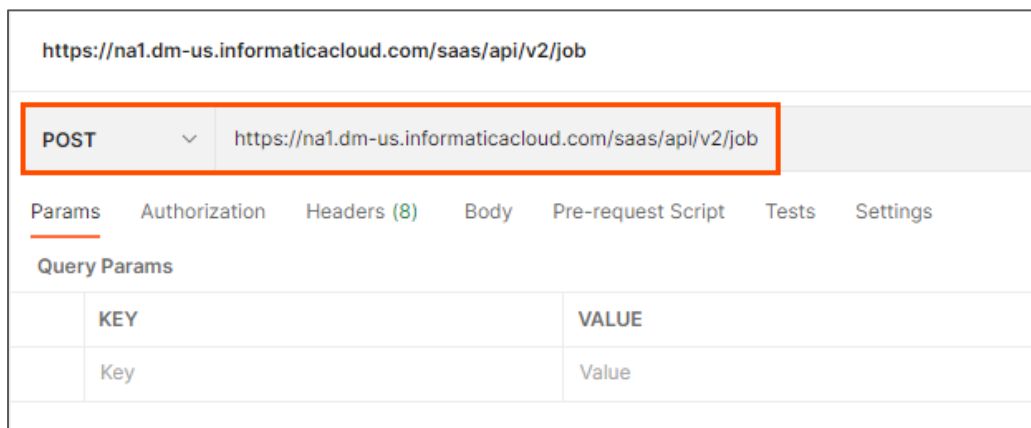
18  "timezone": null,
19  "serverUrl": "https://na1.dm-us.informaticacloud.com/saas",
20  "icSessionId": "anUJQAncBRihY7osYioMQH",
21  "securityQuestion": "In what city was your first job?",
22  "securityAnswer": "*****",
23  "uuid": "bZZngePyraJl5A07mmZIwr",
24  "forceChangePassword": false,

```

Send Request to Run a Job

18. To add a new tab, click .
19. From the drop-down, select **POST**.
20. Enter the URL in the request URL field in the following format:

<Copied ServerUrl>/api/v2/job



https://na1.dm-us.informaticacloud.com/saas/api/v2/job

POST  https://na1.dm-us.informaticacloud.com/saas/api/v2/job

Params Authorization Headers (8) Body Pre-request Script Tests Settings

Query Params

KEY	VALUE
Key	Value

Note: Make sure to remove extra space from the URL after pasting the URL in the request field.

21. Select the **Headers** tab.
22. Enter Key and Value, as shown in the table below:

Key	Value
Accept	application/json
Content-Type	application/json
icSessionId	Enter the icSessionId noted earlier

https://na1.dm-us.informaticacloud.com/saas/api/v2/job

POST ▼ https://na1.dm-us.informaticacloud.com/saas/api/v2/job

Params Authorization Headers (11) Body Pre-request Script Tests Settings

Headers 8 hidden

	KEY	VALUE
☒	Accept	application/json
☒	Content-Type	application/json
☒	icSessionId	anUJQAncBRihY7osYioMQH

23. Select the **Body** tab and select **raw**.

24. Enter the following syntax:

Note: In this syntax, use the exact asset name you have used while creating this request. Replace XX with your student number.

```
{
  "@type": "job",
  "taskName": "mt_XX_MappingTask",
  "taskType": "MTT"
}
```

25. Ensure that the **taskName** is same as the task name in IICS.

Params Authorization Headers (7) Body ● Pre-request Script Tests Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼

```
1 {
2   "@type": "job",
3   "taskName": "mt_XX_MappingTask",
4   "taskType": "MTT"
5 }
6
```

Note: The task name shown in the screenshot is for demonstration purposes and you must specify your exact task name as a value to the taskName parameter.

26. To view the response, click **Send**.

https://na1.dm-us.informaticacloud.com/saas/api/v2/job

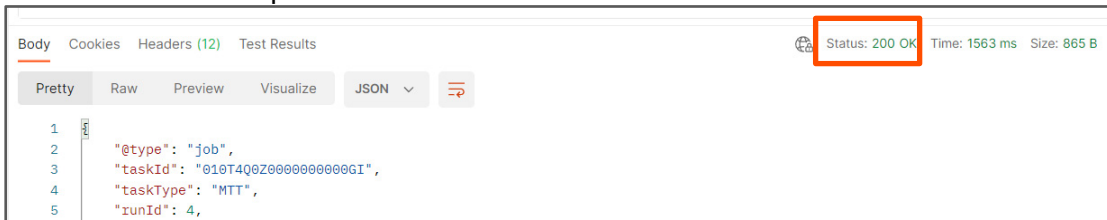
POST ▼ https://na1.dm-us.informaticacloud.com/saas/api/v2/job Send ▼

Params Authorization Headers (11) Body ● Pre-request Script Tests Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼

[Cookies](#)
[Beautify](#)

27. The status of the response is **200 OK**.

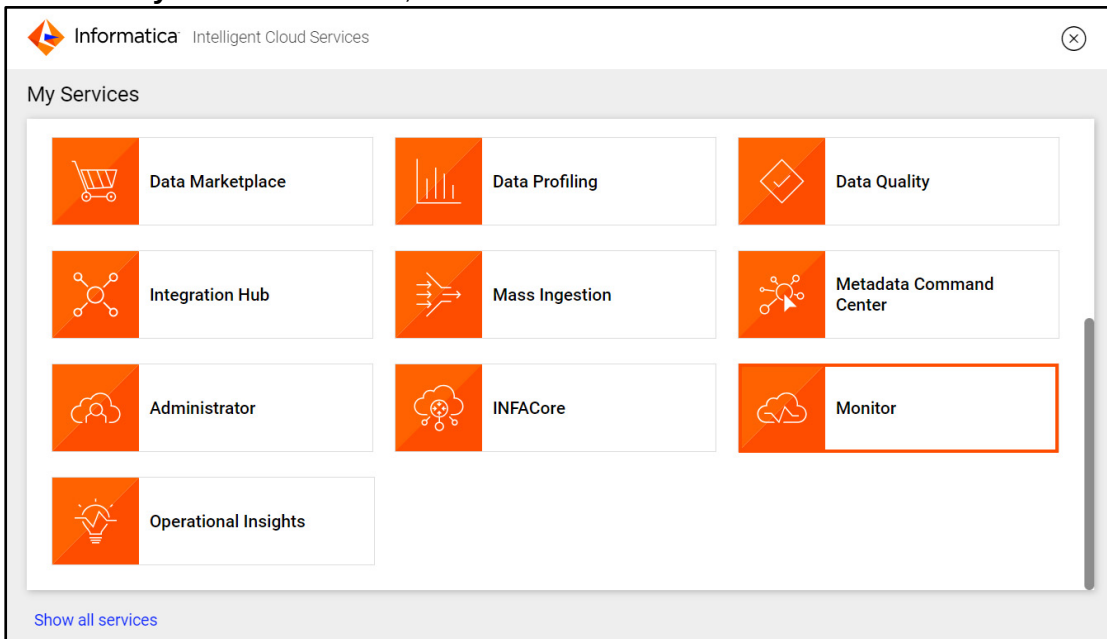


28. Open the IICS Login page from the Bookmarks bar.

29. Log in to your IICS Org.

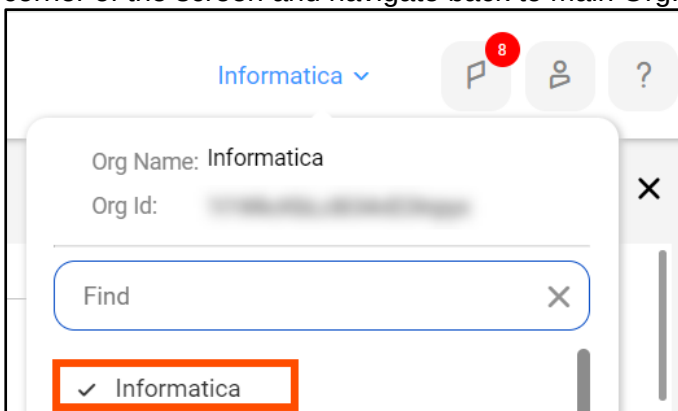
Note: If you are already logged in, log out and log in again.

30. From the **My Services** window, select **Monitor**.



Note: Ensure that you are accessing the service from the main org.

Important: If you are in the sub-org, click on the Org name available on the top right corner of the screen and navigate back to main Org.



31. From the navigation pane, select **All Jobs**.

32. Open the **mt_XX_MappingTask** and verify that the task is started by API.

Up to date		Updated 12:40:58 AM PST				
Name	Location	Subtasks	Start Time	End Time	Rows Processed	Status
MappingTask-4	CDI (LT Development)\CDI...		Jan 27, 2023, 12:35 AM	Jan 27, 2023, 12:35 AM	4	Success

Instance ID:	4	Success Rows:	4
Task Type:	Mapping Task	Errors:	0
Started By:	cdiuser through API	Session Log:	Download Session Log
Start Time:	Jan 27, 2023 12:35:44 AM		
End Time:	Jan 27, 2023 12:35:57 AM		
Duration:	13 seconds		
Runtime Environment:	IICS-XX-FIRSTNAME		
Secure Agent:	IICS-XX-FIRSTNAME		

Send a Request to Logout

33. In Postman, to add a new tab, click .

34. From the drop-down, select **POST**.

35. Enter the following URL in the request URL field:

<Copied ServreUrl>/api/v2/user/logout

Untitled Request

POST
 https://na1.dm-us.informaticacloud.com/saas/api/v2/user/logout

Params
 Authorization
 Headers (7)
 Body
 Pre-request Script
 Tests
 Settings

 Response

Note: Make sure to remove extra space from the URL after pasting the URL in the request field.

36. Select the **Headers** tab.

37. Enter Key and Value, as shown in the table below:

Key	Value
Accept	application/json
Content-Type	application/json
icSessionId	Enter the icSessionId noted earlier

https://na1.dm-us.informaticacloud.com/saas/api/v2/user/logout

POST ▼ https://na1.dm-us.informaticacloud.com/saas/api/v2/user/logout

Params Authorization Headers (11) Body Pre-request Script Tests Settings

Headers 👁 8 hidden

	KEY	VALUE
☒	Accept	application/json
☒	Content-Type	application/json
☒	icSessionId	anUJQAncBRihY7osYioMQH

38. Select the **Body** tab and select **raw**.

39. Enter the following syntax:

```
{"@type":"logout"}
```

POST ▼ https://na1.dm-us.informaticacloud.com/saas/api/v2/user/logout

Params Authorization Headers (11) Body ● Pre-request Script Tests Settings

● none ● form-data ● x-www-form-urlencoded ● raw ● binary ● GraphQL JSON ▼

1 {"@type":"logout"}

40. To view the response, click **Send**.

41. The status of the response is **200 OK**.

Body Cookies Headers (8) Test Results 🌐 Status: 200 OK Time: 568 ms

Pretty Raw Preview Visualize Text ▼ ≡

1

Note: There is no response in body for log out request.

This concludes the lab.

Module 14: IICS REST APIs

Lab 14-4: Monitoring Tasks Using REST API

Overview:

Informatica Intelligent Cloud Services supports the platform REST API version 2 and version 3 resources, and service-specific resources.

In this lab, you will use the REST API to request log information for running jobs from the Monitor service.

Objective:

- Use a REST client application to call the login resource and obtain a session ID
- Get the log information about the running jobs

Duration:

15 minutes

Tasks

Open Postman

1. Navigate to the web Postman application.
2. Click  to open a new tab.

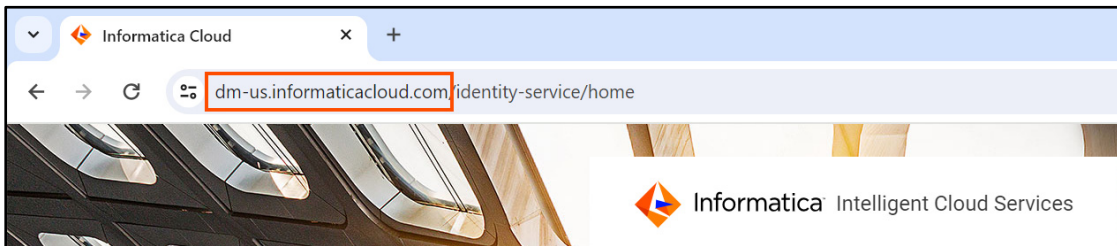
Send a Login Request

3. From the drop-down, select **POST**.
4. Enter the URL in the request URL field in the following format:
https://<Informatica URL>/ma/api/v2/user/login

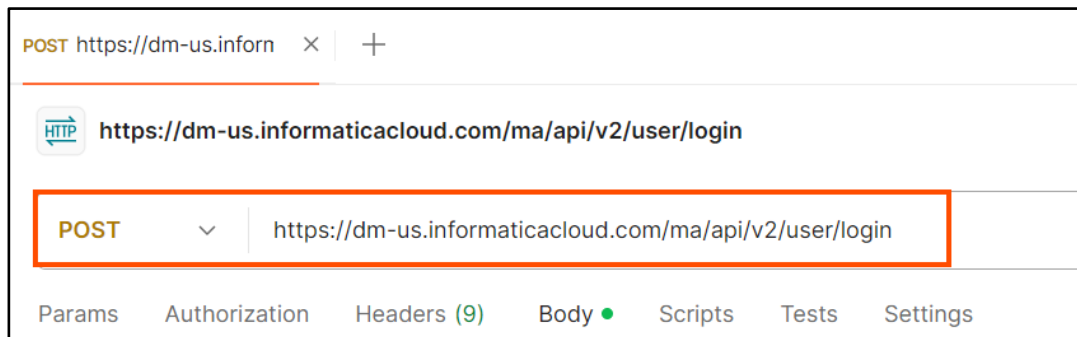
Note: The Informatica URL can change based on the POD (Point of Deployment) region of your IICS Org. Your POD region is based on the location of your Informatica Intelligent Cloud Services data center. Use one of the following POD regions

- For North America, use **us**
- For Europe, use **em**
- For Asia, use **ap**

You can identify the Informatica URL from the IICS login page, as shown in the image below:



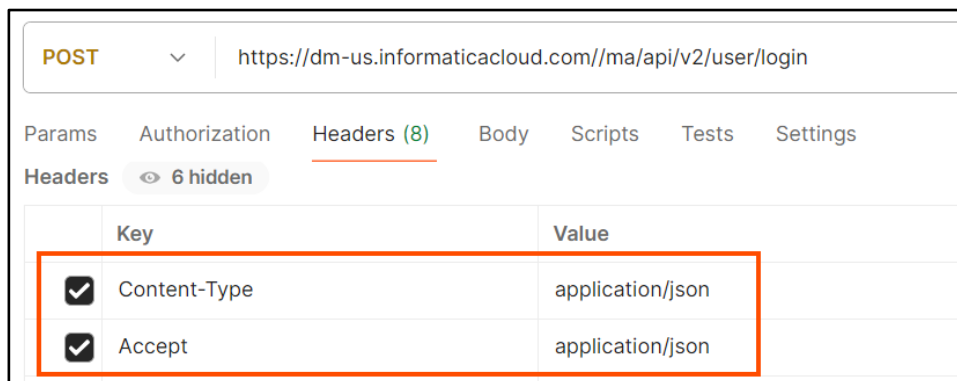
- For the lab environment provided by the instructor, enter the following URL:
https://dm-us.informaticacloud.com/ma/api/v2/user/login



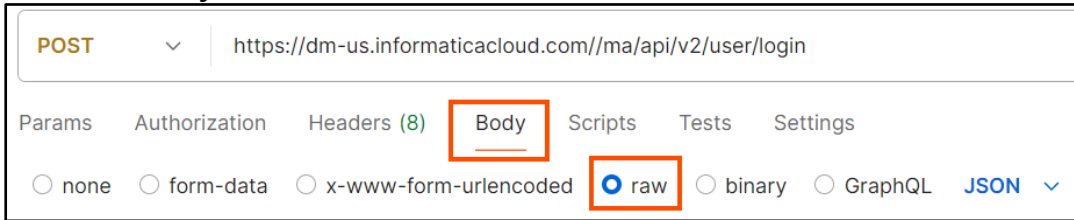
Note: Make sure to remove extra space from the URL after pasting the URL in the request field.

- Select the **Headers** tab.
- Enter Key and Value, as shown in the table below:

Key	Value
Content-Type	application/json
Accept	application/json



8. Select the **Body** tab and select **raw**.



POST ▼ <https://dm-us.informaticacloud.com/ma/api/v2/user/login>

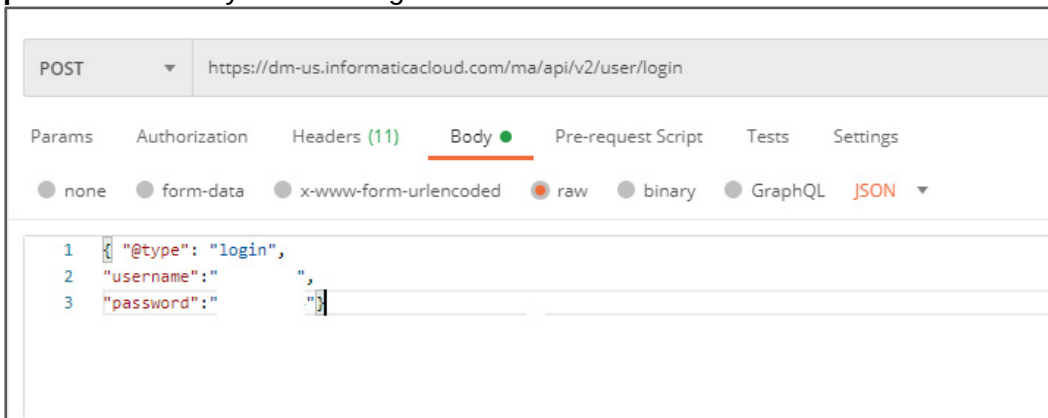
Params Authorization Headers (8) **Body** Scripts Tests Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼

9. Enter the following syntax:

```
{ "@type": "login",
  "username": "< Informatica Cloud username> ",
  "password": "< Informatica Cloud password>"}
```

10. In the syntax, replace < Informatica Cloud username> and < Informatica Cloud password> with your IICS Org credentials.



POST ▼ <https://dm-us.informaticacloud.com/ma/api/v2/user/login>

Params Authorization Headers (11) **Body** Pre-request Script Tests Settings

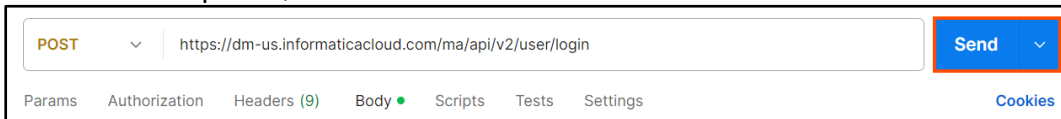
☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼

```
1 { "@type": "login",
2   "username": "
3   "password": "

```

Note: For this lab, we have masked the username and password in the screenshot.

11. To view the response, click **Send**.



POST ▼ <https://dm-us.informaticacloud.com/ma/api/v2/user/login> **Send** ▼

Params Authorization Headers (9) **Body** Scripts Tests Settings

[Cookies](#)

12. The status of the response is **200 OK**.

13. From the response, note down the values for **ServerUrl** and **icSessionId** in a notepad.



Body Cookies Headers (14) Test Results 200 OK • 797 ms • 1.69 KB

Pretty Raw Preview Visualize JSON ▼

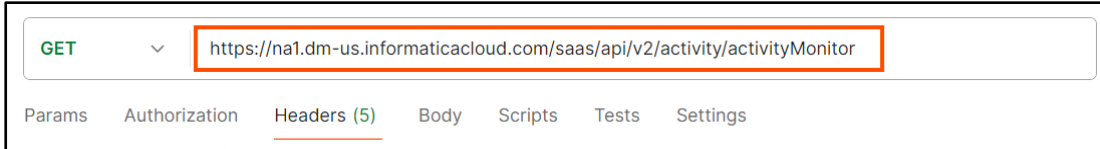
```
18 "timezone": null,
19 "serverUrl": "https://na1.dm-us.informaticacloud.com/saas",
20 "icSessionId": "jLUwDjFD5SkhhfG7BYiCw1",
21 "securityQuestion": "In what city was your first job?",
22 "securityAnswer": "*****",
23 "uuid": "bgc4GdMwK4u19Lkp0lyDQ9",
24 "forceChangePassword": false,

```

Request the information from the activity monitor

14. To add a new tab, click .
15. From the drop-down, select **GET**.
16. Enter the URL in the request URL field in the following format:

<Copied ServerUrl>/api/v2/activity/activityMonitor



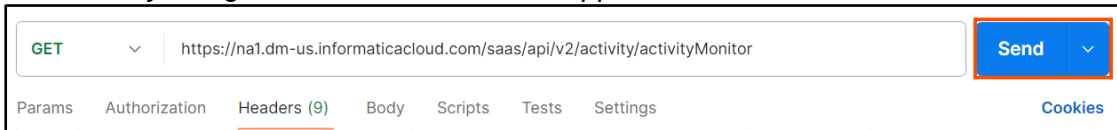
The image shows the Postman interface with a GET request to the URL `https://na1.dm-us.informaticacloud.com/saas/api/v2/activity/activityMonitor`. The 'Headers' tab is selected, showing 5 headers.

17. Select the **Headers** tab.
18. Enter Key and Value, as shown in the table below:

Key	Value
Accept	application/json
Content-Type	application/json
icSessionId	<enter the copied session ID>

Before you send the request, you must ensure that at least 1 job is running in the DI service. This API will then fetch the details of the running jobs. Let us start a job in DI and view the results in Postman.

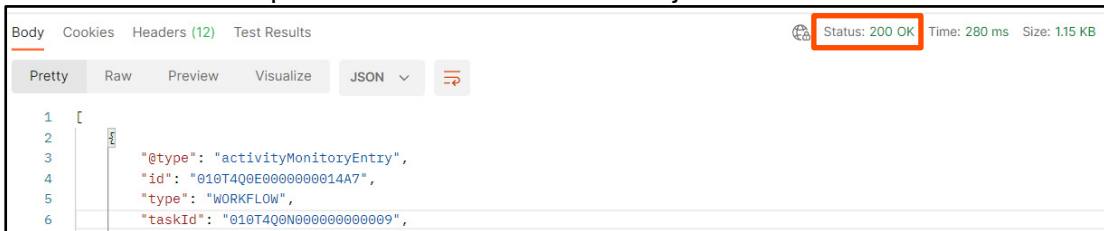
19. Navigate back to IICS > Data Integration service and open your working student folder.
20. Locate the task **tf_XX_BULK_LOAD_PROCESSING** and run it (you can right-click on the task to view the Run option).
21. Immediately navigate back to the Postman application and hit **Send**.



The image shows the Postman interface with the 'Send' button highlighted in blue. The 'Headers' tab is selected, showing 9 headers.

Note: You will view the results in the Body response tab only if you send the request as soon as you hit the run button in DI. Because, the ActivityMonitor API will only fetch the details of running jobs.

22. The status of the response is **200 OK**. Observe the job details.



The image shows the Postman interface with the 'Body' tab selected. The response status is 'Status: 200 OK' and the response body is displayed in JSON format:

```
[
  {
    "@type": "activityMonitoryEntry",
    "id": "010T4Q0E0000000014A7",
    "type": "WORKFLOW",
    "taskId": "010T4Q0N0000000000009"
  }
]
```

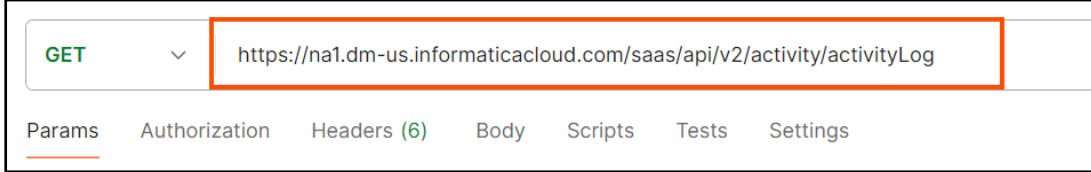
Retrieve the entries from the activityLog REST API resource.

23. To add a new tab, click .

24. From the drop-down, select **GET**.

25. Enter the URL in the request URL field in the following format:

<Copied ServerUrl>/api/v2/activity/activityLog



GET ▼ https://na1.dm-us.informaticacloud.com/saas/api/v2/activity/activityLog

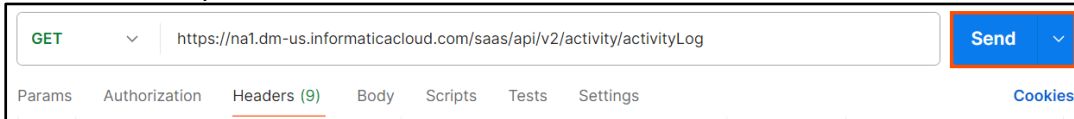
Params Authorization Headers (6) Body Scripts Tests Settings

26. Select the **Headers** tab.

27. Enter Key and Value, as shown in the table below:

Key	Value
Accept	application/json
Content-Type	application/json
icSessionId	<enter the copied session ID>

28. To view the response, click **Send**.



GET ▼ https://na1.dm-us.informaticacloud.com/saas/api/v2/activity/activityLog Send ▼

Params Authorization Headers (9) Body Scripts Tests Settings Cookies

29. The status of the response is **200 OK**. Observe the results.



Body Cookies (3) Headers (13) Test Results 200 OK 3.47 s 502.59 KB 🌐 🔍 ⋮

Pretty Raw Preview Visualize JSON ▼ 🔗

```

1 [
2   {
3     "@type": "activityLogEntry",
4     "id": "010T4QC1000000001PQF",
5     "type": "DSS",
6     "objectId": "010T4Q0I00000000005M",
7     "objectName": "syn_XX_Employee",

```

This concludes the lab.